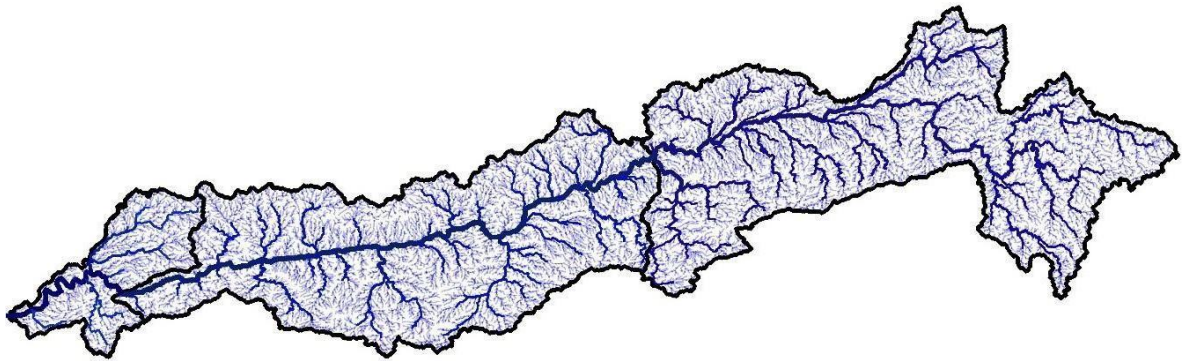




National River Conservation Directorate
Ministry of Jal Shakti,
Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Pollution Load in Narmada River Basin



September 2025



© cNarmada, cGanga and NRCD, 2025

Pollution Load in Narmada River Basin



© cNarmada, cGanga and NRCD, 2025

National River Conservation Directorate (NRCDD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and the Ministry of Jal Shakti, provides financial assistance to the State Government for the conservation of rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of rivers in identified polluted river stretches based on proposals received from the State Governments/ local bodies.

www.nrcd.nic.in

Centres for Narmada River Basin Management Studies (cNarmada)

The Centres for Narmada River Basin Management Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the centre serves as a knowledge wing of the National River Conservation Directorate (NRCDD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement.

www.cnarmada.org

Centre for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in river and water science. The Centre is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as a think-tank in the implementation and dynamic evolution of the Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition to this, it is also responsible for introducing new technologies, innovations, and solutions into India.

www.cganga.org

Acknowledgment

This report is a comprehensive outcome of the project jointly executed by IIT Gandhinagar (Lead Institute) and IIT Indore (Fellow Institute) under the supervision of cGanga at IIT Kanpur. It was submitted to the National River Conservation Directorate (NRCDD) in 2024. We gratefully acknowledge the individuals who provided information and photographs for this report.

Disclaimer

This report is a preliminary version prepared as part of the ongoing Condition Assessment and Management Plan (CAMP) project. The analyses, interpretations and data presented in the report are subject to further validation and revision. Certain datasets or assessments may contain provisional or incomplete information, which will be updated and refined in the final version of the report after comprehensive review and verification.

Team Members

Dr. Pranab Kumar Mohapatra, cNarmada, IIT Gandhinagar
Dr. Ashootosh Mandpe, cNarmada, IIT Indore
Dr. Divyanshu Kumar Dixit, cNarmada, IIT Indore
Adarsh Singh, cNarmada, IIT Indore
Shashank Agrawal, cNarmada, IIT Indore
Shreya Dixit, cNarmada, IIT Indore
Ashok Kumar Parmar, cNarmada, IIT Gandhinagar
Rajesh Kumar, cNarmada, IIT Gandhinagar
Priyal Jasmani, cNarmada, IIT Gandhinagar
Dr. Vinod Tare, cGanga, IIT Kanpur

PREFACE

This report provides an overview of pollution sources, water quality, and waste management systems in the Narmada River Basin, predominantly spanning Madhya Pradesh and supporting inhabitants. It presents detailed assessments of domestic sewage, industrial effluents, agricultural runoff, and solid and hazardous waste management, drawing from national and state monitoring data. The analysis highlights generally high-water quality in much of the basin, with localized pollution challenges in urban and industrial hubs, where gaps in sewage treatment and effluent management remain critical.

The report compiles and organizes previously scattered information from government sources, official reports, and research studies, while acknowledging the challenges of incomplete or uneven data coverage. Particular attention is given to the uneven development of sewage treatment plants, the limited reach of scientific solid waste disposal in smaller towns, and the underdeveloped state of e-waste management, in contrast with successful initiatives such as biomining, composting, and waste-to-energy plants.

We extend our sincere gratitude to the government agencies, research institutions, project staff, and individuals whose contributions of data and insights made this assessment possible. It is our hope that this report inspires informed dialogue and coordinated action, contributing to the restoration and long-term sustainability of the Narmada River Basin for both ecological health and socio-economic development.

**Centre for Narmada River Basin
Management and Studies (cNarmada)
IIT Gandhinagar, IIT Indore**

Table of Contents

1 Introduction.....	1
1.1Background	1
1.2 Methodology and Data Source.....	2
2. Water Quality Status	3
2.1 Monitoring Stations and Sampling Locations	3
2.1.1 National Water Quality Monitoring Stations	3
2.1.2 MPPCB Water Quality Monitoring Stations.....	5
2.1.3 GPCB Water Quality Monitoring Stations	6
2.2 Key Water Quality Parameters	10
2.2.1. Station: Panetha.....	11
2.2.2. Station: Garudeshwar	23
2.2.3. Station: Zonor.....	36
2.2.4. Station: Zadeshwar	50
3. Pollution Sources.....	65
3.1 Domestic Wastewater Load	65
3.1.1 Status of Water Supply Connections	65
3.1.2 Status of Sewerage Network and Household Connections.....	67
3.1.3 Status of Domestic Sewage Load	72
3.2 Industrial Wastewater Load.....	77
3.2.1 Major Industrial Clusters in the Narmada River Basin	77
3.2.2 Status of Industrial Sewage Generation	88
3.3 Agricultural Runoff and Water Quality	91
3.3.1 Fertilizer and Pesticide Use Patterns.....	91
3.3.2 Seasonal Runoff Contribution	94
3.3.3 Impact on Water Quality and Environmental Risk	95

3.4 Sewage and Other Pollution Sources in the Narmada River Basin, Gujarat	97
3.4.1 Untreated Sewage Discharges.....	97
3.4.2 Non-Point & Industrial Pollution Sources	99
4. Solid Waste Management	100
4.1 Quantity of Solid Waste Generation (Urban and Rural)	100
4.1.1 Solid Waste Generation and Processing Status in Narmada River Basin Districts (Nov 2022 vs. Jan 2024)	107
4.2 Composition of Solid Waste.....	114
4.3 Collection, Transportation, and Disposal Facilities.....	121
4.4 Status of Dumping Sites and Legacy Waste	127
4.4.1 Status Legacy Waste in Narmada Basin ULBs	127
4.4.2 Status of Dumping Sites in Narmada Basin ULBs	132
4.5 Waste to Energy	138
4.5.1 Wet Waste Processing Facilities in Urban Local Bodies (ULBs) under Narmada River Basin.....	138
4.5.2 Dry Waste Processing Facilities in ULBs under the Narmada River Basin .	142
4.5.3 Waste-to-Energy Facilities in ULBs of the Narmada River Basin	146
4.5.4 Status of Waste-to-Energy and Composting Facilities in Narmada River Basin Districts	149
4.5.6 Successful Models of Solid Waste Management (Case Studies).....	155
4.5.6.1 Solid Waste Management Initiatives in Narmada Basin Cities	155
4.5.6.2 Indore - A Model for Municipal Solid Waste Management	156
4.5.6.3 Dumpsite Remediation of Legacy Waste – Bhanpur Khanti (Bhopal)	161
4.5.6.4 Best Practice Integrated Solid Waste Management Project at Jabalpur ..	162
4.5.6.5 Best Practice Integrated Solid Waste Management Project, Kanti Cluster	164

4.5.6.6 Best Practice - Integrated Solid Waste Management Project – Sagar Cluster	165
4.5.6.7 Surat – A Model for Integrated Waste Management	166
4.5.6.8 Vadodara – Integration of Composting, Biomethanation, and RDF	167
4.5.8.9 Panchmahal (Godhra) – Decentralized and Market Waste Composting ..	168
5. Wastewater Treatment Facilities	170
5.1 Sewage Treatment Plants (STPs)	170
5.1.1 Operational Sewage Treatment Plants	170
5.1.2 Treatment Technologies Adopted in STPs & Treated Effluent Quality Parameters	171
5.1.3 Under-Construction STPs	173
5.2 Common Effluent Treatment Plants (CETPs) and Final Effluent Treatment Plant (FETPs)	175
6. Solid and Hazardous Waste Disposal	178
6.1 Hazardous waste	178
6.1.1 Hazardous Waste Generation	178
6.1.2. Hazardous Waste Management	180
6.2 Biomedical waste	182
6.2.1. Biomedical Waste Generation	182
6.2.2. Biomedical Waste Management	184
6.3 E-Waste	186
6.3.1. E-Waste Generation	186
6.3.2. E-Waste Management	188
6.4 Plastic waste	190
6.4.1. Plastic Waste Generation	190
6.4.2. Plastic Waste Management	192
6.5 C&D WASTE	194

6.5.1. C&D Waste Generation	194
6.5.2. C&D Waste Management	196
7. Pollution Load Assessment.....	199
7.1 Domestic Pollution Load Status.....	199
7.2 Industrial Pollution Load Status.....	202
8.Summary And Conclusion	205
REFERENCES.....	207
ANNEXURE.....	209

List of Figure

Figure 1 Spatial Distribution of National Water Quality Monitoring Stations in the Narmada Basin, Madhya Pradesh.....	5
Figure 2 Spatial Distribution of National Water Quality Monitoring Stations in the Narmada Basin, Gujarat	5
Figure 3 Spatial Distribution of MPPCB Water Quality Monitoring Stations in the Narmada Basin, Madhya Pradesh.....	6
Figure 4 Spatial Distribution of GPCB Water Quality Monitoring Stations in the Narmada Basin, Gujarat	7
Figure 5 National Water Quality Monitoring Stations in the Narmada Basin, Gujarat.....	10
Figure 6 Use of Nitrogen in Narmada River Basin.....	92
Figure 7 Potash and Phosphate Consumption, Narmada River Basin	93
Figure 8 Legacy waste site in Waraseoni solid waste composition.....	115
Figure 9 Pithampur dumpsite Dhar Solid waste composition	116
Figure 10 Mandideep dumpsite Raisen solid waste composition.....	117
Figure 11 Thandla municipal council dumpsite Jhabua, solid waste composition.....	118
Figure 12 Typical Composition of Municipal Solid Waste in Lower Narmada Basin (Gujarat).....	121
Figure 13 Waste Collection and Segregation Efficiency (Lower Narmada Basin, Gujarat). 127	
Figure 14 Material Recovery Facilities (MRF) & Composting Facilities in ULBs	143
Figure 15 Status of Waste-to-Energy and Composting Facilities in Lower Narmada Basin (Jan 2024).....	154
Figure 16 Six bin concepts for waste generation and segregation in Indore	158
Figure 17 D2D collection fleet expansion and technological advancement.....	160
Figure 18 Indore bioremediation transformation into city forest (2015 to 2020).....	161
Figure 19 Indore bioremediation transformation into city forest (2022 to present)	161
Figure 20 Transformation of Bhanpur Khanti Dumpsite.....	162
Figure 21 Integrated Solid Waste Management Project at Jabalpur.....	163
Figure 22 Waste Processing Plant & Scientific Landfill of Integrated Solid Waste Management, Katni cluster	164
Figure 23 Waste Processing Plant & Scientific Landfill photos of Integrated Solid Waste Management, Sagar cluster.....	165
Figure 24 Successful Solid Waste Management Models in Lower Narmada Basin (Gujarat).	169

List of Table

Table 1 Status of Water Connection in Narmada River Basin Districts.....	66
Table 2 Details of Household Service Connections (Sewerage) in the Narmada Basin Cities68	
Table 3 Sewerage Connections status in Lower Narmada basin	71
Table 4 District Wise Status of Domestic Sewerage Waste Generation, Narmada Basin'	72
Table 5 Status of Sewage Generation and Treatment	76
Table 6 Major Mineral-Based Industries and Their Locations in the Narmada Basin, Madhya Pradesh.....	79
Table 7 Food Parks and Their Coverage in the Narmada Basin, Madhya Pradesh	80
Table 8 Agri Export Zones and Their Coverage in the Narmada Basin, Madhya Pradesh	81
Table 9 Central Public Sector Undertakings and Their Operational Presence in the Narmada Basin, Madhya Pradesh.....	81
Table 10 State Public Sector Undertakings and Their Operational Presence in the Narmada Basin, Madhya Pradesh.....	83
Table 11 Industrial Areas Managed by DTICs and Their Development Status in the Narmada Basin, Madhya Pradesh.....	85
Table 12 MSME Clusters and Their Coverage in the Narmada Basin, Madhya Pradesh	86
Table 13 District Wise Status of Industrial Sewerage Waste Generation, Narmada Basin.....	88
Table 14 Impact of Agrochemical Use on Water Quality	96
Table 15 Solid Waste Generation in the Narmada River Basin.....	100
Table 16 Summary of Waste Generation in Lower Narmada Basin (Gujarat).....	106
Table 17 Status of solid waste management district-wise	108
Table 18 Solid Waste Generation Comparison (Nov 2022 vs Jan 2024) for Lower Narmada Basin Districts, Gujarat.....	111
Table 19 Indicative composition of municipal solid waste in Lower Narmada Basin districts, Gujarat.....	118
Table 20 Door-to-door collection and segregation status in Lower Narmada Basin districts.....	123
Table 21 Status of waste processing and disposal facilities in Lower Narmada Basin districts.....	125
Table 22 Legacy waste in Narmada basin ULBs.....	128
Table 23 District-wise status of legacy waste remediation in Lower Narmada Basin ULBs (as of Jan 2024).....	131
Table 24 Status of Dhar Road side dumpsite.....	134
Table 25 Status of dumping sites in Narmada basin ULBs	134
Table 26 Status of existing dumping sites in Lower Narmada Basin districts (Jan 2024)	136
Table 27 List of waste-to-energy facilities under the Narmada River basin ULBs.....	147
Table 28 Status of waste-to-energy facilities under the Narmada River basin ULBs	150
Table 29 Status of composting, biomethanation, and waste-to-energy facilities in Lower Narmada Basin ULBs (as of Jan 2024).....	153
Table 30 Solid waste management initiatives in Narmada basin cities	156
Table 31 Waste generation and processing mechanisms in Indore	157
Table 32 Operational Sewage Treatment Plants in the Narmada River Basin	170
Table 33 Wastewater treatment technologies adopted in STPs	172
Table 34 Under construction Sewage Treatment Plants in the Narmada River Basin	174
Table 35 Status of CETPs in the Narmada River Basin	175
Table 36 Industrial clusters and CETP in lower Narmada basin gujarat	177
Table 37 Hazardous Waste Generation in the Narmada River Basin	178
Table 38 Hazardous waste generated in lower basin	180

Table 39 Biomedical Waste Generation in the Narmada River Basin.....	182
Table 40 Biomedical Waste generated in lower basin.....	184
Table 41 Electronic Waste Generation in the Narmada River Basin.....	187
Table 42 Plastic Waste Generation in the Narmada River Basin	191
Table 43 Plastic Waste in each Nagarpalika in lower basin6.4.2. Plastic Waste Management	192
Table 44 C&D Waste Generation in the Upper and middle Narmada River Basin	195
Table 45 C&D Waste generated in each Nagarpalika in lower basin.....	196
Table 46 Domestic Sewage Generation and Treatment in Lower Narmada Basin	201
Table 47 Industrial Wastewater Load in Lower Narmada Basin.....	203

Annexure

Table A - 1 List of National Water Quality Monitoring Stations in Narmada Basin, Madhya Pradesh.....	209
Table A - 2 MPPCB Water Quality Monitoring Stations in Narmada River Basin, Madhya Pradesh (2019–2025)	217
Table A - 3 List of Large-Scale Industries in Narmada Basin, Madhya Pradesh	222
Table A - 4 List of Medium Scale Industries in Narmada Basin, Madhya Pradesh	224
Table A - 5 Solid, Plastic and C&D waste generation in Narmada river Basin, Madhya Pradesh.....	228
Table A - 6 District-wise Status of Solid Waste Collection and Segregation Facilities in Narmada Basin, Madhya Pradesh	237
Table A - 7 District-wise Status of Solid Waste Treatment and Disposal Facilities in Narmada Basin, Madhya Pradesh.....	246
Table A - 8 Comprehensive overview of legacy waste dumpsites and projected timelines for complete processing in Narmada basin ULBs	261
Table A - 9 Wet waste processing facilities under the Narmada River basin ULBs	263
Table A - 10 List of dry waste processing facilities under the Narmada River basin ULBs.....	274
Table A - 11 District-wise Status of Hazardous Waste Management in Narmada Basin, Madhya Pradesh.....	294
Table A - 12 District-wise Status of Biomedical Waste Management in Narmada Basin, Madhya Pradesh.....	300
Table A - 13 District-wise Status of E-Waste Management in Narmada Basin, Madhya Pradesh.....	306
Table A - 14 District-wise Status of Plastic Waste Management in Narmada Basin, Madhya Pradesh.....	313
Table A - 15 District-wise Status of C&D Waste Management in Narmada Basin, Madhya Pradesh.....	326
Table A - 16 Operational STPs details in Narmada Basin, Madhya Pradesh.....	340
Table A - 17 Under Construction STPs details in Narmada Basin, Madhya Pradesh	344
Table A - 18 District-wise Status of Domestic Sewage Status in Narmada Basin, Madhya Pradesh.....	347
Table A - 19 District-wise Status of Industrial Wastewater Status in Narmada Basin, Madhya Pradesh.....	350

List of Graph

Graph 1 Monthly variation of Turbidity at Panetha Station (2015-2025)	12
Graph 2 Monthly variation of Total Hardness as CaCo ₃ at Panetha Station (2015-2025).....	13
Graph 3 Monthly variation of Total Kjeldahl Nitrogen at Panetha Station (2015-2025)	13
Graph 4 Monthly variation of Total Dissolved Solids at Panetha Station (2015-2025)	13
Graph 5 Monthly variation of Temperature at Panetha Station (2015-2025)	14
Graph 6 Monthly variation of Suspended Solids at Panetha Station (2015-2025)	14
Graph 7 Monthly variation of Sulphate at Panetha Station (2015-2025)	14
Graph 8 Monthly variation of Sodium at Panetha Station (2015-2025)	15
Graph 9 Monthly variation of Potassium at Panetha Station (2015-2025)	15
Graph 10 Monthly variation of Phosphate at Panetha Station (2015-2025)	15
Graph 11 Monthly variation of pH at Panetha Station (2015-2025)	16
Graph 12 Monthly variation of Nitrite at Panetha Station (2016-2025)	16
Graph 13 Monthly variation of Nitrate at Panetha Station (2015-2025)	16
Graph 14 Monthly variation of Magnesium at Panetha Station (2015-2025)	17
Graph 15 Monthly variation of Fluoride at Panetha Station (2015-2025)	17
Graph 16 Monthly variation of Fixed Dissolved Solids at Panetha Station (2015-2025)	17
Graph 17 Monthly variation of Fecal Coliform at Panetha Station (2015-2025)	18
Graph 18 Monthly variation of Dissolved Oxygen at Panetha Station (2015-2025)	18
Graph 19 Monthly variation of Conductivity at Panetha Station (2015-2025)	18
Graph 20 Monthly variation of Chloride at Panetha Station (2015-2025)	19
Graph 21 Monthly variation of Chemical Oxygen Demand at Panetha Station (2015-2025) ..	19
Graph 22 Monthly variation of Boron at Panetha Station (2016-2025)	19
Graph 23 Monthly variation of B.O.D (3 Days 27°C at Panetha Station (2015-2025)	20
Graph 24	20
Graph 25 Monthly variation of Turbidity at Garudeshwar Station (2015-2025)	24
Graph 26 Monthly variation of Total Kjeldahl Nitrogen at Garudeshwar Station (2015-2025)	24
Graph 27 Monthly variation of Total Hardness as CaCo ₃ at Garudeshwar Station (2015- 2025)	25
Graph 28 Monthly variation of Total Fix Solid at Garudeshwar Station (2015-2025)	25
Graph 29 Monthly variation of Total Dissolved Solids at Garudeshwar Station (2015-2025)	25
Graph 30 Monthly variation of Temperature at Garudeshwar Station (2015-2025)	26
Graph 31 Monthly variation of Suspended Solids at Garudeshwar Station (2015-2025)	26
Graph 32 Monthly variation of Sulphate at Garudeshwar Station (2015-2025)	26
Graph 33 Monthly variation of Sodium at Garudeshwar Station (2015-2025)	27
Graph 34 Monthly variation of Potassium at Garudeshwar Station (2015-2025)	27
Graph 35 Monthly variation of Phosphate at Garudeshwar Station (2015-2025)	27
Graph 36 Monthly variation of pH at Garudeshwar Station (2015-2025)	28
Graph 37 Monthly variation of Nitrite at Garudeshwar Station (2015-2025)	28
Graph 38 Monthly variation of Nitrate at Garudeshwar Station (2015-2025)	28
Graph 39 Monthly variation of Nickel at Garudeshwar Station (2015-2025)	29
Graph 40 Monthly variation of Magnesium at Garudeshwar Station (2015-2025)	29
Graph 41 Monthly variation of Fluoride at Garudeshwar Station (2015-2025)	29
Graph 42 Monthly variation of Fixed Dissolved Solids at Garudeshwar Station (2015-2025)	30
Graph 43 Monthly variation of Fecal Coliform at Garudeshwar Station (2015-2025)	30
Graph 44 Monthly variation of Dissolved Oxygen at Garudeshwar Station (2015-2025)	30
Graph 45 Monthly variation of Conductivity at Garudeshwar Station (2015-2025)	31

Graph 46 Monthly variation of Chloride at Garudeshwar Station (2015-2025).....	31
Graph 47 Monthly variation of C.O.D. at Garudeshwar Station (2015-2025)	31
Graph 48 Monthly variation of Calcium at Garudeshwar Station (2015-2025)	32
Graph 49 Monthly variation of Boron at Garudeshwar Station (2016-2025).....	32
Graph 50 Monthly variation of B.O.D (3 Days 27oC) at Garudeshwar Station (2015-2025).32	
Graph 51 Monthly variation of Alkalinity as CaCo3 at Zantor Station (2015-2025).....	37
Graph 52 Monthly variation of B.O.D. (3 Days 27 °C at Zantor Station (2015-2025)	38
Graph 53 Monthly variation of Boron at Zantor Station (2015-2025)	38
Graph 54 Monthly variation of Calcium at Zantor Station (2015-2025).....	38
Graph 55 Monthly variation of Chemical Oxygen Demand at Zantor Station (2015-2025)....	39
Graph 56 Monthly variation of Chloride at Zantor Station (2015-2025)	39
Graph 57 Monthly variation of Dissolved Oxygen at Zantor Station (2015-2025).....	39
Graph 58 Monthly variation of Fecal Coliform at Zantor Station (2015-2025)	40
Graph 59 Monthly variation of Fluoride at Zantor Station (2015-2025).....	40
Graph 60 Monthly variation of Fixed Dissolved Solids at Zantor Station (2015-2025)	40
Graph 61 Monthly variation of Magnesium at Zantor Station (2015-2025).....	41
Graph 62 Monthly variation of Nitrite at Zantor Station (2015-2025)	41
Graph 63 Monthly variation of Nitrate at Zantor Station (2015-2025)	41
Graph 64 Monthly variation of Phosphate at Zantor Station (2015-2025)	42
Graph 65 Monthly variation of Potassium at Zantor Station (2015-2025)	42
Graph 66 Monthly variation of pH at Zantor Station (2015-2025).....	42
Graph 67 Monthly variation of Sodium at Zantor Station (2015-2025).....	43
Graph 68 Monthly variation of Sulphate at Zantor Station (2015-2025)	43
Graph 69 Monthly variation of Suspended Solids at Zantor Station (2015-2025)	43
Graph 70 Monthly variation of Temperature at Zantor Station (2015-2025).....	44
Graph 71 Monthly variation of Total Dissolved Solids at Zantor Station (2015-2025)	44
Graph 72 Monthly variation of Total Coliform at Zantor Station (2015-2025)	44
Graph 73 Monthly variation of Total Fix Solid at Zantor Station (2015-2025)	45
Graph 74 Monthly variation of Total Kjeldahl Nitrogen at Zantor Station (2015-2025).....	45
Graph 75 Monthly variation of Total Hardness as CaCo3 at Zantor Station (2015-2025).....	45
Graph 76 Monthly variation of Turbidity at Zantor Station (2015-2025)	46
Graph 77 Monthly variation of Conductivity at Zantor Station (2015-2025).....	46
Graph 78 Monthly variation of Ammonical Nitrogen at Zadeshwar Station (2015-2025)	51
Graph 79 Monthly variation of B.O.D. (3 days 27 °C at Zadeshwar Station (2015-2025)	52
Graph 80 Monthly variation of Boron at Zadeshwar Station (2015-2025)	52
Graph 81 Monthly variation of Calcium at Zadeshwar Station (2015-2025).....	52
Graph 82 Monthly variation of Chloride at Zadeshwar Station (2015-2025)	53
Graph 83 Monthly variation of Chemical Oxygen Demand at Zadeshwar Station (2015-2025)	53
Graph 84 Monthly variation of Conductivity at Zadeshwar Station (2015-2025).....	53
Graph 85 Monthly variation of Dissolved Oxygen at Zadeshwar Station (2015-2025).....	54
Graph 86 Monthly variation of Fecal Coliform at Zadeshwar Station (2015-2025)	54
Graph 87 Monthly variation of Fluoride at Zadeshwar Station (2015-2025).....	54
Graph 88 Monthly variation of Magnesium at Zadeshwar Station (2015-2025).....	55
Graph 89 Monthly variation of Nitrate at Zadeshwar Station (2015-2025)	55
Graph 90 Monthly variation of Nitrite at Zadeshwar Station (2015-2025).....	55
Graph 91 Monthly variation of pH at Zadeshwar Station (2015-2025).....	56
Graph 92 Monthly variation of Phosphate at Zadeshwar Station (2015-2025)	56
Graph 93 Monthly variation of Potassium at Zadeshwar Station (2015-2025).....	56
Graph 94 Monthly variation of Alkalinity as CaCo3 at Zadeshwar Station (2015-2025).....	57

Graph 95 Monthly variation of Sodium at Zadeshwar Station (2015-2025)	57
Graph 96 Monthly variation of Sulphate at Zadeshwar Station (2015-2025)	57
Graph 97 Monthly variation of Suspended Solids at Zadeshwar Station (2015-2025)	58
Graph 98 Monthly variation of Temperature at Zadeshwar Station (2015-2025)	58
Graph 99 Monthly variation of Total Coliform at Zadeshwar Station (2015-2025)	58
Graph 100 Monthly variation of Total Dissolved Solids at Zadeshwar Station (2015-2025)	59
Graph 101 Monthly variation of Total Fix Solid at Zadeshwar Station (2015-2025)	59
Graph 102 Monthly variation of Total Hardness at Zadeshwar Station (2015-2025)	59
Graph 103 Monthly variation of Total Kjeldahl Nitrogen at Zadeshwar Station (2015-2025)	60
Graph 104 Monthly variation of Turbidity at Zadeshwar Station (2015-2025)	60
Graph 105 Narmada Basin, ULB-wise water connections	67
Graph 106 Domestic Sewage Generation in Narmada River Basin	74
Graph 107 Distribution of Large-Scale Industries Across Districts	87
Graph 108 Distribution of Medium-Scale Industries Across Districts	87
Graph 109 Comparison of Waste Generation (Nov 22 vs Jan 24)	109
Graph 110 Comparison of Waste Processing (Nov 22 vs Jan 24)	110
Graph 111 Comparison of Processing Gap (Nov 22 vs Jan 24)	110
Graph 112 MSW Generation Comparison (Nov 2022 vs Jan 2024) — Lower Narmada Basin Districts	112
Graph 113 Share of MSW Generation by District — Jan 2024	113
Graph 114 Trend of MSW Generation (Nov 2022 → Jan 2024) — Lower Narmada Basin	113
Graph 115 Legacy Waste Remediation Progress (Lower Narmada Basin, Jan 2024).	130
Graph 116 Status of Dumpsites in Lower Narmada Basin (Gujarat, 2024)	138

1 Introduction

1.1 Background

The Narmada River is one of the major rivers in India. It flows west and supports millions of people in Madhya Pradesh, Gujarat, Maharashtra, and Chhattisgarh. The Narmada River Basin covers about 98,786 square kilometres. Around 20.4 million people live in the basin. The basin has 38 districts, more than 17,000 villages, and nearly 3 million households.

Madhya Pradesh has the largest area and population in the basin. The Upper Basin has big towns like Jabalpur, Narsimhapur, Mandla, and Hoshangabad. The Middle Basin has mostly rural and tribal areas, like Khargone, Dhar, and Barwani. The Lower Basin has towns like Bharuch, Ankleshwar, Rajpipla, and Chhota Udepur, but most people live in villages.

The river provides water for drinking, farming, industries, hydropower, fisheries, and cultural activities. It also supports forests, wildlife, and agriculture. But the river is facing more pollution now. Cities, industries, and farms release untreated sewage, industrial waste, and chemical runoff into the river. Solid waste and hazardous waste also pollute the water and land. During monsoon, water flow reduces pollution temporarily, but in dry seasons, pollution becomes stronger.

The river basin has natural boundaries. The Vindhya ranges are in the north, the Satpura ranges in the south, and the Maikala ranges in the east. The basin has three parts. The Upper Basin has hills and forests. The Middle Basin has fertile plains and farmland. The Lower Basin has flat alluvial land but also faces pressure from towns and industries. The river gets water from 41 tributaries, which adds to its flow and ecology.

The Narmada Basin has a mix of urban, industrial, and rural areas. About 8.6 million people live in the Upper Basin, 9.1 million in the Middle Basin, and 2.7 million in the Lower Basin. Farming, industries, and cities together create pollution challenges. Untreated sewage, industrial effluents, chemical runoff, and solid waste often enter the river. Hazardous waste from large industries, especially in Bharuch, adds to land and water pollution.

Although there are treatment plants for sewage, industrial effluents, and hazardous waste, their capacity is not enough. Polluted water and untreated waste often reach the river and nearby land. This has created pollution hotspots in the basin.

This report studies the pollution load in the Narmada Basin. It looks at natural and administrative boundaries, waste generation, treatment, and areas under high pressure. The report aims to show the current pollution situation and help plan better management of sewage, industrial waste, and solid waste in the future.

1.2 Methodology and Data Source

This report mainly used secondary data from government sources. Important data came from NGT reports, MPPCB reports, GPCB reports, and other official documents. These reports give information on water quality, pollution sources, and waste management in the Narmada River Basin. The data is reliable and collected using standard methods.

In addition to secondary data, the study included three group field visits. The visits were to Indore, Bhopal, Jhabua, Alirajpur, and Dewas. The team met officials from MPPCB, Narmada Valley Development Authority, local Panchayats, and municipalities. The goal was to check data, get unpublished information, and understand problems in pollution monitoring and data management.

Three visits were also made to GPCB in Gandhinagar. These visits helped to get data not available in public reports and to verify secondary data.

Using both secondary data and field visits made the study stronger. It helped to assess water quality, pollution hotspots, and solid waste management in the basin. This method gives a clear understanding of the river's condition and supports planning for better pollution management.

Data Sources and Review Steps

Data collection included gathering datasets and official records from national and state water quality monitoring stations. These included stations run by CPCB, GPCB, and MPPCB. The study also looked at government reports, policy documents, and archives from regional development authorities. These sources helped to understand how water quality and pollution levels change across different areas and over time in the basin..

Field Visits

Three major field visits were conducted to the important urban centres of Indore, Bhopal, Jhabua, Alirajpur, and Dewas. The visits facilitated direct interaction with key governmental

offices such as MPPCB regional offices, Narmada Valley Development Authority, Zila Panchayats, and Collector offices. These engagements focused on securing access to unpublished datasets, clarifying data ownership and management roles, and observing ground realities relating to pollution monitoring and waste management.

Data Acquisition Challenges

The field visits highlighted challenges in acquiring datasets directly from governmental departments, especially in Bhopal, where fragmentation of data responsibility required formal requests and follow-up communications. These insights underscored the complexities of decentralized environmental governance and the need for persistent stakeholder engagement.

In lower basin data made available with multiple field visit, telephonic contacts and email follow ups. However, some data was missing and not available for the current month and year.

Outcomes

The combined methodology of secondary data analysis and group field visits enabled ground-level validation of desk-based findings, identification of institutional barriers to pollution data transparency, and an enriched understanding of water quality and solid waste management challenges in the Narmada River Basin. This comprehensive approach facilitated a nuanced environmental assessment capable of informing future management strategies.

2. Water Quality Status

2.1 Monitoring Stations and Sampling Locations

2.1.1 National Water Quality Monitoring Stations

The **National Water Quality Monitoring Programme (NWMP)** is an initiative of the **Central Pollution Control Board (CPCB)** in collaboration with **State Pollution Control Boards (SPCBs)** and other allied agencies to monitor the quality of water resources across India systematically. The programme aims to assess the health of rivers, lakes, tanks, ponds, reservoirs, and groundwater sources, ensuring the availability of safe water and supporting sustainable water resource management.

In the **Narmada Basin**, the NWMP network is jointly implemented by the **Madhya Pradesh Pollution Control Board (MPPCB)** and the **Gujarat Pollution Control Board (GPCB)**. Within the **Madhya Pradesh region**, more than **80 monitoring stations** have been established under the NWMP framework. These stations encompass diverse water bodies, including the main stem of the Narmada River, its tributaries, lakes, ponds, tanks, reservoirs, and groundwater sources. Monitoring is conducted on a **monthly basis** for surface water and **half-yearly** for groundwater sources.

The stations are strategically distributed from the **source of the Narmada at Amarkantak** in the Maikal ranges, through key urban and rural centers such as **Mandla, Jabalpur, Hoshangabad, Nemawar, Omkareshwar, Maheshwar, and Barwani**, extending up to the **interstate boundary near Kakrana, Alirajpur**. This extensive coverage ensures that the entire upper and middle stretches of the river, as well as important reservoirs and groundwater sources, are comprehensively assessed for water quality status.

The **map of the Narmada Basin in Madhya Pradesh** is shown in **Figure 1**, highlighting the spatial distribution of monitoring stations across the upper and middle basin. The **detailed list of these stations**, including station codes, coordinates, and corresponding districts, is presented in **Annexure Table A1**.

Extending further downstream, the **Lower Narmada Basin** lies within the **Gujarat region**, covering **six districts** — **Narmada, Bharuch, Vadodara, Chhota Udepur, Panchmahal, and Surat**. Under the NWMP, **six surface water monitoring stations** have been established by **GPCB** along the river stretch and associated water bodies. These include key locations such as **Panetha, Garudeshwar, Chandod, Zadeshwar, Zanor, and Navagam**, representing critical monitoring points in the lower reach of the river where industrial, agricultural, and domestic influences are more pronounced.

Monitoring in the Gujarat stretch follows the same frequency pattern — **monthly sampling for surface water** and **biannual sampling for groundwater**. The data generated from these stations help evaluate the influence of urban settlements, industrial clusters, and agricultural runoff on the river's overall water quality before it meets the **Arabian Sea**.

The **map of the Narmada Basin in Gujarat** is shown in **Figure 2**, depicting the spatial distribution of all monitoring stations under the NWMP network. The **detailed list of Gujarat monitoring stations**, along with their coordinates and corresponding districts, is provided in **Annexure Table A1 (A)**.

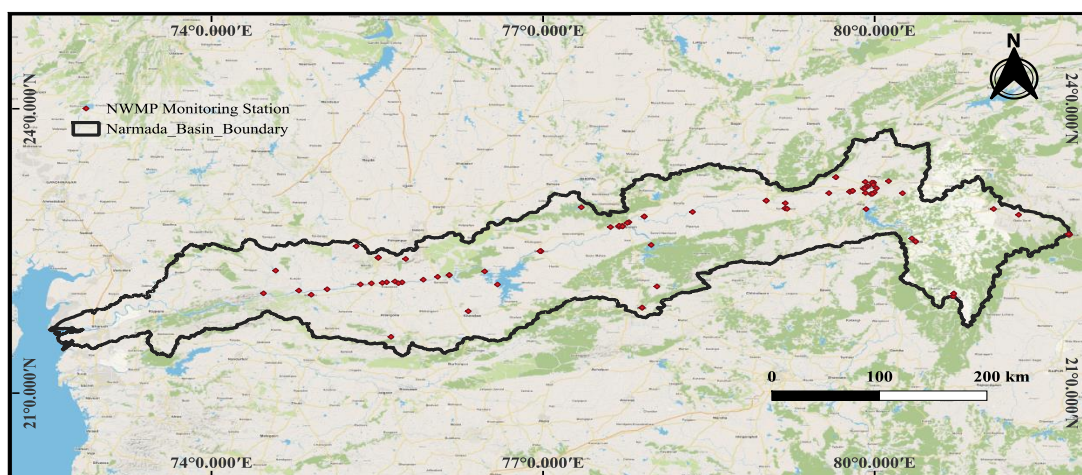


Figure 1 Spatial Distribution of National Water Quality Monitoring Stations in the Narmada Basin, Madhya Pradesh

*(Source: CPCB, List of NWMP Monitoring Network)

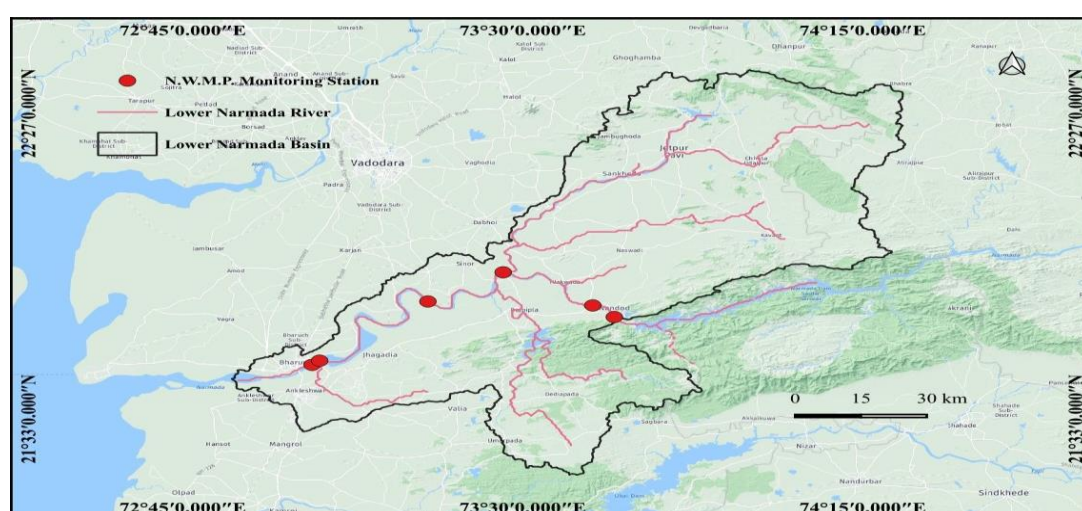


Figure 2 Spatial Distribution of National Water Quality Monitoring Stations in the Narmada Basin, Gujarat

*(Source: CPCB, List of NWMP Monitoring Network)

2.1.2 MPPCB Water Quality Monitoring Stations

The Madhya Pradesh Pollution Control Board (MPPCB) has established an extensive network of 50 water quality monitoring stations along the Narmada River to assess its ecological health and track changes over time. These stations are strategically located from the river's origin at Amarkantak in Anuppur district to its downstream stretches in Alirajpur and Barwani, ensuring coverage of all major confluences, urban centres, and industrially influenced reaches.

The monitoring programme evaluates water quality on an annual basis under the National Water Quality Monitoring Programme (NWMP). Parameters are assessed and categorized into classes A, B, C, D, and E, where Class A represents water fit for drinking after conventional treatment and disinfection, and Class B represents water suitable for outdoor bathing.

The spatial distribution shown in in **Fig. 3** of the 50 monitoring stations is depicted in the accompanying map of the Narmada Basin, which highlights the spread of monitoring efforts across districts. The dataset provides an important basis for water quality management, pollution control enforcement, and sustainable river basin planning in Madhya Pradesh.

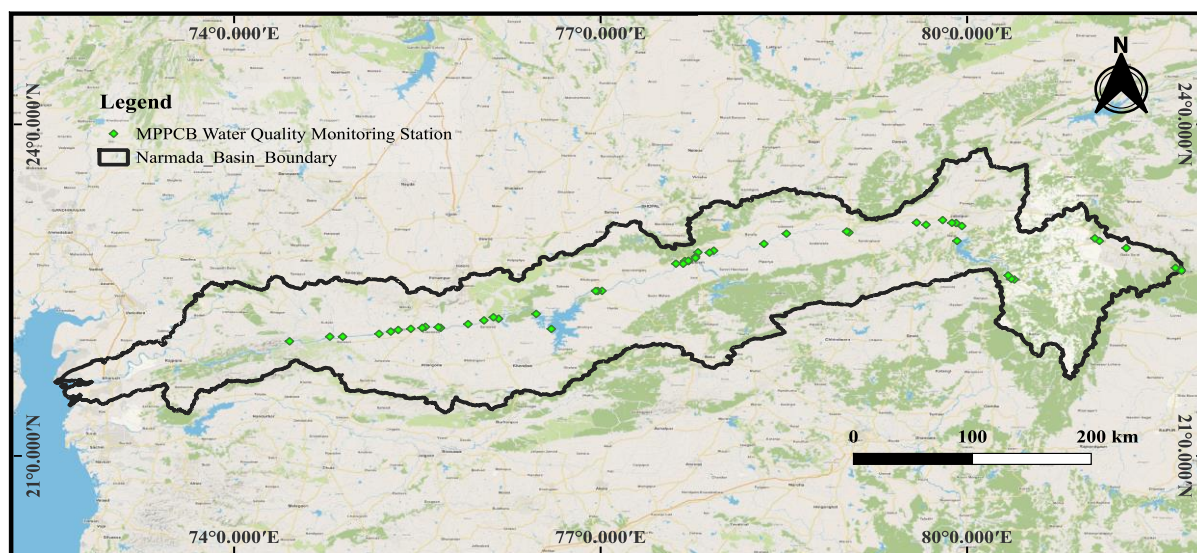


Figure 3 Spatial Distribution of MPPCB Water Quality Monitoring Stations in the Narmada Basin, Madhya Pradesh

**(Source: Locations are digitized using Google Maps after taking reference of locations from the MPPCB Narmada Water Quality Report 2024-25)*

2.1.3 GPCB Water Quality Monitoring Stations

The spatial distribution of the 6 monitoring stations, as shown in **Fig. 4**, is depicted in the accompanying map of the Narmada Basin, which highlights the spread of monitoring efforts across the lower basin.

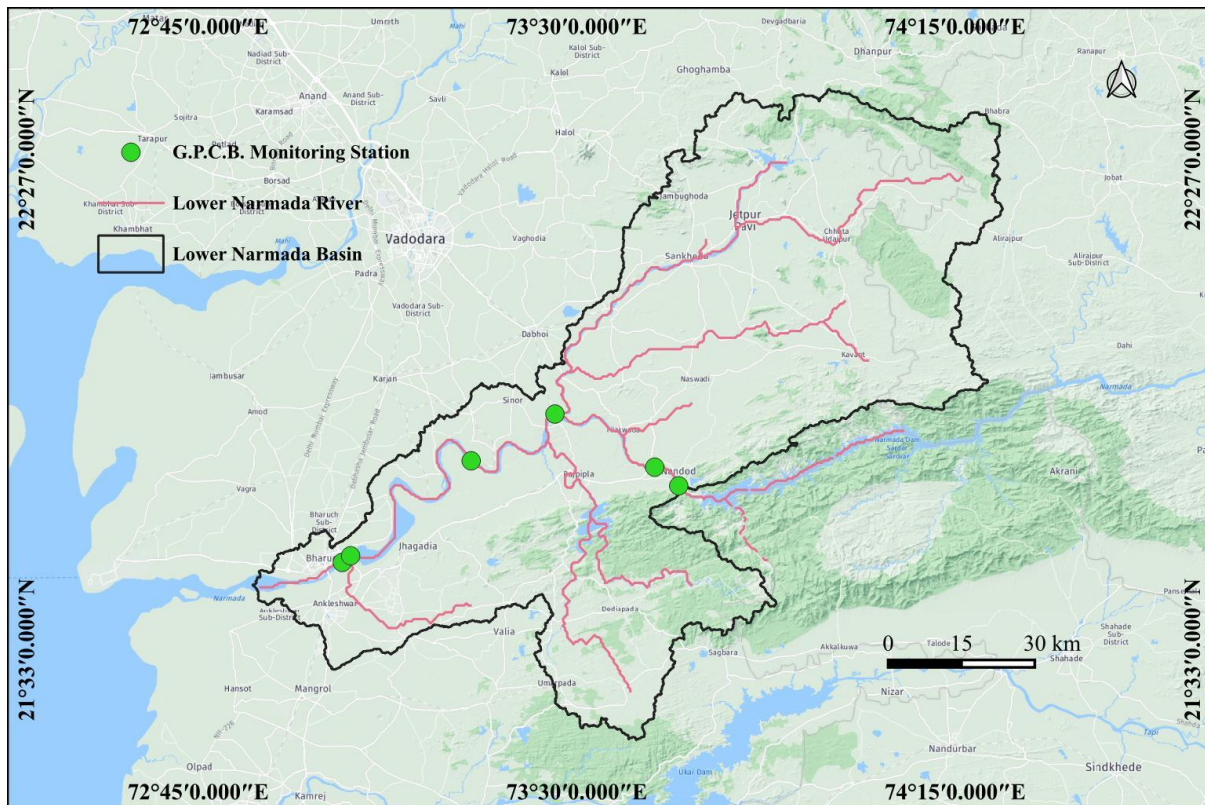


Figure 4 Spatial Distribution of GPCB Water Quality Monitoring Stations in the Narmada Basin, Gujarat
(Source: Locations are digitized using Google Maps after taking reference of locations from the GPCB Narmada Water Quality Report 2024-25)

1. Panetha

Panetha station lies in the upstream portion of the lower basin, before major urban and industrial influences accumulate. It is located in a relatively rural setting, with limited direct discharge from settlements and industries. The hydrology here is influenced by river flow from upstream catchments and seasonal rainfall. Panetha is important as a baseline station representing relatively pristine conditions in the lower basin. Its monitoring provides insight into natural water quality trends and serves as a benchmark to compare impacts at downstream stations.

2. Garudeshwar

Garudeshwar station lies in the middle-lower transitional stretch of the Narmada River. It is located downstream of Panetha, where the river begins to integrate more significant local inputs, including agricultural runoff and small-scale domestic discharge. Garudeshwar is situated near important agricultural zones, and the river here receives runoff rich in nutrients and sediments during the monsoon. This station reflects the transitional phase of water quality,

balancing upstream purity with local anthropogenic impacts. Monitoring here helps understand the influence of agricultural and rural settlements on river health.

3. Zonor

Zonor is further downstream of Garudeshwar, in a region where river flows are influenced by both upstream contributions and local inflows. Zonor Station is strategically located to capture the combined effects of tributary inflows, agricultural runoff, and minor urban discharge. The river stretch at Zonor is characterized by relatively stable flows and natural dilution capacity, which helps moderate pollution loads. This station is significant as it represents a transition point where the river quality generally remains good despite increasing anthropogenic influences.

4. Zadeshwar

Zadeshwar station lies in the lowest stretch of the Central Narmada before it enters the estuarine zone and discharges into the Gulf of Khambhat. This station is closest to urban settlements, industrial zones, and aquaculture activities along the river. Zadeshwar is heavily influenced by domestic sewage inflows, industrial effluents, agricultural runoff, and tidal mixing, making it a key location to assess the cumulative impact of human activities on river water quality. The conditions at this station reflect the stress levels faced by the river before it reaches its estuary and hence provide vital information for environmental management and planning.

5. Chandod

Chandod station is located near the confluence of the Narmada, Orsang, and other minor tributaries, making it a hydrologically significant point in the lower basin. The region holds religious and cultural importance, with regular pilgrimage activities and ritual bathing contributing to localized domestic waste inputs. It can be inferred that the river at Chandod experiences moderate anthropogenic pressure from religious tourism, local settlements, and minor agricultural runoff. Monitoring at this station would provide valuable insights into the influence of cultural and domestic activities on water quality in the lower basin.

6. Navagam

Navagam station lies near the terminal portion of the Narmada River, close to the Sardar Sarovar reservoir backwaters and the estuarine interface. This region is hydrodynamically complex, influenced by reservoir releases, tidal ingress from the Gulf of Khambhat, and saline

water intrusion during low-flow periods. Navagam is critical for understanding the mixing dynamics between freshwater and saline water and assessing the potential impacts of tidal variation, nutrient loading, and industrial activities along the downstream stretches. Inclusion of this station in regular monitoring programs would help track estuarine health and water quality transition at the river–sea interface.

2.1.3.1 Importance of Monitoring These Stations

Monitoring these six stations (**Fig. 5**) along the lower Narmada basin provides a spatial gradient of water quality from relatively clean upstream stretches to more impacted downstream locations. This allows identification of:

- **Natural baseline conditions** – Panetha
- **Transitional impacts of rural and agricultural activities** – Garudeshwar
- **Moderate anthropogenic influence with self-purification capacity** – Zonor
- **High anthropogenic pressure from urban, industrial, and estuarine influences** – Zadeshwar
- **Cultural and domestic influences from religious activities** – Chandod
- **Hydrological and tidal interactions in the estuarine transition zone** – Navagam

By analyzing physico-chemical, microbial, and heavy metal parameters at these stations, the study can assess the health of the river, detect pollution trends, and inform targeted management strategies for the lower basin stretch of the Narmada River in Gujarat.

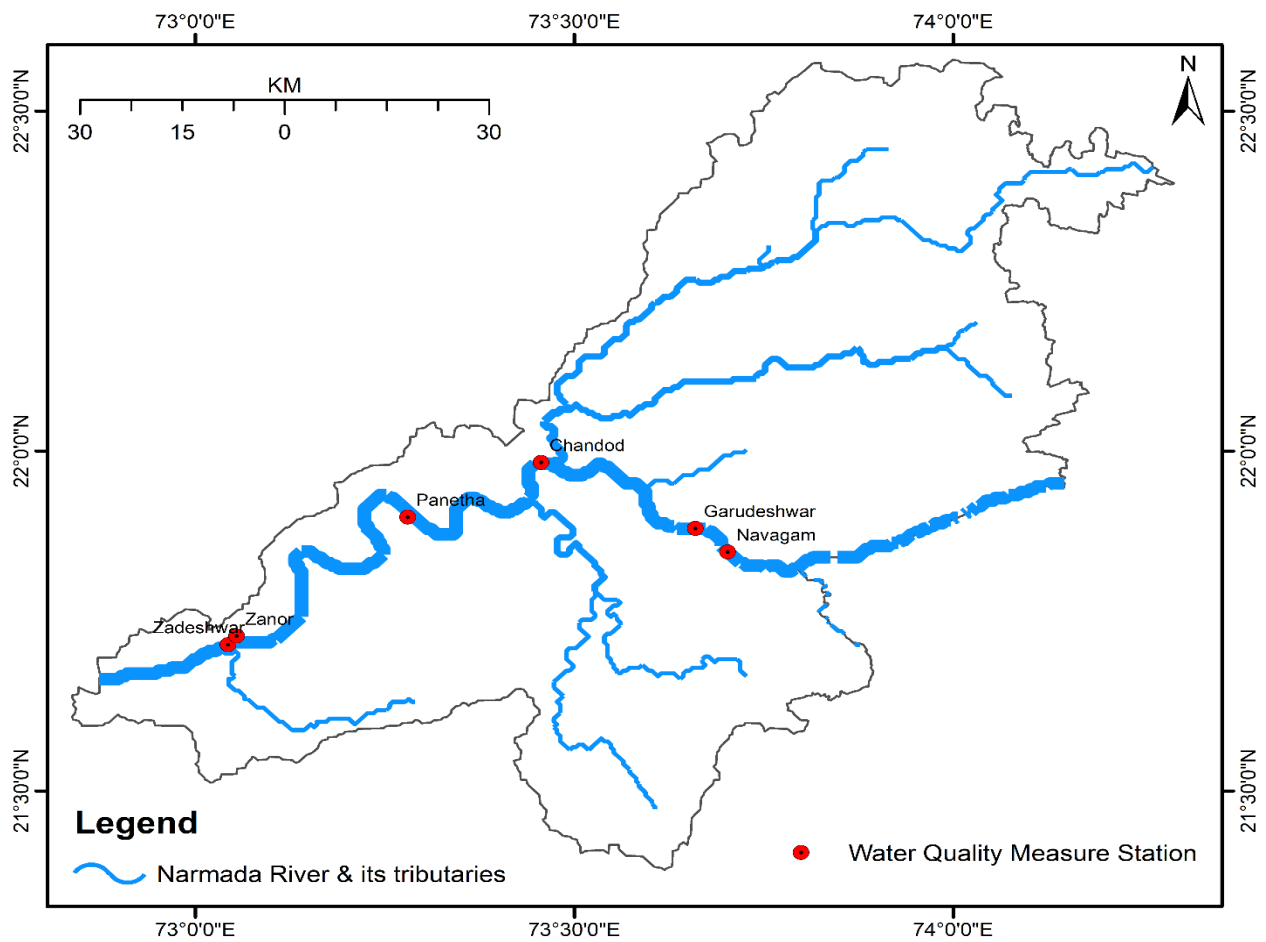


Figure 5 National Water Quality Monitoring Stations in the Narmada Basin, Gujarat
**(Source: CPCB, List of NWMP Monitoring Network)*

2.2 Key Water Quality Parameters

The Madhya Pradesh Pollution Control Board (MPPCB) and Gujarat Pollution Control Board (GPCB) monitor the water quality of the Narmada River through a set of standard physico-chemical and biological parameters measured every month. These include **pH, Alkalinity, Hardness (Calcium, Magnesium, Total), Chloride, Sulphate, Sodium, Potassium, Nitrite, Phosphate, Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Dissolved Oxygen (DO), Suspended Solids, Total Dissolved Solids, Turbidity, Conductivity, Temperature**, and microbial indicators such as **Total Coliform and Fecal Coliform**. Nutrient-related indicators like **Ammonia Nitrogen (NH₄-N) and Total Kjeldahl Nitrogen (TKN)** are also assessed to evaluate organic pollution and nutrient enrichment.

The water quality categorization of monitoring stations further corroborates these observations. Data from **2019-20 to 2024-25** shows that a majority of the **50 monitoring stations** along the Narmada River consistently fall under **Category A**, denoting high water quality suitable for drinking water source after conventional treatment and disinfection. However, some stretches,

particularly in **Sehore and Narmadapuram districts**, were temporarily classified under **Category B** during 2019-2021 and again in 2023-24, primarily due to localized pollution inputs. Importantly, by 2024-25, nearly all stations reverted to Category A, with only one site (Pushkar Dam, Amarkantak) registering a Category B classification. The detailed monitoring results are presented in **Annexure Table A2**, which provides monthly parameter values and station-wise water quality categorization for 2019–2025.

The key parameters influencing water quality in the lower basin of the Narmada River have been analyzed and presented station-wise. Each monitoring station reflects distinct local conditions and contributing factors that affect the overall quality of the river water. The station-wise assessment helps in identifying the specific pollutants and their intensity, thereby providing insight into spatial variations and aiding in the formulation of targeted management strategies for improving water quality in the lower basin region.

2.2.1. Station: Panetha

Panetha lies upstream in the lower basin and benefits from relatively lesser human impact compared to downstream stations. **Graph 1 to Graph 24** show the water quality parameters with time from 2015 to 2025 of Panetha Station.

Organic Pollution (BOD, COD): Both parameters show a declining trend over the monitoring period, suggesting effective natural self-purification and reduced organic discharge in the catchment area. BOD levels consistently remain low, indicating minimal sewage or industrial organic load.

Dissolved Oxygen (DO): DO levels remain high and stable throughout the year (above 6 mg/L), ensuring good ecological conditions for aquatic life. Slight seasonal variations are observed during monsoon due to higher organic load and dilution effects.

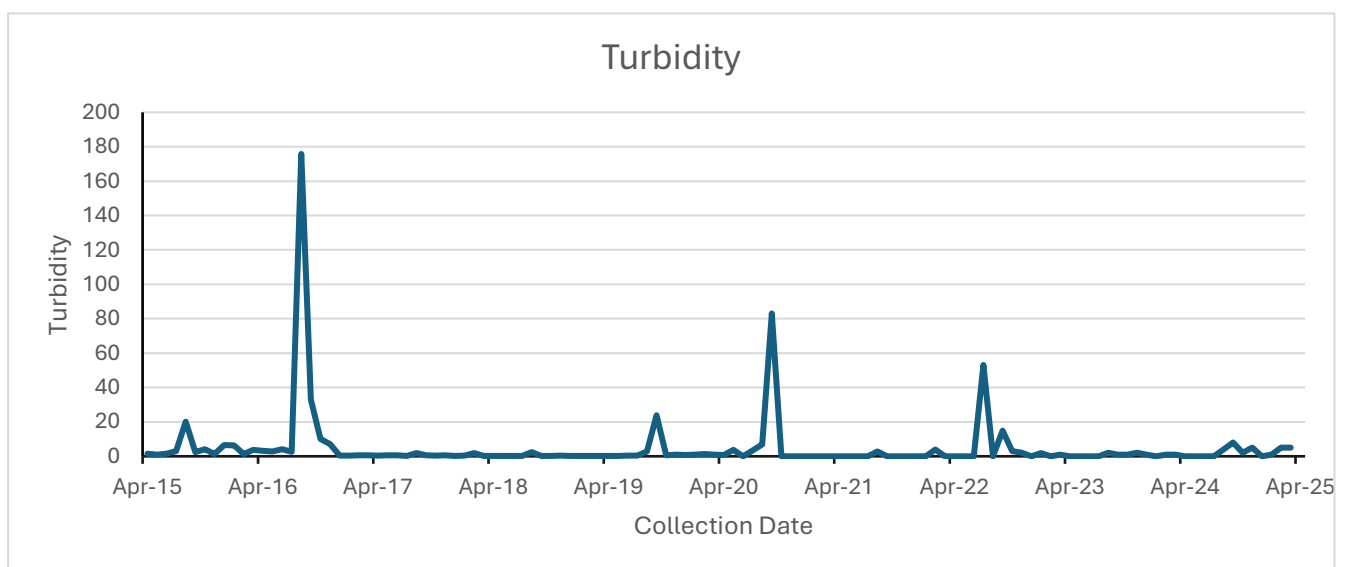
Nutrients (Nitrate, Phosphate, TKN): Nitrate and TKN remain moderate and stable. Phosphate levels show a seasonal rise during the monsoon, attributed to agricultural runoff, but remain within permissible limits.

Microbial Load: Moderate coliform and E. coli presence during monsoon seasons suggest some influence of surface runoff containing domestic wastewater. However, microbial loads are generally within manageable levels for potential water use.

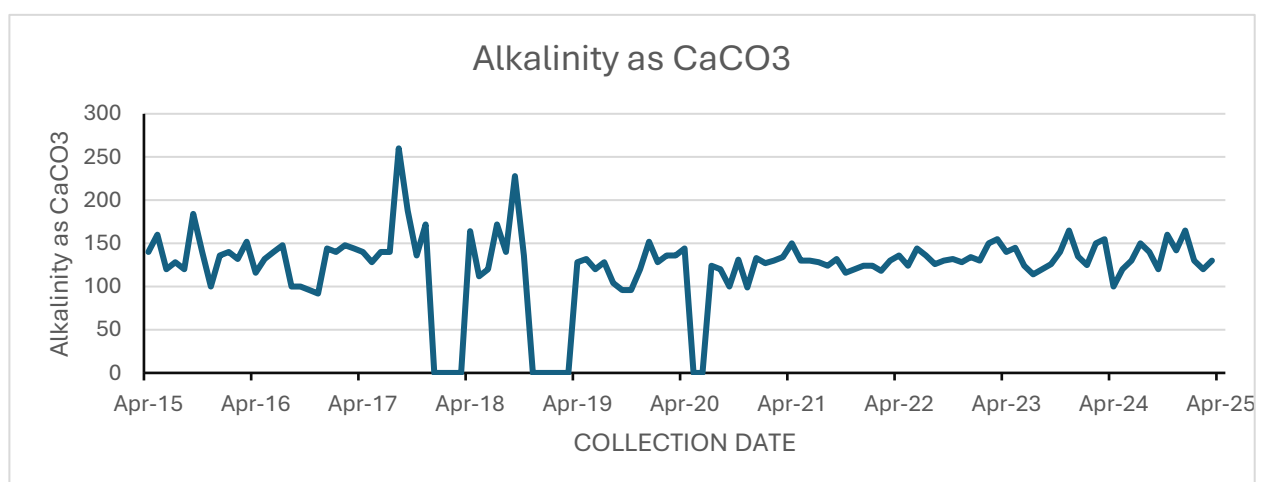
Metals: Iron occasionally exceeds the desirable limit during high-flow monsoon periods, likely due to natural leaching from sediments. Other metals such as manganese, lead, and cadmium remain well within limits.

Salinity & Hardness: Parameters such as chloride, TDS, and hardness are stable and acceptable, with no indication of salinity stress.

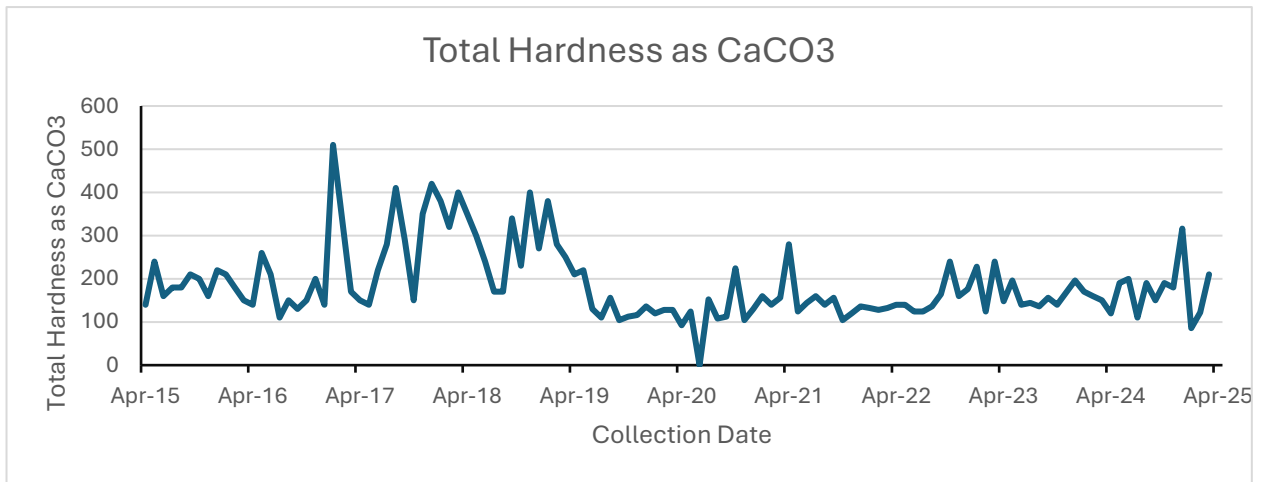
Overall Condition: Panetha reflects good water quality with low organic pollution and high DO. Seasonal nutrient and microbial variations warrant routine monitoring.



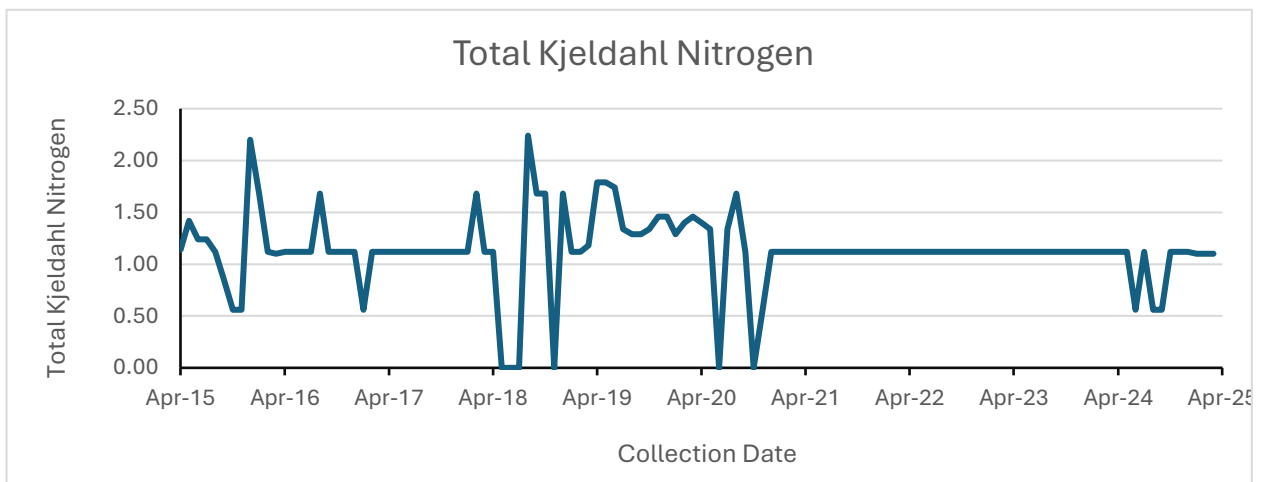
Graph 1 Monthly variation of Turbidity at Panetha Station (2015-2025)



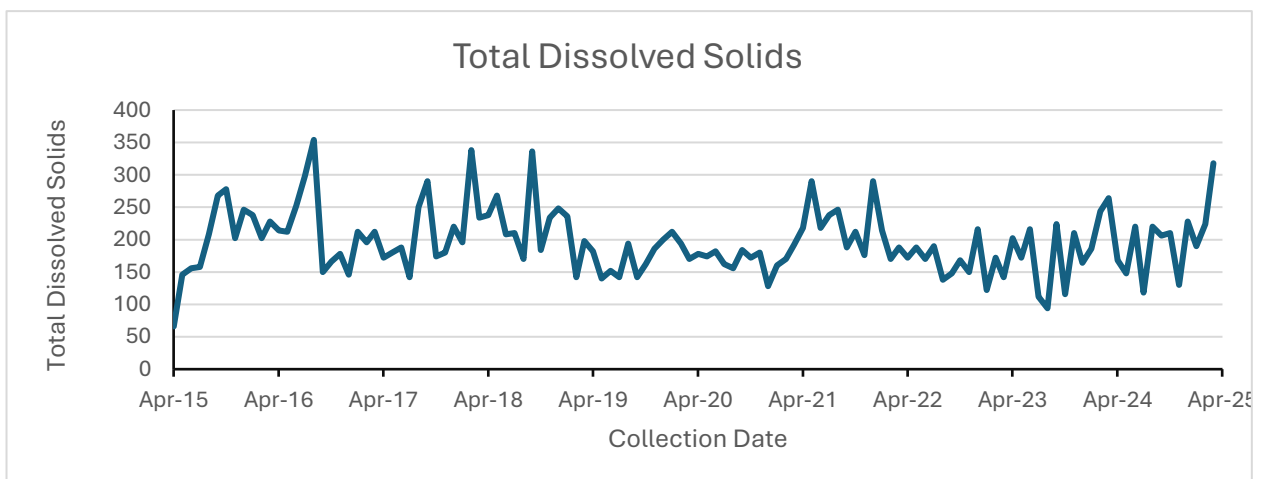
Graph 2 Monthly variation of Alkalinity as Caco3at Panetha Station (2015-2025)



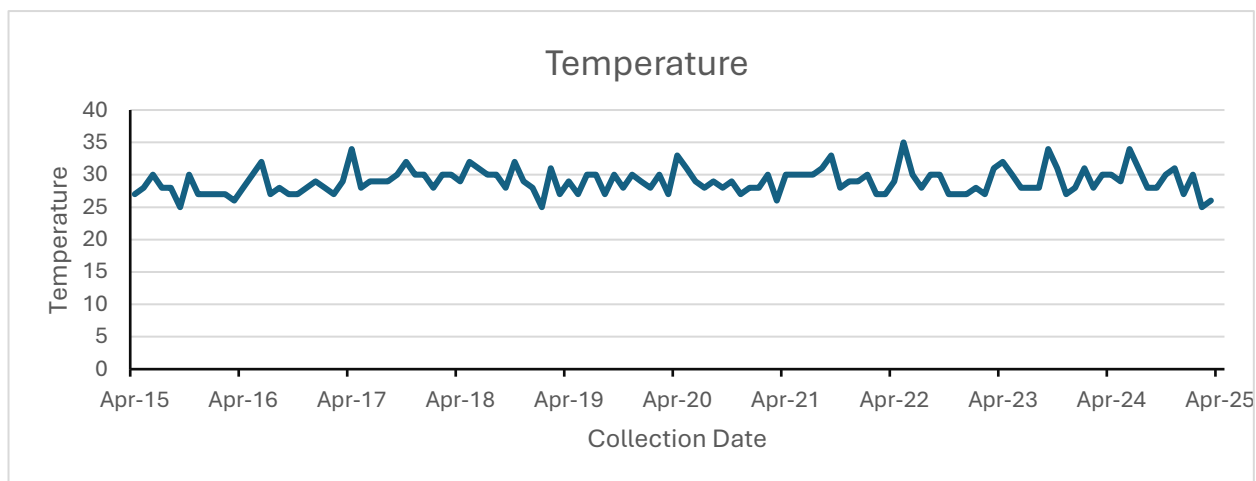
Graph 2 Monthly variation of Total Hardness as CaCo3 at Panetha Station (2015-2025)



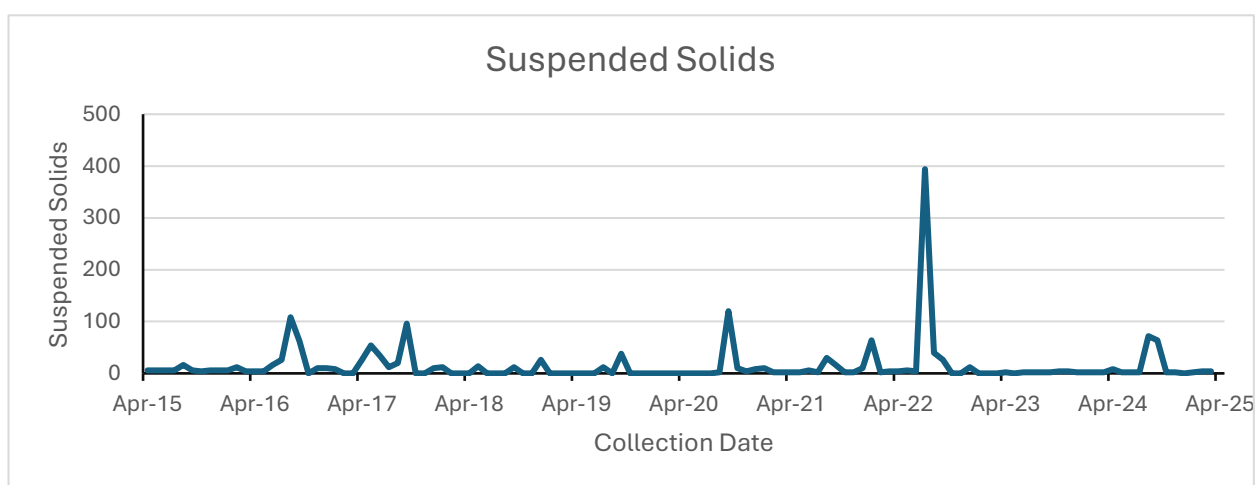
Graph 3 Monthly variation of Total Kjeldahl Nitrogen at Panetha Station (2015-2025)



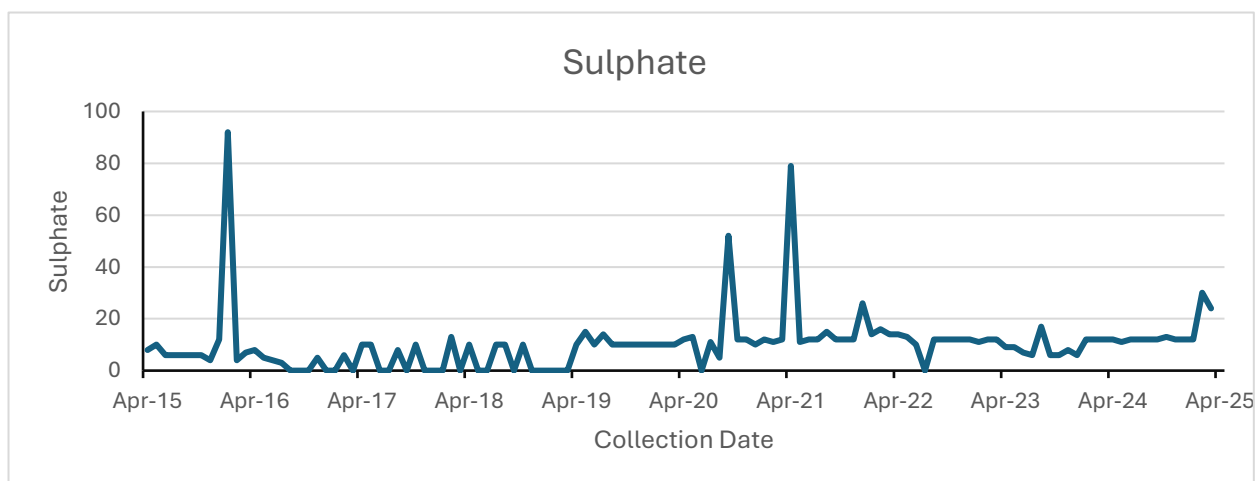
Graph 4 Monthly variation of Total Dissolved Solids at Panetha Station (2015-2025)



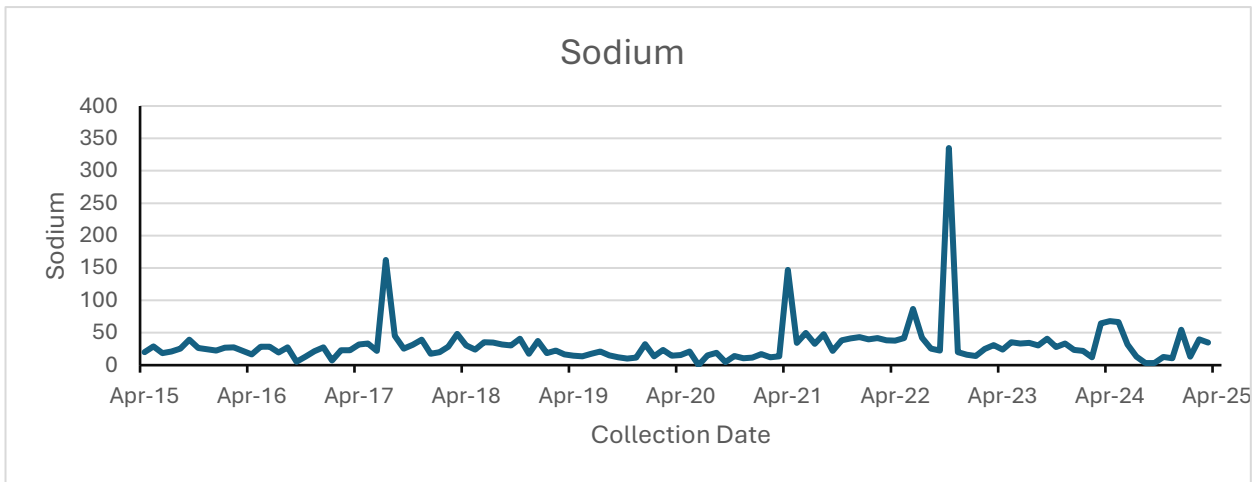
Graph 5 Monthly variation of Temperature at Panetha Station (2015-2025)



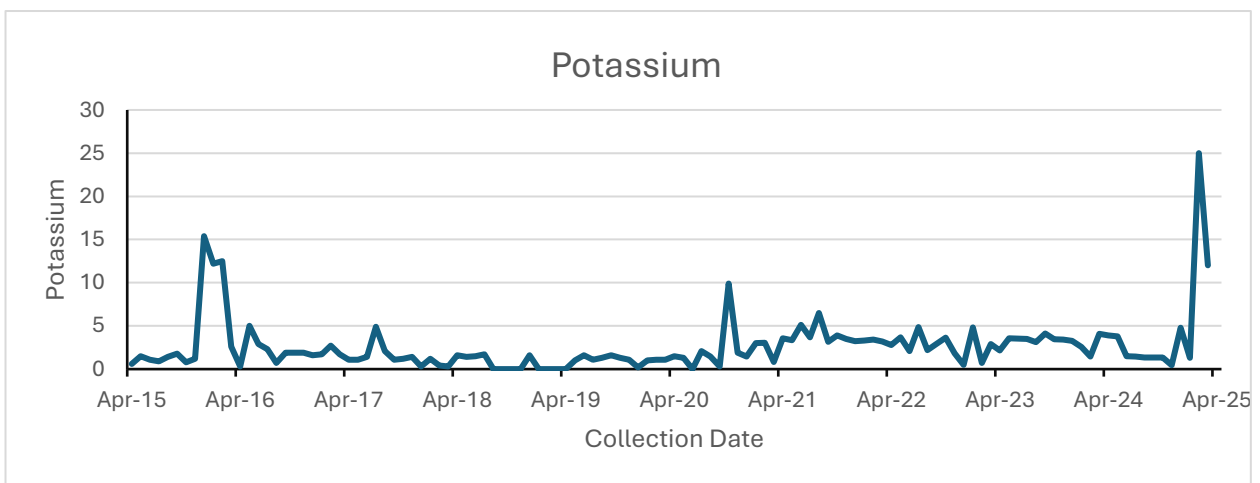
Graph 6 Monthly variation of Suspended Solids at Panetha Station (2015-2025)



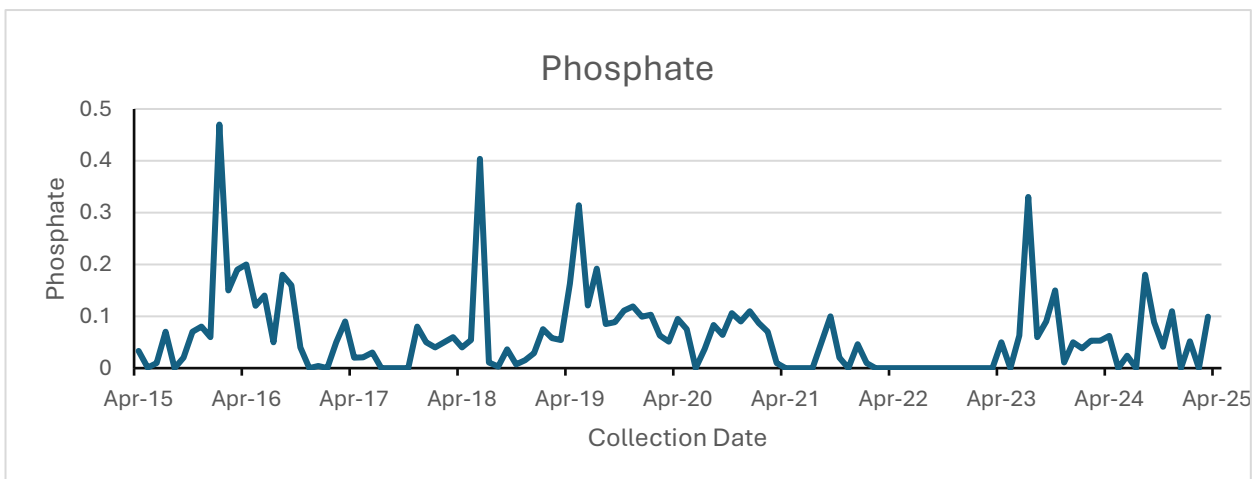
Graph 7 Monthly variation of Sulphate at Panetha Station (2015-2025)



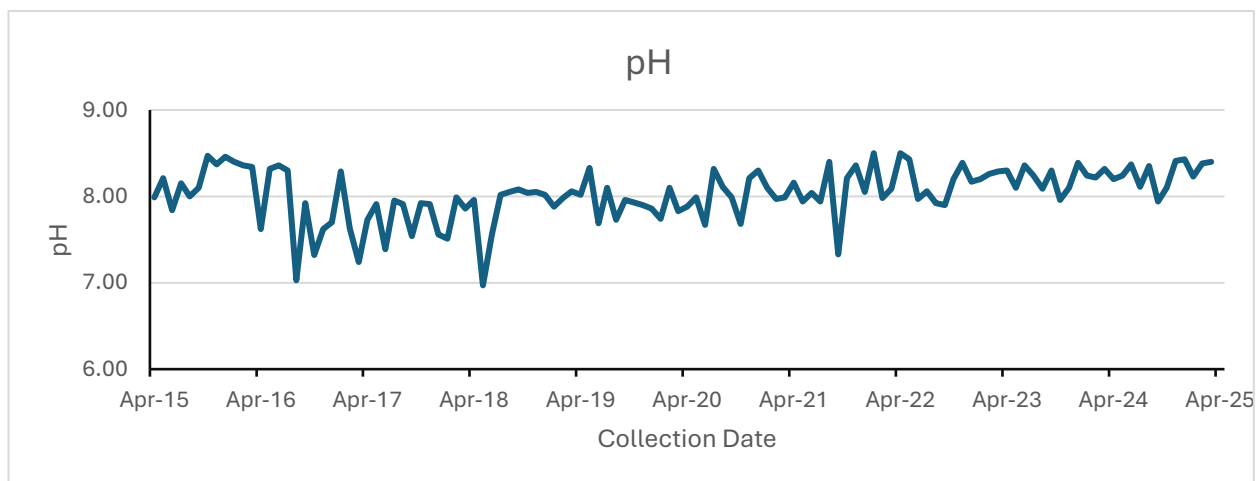
Graph 8 Monthly variation of Sodium at Panetha Station (2015-2025)



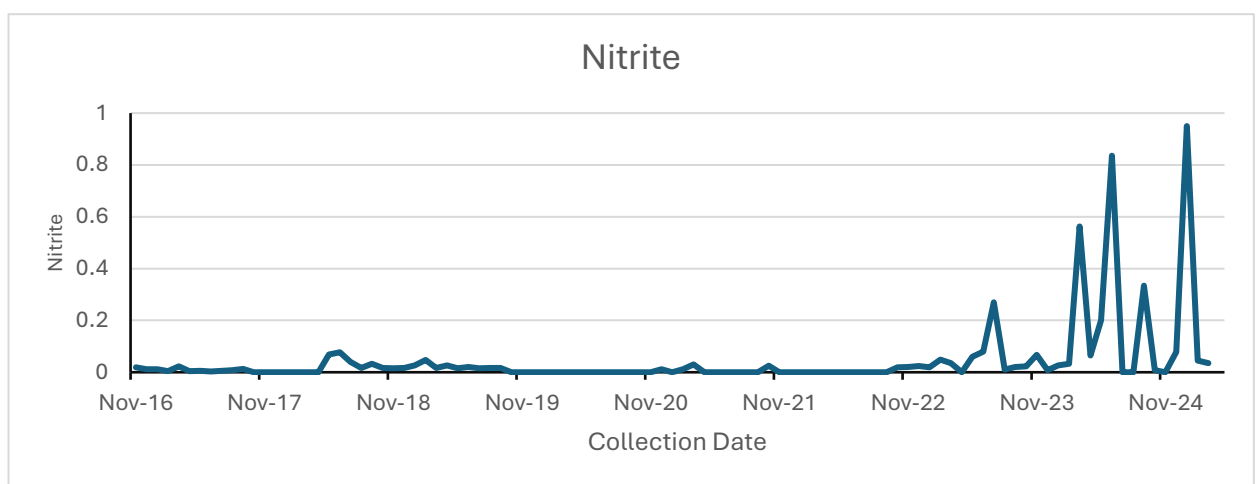
Graph 9 Monthly variation of Potassium at Panetha Station (2015-2025)



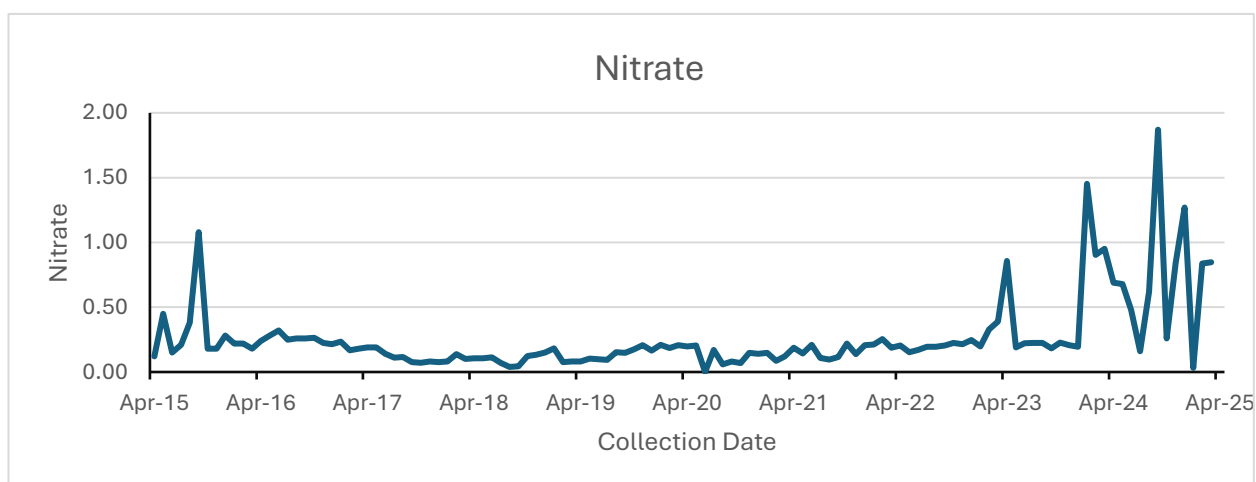
Graph 10 Monthly variation of Phosphate at Panetha Station (2015-2025)



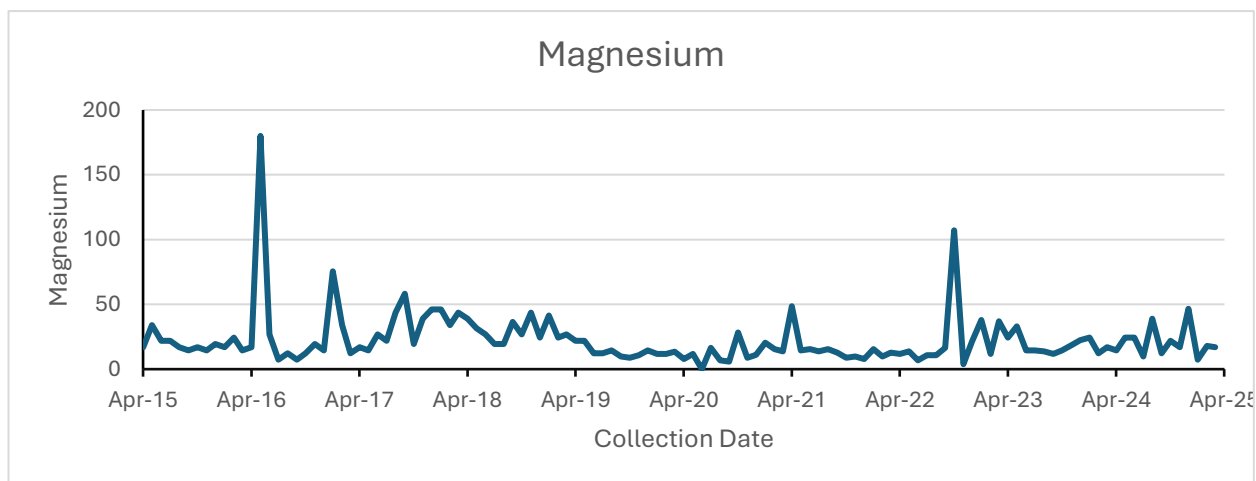
Graph 11 Monthly variation of pH at Panetha Station (2015-2025)



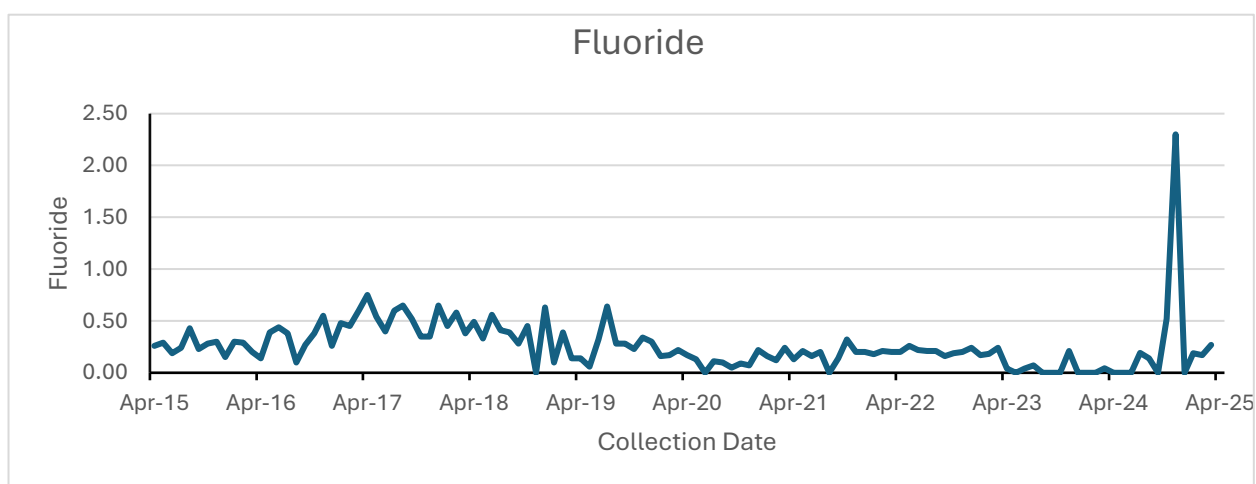
Graph 12 Monthly variation of Nitrite at Panetha Station (2016-2025)



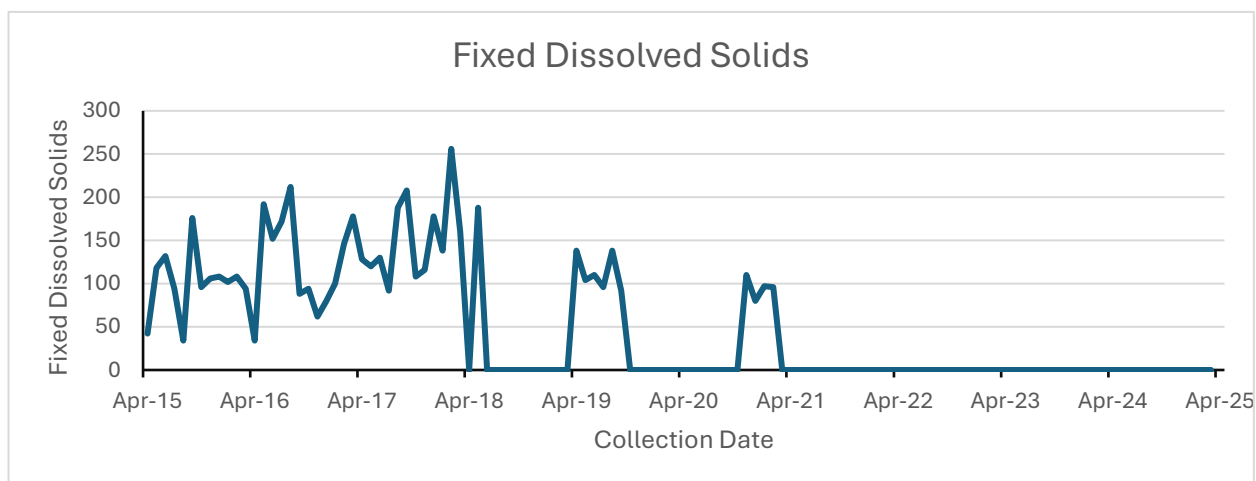
Graph 13 Monthly variation of Nitrate at Panetha Station (2015-2025)



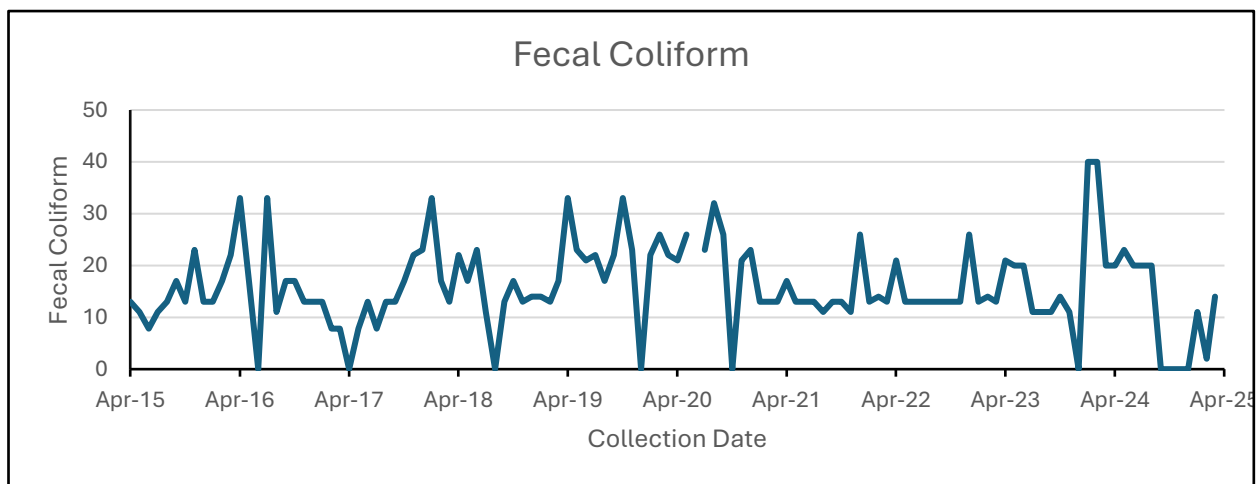
Graph 14 Monthly variation of Magnesium at Panetha Station (2015-2025)



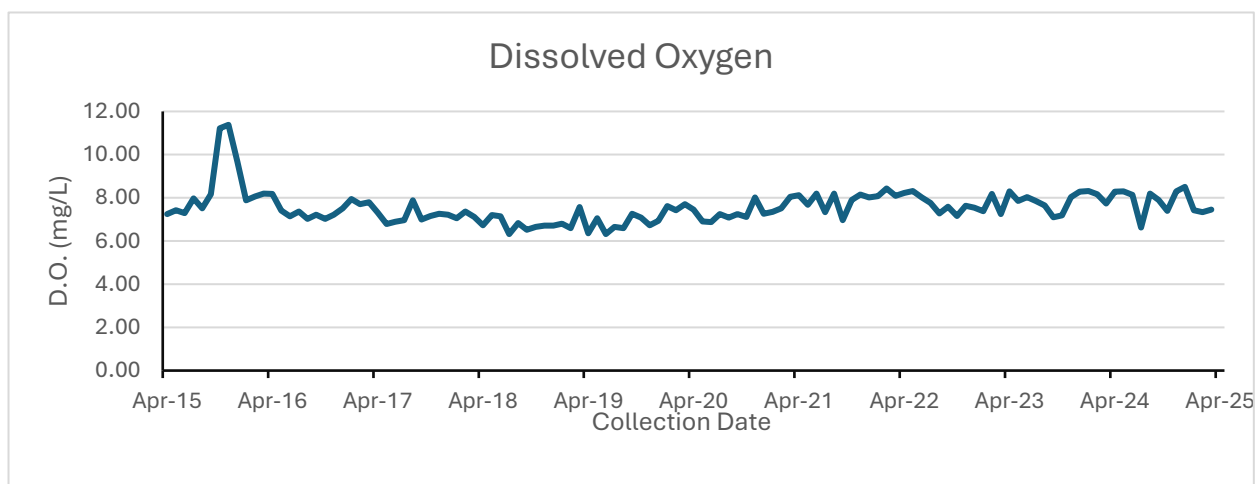
Graph 15 Monthly variation of Fluoride at Panetha Station (2015-2025)



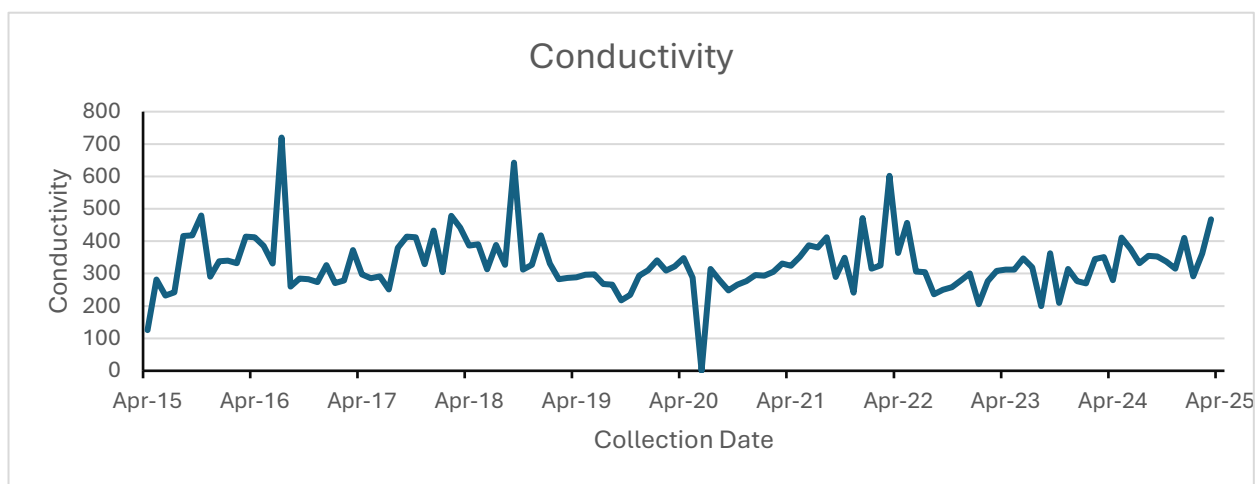
Graph 16 Monthly variation of Fixed Dissolved Solids at Panetha Station (2015-2025)



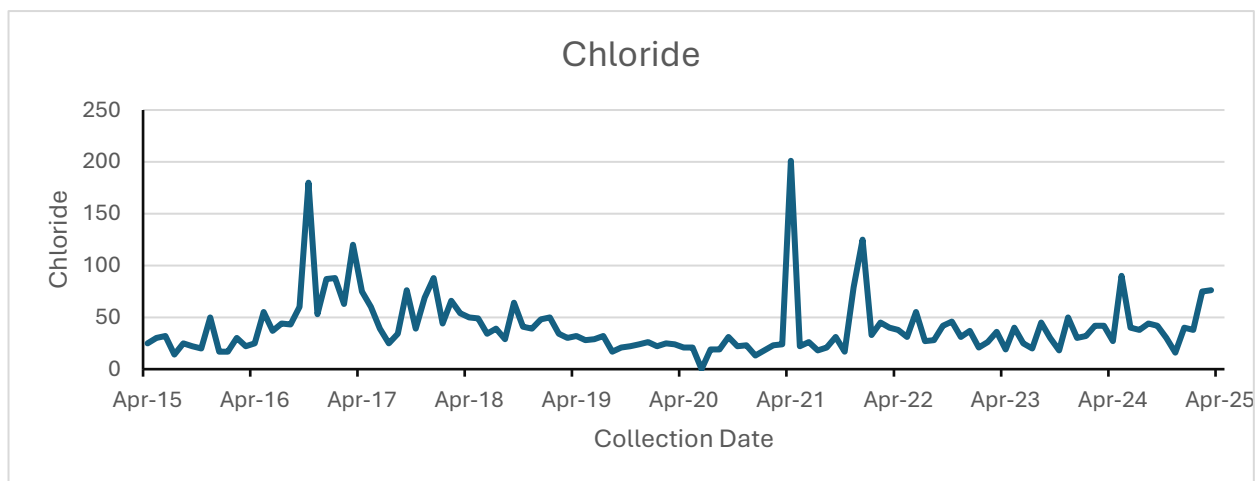
Graph 17 Monthly variation of Fecal Coliform at Panetha Station (2015-2025)



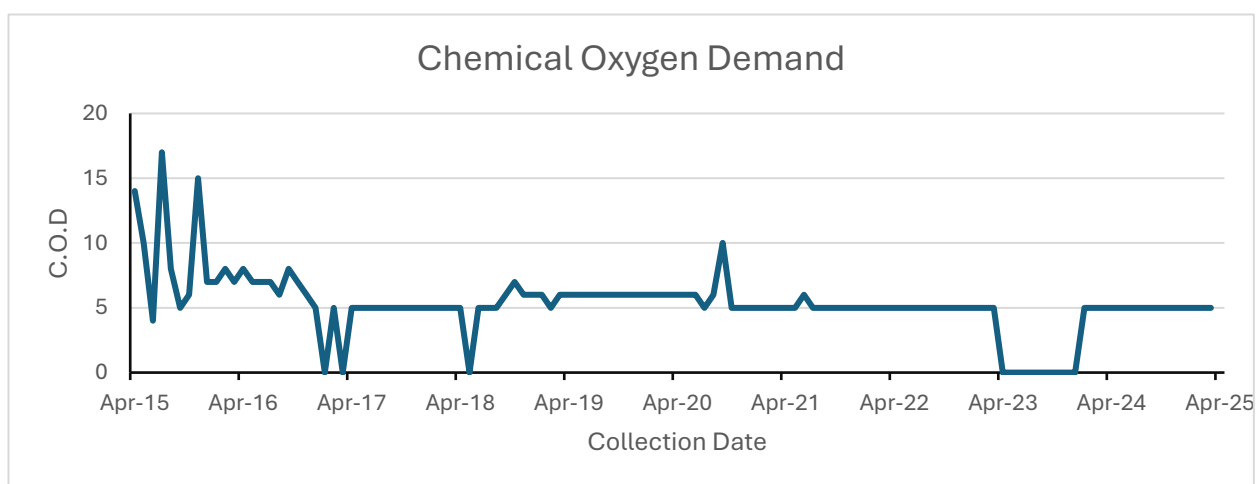
Graph 18 Monthly variation of Dissolved Oxygen at Panetha Station (2015-2025)



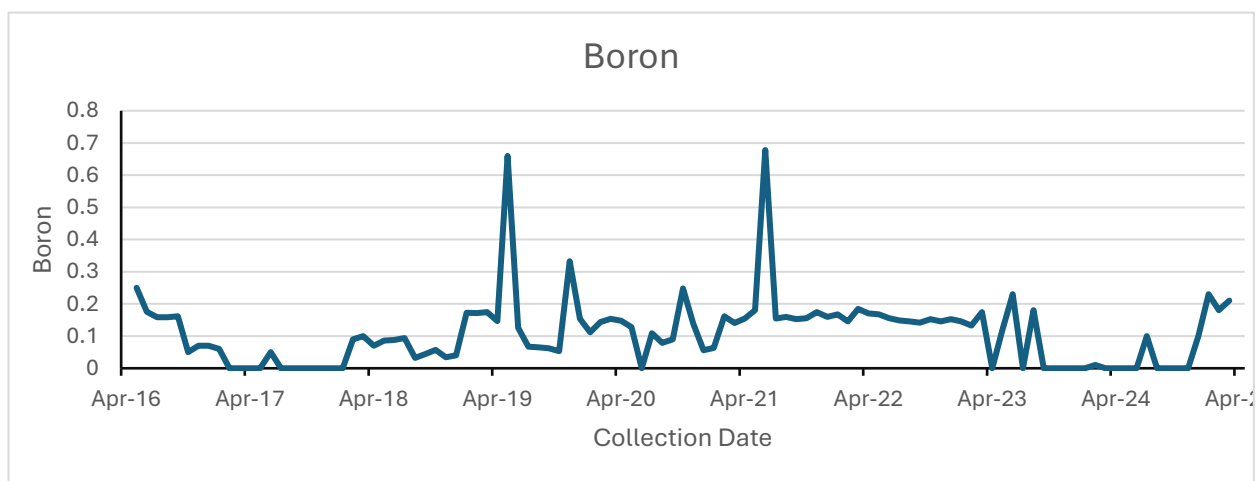
Graph 19 Monthly variation of Conductivity at Panetha Station (2015-2025)



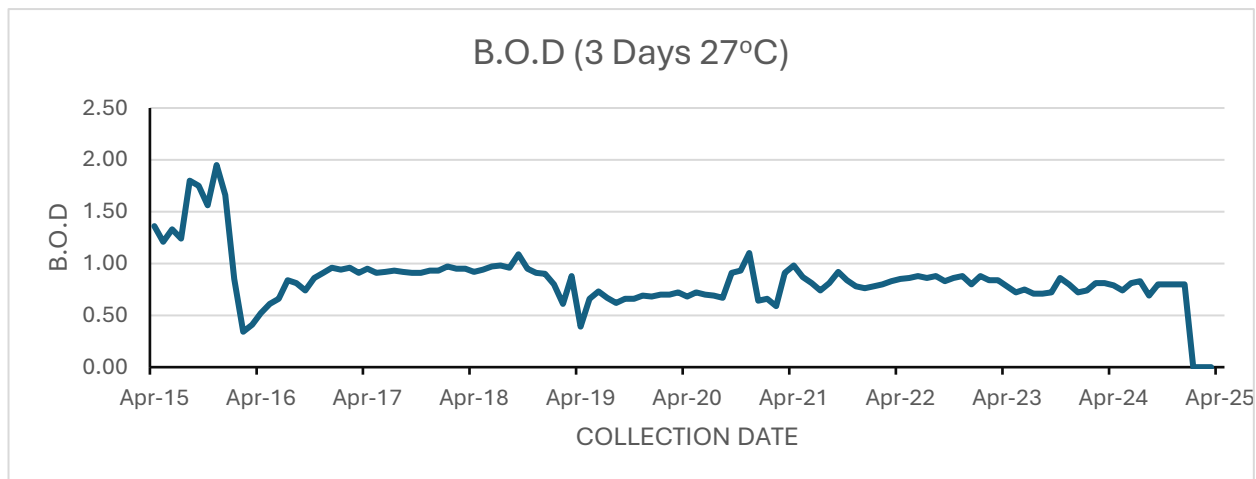
Graph 20 Monthly variation of Chloride at Panetha Station (2015-2025)



Graph 21 Monthly variation of Chemical Oxygen Demand at Panetha Station (2015-2025)



Graph 22 Monthly variation of Boron at Panetha Station (2016-2025)



Graph 23 Monthly variation of B.O.D (3 Days 27°C at Panetha Station (2015-2025))

Graph 1: The graph shows the monthly variation of *Turbidity* at Panetha Station from April 2015 to April 2025. Turbidity ranged from 0.2 to 175.8 NTU, permissible limit (BIS IS10500): 5 NTU (desirable 1 NTU), exceeds the permissible limit and requires attention.

Graph 2: The graph shows the monthly variation of *Total Hardness as CaCo₃* at Panetha Station from April 2015 to April 2025. Total Hardness ranged from 86.0 to 510.0 mg/L, permissible limit (BIS IS10500): 600 mg/L (desirable 200 mg/L), within the permissible limit.

Graph 3: The graph shows the monthly variation of *Total Kjeldahl Nitrogen* at Panetha Station from April 2015 to April 2025. Total Kjeldahl Nitrogen ranged from 0.56 to 2.24 mg/L, no specific BIS drinking-water limit (reported as nutrient), reported; within typical environmental ranges.

Graph 4: The graph shows the monthly variation of *Total Dissolved Solids* at Panetha Station from April 2015 to April 2025. TDS ranged from 66.0 to 354.0 mg/L, permissible limit (BIS IS10500): 2000 mg/L (desirable 500 mg/L), well within permissible limit (but some values above desirable 500 mg/L).

Graph 5: The graph shows the monthly variation of *Temperature* at Panetha Station from April 2015 to April 2025. Temperature ranged from 25.0 to 36.5 °C, no fixed BIS drinking limit (reported), reported seasonal variation consistent with climate.

Graph 6: The graph shows the monthly variation of *Suspended Solids* at Panetha Station from April 2015 to April 2025. Suspended Solids ranged from 2.0 to 394.0 mg/L, no direct BIS drinking-water limit (STP reuse norms often use TSS ≤ 20 mg/L), periodic spikes observed (monsoon); some values above common STP reuse benchmarks.

Graph 7: The graph shows the monthly variation of *Sulphate* at Panetha Station from April 2015 to April 2025. Sulphate ranged from 3.0 to 92.0 mg/L, permissible limit (BIS IS10500): 200 mg/L, within permissible limit.

Graph 8: The graph shows the monthly variation of *Sodium* at Panetha Station from April 2015 to April 2025. Sodium ranged from 3.0 to 335.0 mg/L, no strict BIS numeric limit (reported; guideline values often ~200 mg/L), within typical reporting/guideline ranges.

Graph 9: The graph shows the monthly variation of *Potassium* at Panetha Station from April 2015 to April 2025. Potassium ranged from 0.2 to 25.0 mg/L, with no specific BIS limit (reported), within normal environmental concentrations.

Graph 10: The graph shows the monthly variation of *Phosphate* at Panetha Station from April 2015 to April 2025. Phosphate ranged from 0.01 to 0.45 mg/L, no specific BIS drinking limit (nutrient environmental concern when elevated), levels mostly low; occasional monsoon rises.

Graph 11: The graph shows the monthly variation of *pH* at Panetha Station from April 2015 to April 2025. pH ranged from 6.6 to 8.4, permissible limit (BIS IS10500): 6.5–8.5 (no relaxation), within permissible limits.

Graph 12: The graph shows the monthly variation of *Nitrite* at Panetha Station from April 2016 to April 2025. Nitrite ranged from 0.01 to 0.95 mg/L, permissible limit (BIS IS10500): 0.2 mg/L (permissible), at or below permissible limit (close to limit in some months).

Graph 13: The graph shows the monthly variation of *Nitrate* at Panetha Station from April 2015 to April 2025. Nitrate ranged from 0.03 to 1.87 mg/L, permissible limit (BIS IS10500): 45 mg/L, exceeds the permissible limit in some months and requires attention.

Graph 14: The graph shows the monthly variation of *Magnesium* at Panetha Station from April 2015 to April 2025. Magnesium ranged from 4.0 to 180.0 mg/L, no separate BIS numeric limit (contributes to hardness), exceeds permissible limit in some areas.

Graph 15: The graph shows the monthly variation of *Fluoride* at Panetha Station from April 2015 to April 2025. Fluoride ranged from 0.06 to 2.30 mg/L, permissible limit (BIS IS10500): 1.5 mg/L (permissible; desirable 1.0 mg/L), within permissible limit.

Graph 16: The graph shows the monthly variation of *Fixed Dissolved Solids* at Panetha Station from April 2015 to April 2025. Fixed Dissolved Solids ranged from 34 to 256 (data merged with TDS), see TDS above, reported with TDS, within permissible limits.

Graph 17: The graph shows the monthly variation of *Fecal Coliform* at Panetha Station from April 2015 to April 2025. Fecal Coliform ranged from 2 to 40 MPN/100mL, permissible limit (BIS IS10500 for drinking): 0 MPN/100mL, well above drinking-water limits; indicates microbial contamination (especially monsoon).

Graph 18: The graph shows the monthly variation of *Dissolved Oxygen* at Panetha Station from April 2015 to April 2025. DO ranged from 6.32 to 11.38 mg/L, CPCB ecological guideline: $DO \geq 6$ mg/L for good water, mostly meets ecological criteria though some months dip slightly below.

Graph 19: The graph shows the monthly variation of *Conductivity* at Panetha Station from April 2015 to April 2025. Conductivity ranged from 120 to 758 μ S/cm, no direct BIS limit (reported), shows seasonal/point-source variability; some high values reflect higher ionic load.

Graph 20: The graph shows the monthly variation of *Chloride* at Panetha Station from April 2015 to April 2025. Chloride ranged from 13 to 201 mg/L, permissible limit (BIS IS10500): 250 mg/L (desirable $\sim$ 200 mg/L), within permissible range.

Graph 21: The graph shows the monthly variation of *Chemical Oxygen Demand (COD)* at Panetha Station from April 2015 to April 2025. COD ranged from 4 to 17.0 mg/L, no BIS drinking limit (effluent limits vary), shows periodic organic/chemical loading; some months are elevated.

Graph 22: The graph shows the monthly variation of *Boron* at Panetha Station from April 2016 to April 2025. Boron ranged from 0.02 to 0.68 mg/L, typical BIS guidance: desirable $\sim$ 0.5 mg/L; permissible often 2.0 mg/L, within typical drinking-water guidance.

Graph 23: The graph shows the monthly variation of *B.O.D (3 Days 27°C)* at Panetha Station from April 2015 to April 2025. BOD ranged from 0.3 to 2.0 mg/L, CPCB bathing criterion for Class B (outdoor bathing): $BOD \leq 3$ mg/L, some months exceed the bathing standard and indicate organic pollution.

Graph 24: The graph shows the monthly variation of *Alkalinity as CaCO₃* at Panetha Station from April 2015 to April 2025. Alkalinity ranged from 92 to 260.0 mg/L, no strict BIS numeric limit beyond general desirability (often 200 mg/L desirable), mostly within acceptable ranges though occasionally elevated.

River Narmada at Panetha

The water quality at Panetha station shows moderate variation across parameters during the period April 2015 to April 2025. Panetha Station's water is chemically acceptable for many parameters (TDS, total hardness, sulphate, chloride, boron, pH and generally adequate DO) but shows clear episodic and seasonal problems that compromise safety for direct consumption: large turbidity spikes (up to 175.8 NTU) and persistent faecal coliform counts (2–40 MPN/100 mL vs drinking-water target of 0) are the most serious concerns, while nitrite (to 0.95 mg/L), fluoride (to 2.30 mg/L), elevated suspended solids and occasional high sodium/magnesium values exceed desirable limits in some months, particularly linked to seasonal discharges and runoff.

2.2.2. Station: Garudeshwar

Garudeshwar represents the middle-lower transitional stretch of the river and reflects a moderate water quality profile with gradual improvements. **Graph 25 to Graph 50** show the water quality parameters with time from 2015 to 2025 of Garudeshwar Station.

Organic Pollution: BOD and COD values show a declining trend but remain higher than Panetha, reflecting moderate organic input from tributaries and small-scale anthropogenic sources. Seasonal spikes during monsoon are observed due to stormwater runoff.

Dissolved Oxygen: DO remains consistently high (>5.5 mg/L), showing the river's capacity for self-purification. Slight dips occur during high-flow monsoon months.

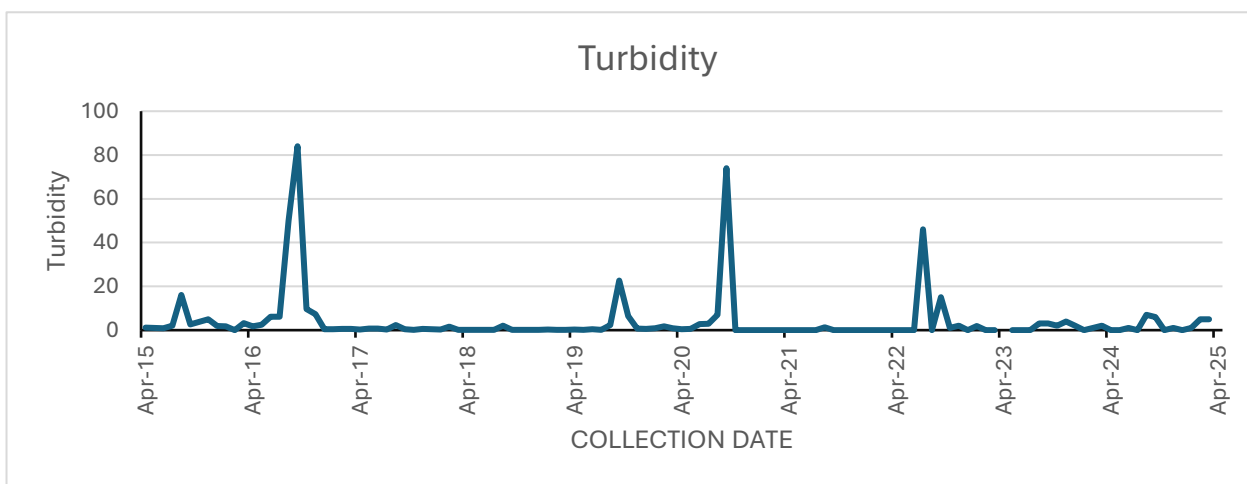
Nutrients: Nitrate, phosphate, and TKN levels are moderate, with phosphate increasing slightly during monsoon due to agricultural and surface runoff.

Microbial Load: Moderate coliform contamination occurs year-round, with higher levels during monsoon due to untreated sewage inflows from nearby settlements.

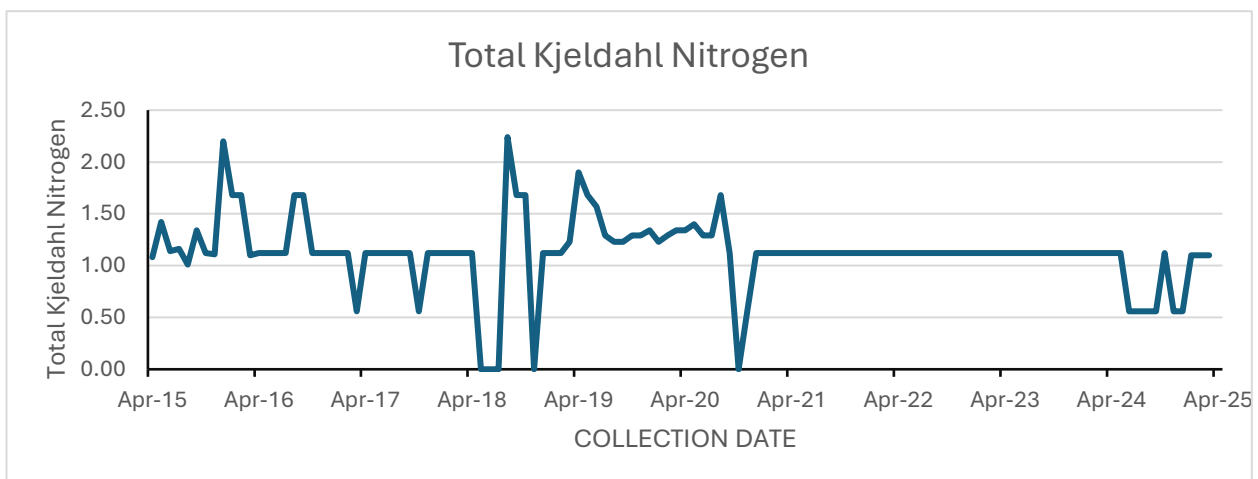
Metals: Most heavy metals remain within safe limits, though occasional iron increases are recorded during rainy season due to sediment disturbance.

Salinity & Hardness: Chloride and TDS levels remain acceptable, though seasonal variations occur.

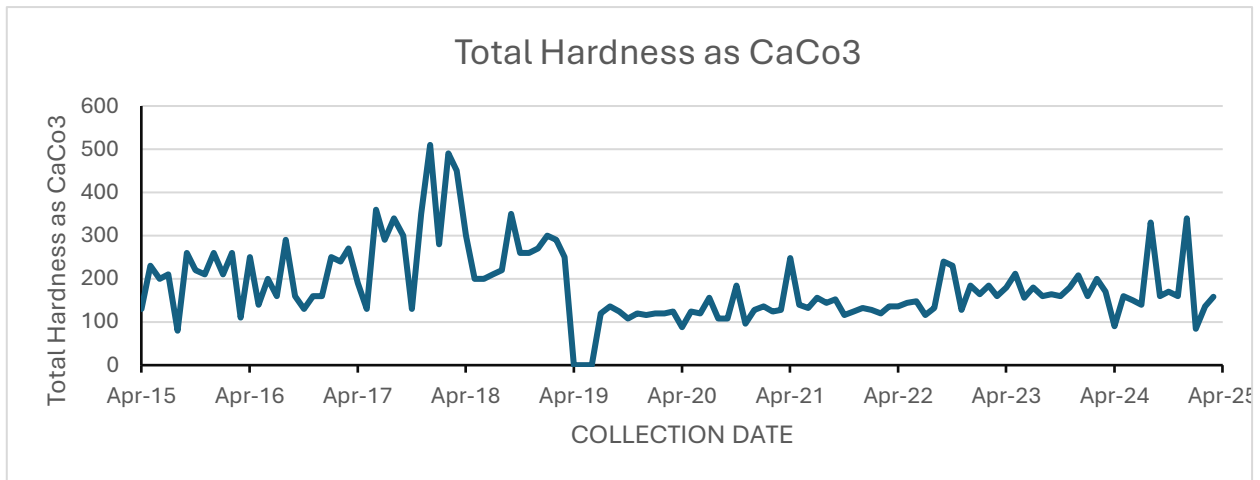
Overall Condition: Garudeshwar exhibits moderate to good water quality, reflecting transitional influences from upstream clean stretches and local anthropogenic inputs. Continued sewage management and nutrient control are needed.



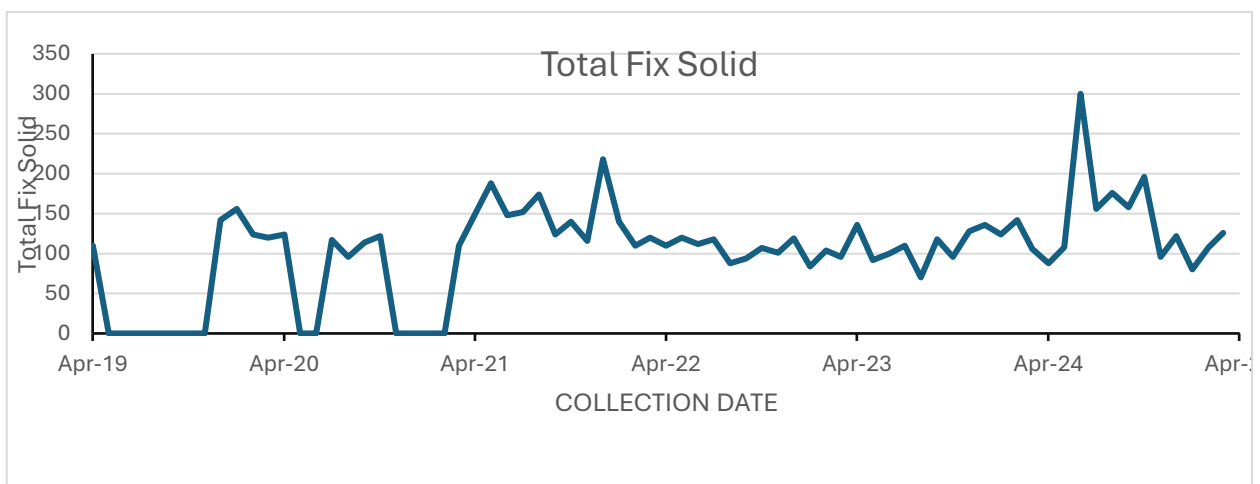
Graph 25 Monthly variation of Turbidity at Garudeshwar Station (2015-2025)



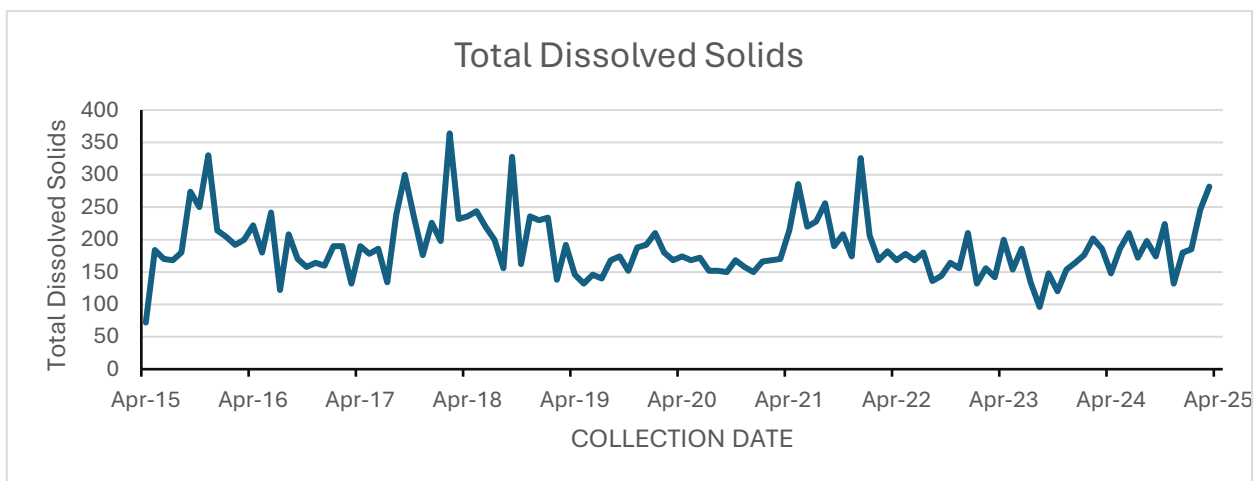
Graph 26 Monthly variation of Total Kjeldahl Nitrogen at Garudeshwar Station (2015-2025)



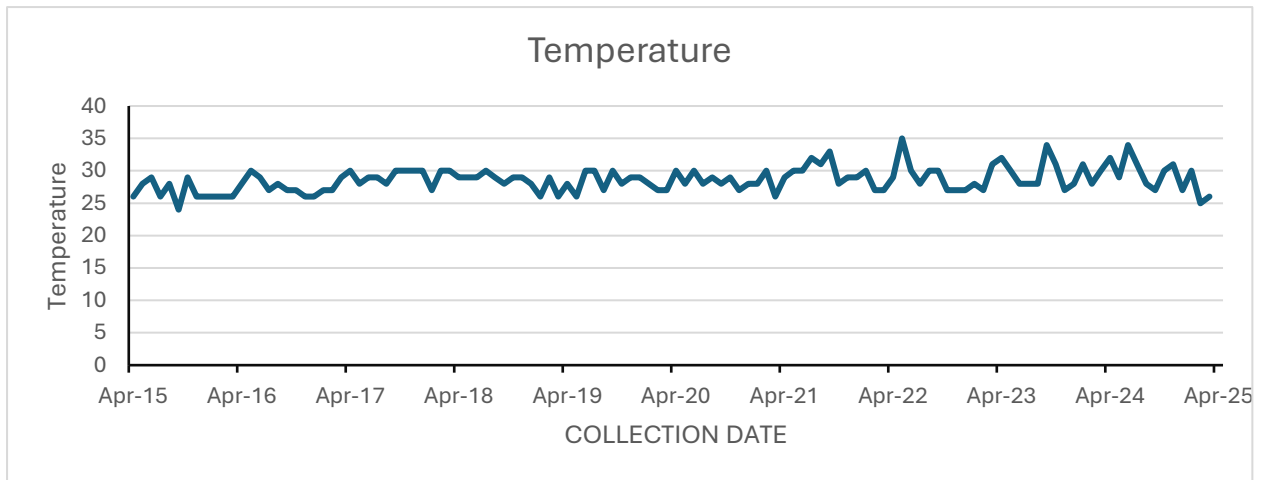
Graph 27 Monthly variation of Total Hardness as CaCo₃ at Garudeshwar Station (2015-2025)



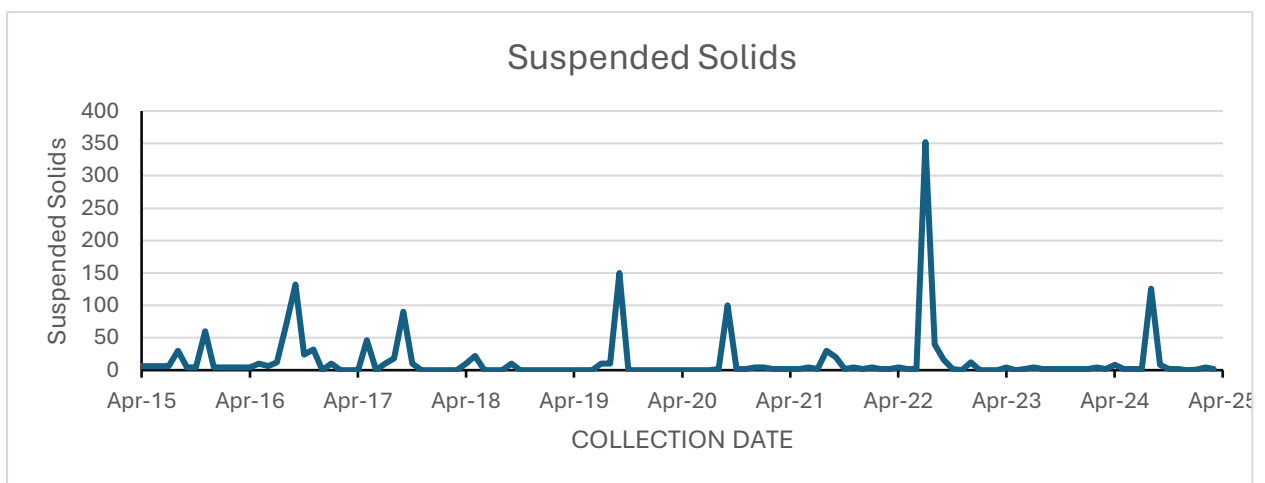
Graph 28 Monthly variation of Total Fix Solid at Garudeshwar Station (2015-2025)



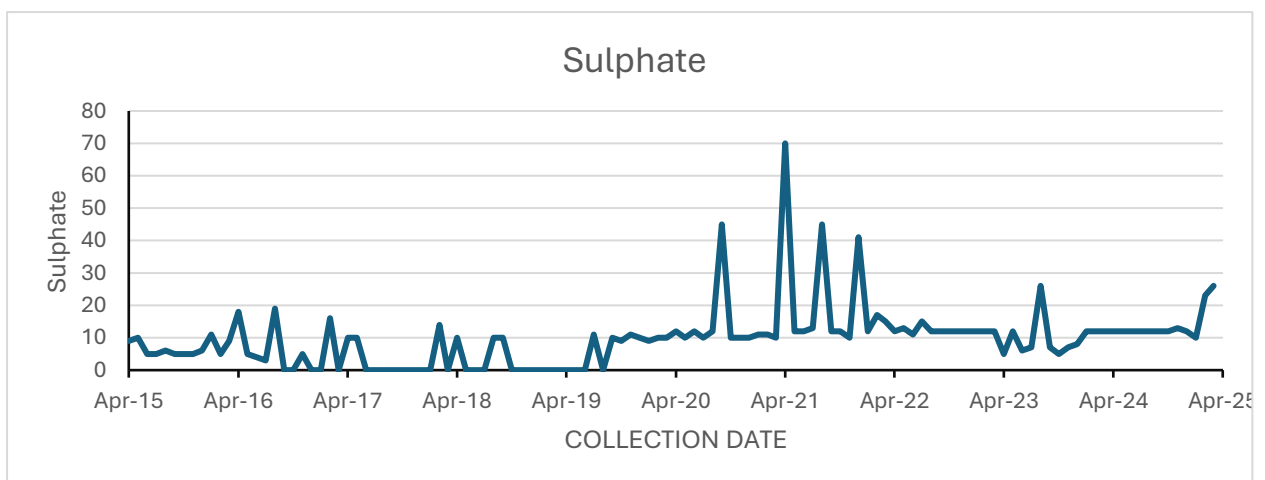
Graph 29 Monthly variation of Total Dissolved Solids at Garudeshwar Station (2015-2025)



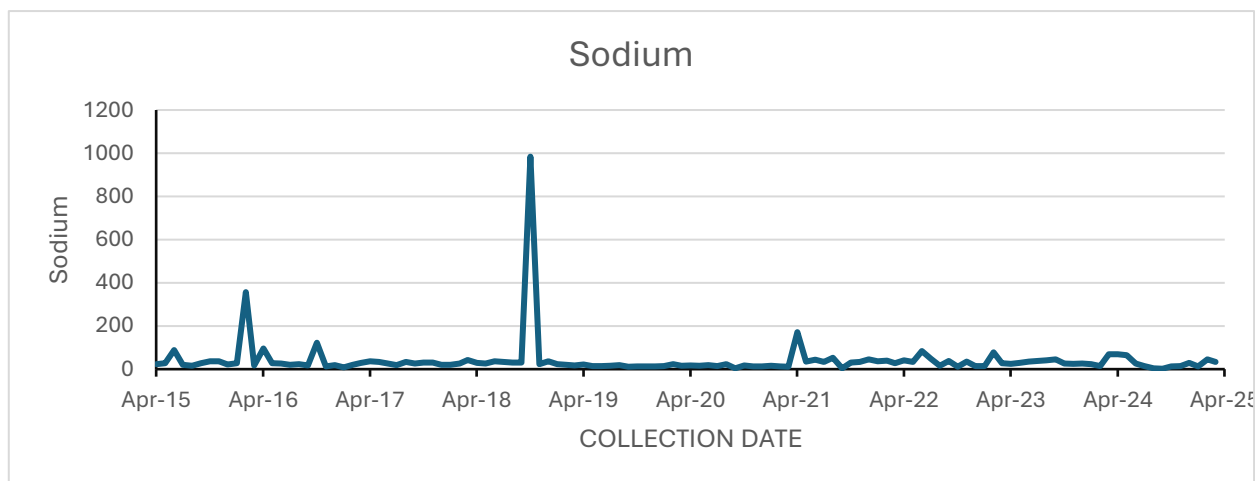
Graph 30 Monthly variation of Temperature at Garudeshwar Station (2015-2025)



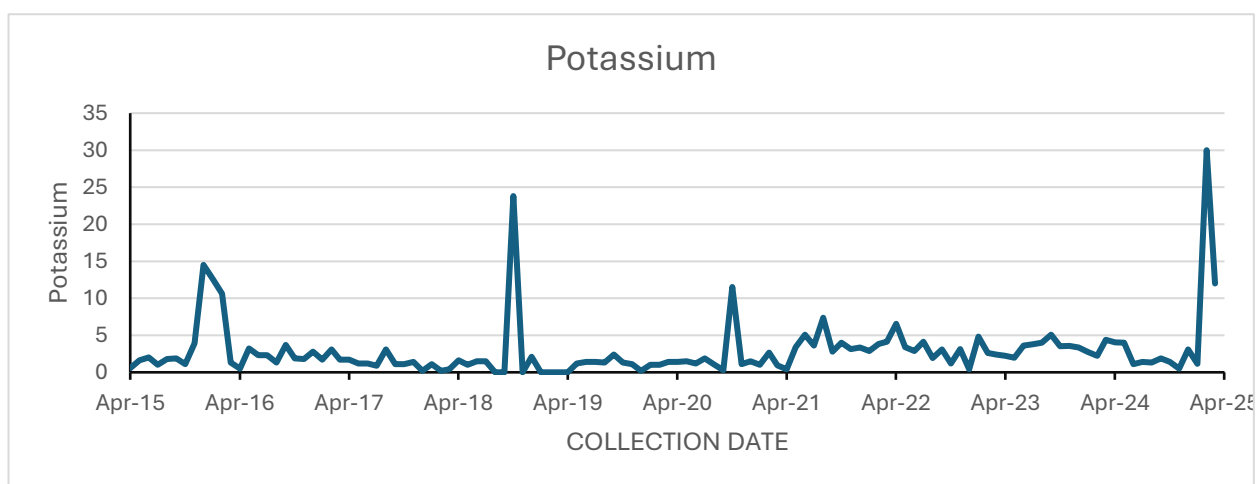
Graph 31 Monthly variation of Suspended Solids at Garudeshwar Station (2015-2025)



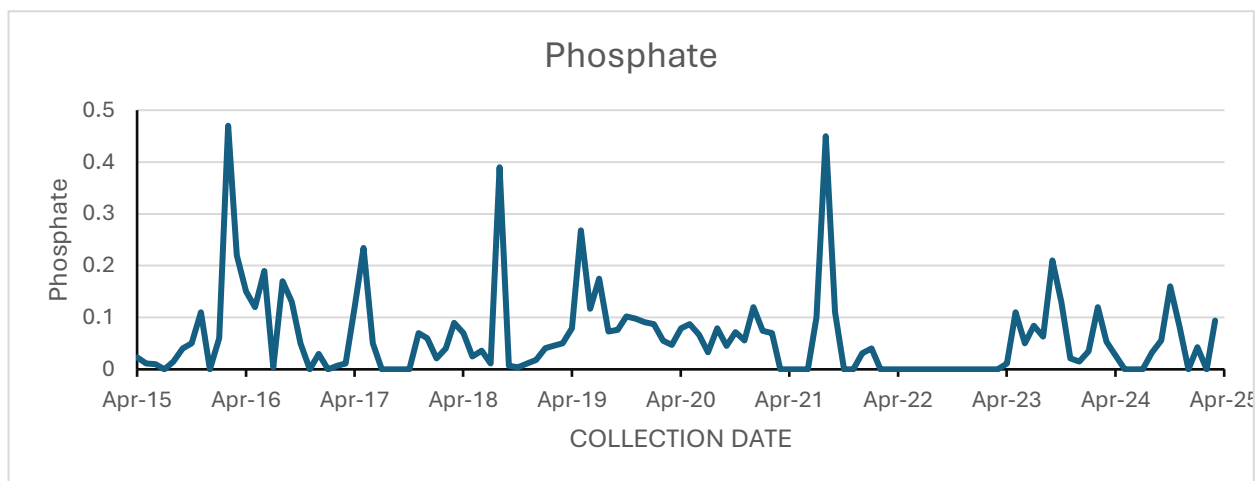
Graph 32 Monthly variation of Sulphate at Garudeshwar Station (2015-2025)



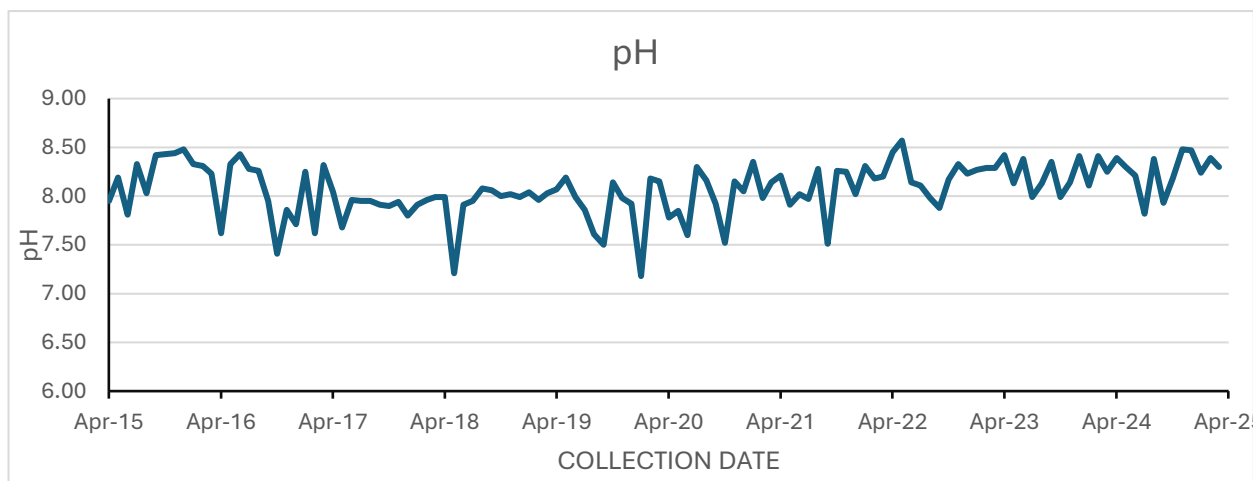
Graph 33 Monthly variation of Sodium at Garudeshwar Station (2015-2025)



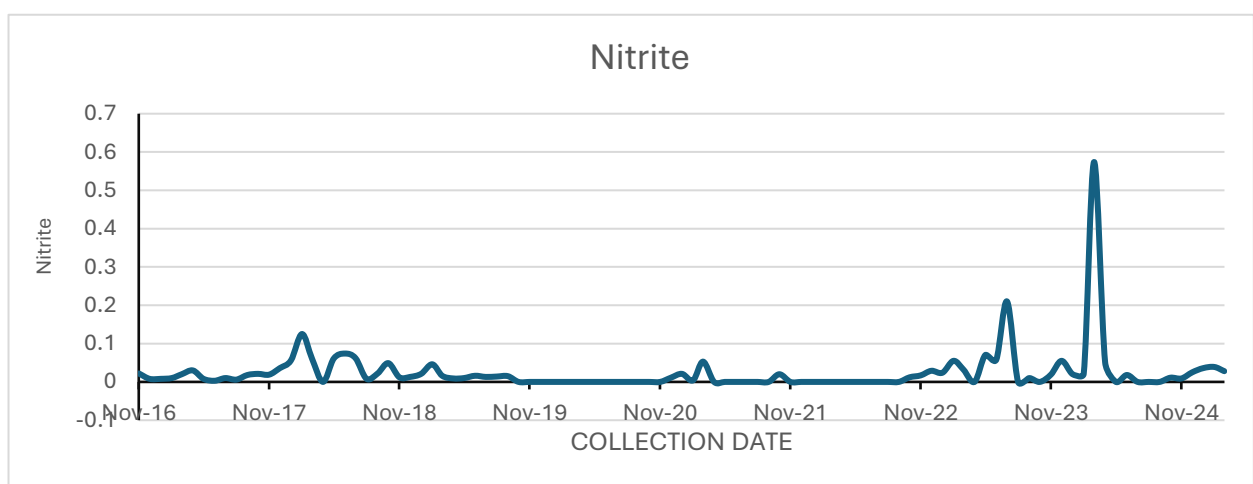
Graph 34 Monthly variation of Potassium at Garudeshwar Station (2015-2025)



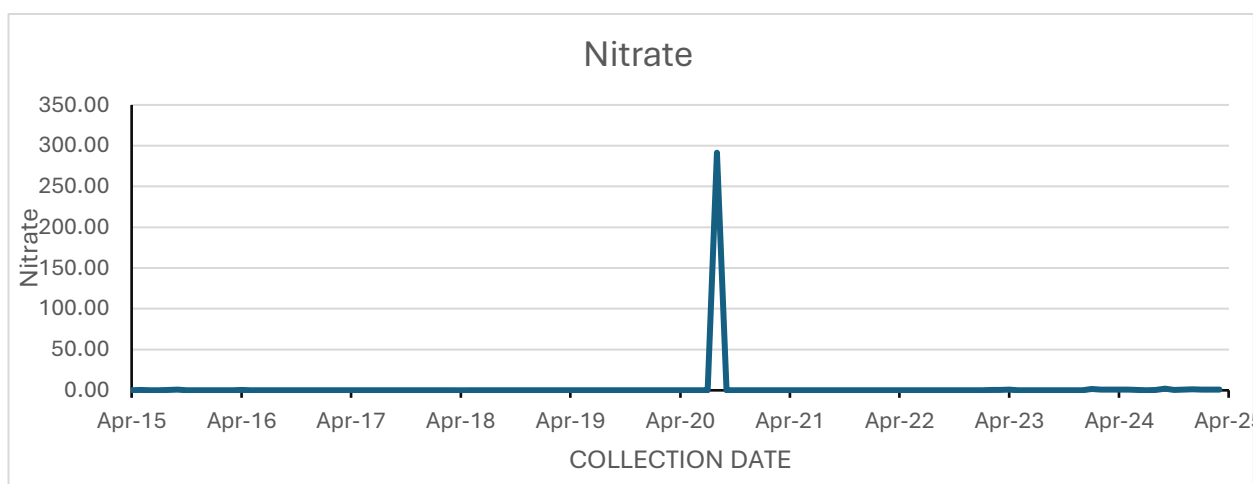
Graph 35 Monthly variation of Phosphate at Garudeshwar Station (2015-2025)



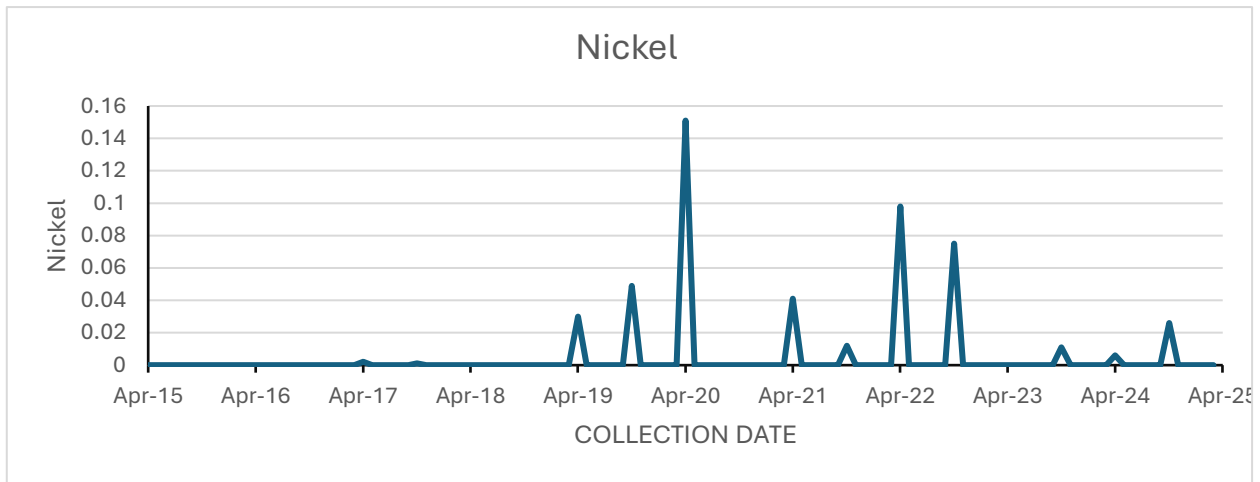
Graph 36 Monthly variation of pH at Garudeshwar Station (2015-2025)



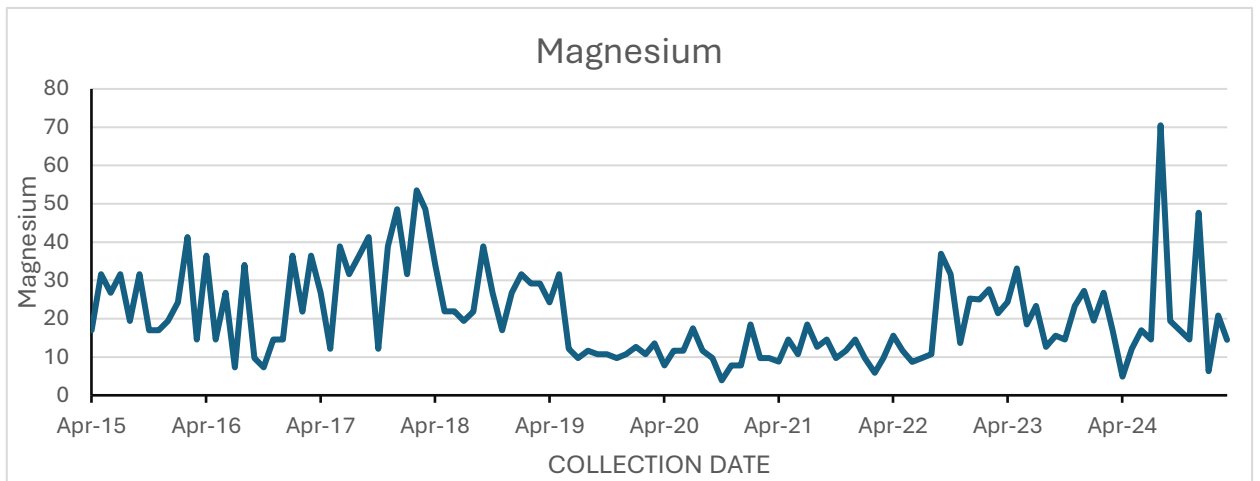
Graph 37 Monthly variation of Nitrite at Garudeshwar Station (2015-2025)



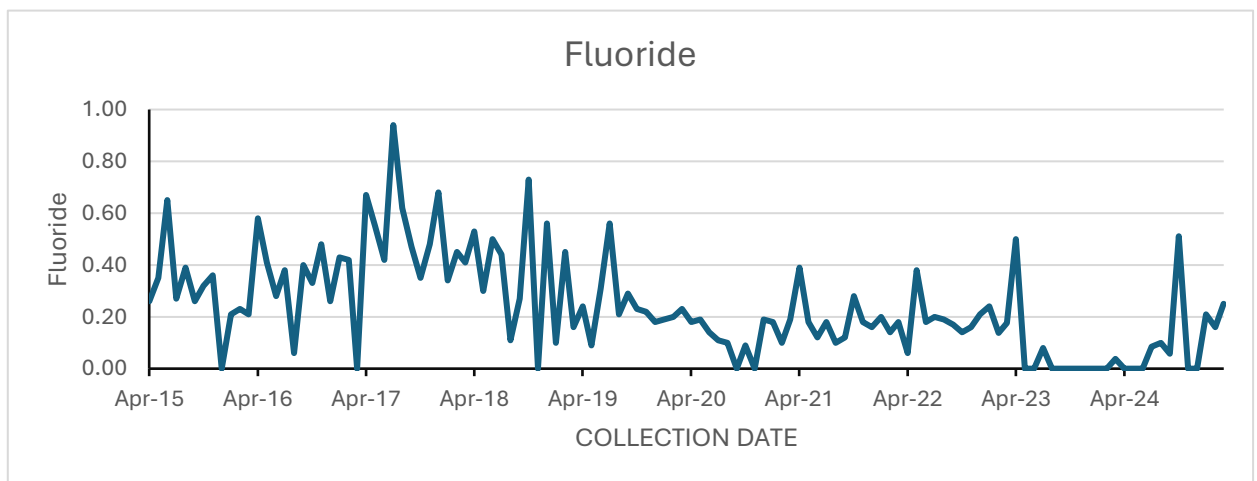
Graph 38 Monthly variation of Nitrate at Garudeshwar Station (2015-2025)



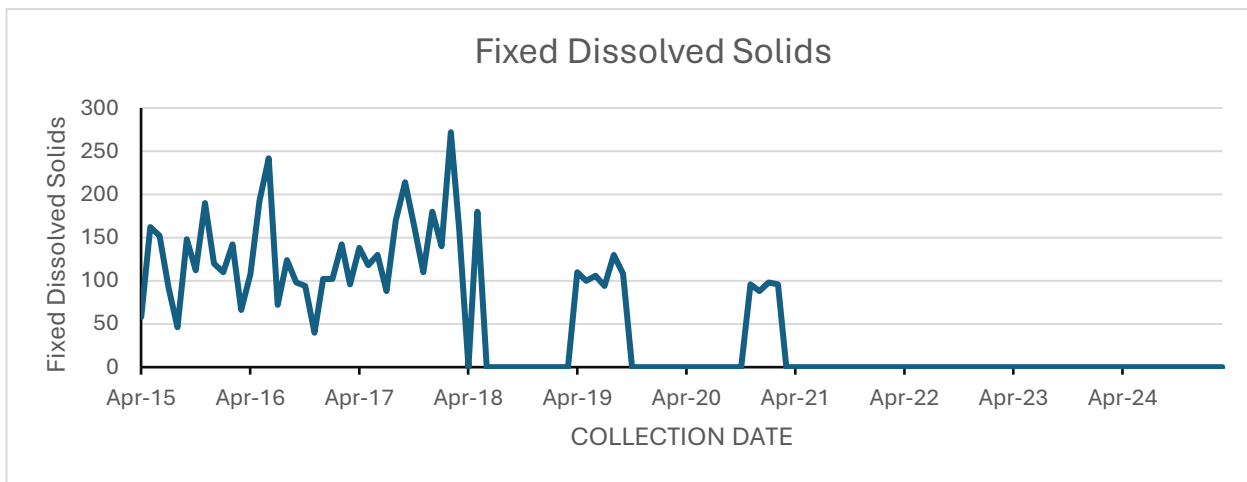
Graph 39 Monthly variation of Nickel at Garudeshwar Station (2015-2025)



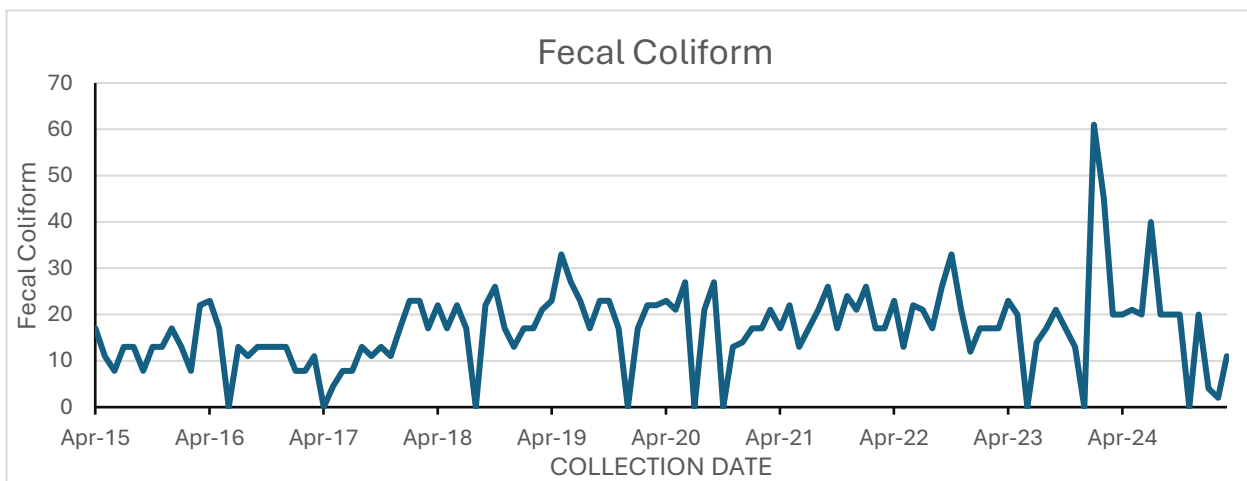
Graph 40 Monthly variation of Magnesium at Garudeshwar Station (2015-2025)



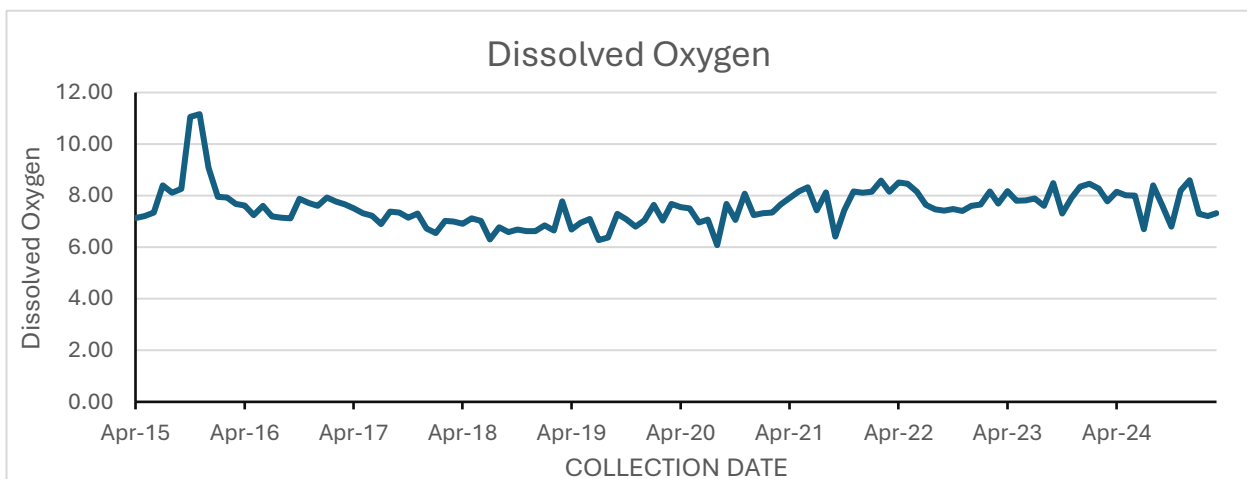
Graph 41 Monthly variation of Fluoride at Garudeshwar Station (2015-2025)



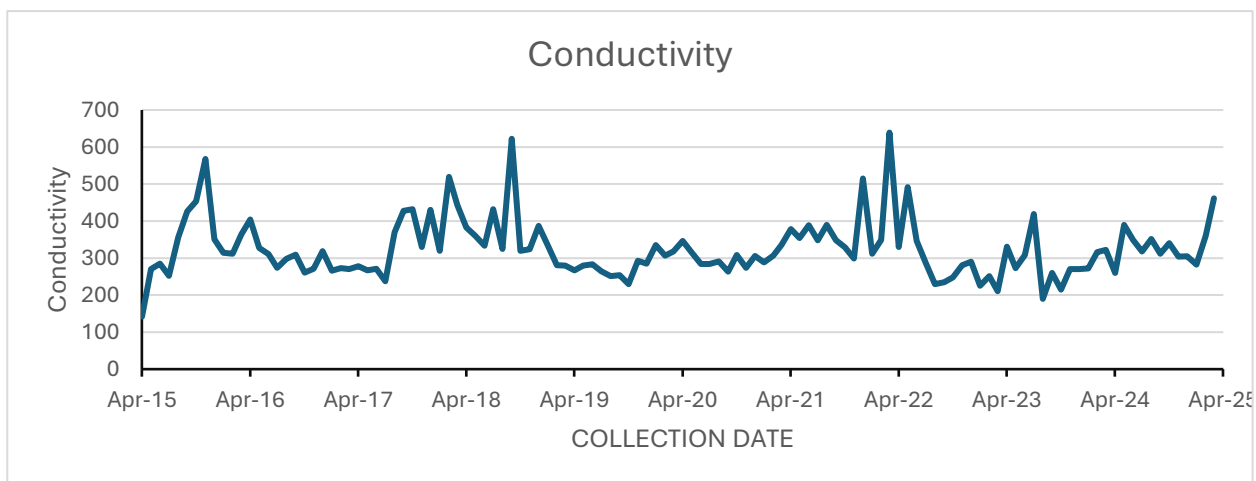
Graph 42 Monthly variation of Fixed Dissolved Solids at Garudeshwar Station (2015-2025)



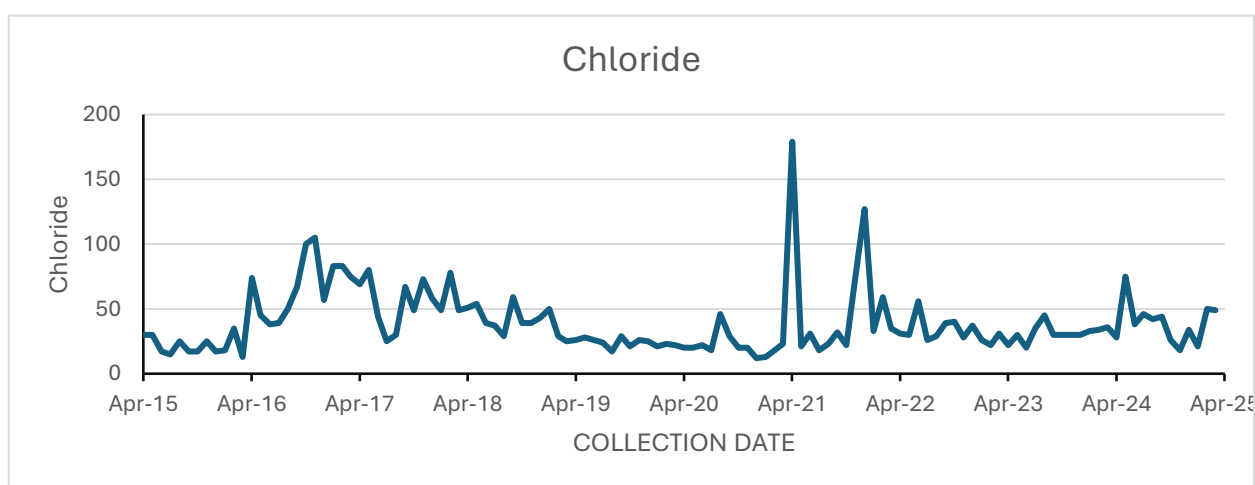
Graph 43 Monthly variation of Fecal Coliform at Garudeshwar Station (2015-2025)



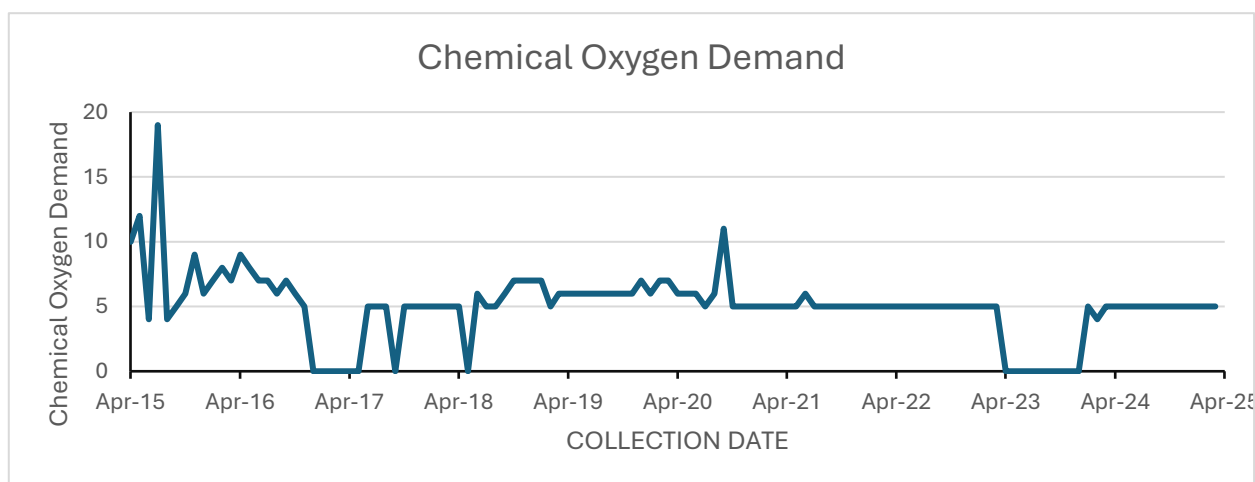
Graph 44 Monthly variation of Dissolved Oxygen at Garudeshwar Station (2015-2025)



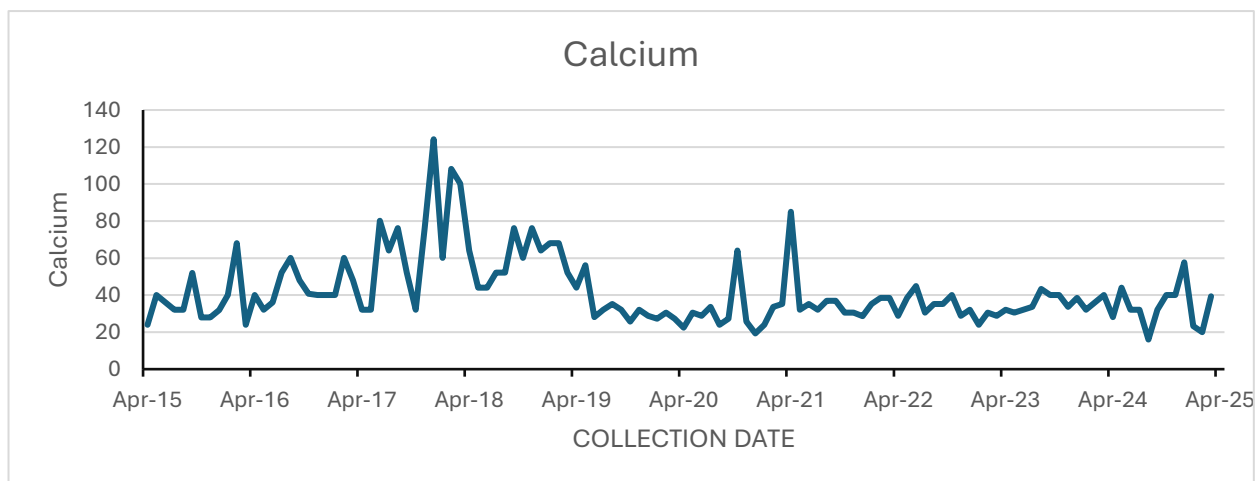
Graph 45 Monthly variation of Conductivity at Garudeshwar Station (2015-2025)



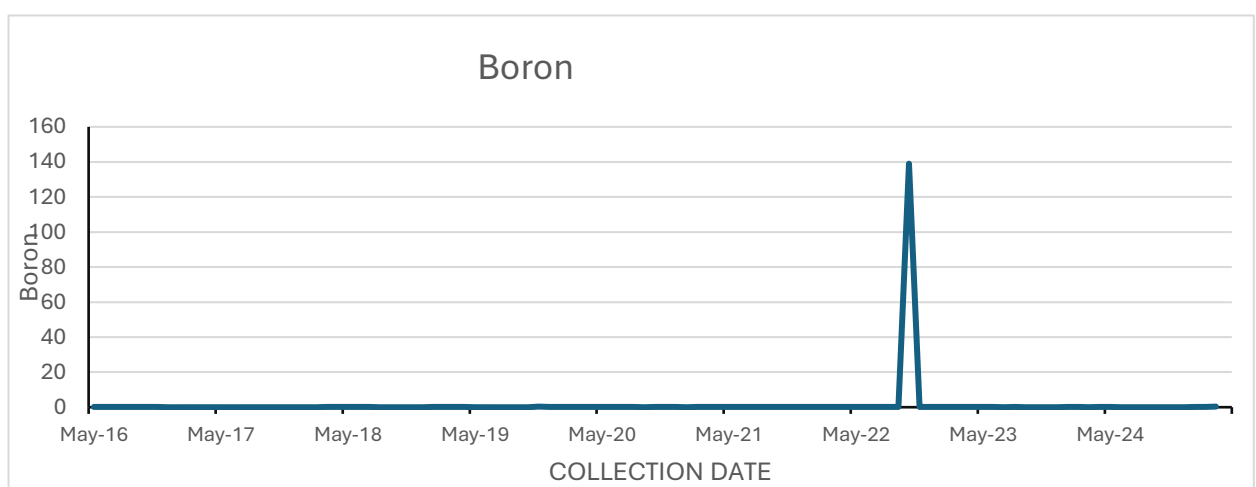
Graph 46 Monthly variation of Chloride at Garudeshwar Station (2015-2025)



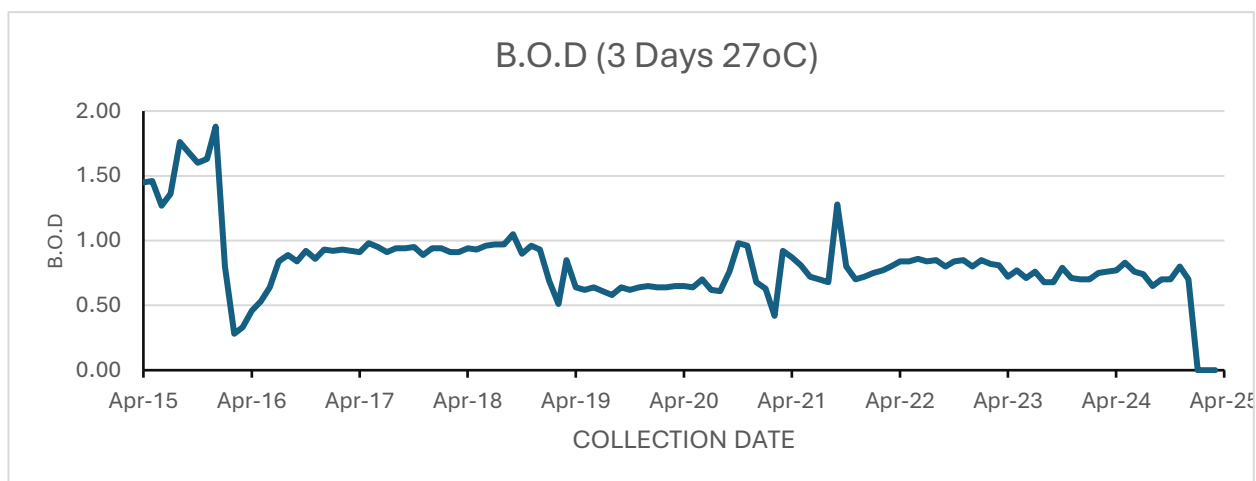
Graph 47 Monthly variation of C.O.D. at Garudeshwar Station (2015-2025)



Graph 48 Monthly variation of Calcium at Garudeshwar Station (2015-2025)



Graph 49 Monthly variation of Boron at Garudeshwar Station (2016-2025)



Graph 50 Monthly variation of B.O.D (3 Days 27oC) at Garudeshwar Station (2015-2025)

Graph 25: The graph shows the monthly variation of *Turbidity* at Garudeshwar Station from April 2015 to April 2025. Turbidity ranged from 0.1 to 84 , permissible limit (BIS IS10500): 5 NTU, very high maxima (monsoon/episodic spikes) exceeds permissible limit and requires attention.

Graph 26: The graph shows the monthly variation of *Total Kjeldahl Nitrogen* at Garudeshwar Station from April 2015 to April 2025. TKN ranged from 0.56 to 2.3 mg/L, no specific BIS drinking limit (reported), within expected environmental ranges.

Graph 27: The graph shows the monthly variation of *Total Hardness as CaCo₃* at Garudeshwar Station from April 2015 to April 2025. Total Hardness ranged from 82.0 to 524.0 mg/L, permissible limit (BIS IS10500): 600 mg/L, within permissible limit.

Graph 28: The graph shows the monthly variation of *Total Fix Solid* at Garudeshwar Station from April 2015 to April 2025. Total Fixed Solids (reported similarly to TDS) ranged from 70 to 300 (see TDS), reported as part of TDS, within typical ranges.

Graph 29: The graph shows the monthly variation of *Total Dissolved Solids* at Garudeshwar Station from April 2015 to April 2025. TDS ranged from 72.0 to 364.0 mg/L, permissible limit (BIS IS10500): 2000 mg/L, well within permissible limit (but above desirable 500 mg/L in some months).

Graph 30: The graph shows the monthly variation of *Temperature* at Garudeshwar Station from April 2015 to April 2025. Temperature ranged from 24.0 to 36.0 °C, seasonal variation expected no drinking-water limit.

Graph 31: The graph shows the monthly variation of *Suspended Solids* at Garudeshwar Station from April 2015 to April 2025. Suspended Solids ranged from 1.0 to 350.0 mg/L, monsoon spikes; some months above typical STP reuse benchmarks.

Graph 32: The graph shows the monthly variation of *Sulphate* at Garudeshwar Station from April 2015 to April 2025. Sulphate ranged from 4.0 to 130.0 mg/L, permissible limit (BIS IS10500): 200 mg/L, within permissible limit.

Graph 33: The graph shows the monthly variation of *Sodium* at Garudeshwar Station from April 2015 to April 2025. Sodium ranged from 2.0 to 1000.0 mg/L, within typical guideline ranges some months above typical reuse benchmarks.

Graph 34: The graph shows the monthly variation of *Potassium* at Garudeshwar Station from April 2015 to April 2025. Potassium ranged from 0.2 to 30.0 mg/L, within typical environmental concentrations.

Graph 35: The graph shows the monthly variation of *Phosphate* at Garudeshwar Station from April 2015 to April 2025. Phosphate ranged from 0.01 to 0.45 mg/L, generally low, with seasonal rises during monsoon.

Graph 36: The graph shows the monthly variation of *pH* at Garudeshwar Station from April 2015 to April 2025. pH ranged from 7.15 to 8.57, permissible limit (BIS IS10500): 6.5–8.5, within permissible limits.

Graph 37: The graph shows the monthly variation of *Nitrite* at Garudeshwar Station from April 2015 to April 2025. Nitrite ranged from 0.01 to 0.57 mg/L, permissible limit (BIS IS10500): 0.2 mg/L, within permissible limits.

Graph 38: The graph shows the monthly variation of *Nitrate* at Garudeshwar Station from April 2015 to April 2025. Nitrate ranged from 0.6 to 291.0 mg/L, permissible limit (BIS IS10500): 45 mg/L, mostly within permissible limit though close to the threshold in some months.

Graph 39: The graph shows the monthly variation of *Nickel* at Garudeshwar Station from April 2015 to April 2025. Nickel ranged from 0.00 to 0.15 mg/L, permissible limit (BIS IS10500 for heavy metals): 0.02 mg/L, around limit in some months; monitor for heavy inputs.

Graph 40: The graph shows the monthly variation of *Magnesium* at Garudeshwar Station from April 2015 to April 2025. Magnesium ranged from 4.0 to 70.0 mg/L, within expected ranges, but some areas need improvement.

Graph 41: The graph shows the monthly variation of *Fluoride* at Garudeshwar Station from April 2015 to April 2025. Fluoride ranged from 0.04 to 1.0 mg/L, permissible limit (BIS IS10500): 1.5 mg/L, within permissible limit.

Graph 42: The graph shows the monthly variation of *Fixed Dissolved Solids* at Garudeshwar Station from April 2015 to April 2025. Fixed Dissolved Solids ranged from 40 to 272 (reported with TDS), within typical ranges.

Graph 43: The graph shows the monthly variation of *Fecal Coliform* at Garudeshwar Station from April 2015 to April 2025. Fecal Coliform ranged from 2 to 61 MPN/100mL, permissible

limit for drinking: 0 MPN/100mL, indicates microbial contamination; not acceptable for drinking without treatment.

Graph 44: The graph shows the monthly variation of *Dissolved Oxygen* at Garudeshwar Station from April 2015 to April 2025. DO ranged from 6.0 to 11 mg/L, CPCB ecological guideline: $\text{DO} \geq 6 \text{ mg/L}$ for good water, often meets ecological guideline but occasional dips noted.

Graph 45: The graph shows the monthly variation of *Conductivity* at Garudeshwar Station from April 2015 to April 2025. Conductivity ranged from 142 to 639 $\mu\text{S/cm}$, within typical freshwater ranges.

Graph 46: The graph shows the monthly variation of *Chloride* at Garudeshwar Station from April 2015 to April 2025. Chloride ranged from 10 to 180 mg/L, permissible limit (BIS IS10500): 250 mg/L, within permissible limit.

Graph 47: The graph shows the monthly variation of *C.O.D.* at Garudeshwar Station from April 2015 to April 2025. COD ranged from 4 to 19 mg/L, shows periodic organic/chemical loading; some months are elevated.

Graph 48: The graph shows the monthly variation of *Calcium* at Garudeshwar Station from April 2015 to April 2025. Calcium ranged from 16 to 124 mg/L, contributes to hardness; within typical ranges while some months are elevated.

Graph 49: The graph shows the monthly variation of *Boron* at Garudeshwar Station from April 2016 to April 2025. Boron ranged from 0.03 to 139 mg/L, within typical drinking-water guidance.

Graph 50: The graph shows the monthly variation of *B.O.D (3 Days 27°C)* at Garudeshwar Station from April 2015 to April 2025. BOD ranged from 0.28 to 1.88 mg/L, CPCB bathing criterion: $\text{BOD} \leq 3 \text{ mg/L}$, within permissible ranges.

At Garudeshwar water quality is broadly acceptable for many chemical indicators (TDS, total hardness, sulphate, chloride, pH, DO and BOD generally within ecological/permissible ranges), but it shows important episodic and chronic risks that preclude direct potable use without treatment — notable are turbidity spikes up to 84 NTU and persistent faecal coliforms (2–61 MPN/100 mL) which indicate surface runoff/sewage contamination and demand urgent attention (filtration/coagulation and disinfection). Several chemical red flags also appear intermittently: very high reported nitrate peaks (up to 291 mg/L) and nickel (to ~0.15 mg/L)

exceed drinking limits and require source-tracing and mitigation, and extreme sodium episodes (to 1000 mg/L) affect usability of the water.

2.2.3. Station: Zantor

Zantor is located further downstream and benefits from dilution effects but is still influenced by localized activities. **Graph 51 to Graph 77** show the water quality parameters with time from 2015 to 2025 of Zantor Station.

Organic Pollution: BOD and COD are low and improving, reflecting reduced pollution loads. Monsoon periods bring temporary increases due to runoff, but these return to baseline quickly.

Dissolved Oxygen: DO levels remain high and stable (>6 mg/L), sustaining aquatic biodiversity.

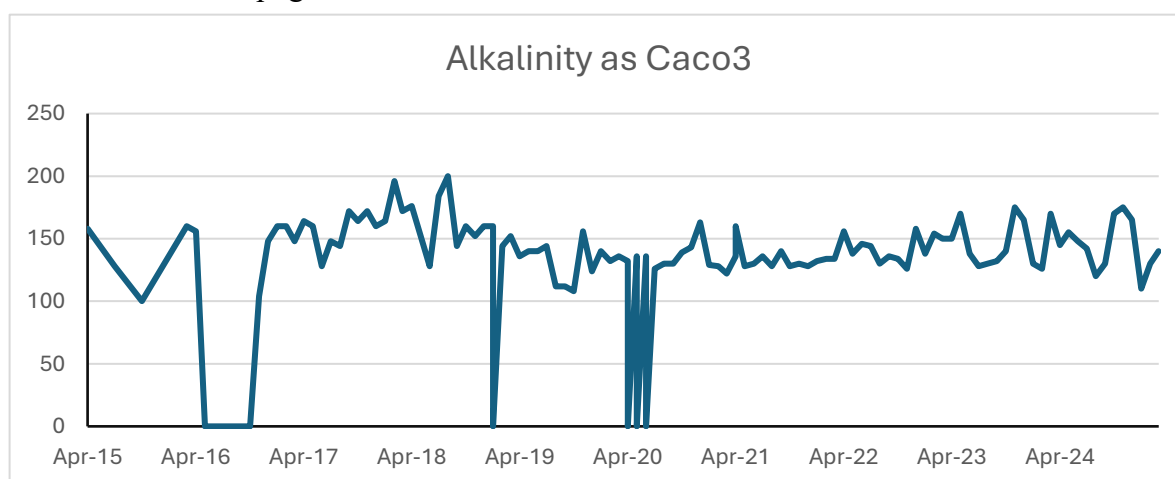
Nutrients: Nitrate and TKN are moderate year-round. Seasonal phosphate peaks occur in monsoon, but concentrations remain within acceptable limits.

Microbial Load: Moderate coliform presence is observed, increasing during monsoon due to inflows from nearby agricultural areas and domestic sewage.

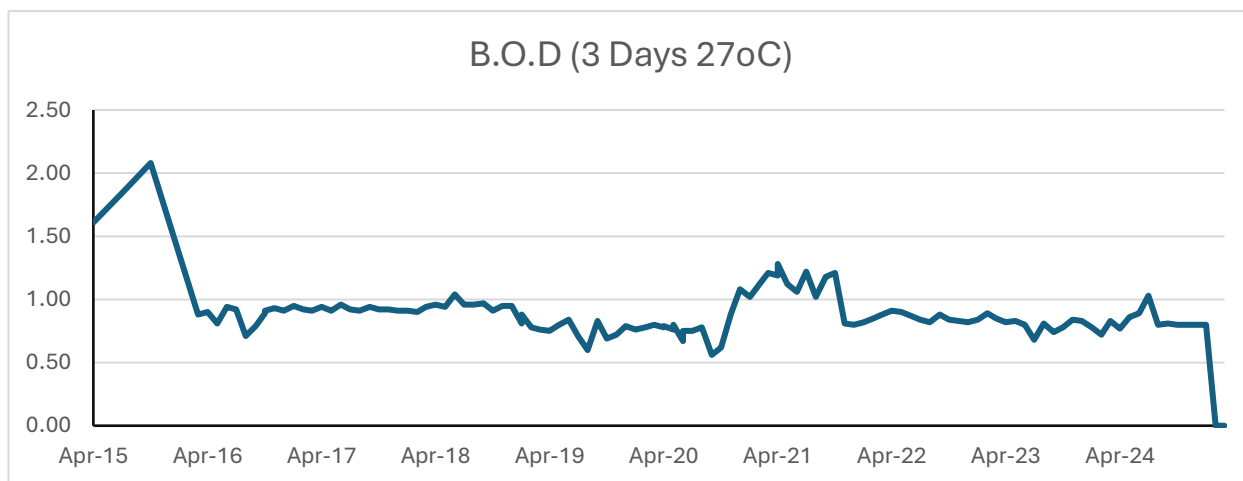
Metals: Heavy metals are generally absent or within safe limits; iron occasionally reaches moderate levels during high flow periods.

Salinity & Hardness: Chloride, conductivity, and hardness are stable and acceptable, with no salinity hazards.

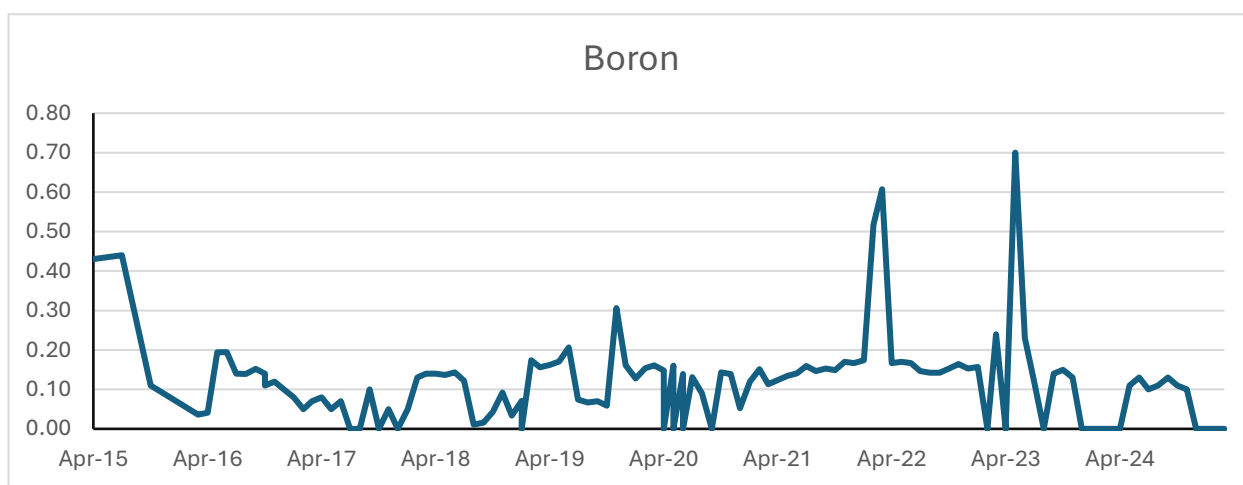
Overall Condition: Zantor demonstrates good water quality, showing better conditions than Garudeshwar, with minor seasonal microbial concerns. It benefits from natural purification and reduced direct anthropogenic stress.



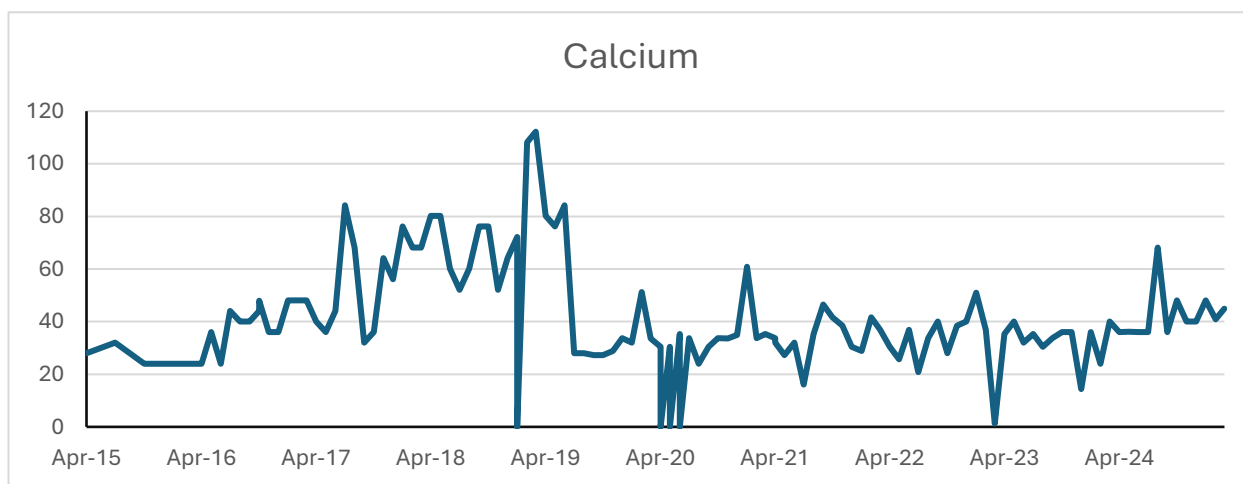
Graph 51 Monthly variation of Alkalinity as CaCo₃ at Zanor Station (2015-2025)



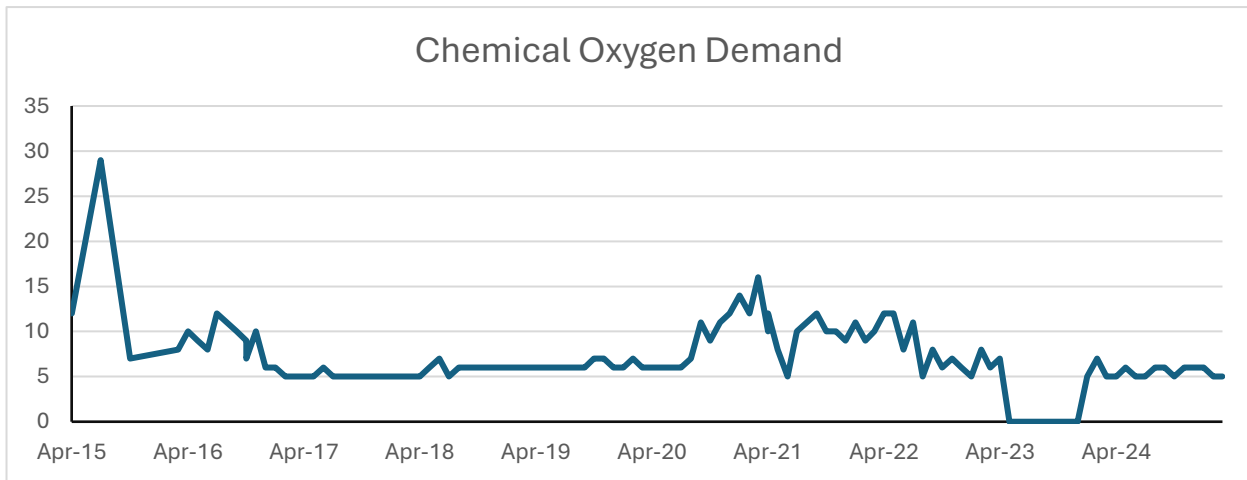
Graph 52 Monthly variation of B.O.D. (3 Days 27 °C at Zanor Station (2015-2025)



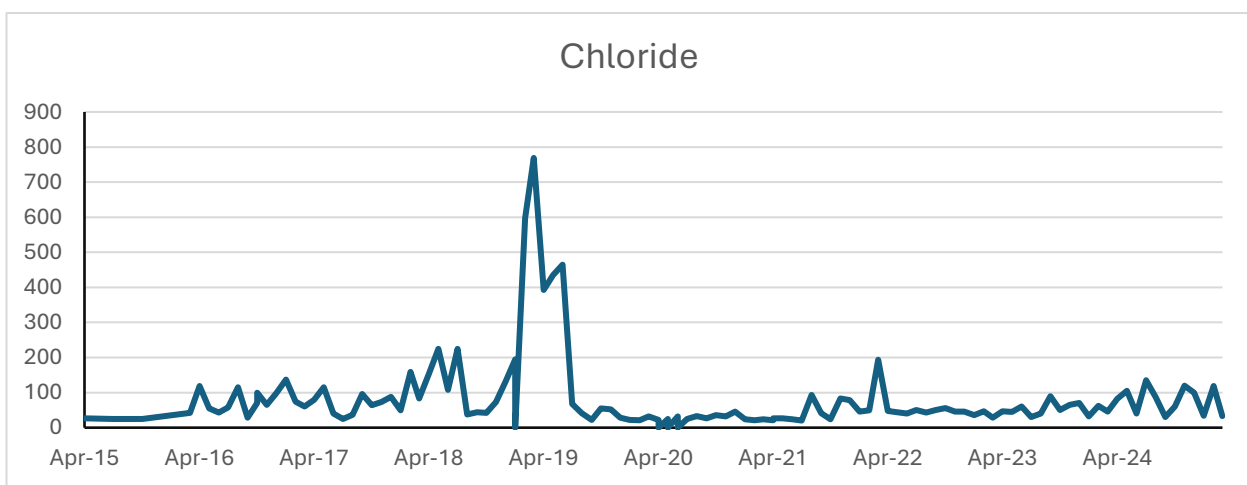
Graph 53 Monthly variation of Boron at Zanor Station (2015-2025)



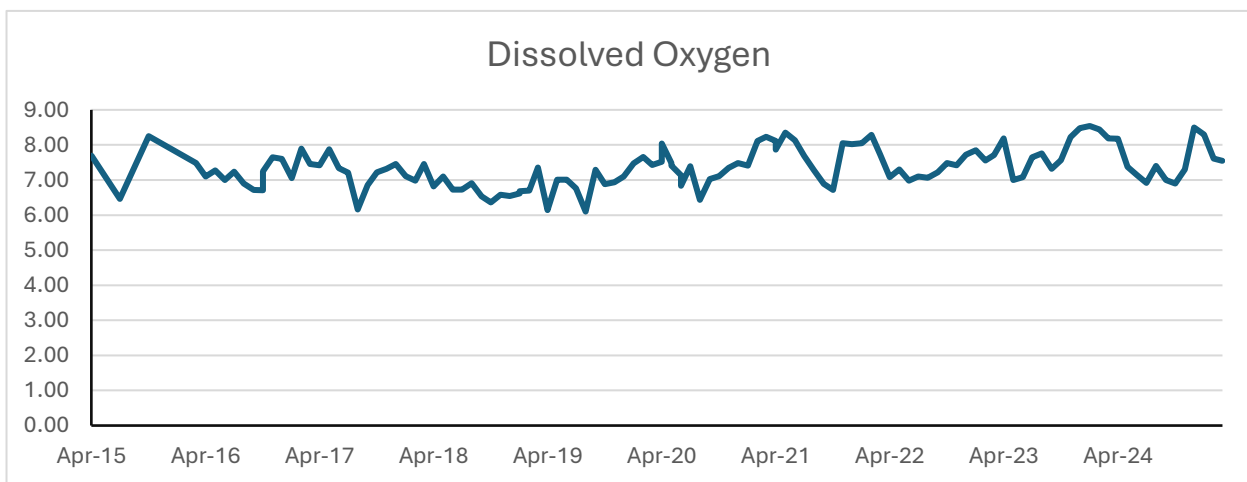
Graph 54 Monthly variation of Calcium at Zanor Station (2015-2025)



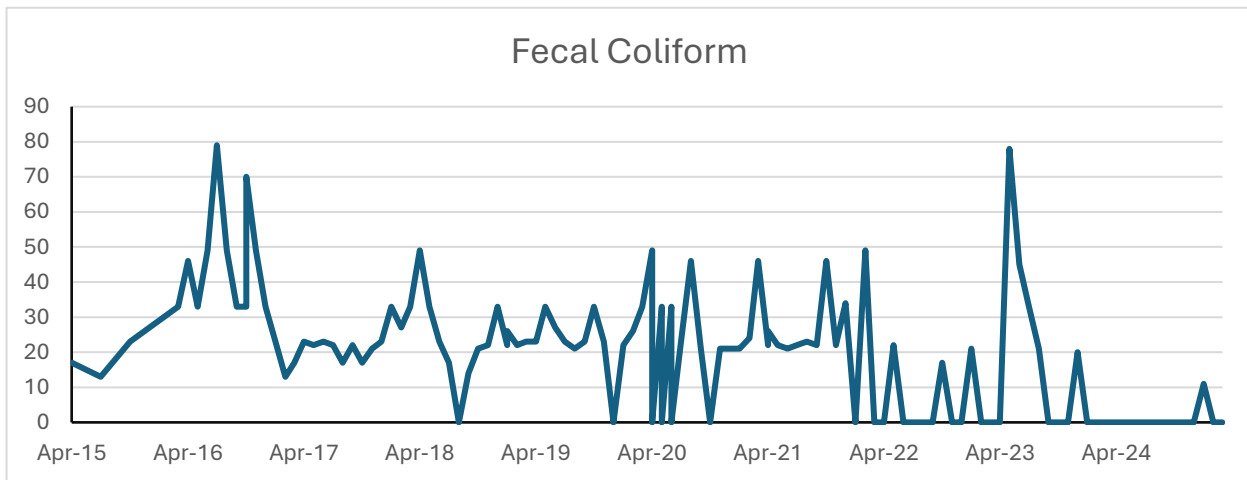
Graph 55 Monthly variation of Chemical Oxygen Demand at Zanor Station (2015-2025)



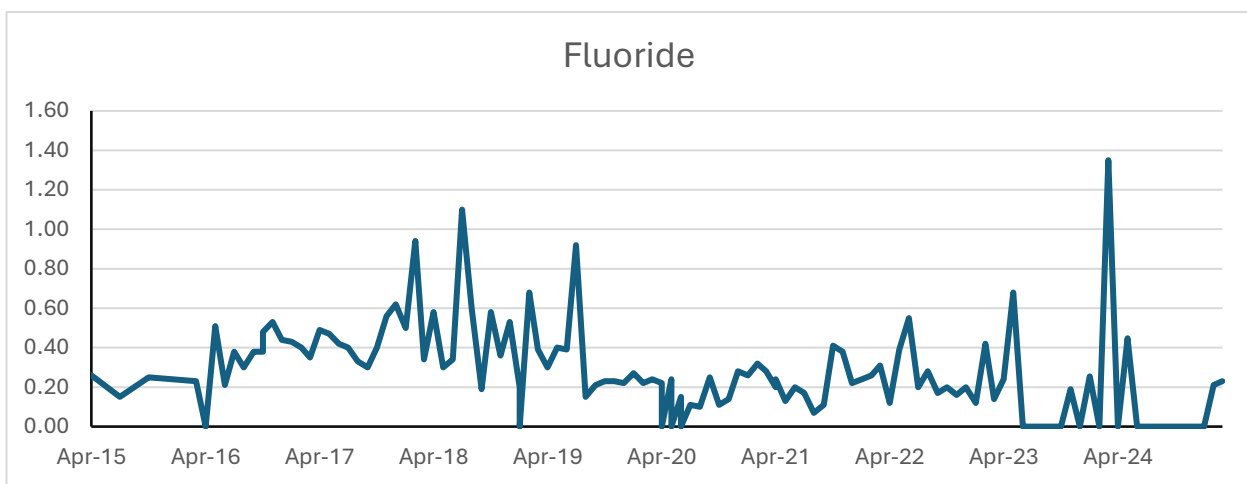
Graph 56 Monthly variation of Chloride at Zanor Station (2015-2025)



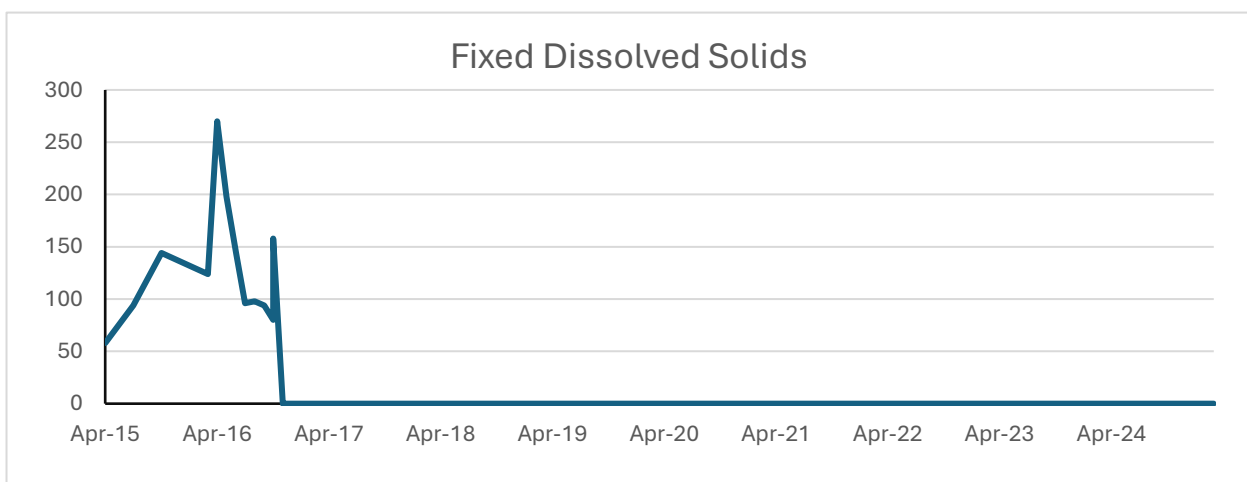
Graph 57 Monthly variation of Dissolved Oxygen at Zanor Station (2015-2025)



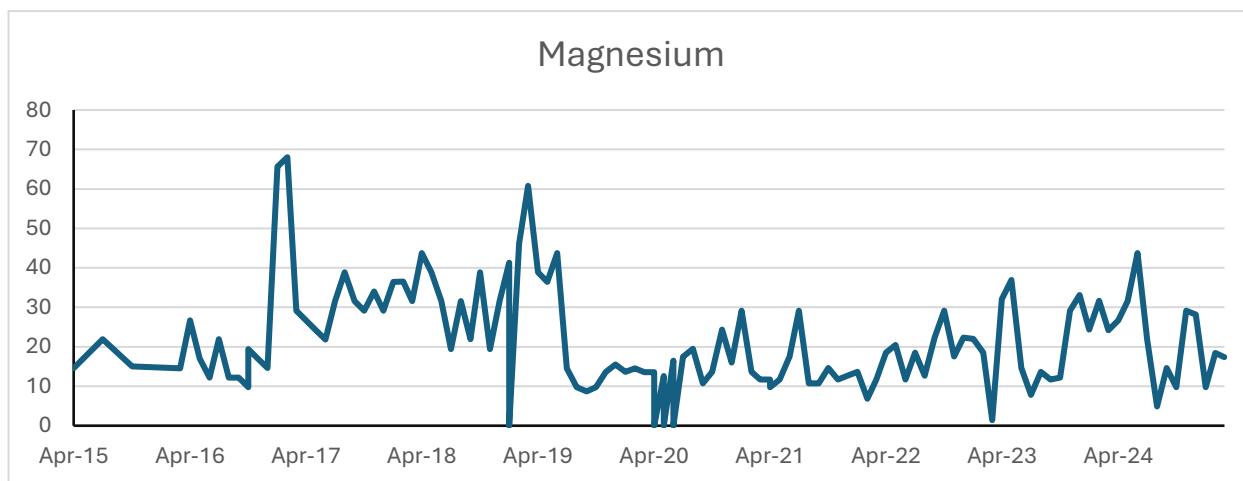
Graph 58 Monthly variation of Fecal Coliform at Zanor Station (2015-2025)



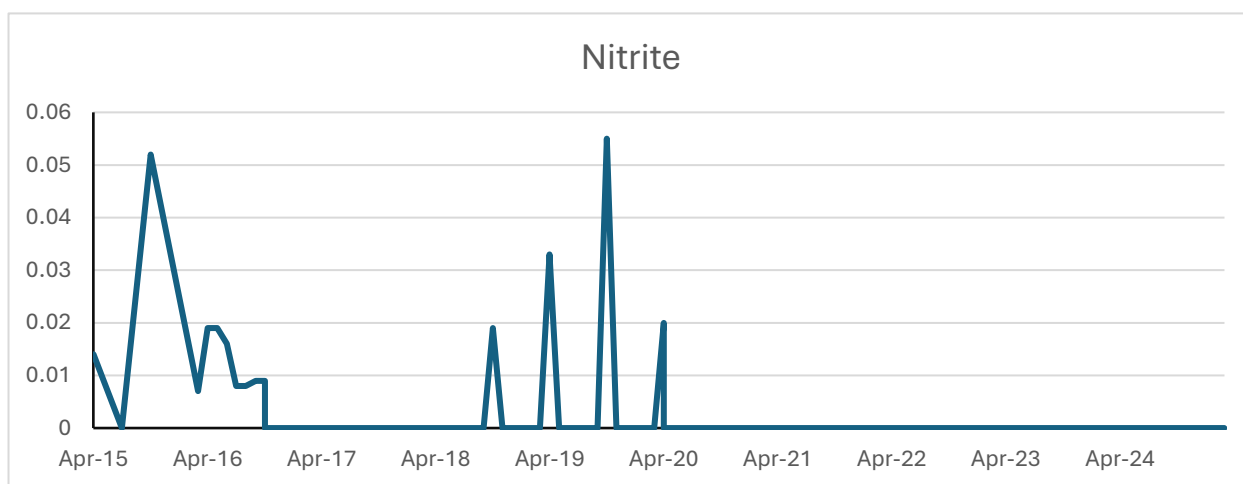
Graph 59 Monthly variation of Fluoride at Zanor Station (2015-2025)



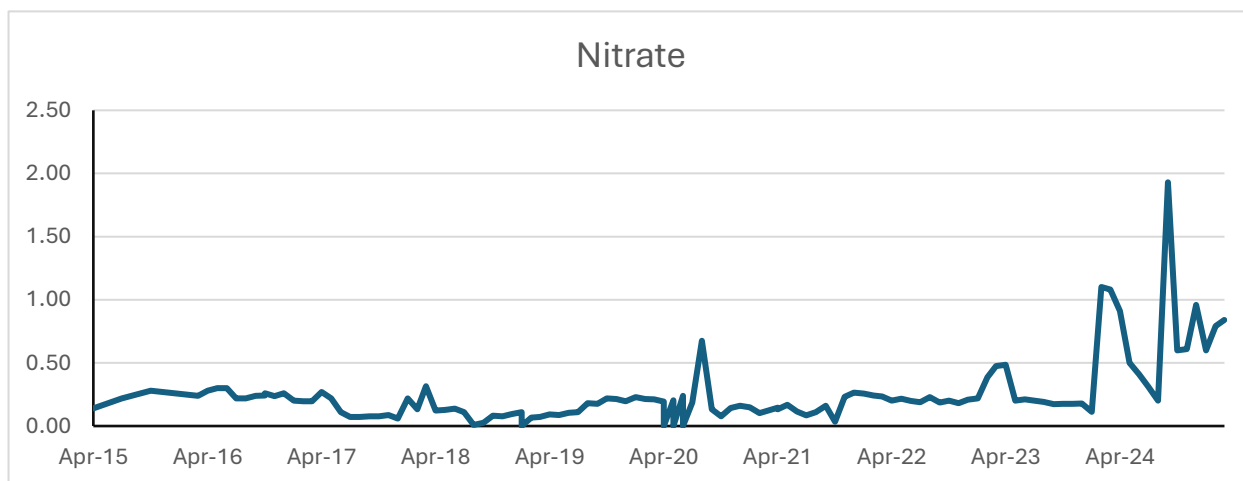
Graph 60 Monthly variation of Fixed Dissolved Solids at Zanor Station (2015-2025)



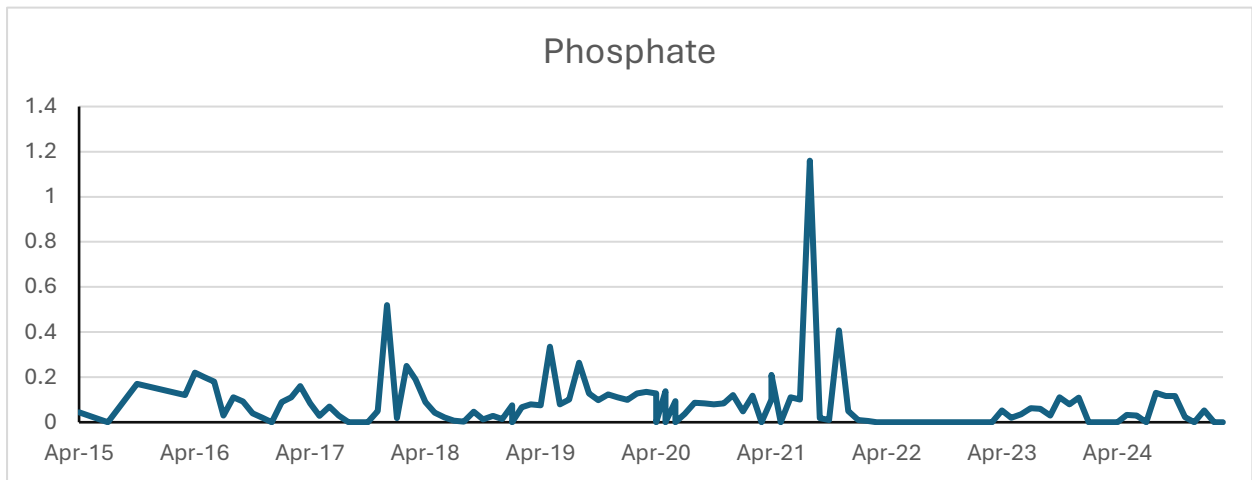
Graph 61 Monthly variation of Magnesium at Zantor Station (2015-2025)



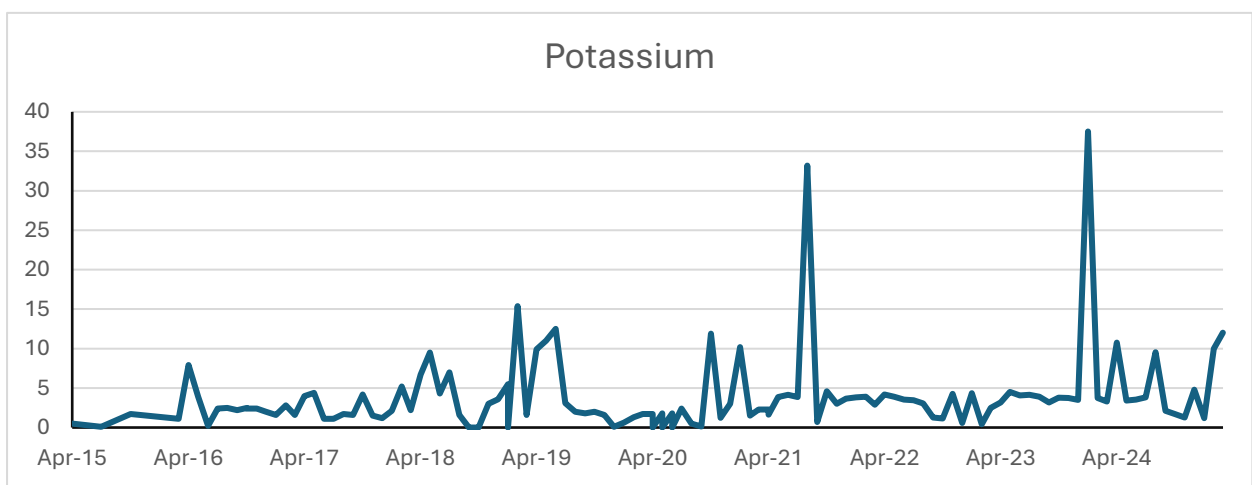
Graph 62 Monthly variation of Nitrite at Zantor Station (2015-2025)



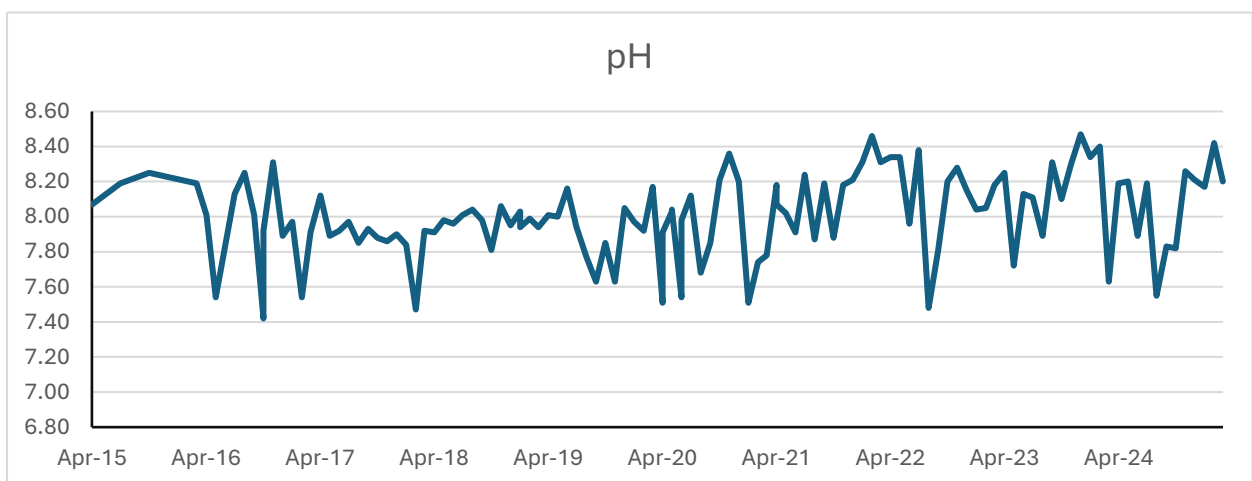
Graph 63 Monthly variation of Nitrate at Zantor Station (2015-2025)



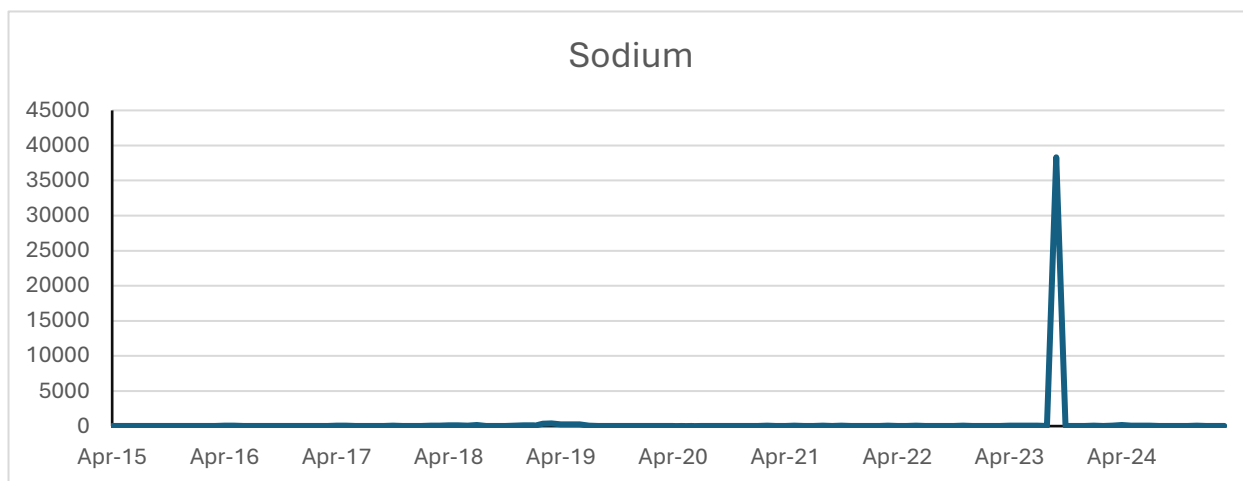
Graph 64 Monthly variation of Phosphate at Zanor Station (2015-2025)



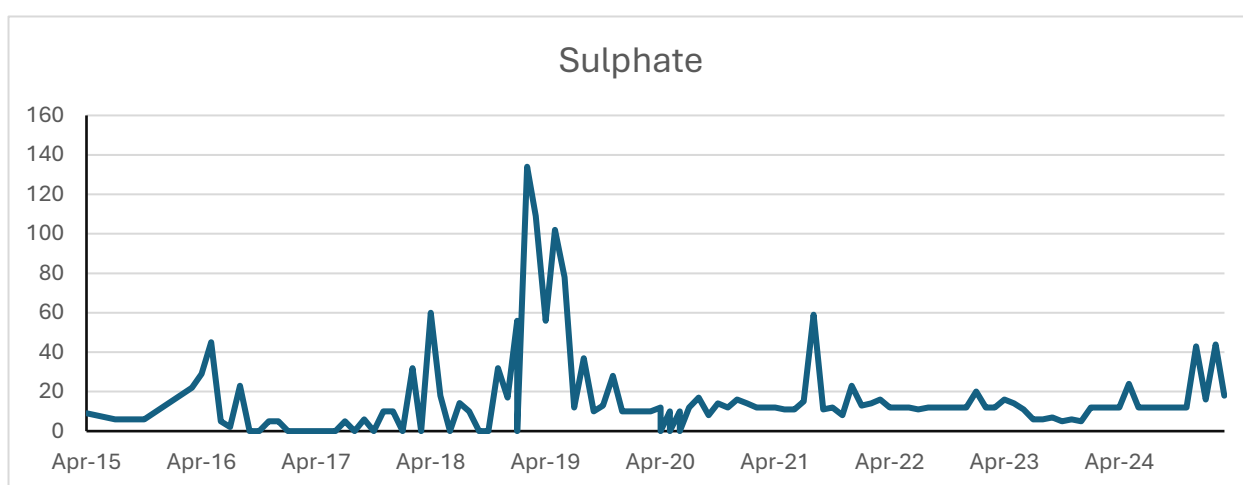
Graph 65 Monthly variation of Potassium at Zanor Station (2015-2025)



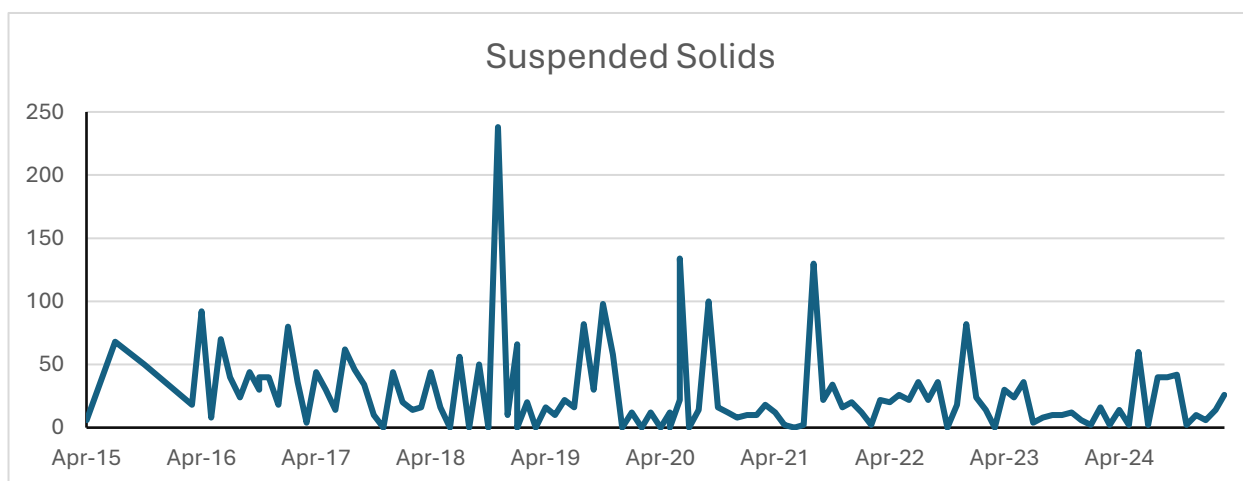
Graph 66 Monthly variation of pH at Zanor Station (2015-2025)



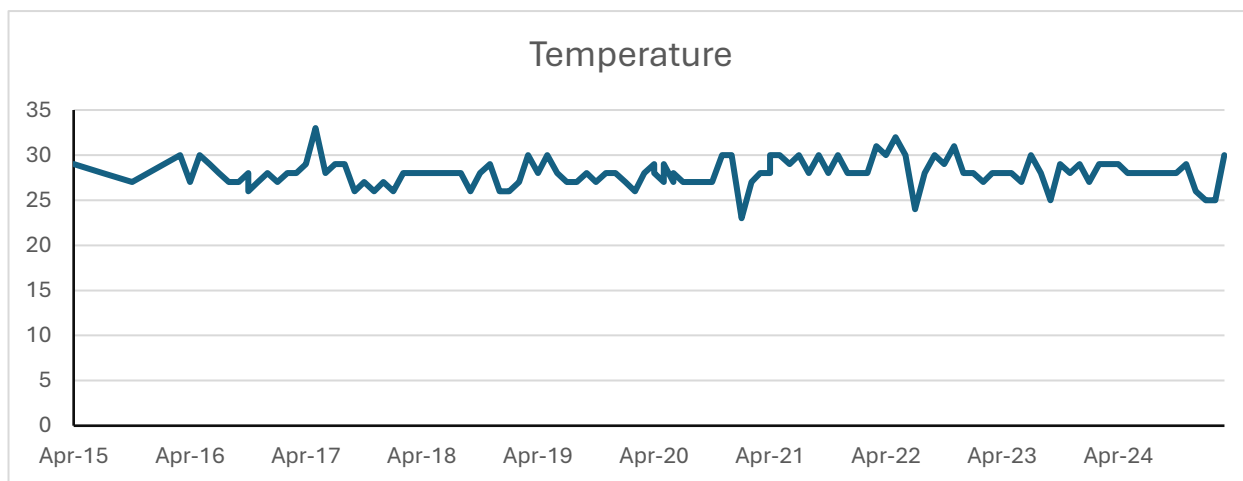
Graph 67 Monthly variation of Sodium at Zanor Station (2015-2025)



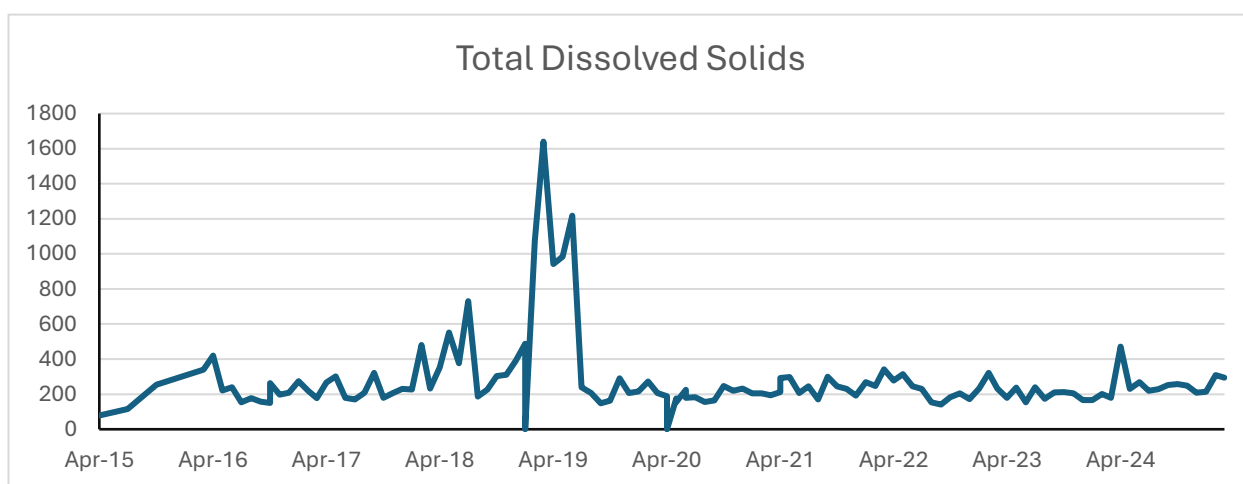
Graph 68 Monthly variation of Sulphate at Zanor Station (2015-2025)



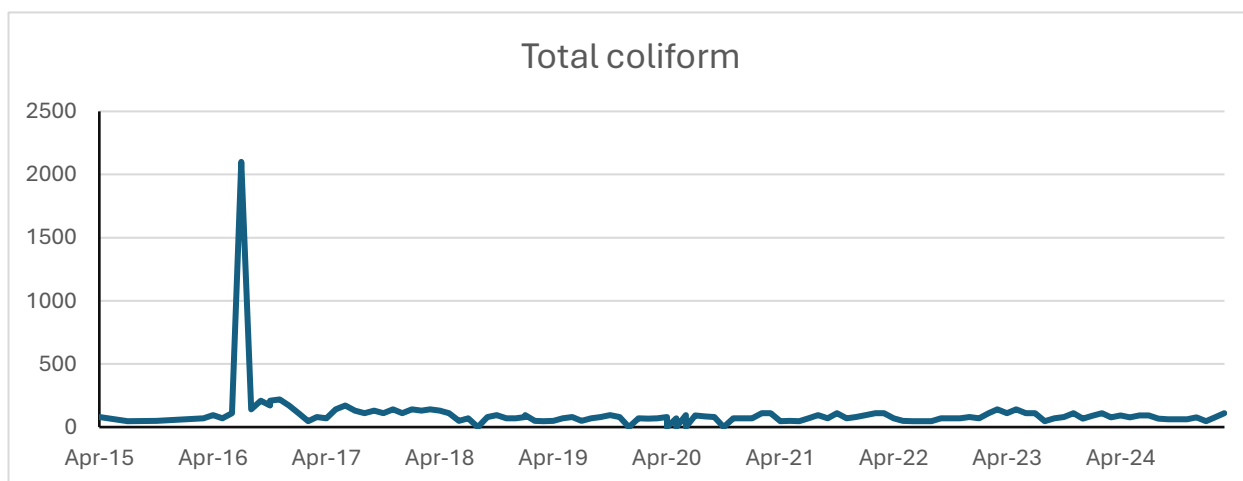
Graph 69 Monthly variation of Suspended Solids at Zanor Station (2015-2025)



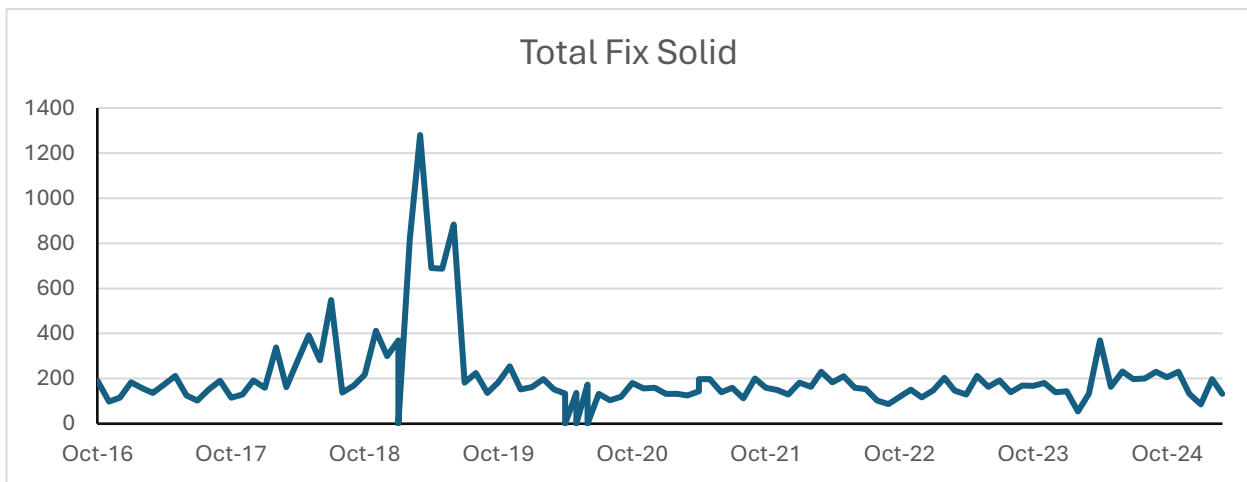
Graph 70 Monthly variation of Temperature at Zantor Station (2015-2025)



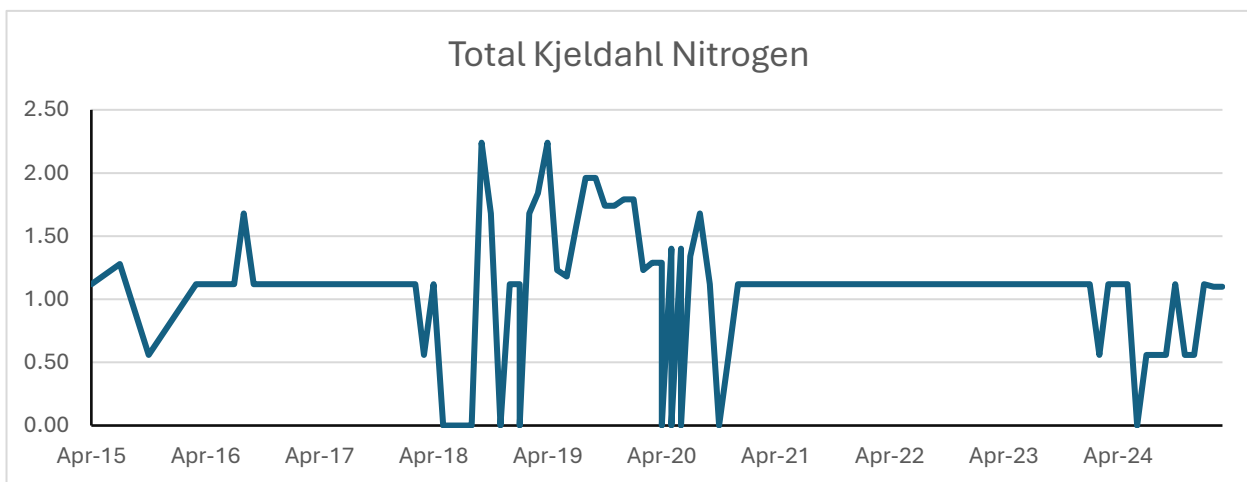
Graph 71 Monthly variation of Total Dissolved Solids at Zantor Station (2015-2025)



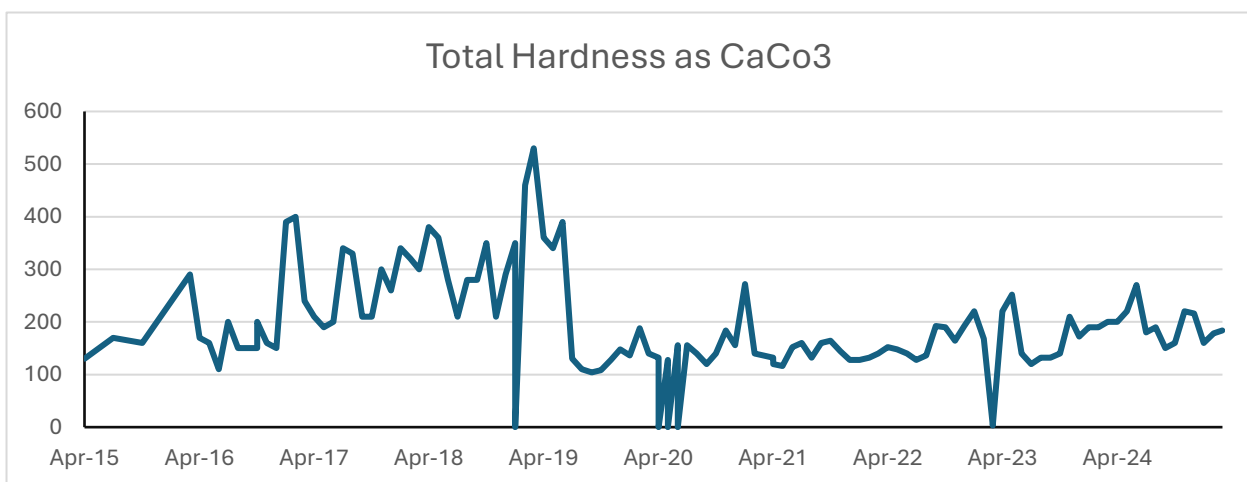
Graph 72 Monthly variation of Total Coliform at Zantor Station (2015-2025)



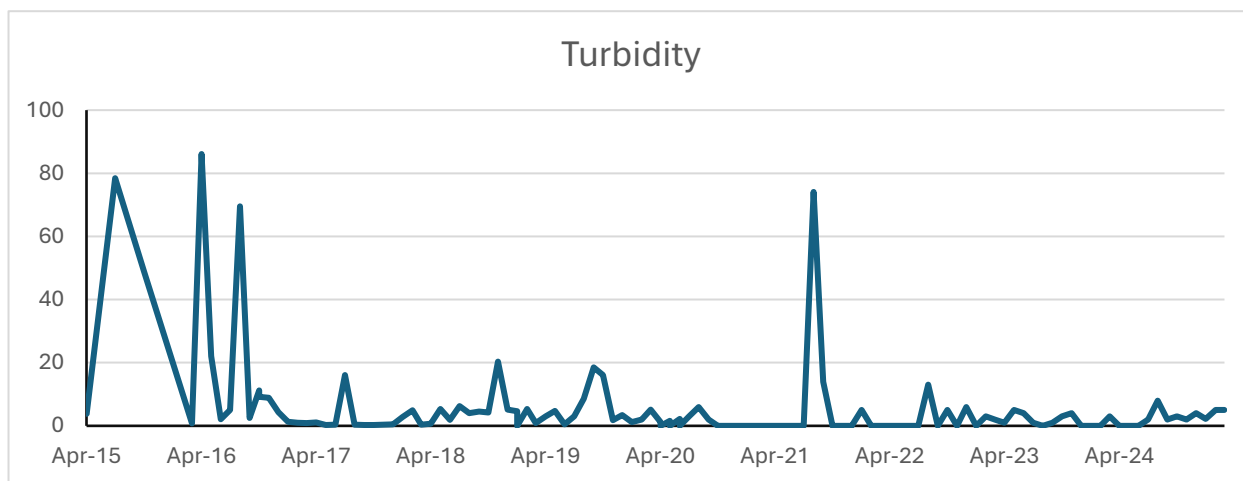
Graph 73 Monthly variation of Total Fix Solid at Zanor Station (2015-2025)



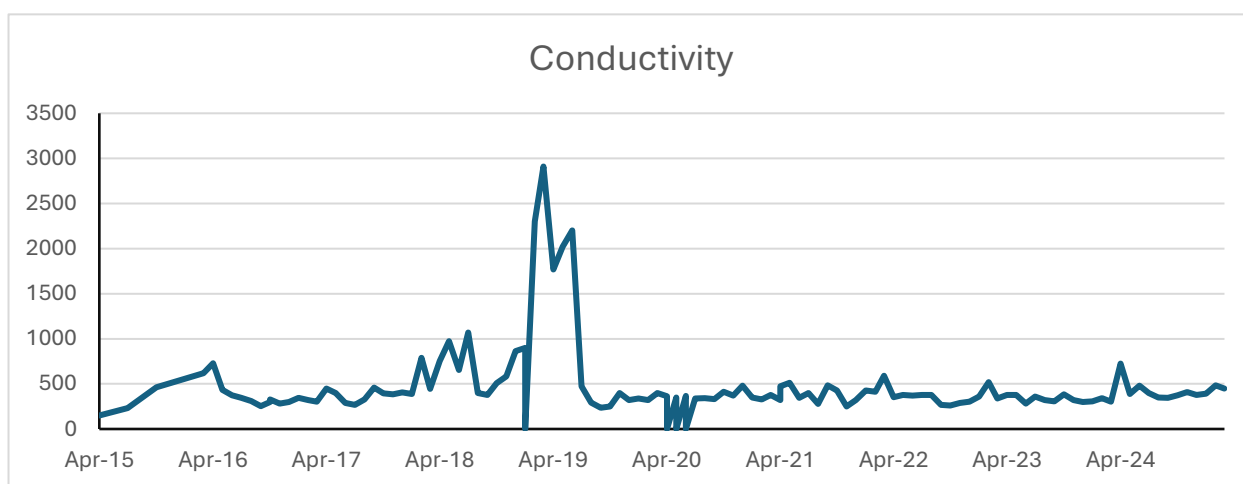
Graph 74 Monthly variation of Total Kjeldahl Nitrogen at Zanor Station (2015-2025)



Graph 75 Monthly variation of Total Hardness as CaCo₃ at Zanor Station (2015-2025)



Graph 76 Monthly variation of Turbidity at Zanol Station (2015-2025)



Graph 77 Monthly variation of Conductivity at Zanol Station (2015-2025)

Graph 51: The graph shows the monthly variation of *Alkalinity as CaCo₃* at Zanol Station from April 2015 to April 2025. Alkalinity ranged from 100 to 180 mg/L, within typical ranges.

Graph 52: The graph shows the monthly variation of *B.O.D. (3 Days 27 °C)* at Zanol Station from April 2015 to April 2025. BOD ranged from 0.56 to 2.5 mg/L, CPCB bathing criterion: $BOD \leq 3$ mg/L, within permissible ranges.

Graph 53: The graph shows the monthly variation of *Boron* at Zanol Station from April 2015 to April 2025. Boron ranged from 0.01 to 0.8 mg/L, within drinking-water guidance.

Graph 54: The graph shows the monthly variation of *Calcium* at Zanol Station from April 2015 to April 2025. Calcium ranged from 1.18 to 112 mg/L, some months above typical benchmarks.

Graph 55: The graph shows the monthly variation of *Chemical Oxygen Demand* at Zantor Station from April 2015 to April 2025. COD ranged from 6 to 30 mg/L, moderate; occasional increases reflect organic inputs.

Graph 56: The graph shows the monthly variation of *Chloride* at Zantor Station from April 2015 to April 2025. Chloride ranged from 20 to 780 mg/L, permissible limit (BIS IS10500): 250 mg/L, exceeds permissible limit in some areas.

Graph 57: The graph shows the monthly variation of *Dissolved Oxygen* at Zantor Station from April 2015 to April 2025. DO ranged from 5.8 to 8.8 mg/L, CPCB ecological guideline: $DO \geq 6$ mg/L, mostly meets ecological criteria, supporting aquatic life.

Graph 58: The graph shows the monthly variation of *Fecal Coliform* at Zantor Station from April 2015 to April 2025. Fecal Coliform ranged from 15 to 80 MPN/100mL, permissible limit for drinking: 0 MPN/100mL, significant microbial contamination in parts of dataset, needs treatment before use.

Graph 59: The graph shows the monthly variation of *Fluoride* at Zantor Station from April 2015 to April 2025. Fluoride ranged from 0.06 to 1.35 mg/L, permissible limit (BIS IS10500): 1.5 mg/L, within permissible limit.

Graph 60: The graph shows the monthly variation of *Fixed Dissolved Solids* at Zantor Station from April 2015 to April 2025. Fixed Dissolved Solids ranged from 58 to 270 (reported with TDS), within typical ranges.

Graph 61: The graph shows the monthly variation of *Magnesium* at Zantor Station from April 2015 to April 2025. Magnesium ranged from 1.44 to 68.0 mg/L, exceeds permissible limit in some areas.

Graph 62: The graph shows the monthly variation of *Nitrite* at Zantor Station from April 2015 to April 2025. Nitrite ranged from 0.01 to 0.06 mg/L, permissible limit (BIS IS10500): 0.2 mg/L, within permissible limit.

Graph 63: The graph shows the monthly variation of *Nitrate* at Zantor Station from April 2015 to April 2025. Nitrate ranged from 0.01 to 1.93 mg/L, permissible limit (BIS IS10500): 45 mg/L, within permissible limit.

Graph 64: The graph shows the monthly variation of *Phosphate* at Zantor Station from April 2015 to April 2025. Phosphate ranged from 0 to 1.16 mg/L, low to moderate: occasional monsoon peaks.

Graph 65: The graph shows the monthly variation of *Potassium* at Zantor Station from April 2015 to April 2025. Potassium ranged from 0.1 to 38 mg/L, within normal environmental concentrations.

Graph 66: The graph shows the monthly variation of *pH* at Zantor Station from April 2015 to April 2025. pH ranged from 7.42 to 8.5, permissible limit (BIS IS10500): 6.5–8.5, within permissible limits.

Graph 67: The graph shows the monthly variation of *Sodium* at Zantor Station from April 2015 to April 2025. Sodium ranged from 2.0 to 390.0 mg/L, within typical guideline ranges.

Graph 68: The graph shows the monthly variation of *Sulphate* at Zantor Station from April 2015 to April 2025. Sulphate ranged from 2.0 to 140.0 mg/L, permissible limit (BIS IS10500): 200 mg/L, within permissible limit.

Graph 69: The graph shows the monthly variation of *Suspended Solids* at Zantor Station from April 2015 to April 2025. Suspended Solids ranged from 2.0 to 238.0 mg/L, seasonal variability, some monsoon increases.

Graph 70: The graph shows the monthly variation of *Temperature* at Zantor Station from April 2015 to April 2025. Temperature ranged from 23.0 to 33.0 °C, seasonal variation.

Graph 71: The graph shows the monthly variation of *Total Dissolved Solids* at Zantor Station from April 2015 to April 2025. TDS ranged from 78 to 1680 mg/L, permissible limit (BIS IS10500): 2000 mg/L, within permissible limit but some months above desirable 500 mg/L.

Graph 72: The graph shows the monthly variation of *Total Coliform* at Zantor Station from April 2015 to April 2025. Total Coliform ranged from 50 to 10,000 MPN/100mL, permissible limit for drinking: 0 MPN/100mL, CPCB criteria for bathing differ, indicates substantial microbial contamination in parts of the dataset.

Graph 73: The graph shows the monthly variation of *Total Fix Solid* at Zantor Station from April 2015 to April 2025. Total Fixed Solid (reported with TDS) ranged from 54 to 1280 (see TDS), within typical ranges.

Graph 74: The graph shows the monthly variation of *Total Kjeldahl Nitrogen* at Zantor Station from April 2015 to April 2025. TKN ranged from 0.6 to 2.3 mg/L, within expected environmental ranges.

Graph 75: The graph shows the monthly variation of *Total Hardness as CaCo₃* at Zantor Station from April 2015 to April 2025. Total Hardness ranged from 3.3 to 530 mg/L, permissible limit (BIS IS10500): 600 mg/L, within permissible limit.

Graph 76: The graph shows the monthly variation of *Turbidity* at Zantor Station from April 2015 to April 2025. Turbidity ranged from 0.3 to 86, permissible limit (BIS IS10500): 5 NTU, episodic spikes (monsoon/loads) exceed permissible limits and require attention.

Graph 77: The graph shows the monthly variation of *Conductivity* at Zantor Station from April 2015 to April 2025. Conductivity ranged from 60 to 1500 $\mu\text{S}/\text{cm}$, shows ionic variability; some high values reflect elevated dissolved salts in parts of the record.

Zantor's water station parameters show many chemical parameters generally within BIS ecological/permissible ranges (TDS mostly ≤ 1680 mg/L and below the 2000 mg/L drinking threshold, fluoride within limits, hardness usually acceptable), but show alarmingly large microbial contamination (total coliforms 50–10,000 MPN/100 mL; fecal coliforms 15–80 MPN/100 mL) and turbidity spikes to ~86 NTU (BIS desirable ≤ 1 , permissible 5 NTU), plus occasional high chloride (to ~780 mg/L, above the 250 mg/L desirable/limit) and very high conductivity/TDS episodes (conductivity to 1500 $\mu\text{S}/\text{cm}$, TDS up to 1680 mg/L) and elevated COD (to 30 mg/L) that point to organic and saline/ionic inputs.

2.2.4. Station: Zadeshwar

Zadeshwar lies closer to the estuarine zone and is more influenced by human activity, effluent discharge, and tidal mixing. **Graph 78 to Graph 104** show the water quality parameters with time from 2015 to 2025 of Zadeshwar Station.

Organic Pollution: BOD levels are moderate, and COD shows elevated values compared to upstream stations, indicating higher organic and chemical loading from urban and industrial sources.

Dissolved Oxygen: DO is adequate (>5 mg/L) but slightly lower compared to upstream stations, reflecting the combined effect of organic load and reduced re-aeration.

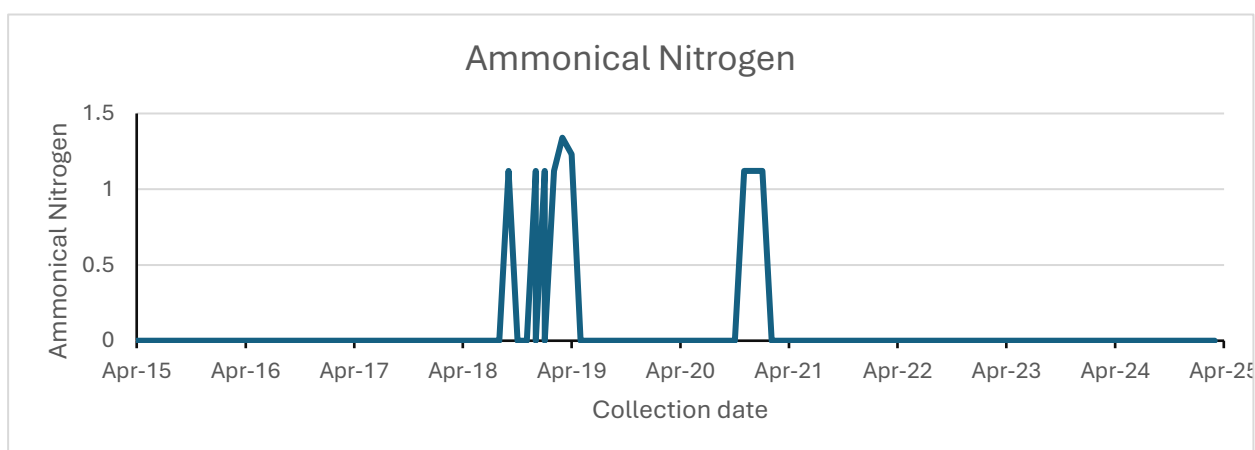
Nutrients: Nitrate and phosphate are moderate to high, reflecting agricultural runoff and untreated sewage inflows. Seasonal spikes during monsoon are notable.

Microbial Load: Coliform counts are high, particularly during monsoon, pointing to substantial sewage and stormwater inflows. This raises concerns for public health and water usability.

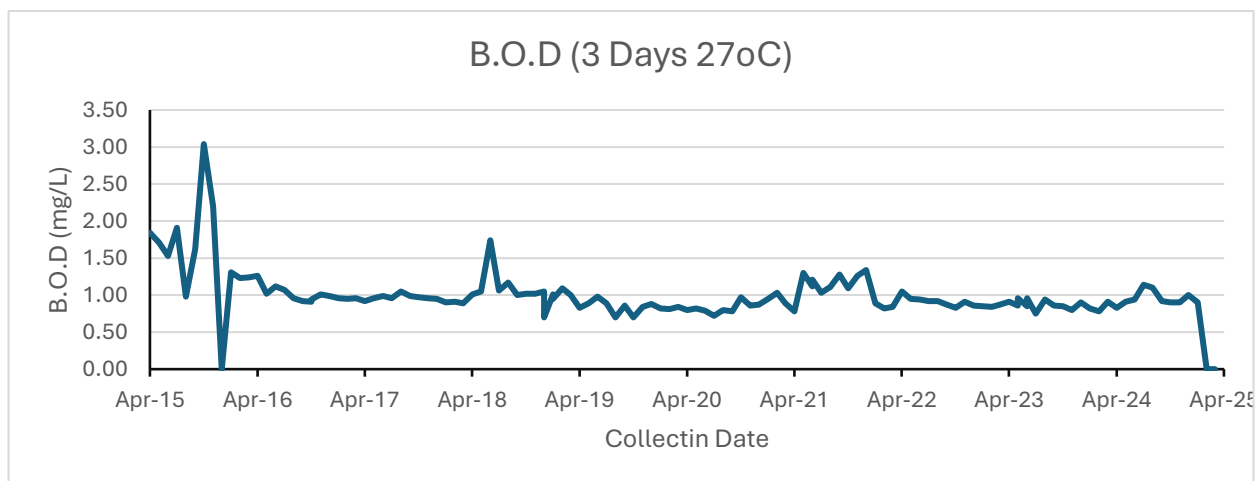
Metals: Iron and manganese occasionally exceed safe limits during monsoon and high tide conditions, likely from industrial discharge and sediment disturbance. Other heavy metals remain within safe ranges.

Salinity & Hardness: Higher chloride and TDS levels are recorded compared to upstream stations, influenced by tidal mixing and estuarine effects.

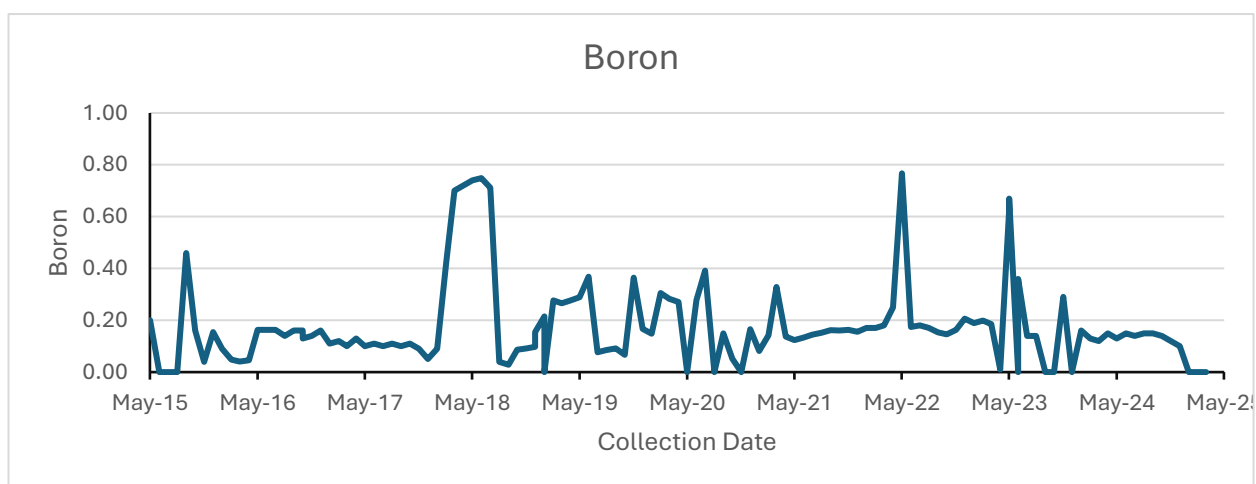
Overall Condition: Zadeshwar shows moderate water quality with higher anthropogenic stress. Microbial contamination and nutrient enrichment are key concerns requiring targeted interventions.



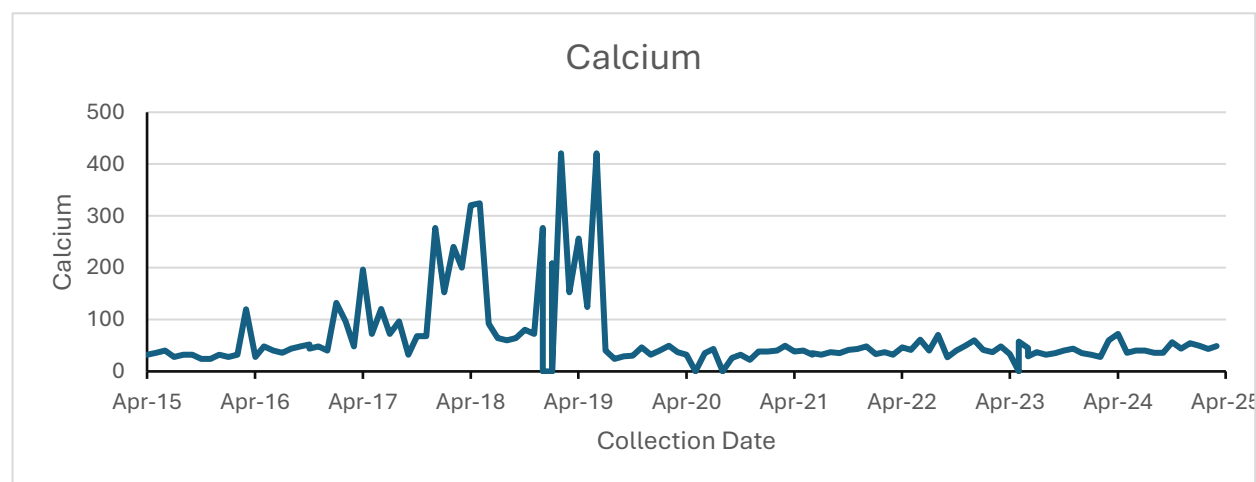
Graph 78 Monthly variation of Ammonical Nitrogen at Zadeshwar Station (2015-2025)



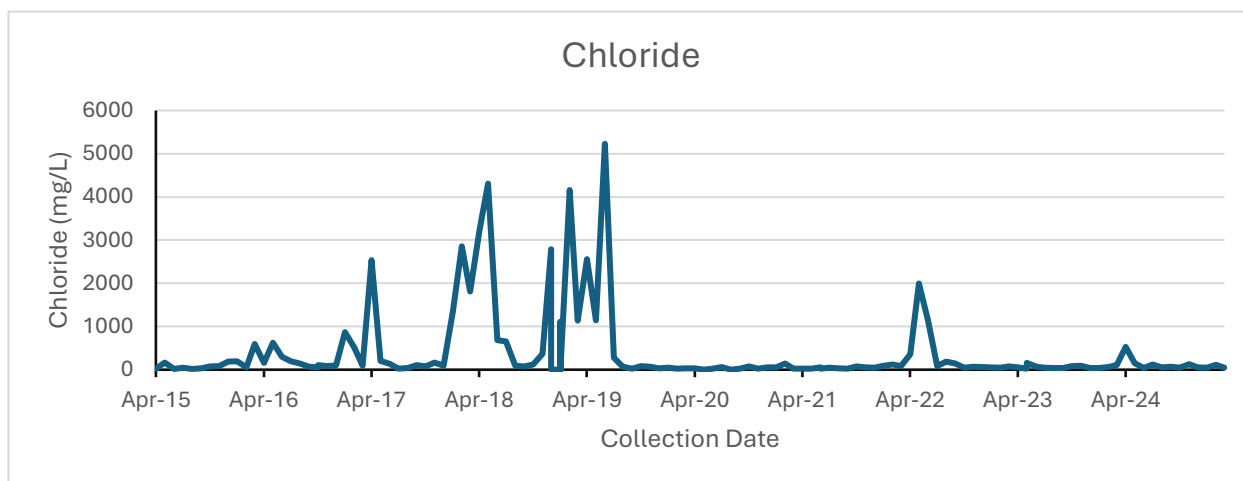
Graph 79 Monthly variation of B.O.D. (3 days 27 °C at Zadeshwar Station (2015-2025))



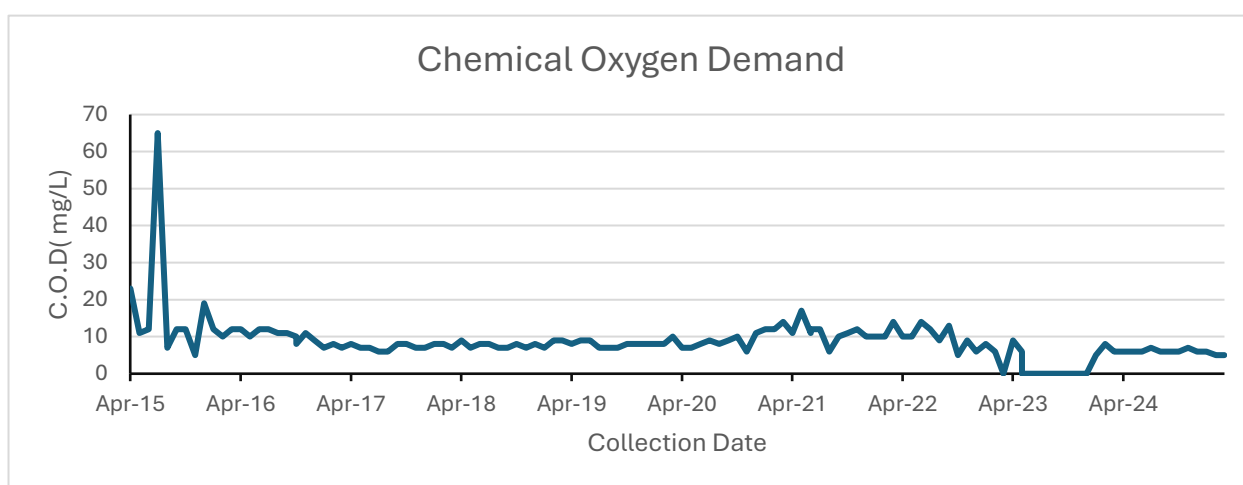
Graph 80 Monthly variation of Boron at Zadeshwar Station (2015-2025)



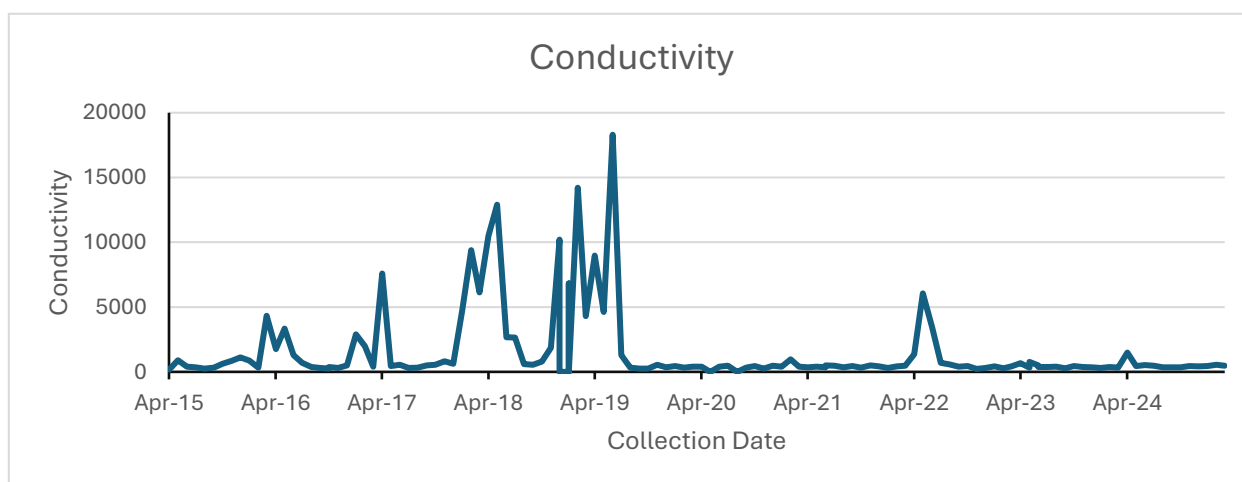
Graph 81 Monthly variation of Calcium at Zadeshwar Station (2015-2025)



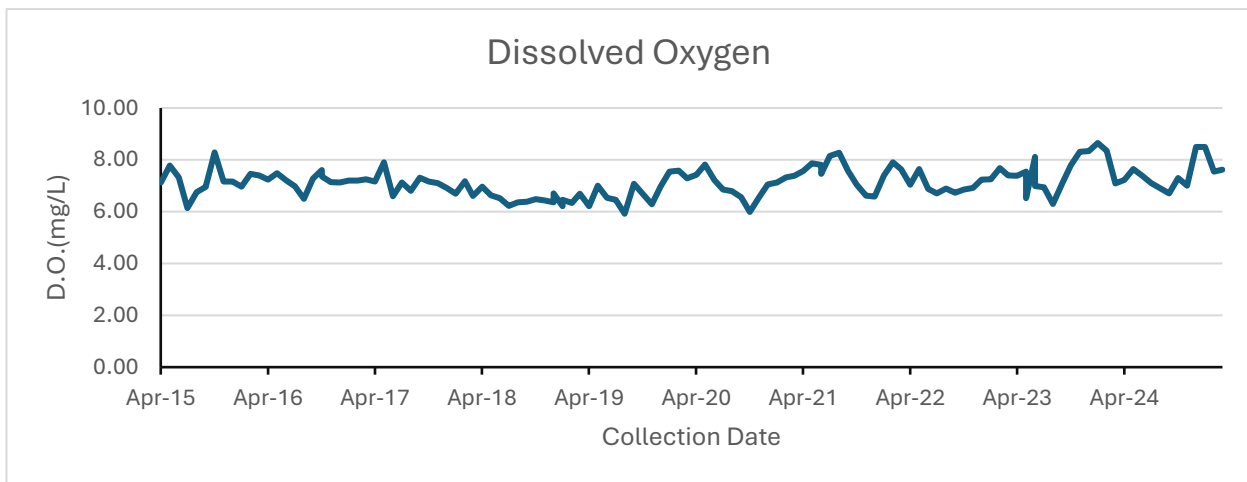
Graph 82 Monthly variation of Chloride at Zadeshwar Station (2015-2025)



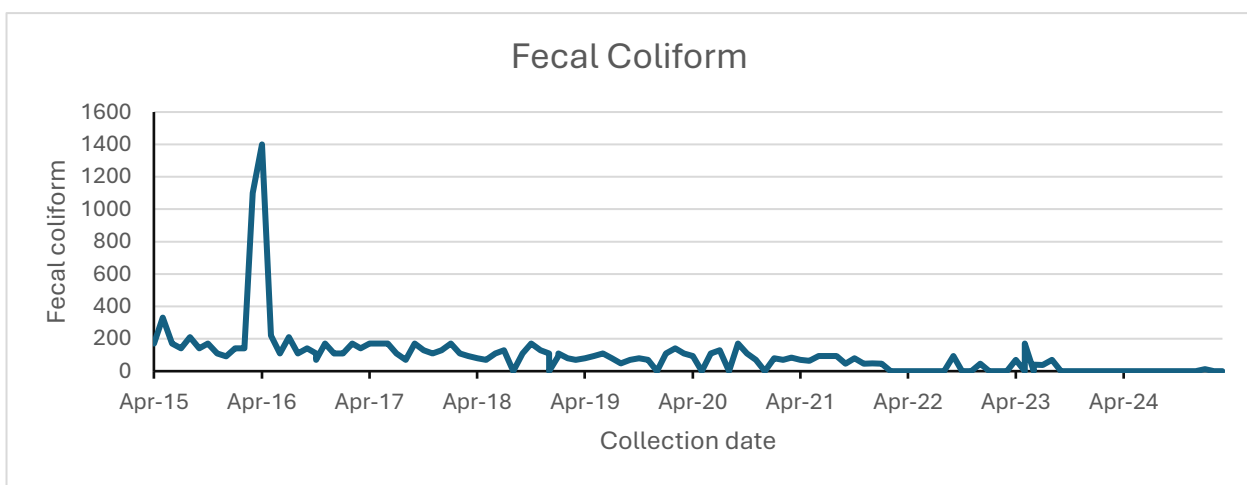
Graph 83 Monthly variation of Chemical Oxygen Demand at Zadeshwar Station (2015-2025)



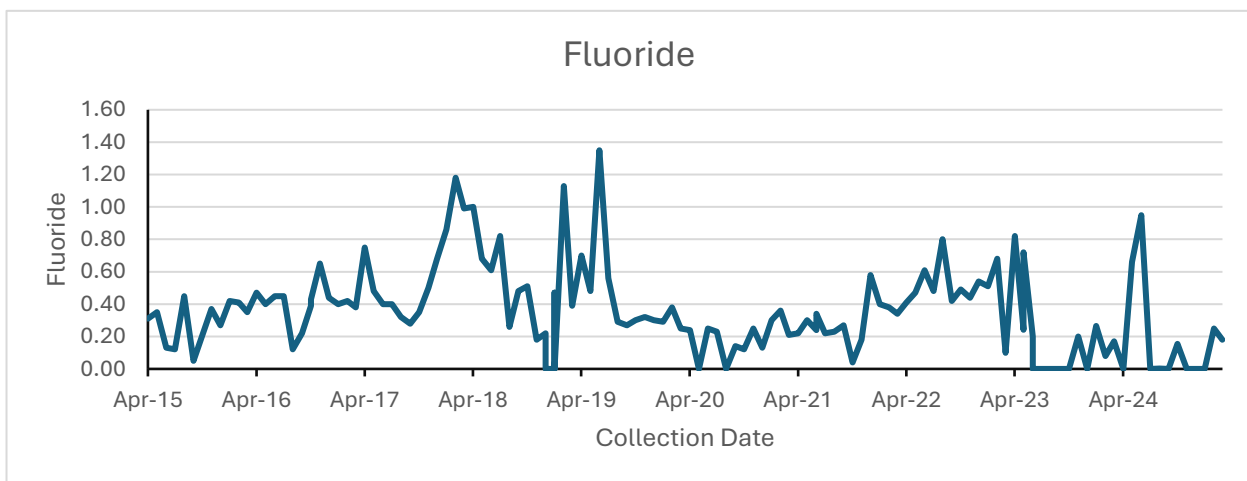
Graph 84 Monthly variation of Conductivity at Zadeshwar Station (2015-2025)



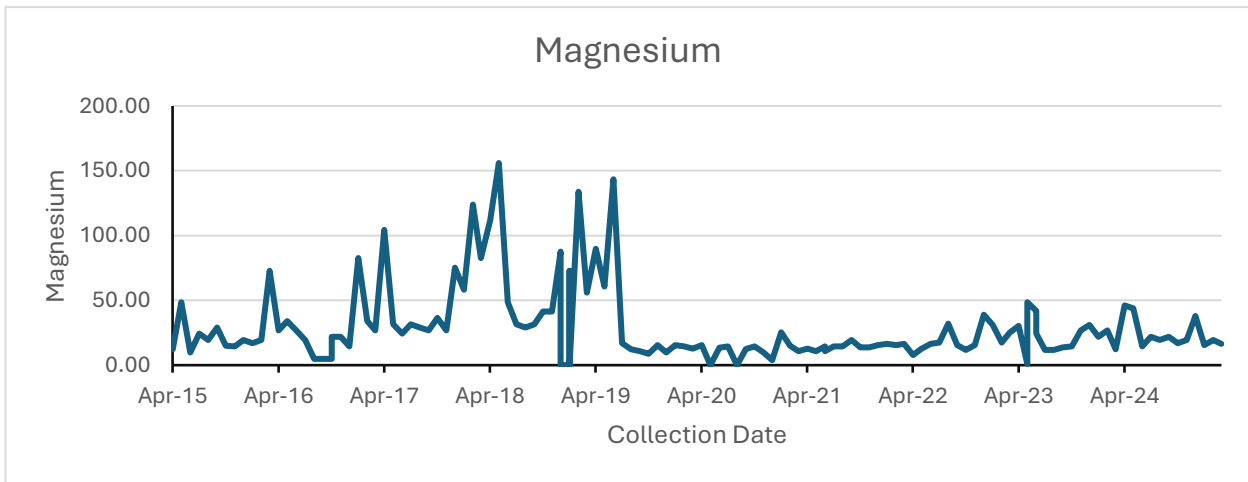
Graph 85 Monthly variation of Dissolved Oxygen at Zadeshwar Station (2015-2025)



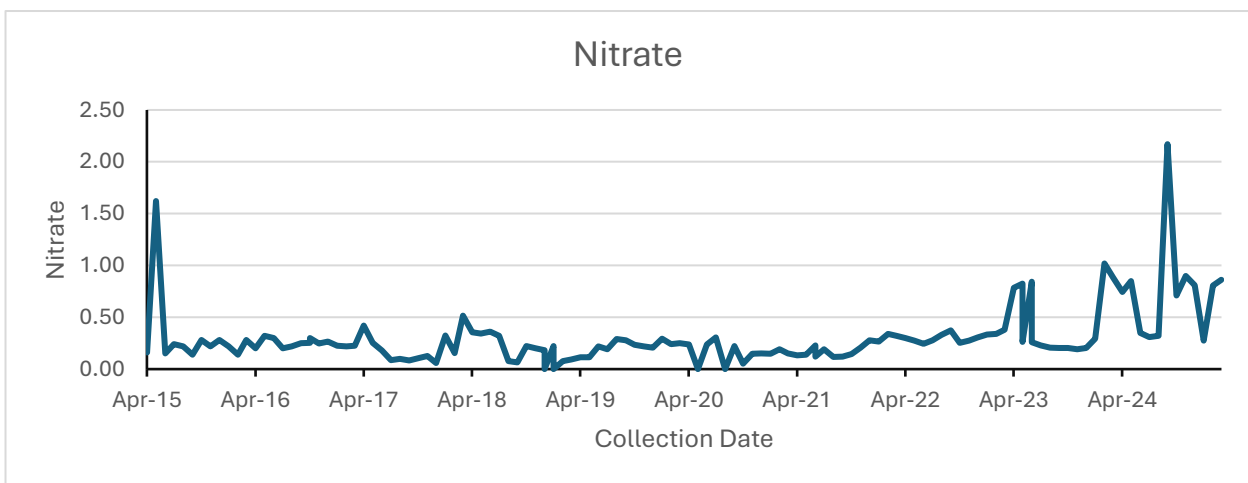
Graph 86 Monthly variation of Fecal Coliform at Zadeshwar Station (2015-2025)



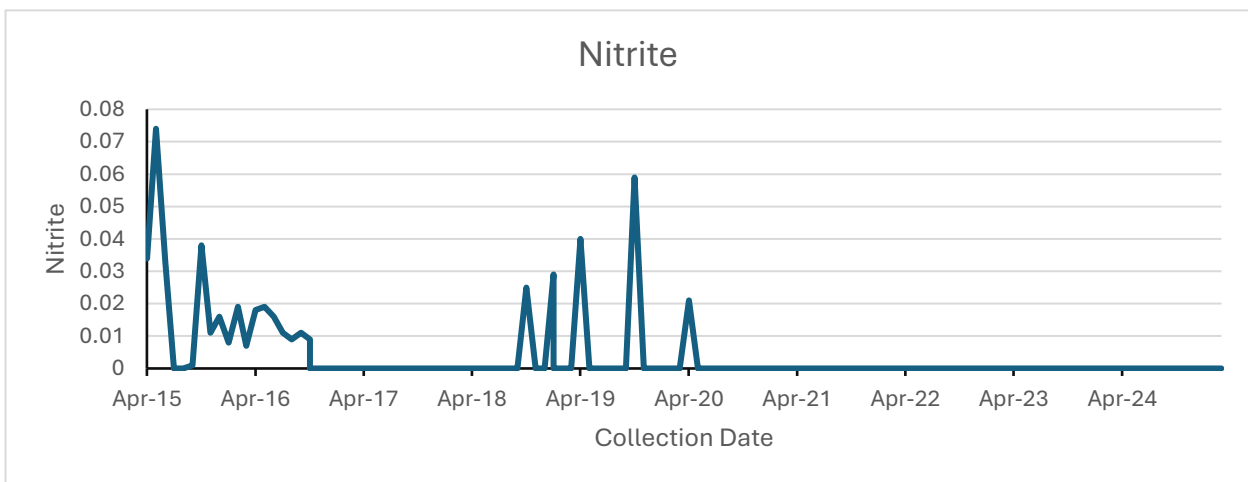
Graph 87 Monthly variation of Fluoride at Zadeshwar Station (2015-2025)



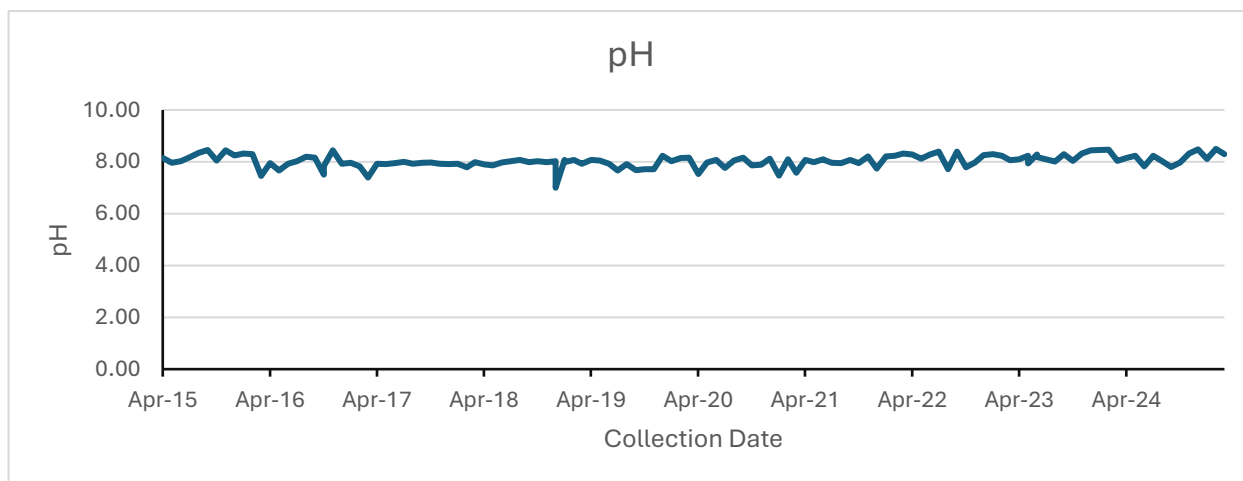
Graph 88 Monthly variation of Magnesium at Zadeshwar Station (2015-2025)



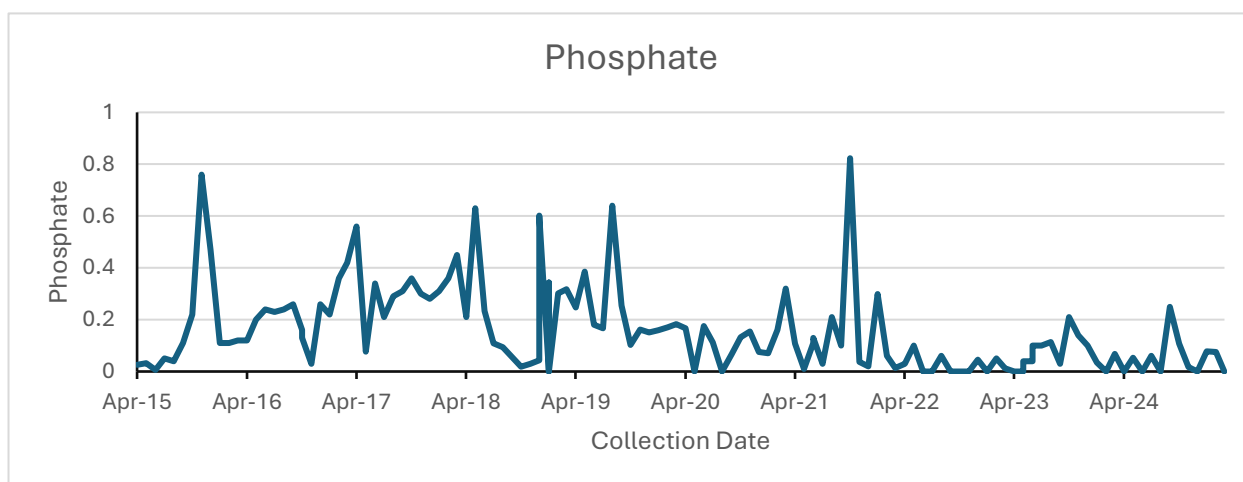
Graph 89 Monthly variation of Nitrate at Zadeshwar Station (2015-2025)



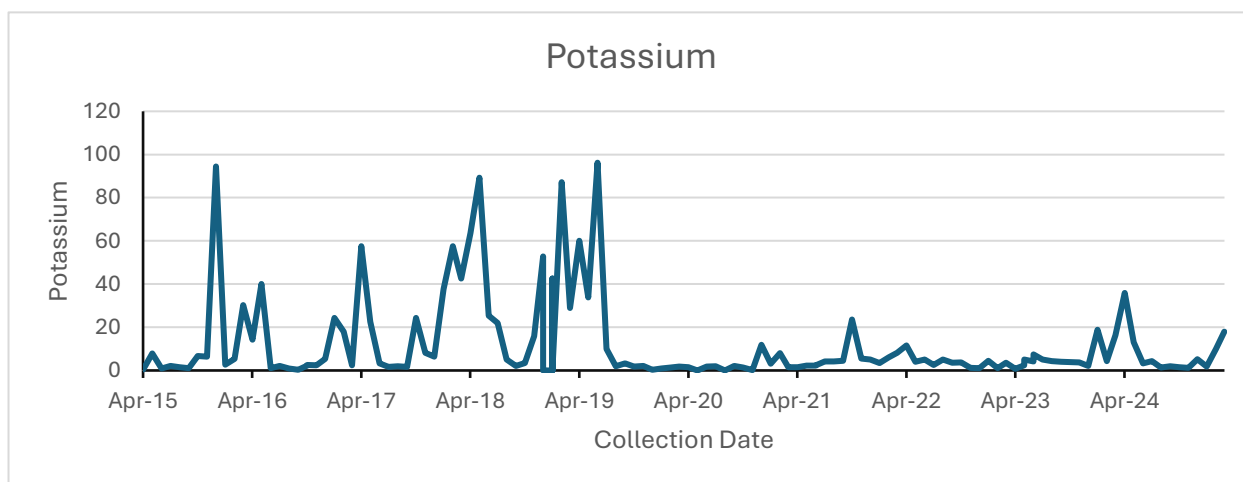
Graph 90 Monthly variation of Nitrite at Zadeshwar Station (2015-2025)



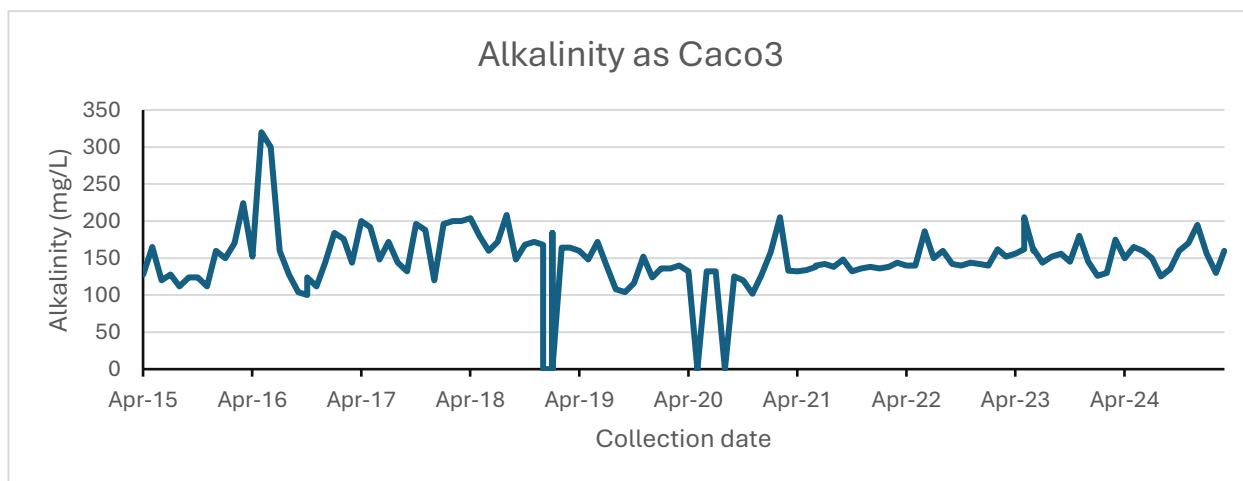
Graph 91 Monthly variation of pH at Zadeshwar Station (2015-2025)



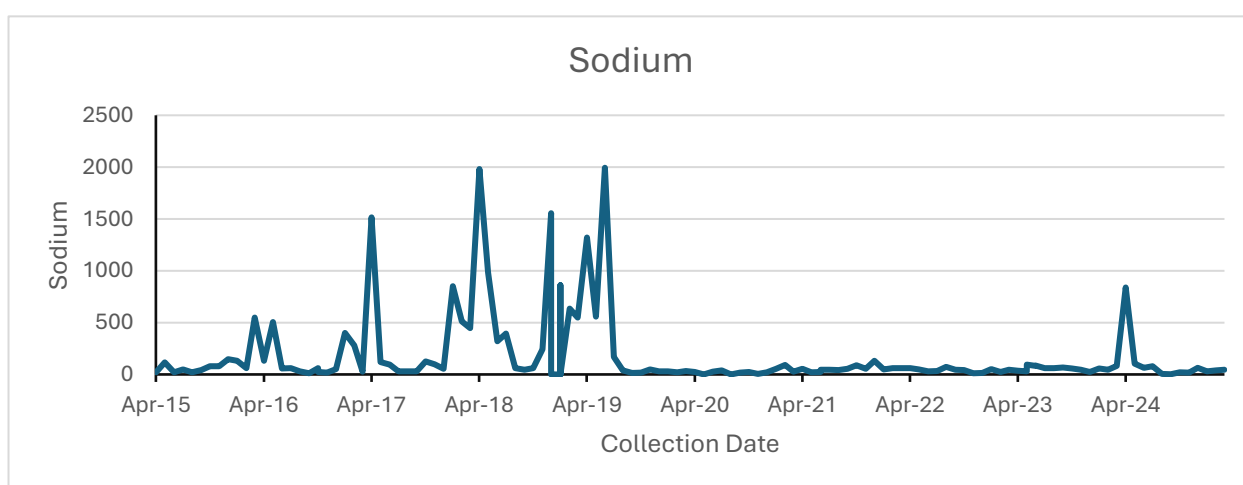
Graph 92 Monthly variation of Phosphate at Zadeshwar Station (2015-2025)



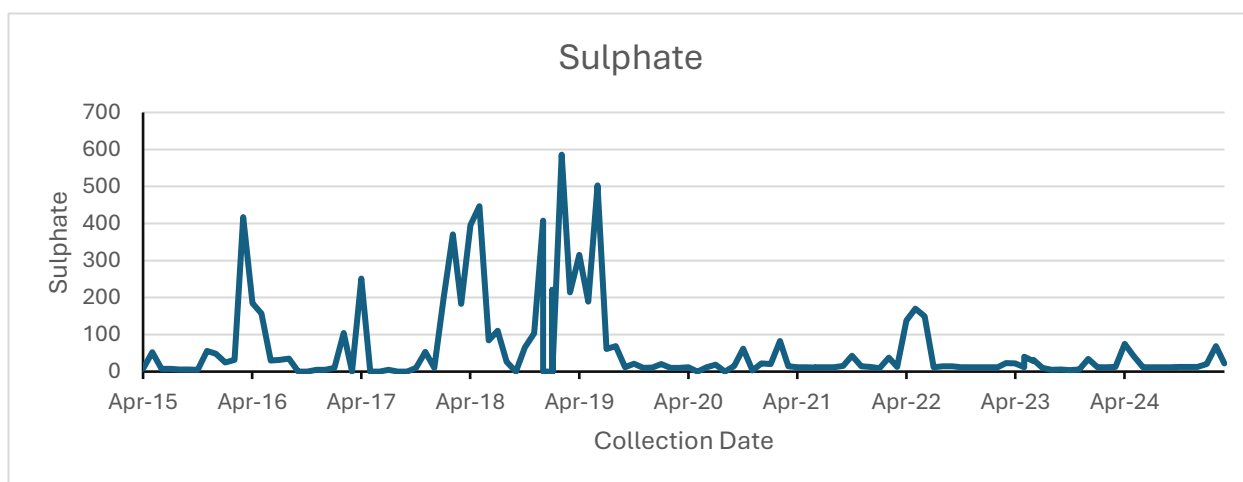
Graph 93 Monthly variation of Potassium at Zadeshwar Station (2015-2025)



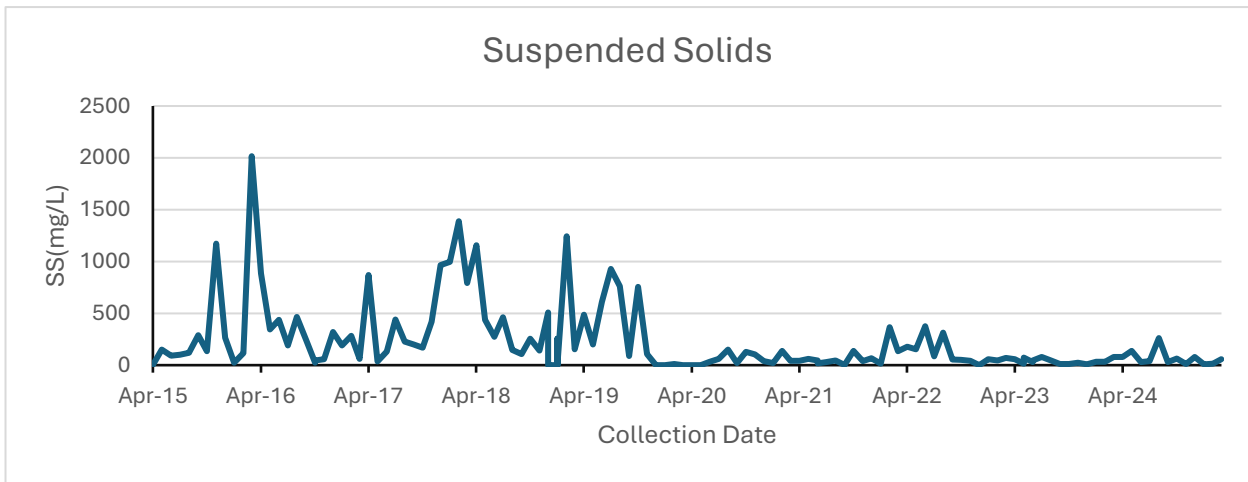
Graph 94 Monthly variation of Alkalinity as CaCo3 at Zadeshwar Station (2015-2025)



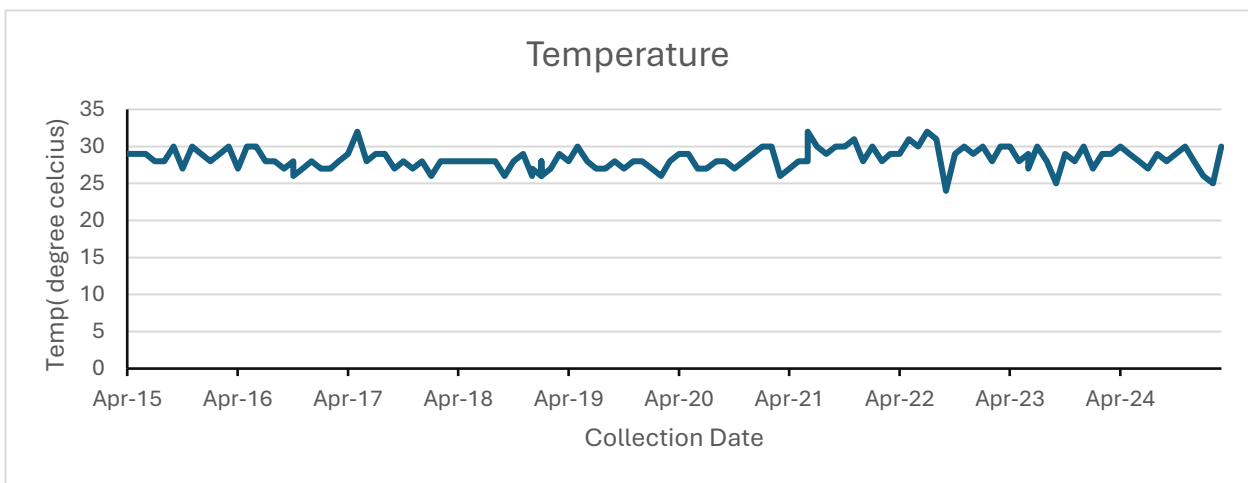
Graph 95 Monthly variation of Sodium at Zadeshwar Station (2015-2025)



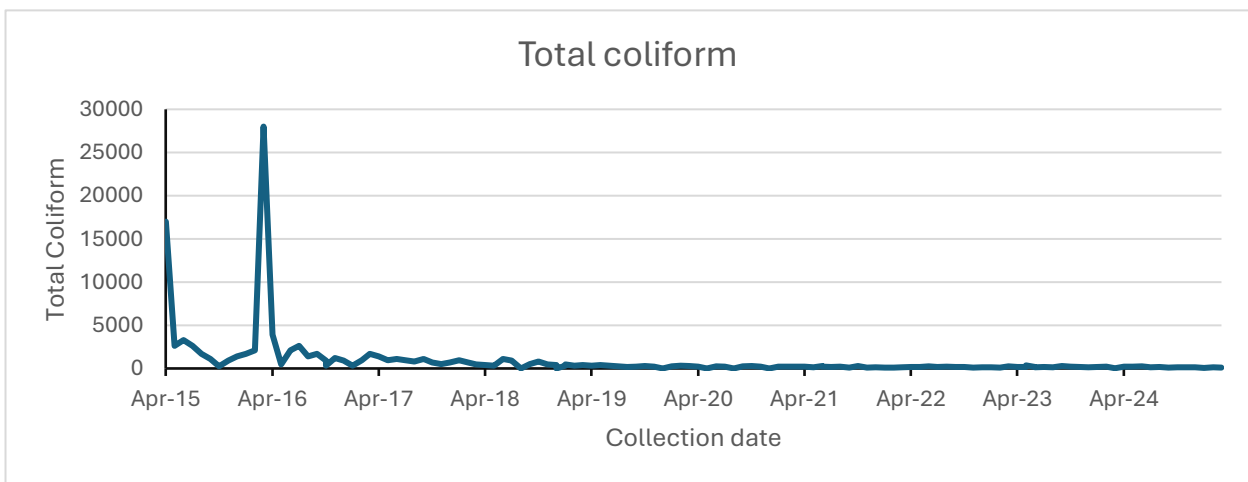
Graph 96 Monthly variation of Sulphate at Zadeshwar Station (2015-2025)



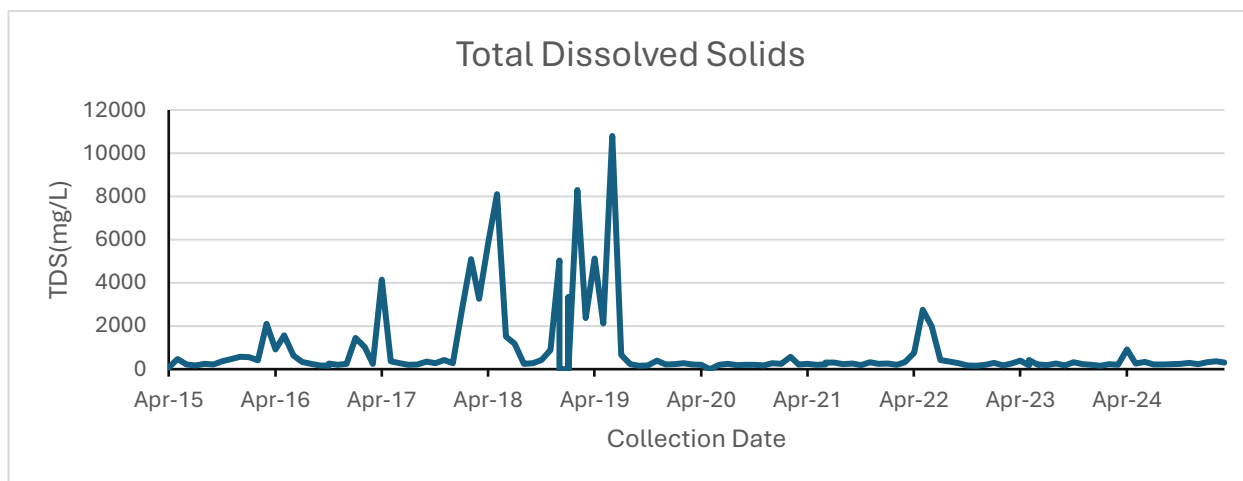
Graph 97 Monthly variation of Suspended Solids at Zadeshwar Station (2015-2025)



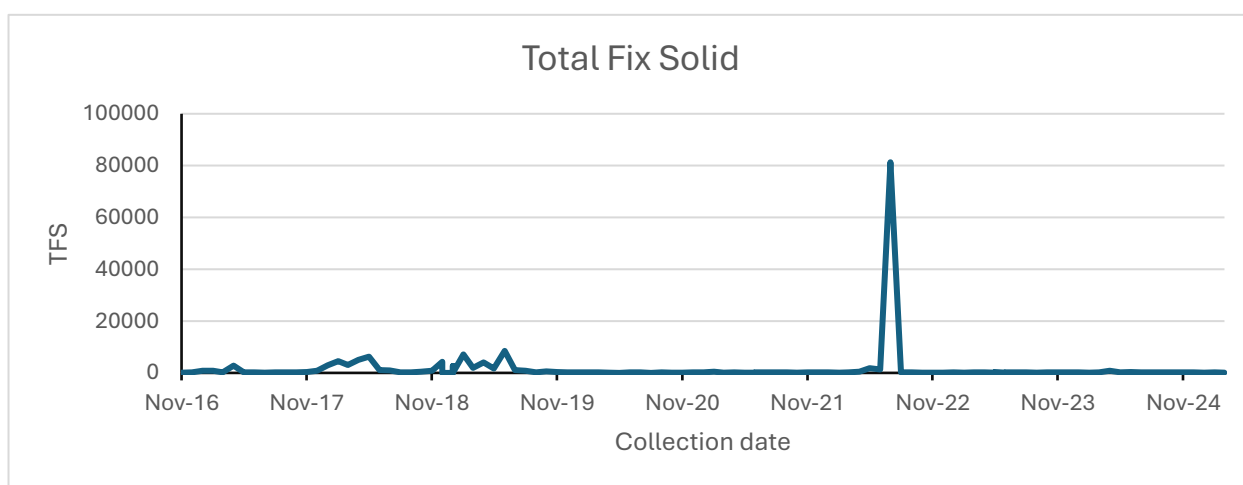
Graph 98 Monthly variation of Temperature at Zadeshwar Station (2015-2025)



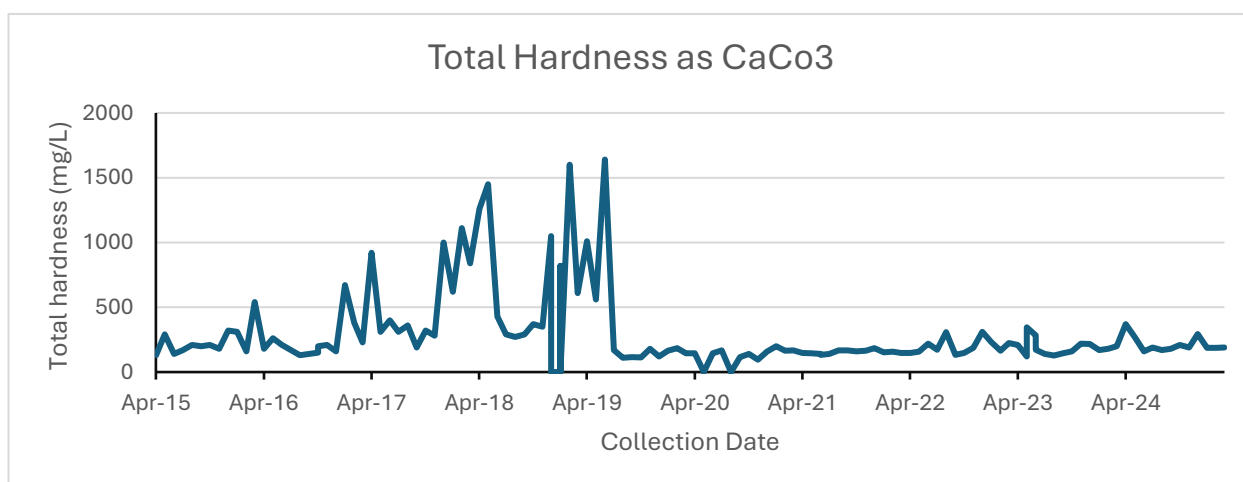
Graph 99 Monthly variation of Total Coliform at Zadeshwar Station (2015-2025)



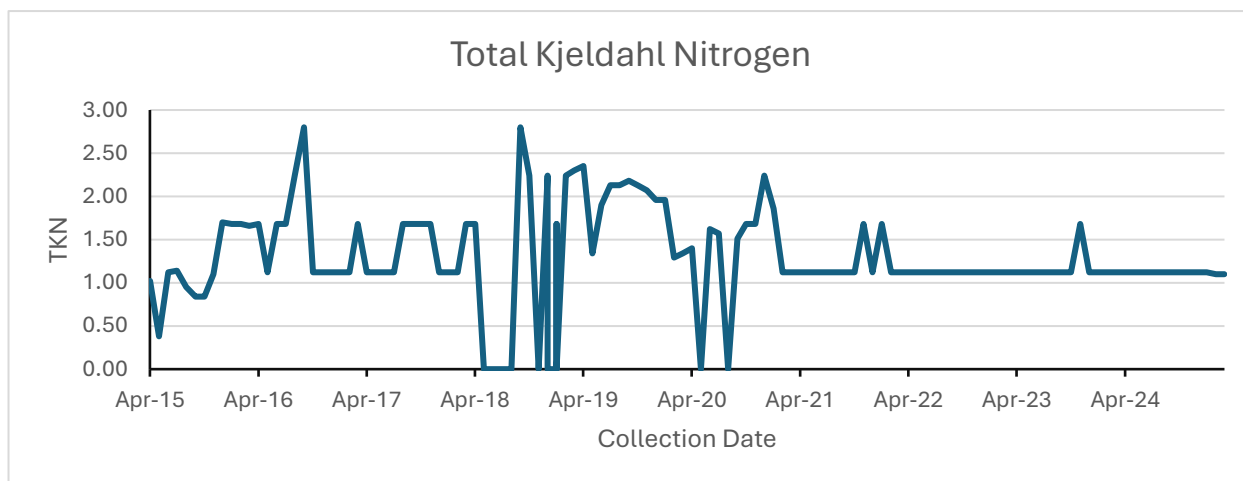
Graph 100 Monthly variation of Total Dissolved Solids at Zadeshwar Station (2015-2025)



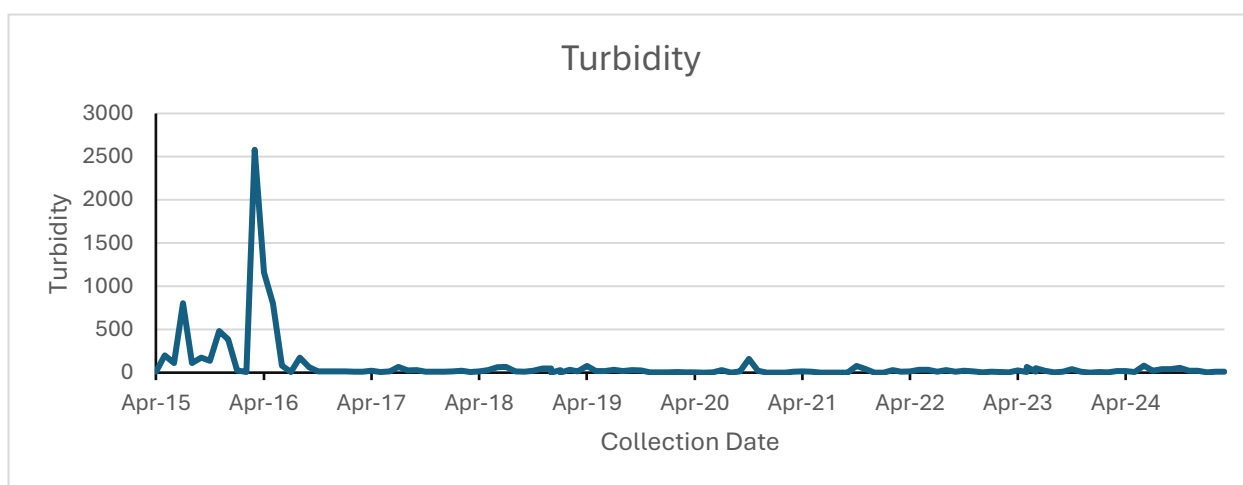
Graph 101 Monthly variation of Total Fix Solid at Zadeshwar Station (2015-2025)



Graph 102 Monthly variation of Total Hardness at Zadeshwar Station (2015-2025)



Graph 103 Monthly variation of Total Kjeldahl Nitrogen at Zadeshwar Station (2015-2025)



Graph 104 Monthly variation of Turbidity at Zadeshwar Station (2015-2025)

Graph 78: The graph shows the monthly variation of *Ammonical Nitrogen* at Zadeshwar Station from April 2015 to April 2025. Ammonical Nitrogen ranged from 1.12 to 1.34 mg/L, periodic elevations suggest organic inputs/effluent influence; no specific BIS drinking limit for ammonical Nitrogen but environmental concern when elevated.

Graph 79: The graph shows the monthly variation of *B.O.D. (3 days 27 °C)* at Zadeshwar Station from April 2015 to April 2025. BOD ranged from 0.7 to 3.0 mg/L, CPCB bathing criterion: $BOD \leq 3$ mg/L, within permissible ranges.

Graph 80: The graph shows the monthly variation of *Boron* at Zadeshwar Station from April 2015 to April 2025. Boron ranged from 0.01 to 0.8 mg/L, typical BIS guidance: desirable ~0.5 mg/L; permissible often ~2.0 mg/L, within permissible ranges.

Graph 81: The graph shows the monthly variation of *Calcium* at Zadeshwar Station from April 2015 to April 2025. Calcium ranged from 22 to 420 mg/L, typical BIS guidance: desirable ~75 mg/L major concern for direct use.

Graph 82: The graph shows the monthly variation of *Chloride* at Zadeshwar Station from April 2015 to April 2025. Chloride ranged from 20 to 520 mg/L, permissible limit (BIS IS10500): 250 mg/L, exceeds permissible limit in some months; estuarine/tidal influence likely contributes.

Graph 83: The graph shows the monthly variation of *Chemical Oxygen Demand* at Zadeshwar Station from April 2015 to April 2025. COD ranged from 5 to 65 mg/L, elevated values in some months indicate chemical/organic loading.

Graph 84: The graph shows the monthly variation of *Conductivity* at Zadeshwar Station from April 2015 to April 2025. Conductivity ranged from 170 to 1830 $\mu\text{S}/\text{cm}$, within permissible ranges. (tidal/reservoir influence).

Graph 85: The graph shows the monthly variation of *Dissolved Oxygen* at Zadeshwar Station from April 2015 to April 2025. DO ranged from 5.8 to 8.7 mg/L, CPCB ecological guideline: $\text{DO} \geq 6$ mg/L for good water, some months fall below ecological threshold, indicating stress.

Graph 86: The graph shows the monthly variation of *Fecal Coliform* at Zadeshwar Station from April 2015 to April 2025. Fecal Coliform ranged from 13 to 1400, permissible limit for drinking: 0 MPN/100mL, very high microbial contamination, major health concern for direct use.

Graph 87: The graph shows the monthly variation of *Fluoride* at Zadeshwar Station from April 2015 to April 2025. Fluoride ranged from 0.0 to 1.4 mg/L, permissible limit (BIS IS10500): 1.5 mg/L, - within permissible ranges.

Graph 88: The graph shows the monthly variation of *Magnesium* at Zadeshwar Station from April 2015 to April 2025. Magnesium ranged from 0.96 to 156.0 mg/L, contributes to hardness: concerns over some elevated months.

Graph 89: The graph shows the monthly variation of *Nitrate* at Zadeshwar Station from April 2015 to April 2025. Nitrate ranged from 0.0 to 2.0 mg/L, permissible limit (BIS IS10500): 45 mg/L, within permissible ranges.

Graph 90: The graph shows the monthly variation of *Nitrite* at Zadeshwar Station from April 2015 to April 2025. Nitrite ranged from 0.01 to 0.07mg/L, permissible limit (BIS IS10500): 0.2 mg/L, within permissible ranges.

Graph 91: The graph shows the monthly variation of *pH* at Zadeshwar Station from April 2015 to April 2025. pH ranged from 6.2 to 8.6, permissible limit (BIS IS10500): 6.5–8.5 (no relaxation), within permissible ranges.

Graph 92: The graph shows the monthly variation of *Phosphate* at Zadeshwar Station from April 2015 to April 2025. Phosphate ranged from 0.02 to .82 mg/L, higher nutrient peaks in monsoon; potential eutrophication risk locally.

Graph 93: The graph shows the monthly variation of *Potassium* at Zadeshwar Station from April 2015 to April 2025. Potassium ranged from 0.2 to 96 mg/L, within environmental ranges.

Graph 94: The graph shows the monthly variation of *Alkalinity as CaCo₃* at Zadeshwar Station from April 2015 to April 2025. Alkalinity ranged from 100 to 320 mg/L, occasionally elevated values but generally within operational ranges.

Graph 95: The graph shows the monthly variation of *Sodium* at Zadeshwar Station from April 2015 to April 2025. Sodium ranged from 10 to 1980 mg/L, high peaks consistent with estuarine/tidal saline influence; some months exceed common guideline levels (200 mg/L), raising concerns.

Graph 96: The graph shows the monthly variation of *Sulphate* at Zadeshwar Station from April 2015 to April 2025. Sulphate ranged from 4 to 586 mg/L, permissible limit (BIS IS10500): 200 mg/L, exceeds permissible limit in some months.

Graph 97: The graph shows the monthly variation of *Suspended Solids* at Zadeshwar Station from April 2015 to April 2025. Suspended Solids ranged from 2 to 2000 mg/L, considerable variability and high monsoon peaks.

Graph 98: The graph shows the monthly variation of *Temperature* at Zadeshwar Station from April 2015 to April 2025. Temperature ranged from 24 to 32 °C, seasonal.

Graph 99: The graph shows the monthly variation of *Total Coliform* at Zadeshwar Station from April 2015 to April 2025. Total Coliform ranged from 100 to 100,000 MPN/100mL, permissible limit (drinking): 0 MPN/100mL, very high contamination; public-health risk.

Graph 100: The graph shows the monthly variation of *Total Dissolved Solids* at Zadeshwar Station from April 2015 to April 2025. TDS ranged from 60 to 3000 mg/L, permissible limit (BIS IS10500): 2000 mg/L, exceeds permissible limit in some months, consistent with estuarine/tidal saline influence.

Graph 101: The graph shows the monthly variation of *Total Fix Solid* at Zadeshwar Station from April 2015 to April 2025.

Graph 102: The graph shows the monthly variation of *Total Hardness* at Zadeshwar Station from April 2015 to April 2025. Total Hardness ranged from 96 to 1720 mg/L, permissible limit (BIS IS10500): 600 mg/L, exceeds permissible limit in some months, indicating hard water episodes.

Graph 103: The graph shows the monthly variation of *Total Kjeldahl Nitrogen* at Zadeshwar Station from April 2015 to April 2025. TKN ranged from 0.38 to 2.8 mg/L, within reported environmental ranges though some months elevated.

Graph 104: The graph shows the monthly variation of *Turbidity* at Zadeshwar Station from April 2015 to April 2025. Turbidity ranged from 2.0 to 2200.0 NTU, permissible limit (BIS IS10500): 5 NTU, episodic very high turbidity (monsoon/tidal resuspension); exceeds permissible limit and requires attention.

Downstream at Zadeshwar, the water quality reflects the highest pollution load among all monitored stations. Zadeshwar's April 2015–April 2025 record shows severe and frequent water-quality problems that make the water unsafe for direct human consumption and raise ecological concern — most striking are extreme turbidity spikes (up to ~2,200 NTU) and very high microbial loads (total coliforms 100–100,000 MPN/100 mL; fecal coliforms up to 1,400 MPN/100 mL), accompanied by saline/ionic surges (TDS to ~3,000 mg/L, sodium to ~1,980 mg/L, sulphate to ~586 mg/L), very hard-water episodes (hardness to ~1,720 mg/L), large suspended-solid events (to ~2,000 mg/L) and intermittent elevated COD (to ~65 mg/L) and low DO months — patterns consistent with estuarine/tidal influence, monsoon-driven resuspension and likely sewage/industrial inputs. Zadeshwar experiences considerable anthropogenic pressure, requiring enhanced treatment and pollution-control interventions.

Overall lower Basin Status

Across April 2015–April 2025 the Narmada shows a clear downstream deterioration. Upstream at Panetha and Garudeshwar many chemical parameters remain broadly within permissible

limits, but both sites show recurrent seasonal/monsoon-driven turbidity spikes and persistent microbial contamination (faecal/total coliforms) that preclude safe drinking without treatment, while Garudeshwar also records intermittent chemical outliers (nitrate, nickel, high sodium) that need source-tracing. Zanor exhibits higher microbial loads, episodic salinity/ionic excursions and turbidity/COD peaks pointing to sewage and saline/industrial influence. Downstream at Zadeshwar the situation is worst: extreme turbidity and suspended solids, massive microbial contamination, large salinity/TDS and hardness episodes and elevated COD — consistent with estuarine/tidal resuspension plus sewage/industrial inputs. Overall, the trend indicates that water quality remains good upstream and progressively declines downstream.

3. Pollution Sources

The Narmada River Basin is subject to multiple pollution pressures from domestic, industrial, agricultural, and non-point sources. Each of these contributes differently in terms of pollutant type, intensity, and spatial distribution. Understanding their nature and quantum is essential for assessing the overall pollution load.

3.1 Domestic Wastewater Load

Domestic wastewater load in the Narmada Basin towns is mainly linked with the level of sewerage coverage, household connections, and the ability of pumping stations and drains to handle the generated sewage.

3.1.1 Status of Water Supply Connections

Domestic wastewater is one of the major contributors to river pollution in the basin due to the growing urban population, limited sewage treatment capacity, and incomplete sewerage networks. However, it is also important to note the status of water supply connections in the Narmada Basin district. **Table 1** provides ULB wise perspective on status of water connections in the basin.

The table presents the number of water supply connections across various Urban Local Bodies (ULBs) situated in districts of the Narmada Basin, spanning Madhya Pradesh and parts of Gujarat. In the upper basin, smaller towns like Amarkantak in Anuppur and Mandla in Mandla district each report 2,100 connections, while Dindori has a slightly higher figure of 3,112 connections. Moving to larger urban centers, Jabalpur stands out with 171,228 connections, making it the dominant city in the basin for water supply, while nearby Bhedaghat contributes a modest 1,167 connections. In Narsinghpur district, the main town has 7,500 connections, complemented by 2,478 connections in Saikheda. Narmadapuram (Hoshangabad) adds a significant 24,000 connections, reinforcing its role as a major hub, with Budhni and Bherunda (Nasrullaganj) in Sehore district contributing 1,756 and 6,236 connections, respectively.

In the middle basin, smaller settlements like Nemawar in Dewas account for 1,198 connections, while Omkareshwar in Khargone reports 1,400 connections. The Khargone district shows a cluster of medium-sized towns with substantial coverage: Barwah (5,082), Sanawad (8,000), Mandleshwar (3,340), Maheshwar (5,400), and Anjad (5,000), indicating a relatively well-distributed water supply network. Neighboring Dhar district contributes with Dhamnod (6,417)

and Dharampuri (3,300) connections. In Barwani district, Barwani itself records 11,232 connections, while Sendhwa leads with 12,000 connections, reflecting stronger urban water access in this part of the basin.

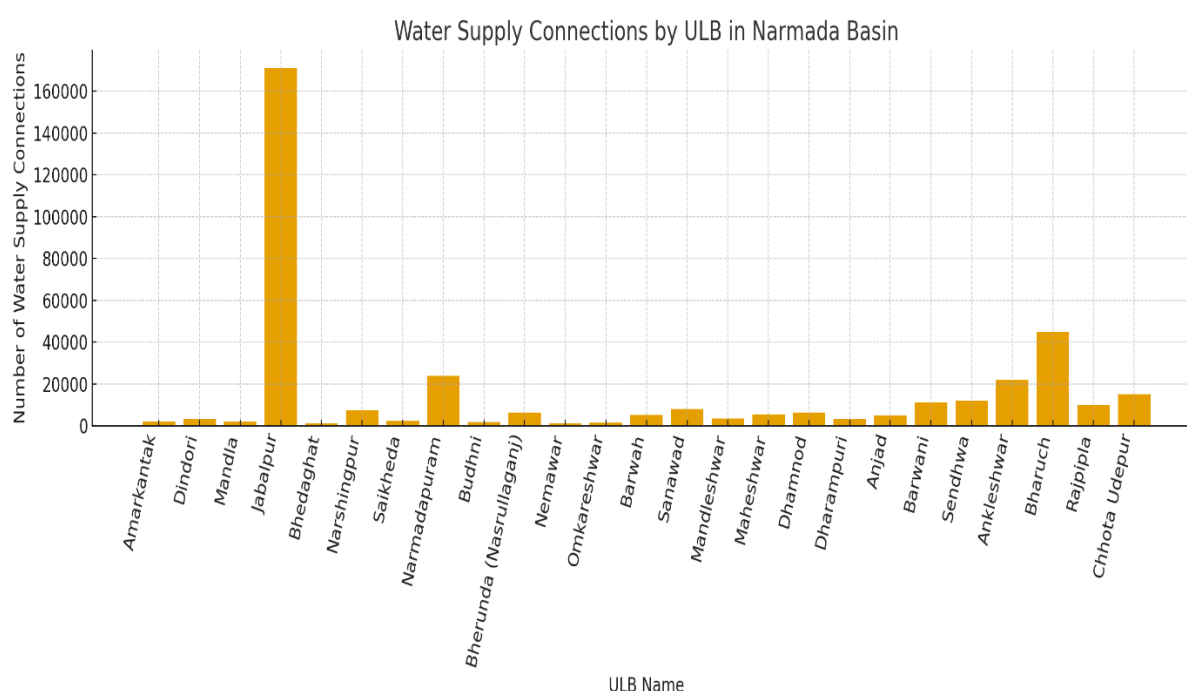
Table 1 Status of Water Connection in Narmada River Basin Districts

Name of the ULB	District	Number of Water Supply Connections
Amarkantak	Anuppur	2,100
Dindori	Dindori	3,112
Mandla	Mandla	2,100
Jabalpur	Jabalpur	171,228
Bhedaghat	Jabalpur	1,167
Narshingpur	Narshingpur	7,500
Saikheda	Narshingpur	2,478
Narmadapuram	Narmadapuram	24,000
Budhni	Sehore	1,756
Bherunda (Nasrullaganj)	Sehore	6,236
Nemawar	Dewas	1,198
Omkareshwar	Khargone	1,400
Barwah	Khargone	5,082
Sanawad	Khargone	8,000
Mandleshwar	Khargone	3,340
Maheshwar	Khargone	5,400
Dhamnod	Dhar	6,417
Dharampuri	Dhar	3,300
Anjad	Khargone	5,000
Barwani	Barwani	11,232
Sendhwa	Barwani	12,000
Ankleshwar	Bharuch	22,000
Bharuch	Bharuch	45,000
Rajpipla	Narmada	10,000
Chhota Udepur	Chhota Udepur	15,000

**Source: Urban Development and Housing Department (from Infrastructure Development Report, cNarmada)*

The table also includes towns from the lower basin in Gujarat. Ankleshwar (22,000) and Bharuch (45,000) emerge as major centers, along with Rajpipla in Narmada district (10,000), Chhota Udepur reports 15,000 connections.

In summary, the data on water connections highlights contrasts between large cities like Jabalpur and Bharuch, which dominate in terms of water supply coverage, and smaller towns such as Bhedaghat, Omkareshwar, and Amarkantak, where connections remain relatively limited which is shown in **Graph 105**. The distribution reflects both population concentration and administrative priorities across different districts of the Narmada Basin.



Graph 105 Narmada Basin, ULB-wise water connections

**Source: Urban Development and Housing Department (from Infrastructure Development Report, cNarmada)*

3.1.2 Status of Sewerage Network and Household Connections

The status of sewerage networks and household connections reflects the level of urban infrastructure and plays a vital role in safeguarding river health in the Narmada Basin. The data presented in **Table 2** shows wide variation in progress across different cities and towns, with some demonstrating strong outcomes while others remain stagnant.

Among the leading performers, Bhopal stands out with over 201,900 complete household connections, including more than 80,000 through AMRUT and 30,000 via convergence, making it the most advanced city in the basin. Indore follows closely with 268,812 connections, though only a fraction of these (5,000) were supported under AMRUT. Burhanpur has also

achieved remarkable progress with 26,613 total connections, while Dewas has added over 22,800 connections through convergence, bringing its total to 30,352. These districts represent the strongest achievements in the basin.

Moderate progress is observed in Sagar (28,000 total connections), Khargone (9,649), and Sehore (14,700). Narsinghpur and Mandla have also shown notable outcomes, with 13,100 and 12,842 connections respectively, indicating steady improvements in sewerage connectivity. Smaller but measurable contributions are seen in places like Budhni (2,804), Dharampuri (2,300), Nasrullaganj (2,833), and Mandleshwar (3,846), reflecting localized development efforts.

Some smaller towns in Barwani district—Badwani (1,743), Sendhwa (4,194), and Anjad (1,463)—show incremental improvements, while settlements such as Saikheda (2,721), Barwaha (1,677), and Maheshwar (2,007) have added modest numbers. Even religious and heritage towns like Amarkantak (1,518) and Bhedaghat (1,084) have made contributions, though on a limited scale.

By contrast, several districts such as Betul, Damoh, and Khandwa show no progress, with total sewerage connections remaining at zero. Similarly, Dhamnod has also reported no development under AMRUT or convergence.

Overall, the analysis highlights that Bhopal, Indore, Burhanpur, Dewas, Sagar, Narsinghpur, and Mandla are the front-runners in sewerage connectivity, while districts such as Betul, Khandwa, Damoh, and Dhamnod remain at the lower end with negligible or no coverage. This uneven distribution underscores the need for targeted interventions to strengthen sewerage infrastructure and expand household connections across underserved areas of the Narmada Basin.

Table 2 Details of Household Service Connections (Sewerage) in the Narmada Basin Cities

S. No.	Name of City	Households Covered at Start of Mission (2015-	Household Connections	Total Connections Done Through	Balance	Connections in	Total Connections Done Through	Convergence	Total Connections Done Through	AMRUT &	Total Connections Done
1	Jabalpur	0	96,900	62,825	34,075		5,913		68,738		68,738
2	Sagar	0	60,173	28,000	32,173		0		28,000		28,000

3	Katni	9,000	29,000	0	29,000	0	0	9,000
4	Chhindwara	0	0	0	0	28,000	21,930	21,930
5	Damoh	0	0	0	0	0	0	0
6	Betul	0	0	0	0	0	0	0
7	Bhopal	91,717	76,820	80,206	0	30,000	110,206	201,923
8	Burhanpur	1,593	23,520	23,520	0	1,500	25,020	26,613
9	Dewas	7,500	0	0	0	22,852	22,852	30,352
10	Hoshangabad	1,100	0	0	0	459	459	1,559
11	Indore	263,812	5,000	5,000	0	0	5,000	268,812
12	Khandwa	0	0	0	0	0	0	0
13	Khargone	300	9,349	9,349	0	0	9,349	9,649
14	Omkareshwar	0	0	0	0	1,818	1,818	1,818
15	Pithampur	6,455	0	0	0	0	0	6,455
16	Sehore	500	14,200	14,200	0	0	14,200	14,700
17	Bhedaghat	0	0	0	0	1,084	1,084	1,084
18	Nasrullaganj	0	0	0	0	2,833	2,833	2,833
19	Mandla	0	0	0	0	12,842	12,842	12,842
20	Narsinghpur	0	0	0	0	13,100	13,100	13,100
21	Badwani	0	0	0	0	1,743	1,743	1,743
22	Sendhwa	0	0	0	0	7,162	4,194	4,194
23	Barwaha	0	0	0	0	1,677	1,677	1,677
24	Anjad	0	0	0	0	1,463	1,463	1,463
25	Saikheda	0	0	0	0	2,721	2,721	2,721
26	Budhni	0	0	0	0	2,804	2,804	2,804
27	Dharampuri	0	0	0	0	2,300	2,300	2,300
28	Nemawar	0	0	0	0	1,226	1,226	1,226

29	Amarkantak	0	0	0	0	1,518	1,518	1,518
30	Mandleshwar	0	0	0	0	3,846	3,846	3,846
31	Dindori	0	0	0	0	2,000	2,000	2,000
32	Sanawad	0	0	0	0	1,021	1,021	1,021
33	Maheshwar	0	0	0	0	2,007	2,007	2,007
34	Dhamnod	0	0	0	0	0	0	0

**Source: MPPCB, Progress Report of June 2025 for the compliance of NGT*

The sewerage infrastructure across towns in the Lower Narmada Basin shows varied levels of development shown in **Table 3**. In **Rajpipla**, the network is only partially developed, comprising a mix of combined and separate sewer lines depending on locality. Many parts of the city are still not connected to the sewerage system, leading to untreated sewage being discharged into open drains and nearby natural water bodies. In earlier years, a significant portion of wastewater was directly discharged into the Karjan River. However, the Rajpipla Nagarpalika has discontinued this practice and instead reuses wastewater for non-potable purposes such as gardening, road cleaning, and general maintenance. Sewage flow at the treatment plant is measured at the Parshall flume (FIT-101), and the plant operations are automated through SCADA (Supervisory Control and Data Acquisition) for monitoring pumps and treatment units.

In **Ankleshwar** city a comprehensive underground drainage system covering approximately 146 km is under construction. This network serves residential, commercial, and some industrial areas and comprises pipelines of different materials and diameters, including SWR, ductile iron, RCC(Reinforced Cement Concrete), asbestos cement pressure pipes, and cast iron, ranging from 150 mm to 600 mm. Sewage pumping is managed by Ankleshwar Nagarpalika, with multiple stations distributed across the city to handle both sewage and stormwater. The Modinagar station on Hansot Road operates with 70–100 HP(Horsepower) motors, a head of 25 m, and a flow capacity between 300,000 and 600,000 LPH(Litres per Hour), supported by SCADA systems and a 100 kVA DG set(Diesel Generator Set). At Chautabazar, pumping is managed with 30–50 HP units, handling up to 250,000 LPH. Sapna Sarvodaya station has 30–75 HP capacity and handles between 165,000 and 300,000 LPH. Smaller stations exist in Rajeshnagar, Ramnagar, Janakvatika/Radhe Park, Gujarat Housing Board, Tirth Nagar, Anas Motors, and Punit Nagar, with motor ratings typically ranging from 5–30 HP and varying flows, all equipped with panels such as DOL or star-delta and connected to DGVCL(Dakshin Gujarat

Vij Company Limited) approved loads. Together, these stations ensure that sewage from low-lying zones is lifted and conveyed to the STP.

The sewerage network in **Chhotaudepur** is only partially developed and not fully functional due to problems in the rising main pipeline..

In **Bharuch**, the city is divided into five sewerage zones for systematic wastewater collection and management.. The city has an extensive network with pipelines totaling 118,138 m in length, built primarily of RCC NP3 and NP4 pipes of diameters ranging from 200 mm to 1200 mm. There are 5,910 manholes of various types and 166 vertical drops across the system, which ensure proper flow and collection. Household connectivity to the sewer network is provided in all five zones. Sewage conveyance is achieved primarily through gravity flow. Pumping arrangements exist mainly at the STP, where Return Activated Sludge (RAS) pumps run in continuous auto mode with manual options, while Surplus/Waste Activated Sludge (SAS/WAS) pumps are operated on PLC(Programmable Logic Controller)-based timed and totalized modes. Thickener and dewatering feed pumps are operated in both auto and manual modes. The system is fully automated with SCADA and PLC controls, supported by sensors for flow, DO, ORP(Oxidation Reduction Potential), and level, along with valves equipped with actuators and blowers with variable frequency drives. This ensures continuous and reliable operation.

In rural parts of the lower Narmada basin, most settlements rely on on-site sanitation systems such as household toilets connected to septic tanks or leaching pits, with coverage of individual household latrines significantly improved under the Swachh Bharat Mission. However, piped sewer networks are rare, with only small towns or larger villages having partial underground drains or open drains.

Table 3 Sewerage Connections status in Lower Narmada basin

Town/Area	Sewerage Connection Status
Rajpipla	Partial network
Ankleshwar	146 km network
Chhotaudepur	Partial, rising main issues
Bharuch	5 zones, 118 km pipes
Rural Areas	Mostly on-site

**Source: Municipality data, GPCB District environment plan*

3.1.3 Status of Domestic Sewage Load

The **Table 4 and 5** provides a detailed overview of domestic sewage generation across districts which also shown in **graph 106**. It is expressed in Million Litres per Day (MLD). It reflects the interaction between population size, urbanization, water supply infrastructure, and socio-economic development in each region. The most notable observation is that Jabalpur, with 230 MLD, emerges as the largest contributor to domestic sewage, which can be attributed to its status as a major urban center hub in Madhya Pradesh. The high-volume highlights both the density of its population and the relatively wide coverage of piped water supply that results in greater wastewater discharge.

Similarly, Damoh (136 MLD), Betul (100 MLD), and Chhindwara (100 MLD) are also significant contributors despite being smaller than Jabalpur in population size, indicating the rapid pace of urban growth and rising domestic water consumption in these towns.

A second category of districts can be identified as medium-level sewage generators, with values ranging between 20–60 MLD. This group includes Bharuch (~35.5 MLD), Dhar (47.4 MLD), Sagar (40 MLD), Raisen (39 MLD), and Sehore (38 MLD). These districts are characterized by medium size urban settlements, expanding municipal coverage, and growing economic activity. Within this category, Burhanpur (27.76 MLD), Khargone (26 MLD), Katni (23 MLD), and Dewas (22 MLD) reflect smaller but steadily urbanizing regions, where sewage generation is already considerable and is expected to grow further in the coming years (Fig. 4).

The low sewage generation districts form the third category, producing less than 20 MLD of wastewater. These include Seoni (17.5 MLD), Harda (11.8 MLD), Barwani (12.31 MLD), and

Table 4 District Wise Status of Domestic Sewerage Waste Generation, Narmada Basin'

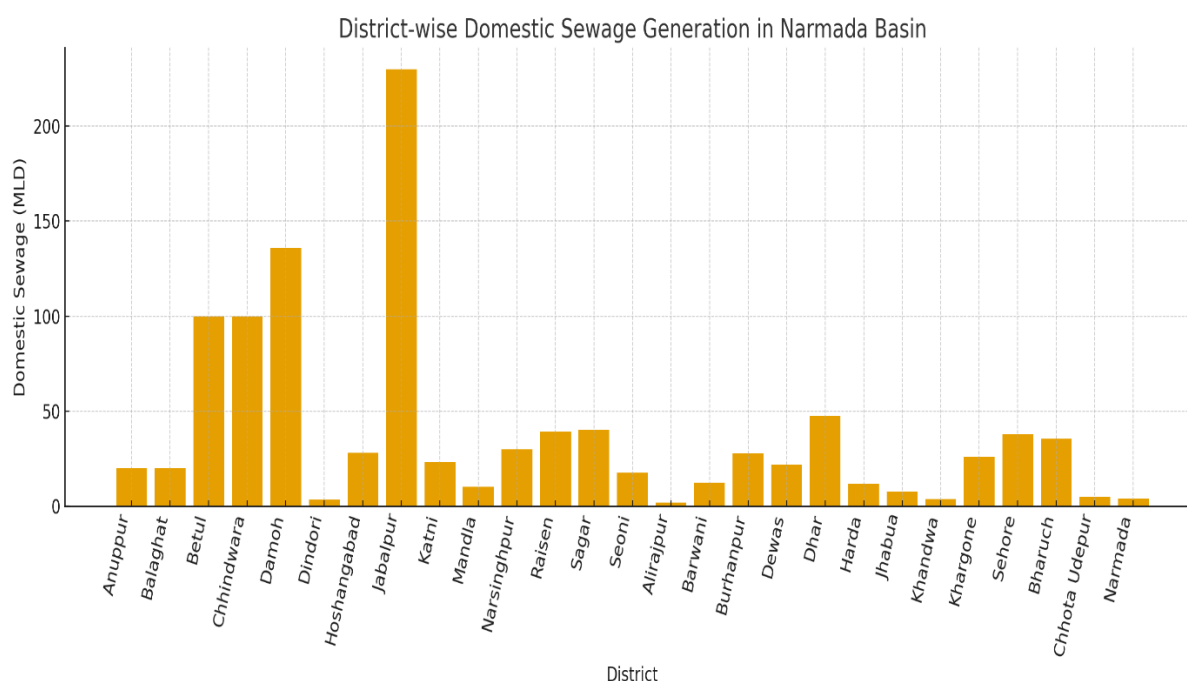
District	Domestic Sewage (MLD)
Anuppur	20.00
Balaghat	20.00
Betul	100.00
Chhindwara	100.00

District	Domestic Sewage (MLD)
Damoh	136.00
Dindori	3.49
Hoshangabad	28.00
Jabalpur	230.00
Katni	23.00
Mandla	10.25
Narshimapur	30
Raisen	39
Sagar	40
Seoni	17.50
Alirajpur	1.80
Barwani	12.31
Burhanpur	27.76
Dewas	22
Dhar	47.40
Harda	11.80
Jhabua	7.64
Khandwa	3.64
Khargone	26
Sehore	38

District	Domestic Sewage (MLD)
Bharuch	~35.5
Chhota udepur	5
Narmada (District)	~4

**Source: District Environment Plan, MPPCB & GPCB (from Narmada River Basin Infrastructure and Planning Report*

Mandla (10.25 MLD), which are typically semi-urban districts with modest population bases and lower levels of water infrastructure coverage compared to the larger cities. At the bottom end of the scale are predominantly tribal and rural districts such as Jhabua (7.64 MLD), Chhota Udepur(5MLD), Narmada(4MLD), Dindori (3.49 MLD), Khandwa (3.64 MLD), and Alirajpur (1.8 MLD). In these areas, the combination of sparse urban settlements, limited household water connections, and reliance on non-piped sources of water results in lower sewage generation.



Graph 106 Domestic Sewage Generation in Narmada River Basin

Thus, the status of district wise sewage generation shows a clear difference of from very high loads in major urban centers like Jabalpur and Damoh, to medium loads in growing towns such as Dhar, and Sagar, down to very small loads in tribal and rural districts like Alirajpur and Dindori. These insights are crucial for planning wastewater treatment facilities: large,

centralized treatment plants will be necessary in high-load districts, while smaller decentralized systems or on-site sanitation measures may be more appropriate in low-load, dispersed regions.

- **Observations from the Kirtikumar Sadashiv Bhatt vs Narmada Water Resources Water Supply Department** (No.71/2023(CZ) (O.A.No.97/2021-WZ)

- 1. Pollution Hotspots Identified:**

- CPCB declared stretches from Mandla–Bhedaghat and Sethani Ghat–Nemawar as highly polluted.
- Jabalpur stretch is marked critically polluted.

- 2. STP Gaps:**

- Most basin towns lack functional STPs.
- Even where STPs exist, several are non-operational or outdated (UASB technology) and need upgrading to SBR or advanced systems.

- 3. NGT Directives (Oct 2023 Order):**

- District Collectors of Narmada towns directed to submit sewage generation & STP status within 30 days.
- Encroachment on riverbanks/floodplains must be removed.
- Old/non-functional STPs to be upgraded.
- Functional STPs, CETPs, and ETPs must be installed to address treatment gaps.

- 4. Enforcement & Accountability:**

- MPPCB filed 9 criminal prosecutions against defaulting municipalities discharging untreated sewage.
- ₹12 crore environmental compensation imposed on 10 municipalities.
- Industries near the catchment ordered to adopt Zero Liquid Discharge (ZLD) using ETPs, MEEs, and ATFDs.

- 5. Supreme Court & NGT Benchmarks:**

- 31 March 2018 was the original deadline for STP/ETP completion (missed).
- Delays beyond 31 March 2021 attract ₹10 lakh/month penalty per STP.
- Untreated sewage discharge is a violation of Water Act, 1974 and attracts strict penalties.

- 6. Key Concern:** Despite monitoring for years, large-scale gaps persist. Around 51 MLD untreated sewage daily still reaches the Narmada, degrading drinking water quality and public health.

Table 5 Status of Sewage Generation and Treatment

City/Town	Sewage Generation (MLD)	Treatment Available	Remarks
Amarkantak	0.98 MLD	No STP; 1.2 MLD STP under construction	Untreated sewage currently discharged
Dindori	2.49 MLD	No STP; 3.8 MLD STP under construction	Untreated sewage
Mandla	10.25 MLD	No STP; 2 STPs (2 MLD & 7.75 MLD) under construction, 1 STP (0.5 MLD) not working	Large untreated discharge
Jabalpur	143.68 MLD	Partial treatment: 83.43 MLD by 8 operational STPs; 60 MLD in-situ bioremediation till 34 & 29 MLD STPs are ready	Major treatment hub, but still discharges untreated sewage
Narmadapuram	12 MLD	No STP; 21 MLD STP proposed	Entire load untreated
Budhni	3.295 MLD	Yes; 3 STPs operational	100% treated
Nemawar	4.913 MLD	No STP; 1 MLD proposed	Untreated
Omkareshwar	1.0866 MLD	Yes; 4 STPs (0.65 MLD total)	Partially treated
Barwaha	3.481 MLD	No STP; 3.75 MLD under construction	Untreated
Sanawad	5.097 MLD	No STP; 7.02 MLD under construction	Untreated
Mandleshwar	1.624 MLD	No STP; 3 MLD under construction	Untreated
Maheshwar	3.212 MLD	No STP; 4.91 MLD under construction	Untreated
Badwani	7.303 MLD	No STP; 9 MLD under construction	Untreated
Dhar	1.5 MLD	Yes, STP of 2 MLD operational	Treated

Bharuch	28MLD	Yes, STP of 29.3 MLD operational	Treated
Ankleshwar	7.5MLD	No, STP of 14 MLD under construction	Untreated
Rajpipla	4.5MLD	Yes, STP of 5.5 MLD operational	Treated
Chhota udepur	4MLD	Yes, STP of 5 MLD operational	Treated

**Source: Kirtikumar Sadashiv Bhatt vs Narmada Water Resources Water Supply Department (No.71/2023(CZ) (O.A.No.97/2021-WZ) before NGT and GPCB District Environment Plan*

3.2 Industrial Wastewater Load

Industrial wastewater load represents one of the major sources of pollution in urban and peri-urban regions as well as in dedicated industrial areas. It is generated from various sectors such as manufacturing, textiles, chemicals, pharmaceuticals, food processing, and mining, often containing high concentrations of organic matter, heavy metals, oils, and toxic chemicals. Unlike domestic wastewater, industrial effluents are more complex in nature and require specialized treatment before discharge. However, in many areas, industries either lack adequate treatment facilities or operate them inefficiently, leading to the release of untreated or partially treated wastewater into rivers, streams, and open lands. This not only degrades water quality but also poses risks to aquatic ecosystems, agriculture, and human health. Effective regulation, adoption of cleaner technologies, and enforcement of effluent treatment standards are critical to reducing the industrial wastewater load and its environmental impacts.

The lower Narmada basin has many industries. These industries produce large amounts of wastewater every day. Most of the wastewater comes from chemical, pharmaceutical, textile, pulp and paper, and petrochemical units. Some areas like Bharuch and Ankleshwar have very high wastewater generation, while smaller towns like Rajpipla and Chhota Udepur produce much less. Proper treatment and management of this wastewater is very important for protecting the water quality of the river.

3.2.1 Major Industrial Clusters in the Narmada River Basin

1. Jabalpur (Madhya Pradesh)

Jabalpur hosts a diverse industrial base grounded in both defence and manufacturing. Its industrial profile includes the Ordnance Factory units—the Vehicle Factory Jabalpur, Gun

Carriage Factory, and Grey Iron Foundry—supporting defence production. Beyond this, the district hosts a range of industries such as ready-made garments, poultry, electrical goods, sawmills, limestone and building materials, glassware, furniture, cement, and food processing, complemented by IT infrastructure like Techno Park.

2. Narsinghpur (Madhya Pradesh)

Industrial activity in Narsinghpur remains predominantly agro based. Key industries include agricultural equipment, pulse (dal) mills, oil mills, gur (jaggery) and sugar production, beedi manufacturing, and ancillary sectors like paper, plastic, leather goods, and pottery. The district also holds mineral resources such as limestone, fireclay, and soapstone, fuelling cement and utensil production.

3. Khandwa – East Nimar (Madhya Pradesh)

Notable industries include edible oil mills, cotton ginning, and power generation—specifically the Sant Singaji and Dada Dhuniwale Thermal Power Plants, supported by supplementary solar power infrastructure.

4. Hoshangabad (Madhya Pradesh)

The town houses a singular significant industrial entity—the Security Paper Mill—which discharges treated industrial effluent into the Narmada.

5. Bharuch, Dahej & Ankleshwar (Gujarat)

- **Bharuch district**, situated at the Narmada River’s mouth, is one of the most heavily industrialized regions in India. It includes sectors such as chemicals and petrochemicals, pharmaceuticals, textiles, engineering, dairy products, and agro processing like peanut processing. It also hosts substantial minerals including lignite, silica sand, and agate.
- **Ankleshwar**, within Bharuch, is a major chemical industrial estate. It contributes over 5% of India’s chemical output, specializing in dyes, pharmaceuticals, and industrial chemicals.
- **Dahej**, near Bharuch, is part of a major Petroleum, Chemicals, and Petrochemicals Investment Region (PCPIR), produces polymers, petrochemicals, and intermediates.

Major companies like Reliance, Adani, ONGC, Torrent, and Birla Copper are established here; it also features a significant port infrastructure.

- Additionally, **Safex Chemicals** inaugurated its largest manufacturing facility in Bharuch in April 2025, expanding its production of specialty chemicals (Business Standard, 29th May 2025)
- A **bulk drug park** is under development at Jambusar (in Bharuch district), covering 815 hectares (cmogujrat.gov.in, 4th August 2025)

3.2.1.1 Mineral-Based Industries

The Narmada Basin in Madhya Pradesh hosts a variety of mineral-based industries that

Table 6 Major Mineral-Based Industries and Their Locations in the Narmada Basin, Madhya Pradesh

Sr. No.	Industries	Location
1	Cement	Damoh
2	Thermal Power	Betul, Damoh
3	Coal Ossuary	Narsinghpur, Annuppur, Chindwara
4	Asbestos Cement Sheet	Katni
5	Ceramic	Jabalpur
6	Hydrated Lime	Katni
7	Refractories	Katni, Jabalpur
8	Roofing Tiles	Hoshangabad, Balaghat
9	Marble cutting, Polishing	Katni

**Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17,*

leverage the region's rich mineral resources listed in **Table 6**. Cement industries are concentrated in Damoh, supporting the construction sector. Thermal power plants operate in Betul, Damoh contributing to regional energy generation. Coal-related activities, including ossuaries, are present in Narsinghpur, Anuppur, and Chhindwara, while Katni hosts asbestos cement sheet and hydrated lime production. Ceramic industries are in Jabalpur. Refractory materials are produced in Katni, and Jabalpur, whereas roofing tiles industries are found in

Hoshangabad and Balaghat. Additionally, marble cutting and polishing activities are carried out in Katni. These industries collectively play a significant role in the economic development and industrial landscape of the Narmada Basin.

In Lower basin, many industries are linked to minerals. At Dahej, big chemical and petrochemical plants make caustic soda, soda ash, chlorine and fertilizers. These use minerals like salt, limestone and gypsum. Vilayat has Grasim/Birla Cellulose, which makes viscose fibre using limestone and caustic soda. Ankleshwar, Panoli and Jhagadia also have many chemical and pharma industries that depend on minerals as feedstock. Smaller towns like Tilakwada and Rajpipla have stone crushing, brick kilns and building material units that use local stone and clay. Sankheda, on the other hand, is known for wooden furniture and handicrafts, and does not depend much on minerals.

3.2.1.2 Food Parks

The Narmada Basin hosts several food parks that promote agro-based industries and food processing listed in **Table 7**, supporting local economies and enhancing value addition to agricultural products. These parks are strategically located across different districts, with varying coverage of the basin.

Table 7 Food Parks and Their Coverage in the Narmada Basin, Madhya Pradesh

Location	District	Narmada Basin Coverage
Borgaon	Chhindwara	Partly
Nimrani	Khargone	Fully
Babai – Pipariya	Hoshangabad	Fully
Maneri	Mandla	Fully

**Source: Industrial Profile of Madhya Pradesh 2016–17*

In Lower basin Food and agro industries are also present in some estates. Bharuch GIDC(Gujarat Industrial Development Corporation) has several food units such as Nishant Food Industries, Bharuch Food Products, Jabsons Foods and Jay Bhavani Food Products. These make snacks, namkeen and sweets. Dahej SEZ(Special Economic Zone) does not have large food-processing units, but one company makes synthetic food-grade colours. Most estates in Ankleshwar, Panoli, Dahej and Vilayat are mainly chemical and do not have big food parks. Smaller estates in Rajpipla, Dediapada and Tilakwada have agro and food-based trade such as edible oil, pulses and spices, but they are small in scale.

3.2.1.3 Agri Export Zones

The Narmada Basin in Madhya Pradesh supports several Agricultural Export Zones (AEZs) listed in **Table 8**, which focus on high-value crops for national and international markets. These zones promote agro-based industrial growth, enhance farmers' income, and facilitate the supply of quality produce from the region.

Table 8 Agri Export Zones and Their Coverage in the Narmada Basin, Madhya Pradesh

Produce	Coverage (Narmada Basin Districts)
Potato, Onion & Garlic	Dhar, Indore, Dewas
Wheat (Durum & Sharbati)	Dewas, Indore, Dhar, Sehore, Raisen, Hoshangabad, Harda, Narsinghpur
Pulses (Lentil & Gram)	Raisen, Narsinghpur, Chhindwara
Oranges	Chhindwara, Hoshangabad, Betul

**Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17*

Many GIDC estates in the lower basin are located inside Agri Export Zones (AEZ). These AEZs mainly focus on mango and vegetables. The estates that fall in AEZs include Vagra, Panoli, Nabipur, Valia, Naldhari, Bharuch, Dahej, Atali-Kaladra, Palej, Ankleshwar, Jhagadia, Pavi-Jetpur, Tilakwada, Dediapada and Rajpipla. In these places, small agro trade and some agro-processing exist. However, large food export parks are limited, and most of the AEZ areas are still dominated by chemical and engineering industries.

3.2.1.4 Central Public Sector Undertaking

The Narmada Basin hosts several Central Public Sector Undertakings (CPSUs) and Central Government ministerial units listed in **Table 9**, contributing to industrial development, defence production, and regional employment. These undertakings operate across various districts, providing products, services, and infrastructure support to the basin.

Table 9 Central Public Sector Undertakings and Their Operational Presence in the Narmada Basin, Madhya Pradesh

Category	Undertaking	HQ / District	Operational Presence in Narmada Basin Districts
Central Public Sector Undertaking	Bharat Heavy Electricals Ltd (BHEL)	Bhopal	Units/projects in Bhopal, supply to Hoshangabad, Harda

Central Govt. Ministerial Undertaking	Manganese Ore	Balaghat	Mining operations in Balaghat (Tirodi mines)
	India Ltd (MOIL)		
	Hindustan Copper Ltd, Malajkhanda	Balaghat	Copper mining at Malajkhanda (Balaghat)
	Security Mill	Paper Hoshangabad	Plant at Hoshangabad (now Narmadapuram)
	Bank Note Press	Dewas	Printing currency notes, serves Indore/Ujjain region
	Ordnance Factory	Itarsi, Hoshangabad	Ammunition/defence supplies to basin districts
	Ordnance Factory	Katni	Defence manufacturing, Katni area
	Ordnance Factory, Khamariya	Jabalpur	Major ordnance facility in Narmada Basin
	Defence Vehicle Factory	Jabalpur	Production/maintenance of defence vehicles
	Gun Carriage Factory	Jabalpur	Artillery systems, defence industry hub
	Telecom Factory	Jabalpur	Telecom equipment, supplies within basin
	Heavy Vehicle Factory	Jabalpur	Heavy defence vehicle production
	Rail Coach Repair Workshop	Bhopal	Overhauls serving Narmada region railways
	Railway Zone (HQ)	Jabalpur	Southeast Central Railway HQ, full basin coverage
	Gray Iron Foundry	Jabalpur	Foundry products, defence & civil sector supplies

**Source: Industrial Profile of Madhya Pradesh 2016–17,*

Some central public sector companies are also located in the lower basin. ONGC(Oil and Natural Gas Corporation) has a role in developing Dahej SEZ and is one of its promoters. OPaL(ONGC Petro Additions Limited), which is a joint venture of ONGC, GAIL and GSPC, has a large petrochemical complex at Dahej. Petronet LNG Ltd. has its LNG terminal in the

Dahej port area. These public sector undertakings play a major role in petrochemical and energy industries in the region.

3.2.1.5 State's Public Sector Undertakings

The Narmada Basin is supported by numerous State Public Sector Undertakings (PSUs) listed in **Table 10**, that contribute to industrial development, agro-processing, energy, tourism, and social welfare. These undertakings operate across multiple districts, providing infrastructure, financial support, and sectoral services to strengthen regional development.

Table 10 State Public Sector Undertakings and Their Operational Presence in the Narmada Basin, Madhya Pradesh

Corporation / Board	Headquarters	Operational Presence in Narmada Basin Districts
M.P. Financial Corporation	Indore	Dhar, Dewas, Khargone, Khandwa, Barwani
M.P. Small Industries Corporation (Laghu Udyog)	Bhopal	Jabalpur, Hoshangabad, Harda, Narsinghpur
M.P. State Electronics Development Corp. Ltd.	Bhopal	Jabalpur, Indore, Hoshangabad
M.P. State Industrial Development Corporation	Bhopal	Jabalpur (AKVN), Indore (AKVN), Dhar, Dewas
M.P. Textbook Corporation	Bhopal	Distribution depots in Jabalpur, Hoshangabad
M.P. Tourism Development Corporation	Bhopal	Mandla (Kanha), Jabalpur (Bhedaghat), Omkareshwar (Khandwa), Hoshangabad (Satpura)
M.P. Women Finance & Development Corporation	Bhopal	Training / finance cells in Jabalpur, Narsinghpur
M.P. Agency for Promotion of IT (MAP_IT)	Bhopal	Indore, Jabalpur IT parks
M.P. Housing Board	Bhopal	Housing projects in Jabalpur, Hoshangabad, Harda, Khandwa
M.P. Paschim Kshetra Vidyut Vitran	Indore	Khargone, Khandwa, Barwani, Dhar, Harda

M.P. Poorv Kshetra Vidyut	Jabalpur	Narsinghpur, Hoshangabad, Seoni, Balaghat
M.P. Audhyogik Kendra	Indore	/
Vikas Nigam (AKVN)	Bhopal	Regional AKVN Jabalpur (Narmada Basin coverage)
M.P. Dugdh Vikas Nigam	Bhopal	Dairy units in Jabalpur, Hoshangabad, Narsinghpur
M.P. Energy Development Corporation (Urja Nigam)	Bhopal	Renewable energy projects in Jabalpur, Mandla, Hoshangabad
M.P. Khadi & Village Industries Board	Bhopal	Khadi units in Jabalpur, Hoshangabad, Sehore
M.P. Handicrafts & Handlooms Dev. Corp.	Bhopal	Crafts clusters in Jabalpur, Mandla, Narsinghpur
M.P. State Agro-Industries Dev. Corporation	Bhopal	Agro service centers in Narsinghpur, Hoshangabad, Harda
M.P. State Civil Supplies Corporation	Bhopal	Godowns in Jabalpur, Hoshangabad, Harda, Sehore
M.P. State Forest Development Corporation	Bhopal	Plantation areas in Mandla, Jabalpur, Narsinghpur
M.P. State Mineral Development Corporation	Bhopal	Mining sites in Balaghat, Jabalpur, Katni
M.P. State Pashudhan Evam Kukkut Vikas Nigam	Bhopal	Poultry / livestock units in Hoshangabad, Jabalpur
M.P. State Seed & Farms Development Corporation	Bhopal	Seed farms in Hoshangabad, Narsinghpur, Harda
M.P. State Warehousing Corporation	Bhopal	Warehouses in Jabalpur, Hoshangabad, Harda, Sehore

**Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17*

The state public sector companies are also important in the lower basin. Gujarat Alkalies and Chemicals Limited (GACL) has a large unit at Vilayat. Gujarat State Fertilizers & Chemicals Ltd. (GSFC) and Gujarat Narmada Valley Fertilizers & Chemicals Ltd. (GNFC) are both located in Bharuch. Dahej SEZ Ltd. is another important state company. These state undertakings are linked with chemicals, fertilizers and petrochemicals and add to the wastewater load in the basin.

3.2.1.6 Industrial Areas Managed by DTICs and GIDC

The Narmada Basin hosts numerous industrial areas managed by the District Trade and Industry Centres (DTICs) to promote regional industrialization and economic growth listed in **Table 11**. These areas provide developed land for industries, supporting manufacturing, trade, and services. The industrial land development varies across districts, with some areas fully developed and others partially developed, reflecting ongoing expansion and infrastructure improvement. These industrial zones play a significant role in attracting investment, generating employment, and boosting the industrial base of the basin.

Table 11 Industrial Areas Managed by DTICs and Their Development Status in the Narmada Basin, Madhya Pradesh

Sr. No.	District	Industrial Land Acquired (ha)	Land Developed (ha)	Coverage
1	Barwani	33.213	8.168	Fully
2	Betul	240.210	72.270	Partly
3	Bhopal	360.696	406.060	Partly
4	Dewas	438.859	286.717	Fully
5	Dhar	52.763	5.128	Fully
6	Harda	49.440	98.000	Fully
7	Hoshangabad	97.936	97.936	Fully
8	Indore	576.450	415.020	Partly
9	Jabalpur	249.160	169.630	Fully
10	Khandwa	172.410	12.900	Fully
11	Khargone	202.650	200.630	Fully
12	Mandla	517.210	162.360	Fully
13	Narsinghpur	3.246	2.000	Fully
14	Raisen	44.537	44.480	Partly
15	Sehore	54.290	20.250	Fully

**Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17*

The Gujarat Industrial Development Corporation (GIDC) manages many industrial estates in the lower basin. Vagra and Vilayat have large chemical and fibre industries. Panoli is famous for chemicals, dyes and pharmaceuticals. Nabipur has smaller units like plastics and engineering. Valia has metals, packaging and some chemicals. Naldhari has small industries like plastics, packaging and pharma. Bharuch GIDC has a mix of chemicals, textiles, food

processing and engineering. Dahej has very large petrochemical and chemical industries and a big SEZ. Palej is known for dyes, drugs and plastics. Ankleshwar is one of the largest chemical and pharmaceutical estates in India. Jhagadia is also a big chemical estate. Sankheda is special for wood craft and furniture. Pavi-Jetpur has small engineering and agro units. Jambughoda has small agro and forest-based units. Tilakwada, Dediapada and Rajpipla have small agro, food and textile industries.

3.1.1.7 Clusters by MSME-DI Indore

The Narmada Basin hosts several Micro, Small, and Medium Enterprise (MSME) clusters promoted by the MSME Development Institute (DI) in Indore depicted in **Table 12**. These clusters focus on specific industrial or agro-based activities, facilitating production, skill development, and entrepreneurship. While some clusters partially cover the basin, others are fully located within it, supporting local employment and contributing to the regional economy.

Table 12 MSME Clusters and Their Coverage in the Narmada Basin, Madhya Pradesh

Sr. No.	Cluster	District	Coverage in Basin
1	Engineering Cluster	Bhopal	Partly
2	Namkeen Cluster	Indore	Partly
3	MPPMO Pharma Cluster	Indore	Partly
4	Indore Pharma Cluster	Indore	Partly
5	Plastic Packaging Cluster	Indore	Partly
6	Rechargeable Torch Cluster	Indore	Partly
7	Readymade Garment Cluster	Indore	Partly
8	Power Loom Cluster	Jabalpur	Fully

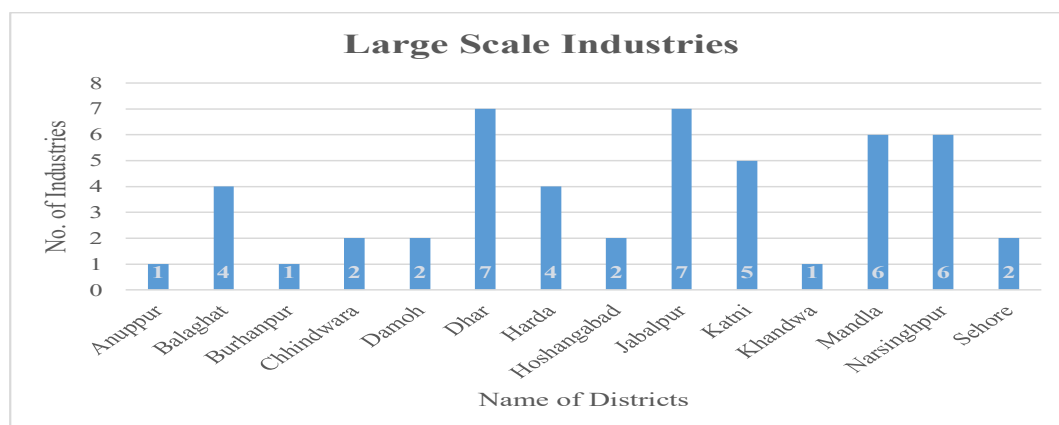
**Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17*

3.2.1.8 Large and Medium Scale Industries

The Narmada Basin hosts a diverse industrial landscape, encompassing large-scale (**Graph 107**) and medium-scale industries (**Graph 108**) across multiple districts. These industries span sectors such as cement, power, mining, pharmaceuticals, textiles, sugar, steel, and agro processing. Large-scale industries, including Bharat Heavy Electricals, Hindustan Copper, Diamond Cement, Trident Ltd., and Hindustan Liver Limited, serve as key drivers of economic growth and employment. Medium-scale industries, particularly concentrated in industrial

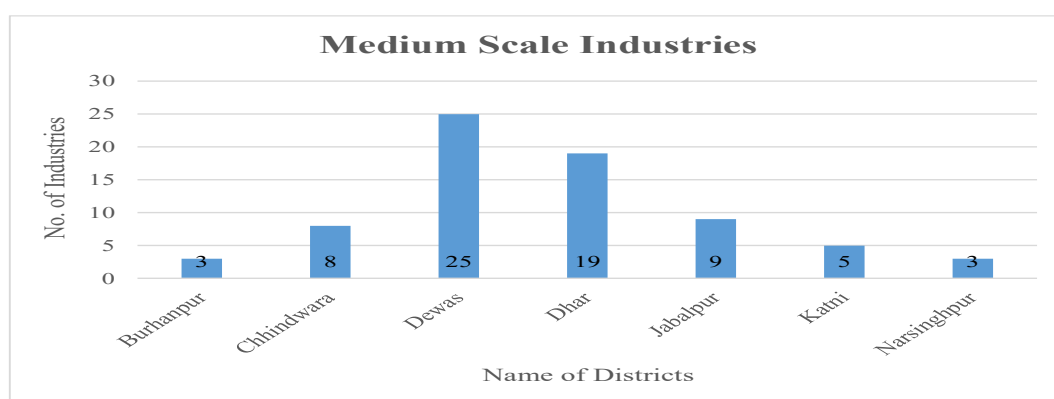
growth centres like Dhar (Pithampur), Dewas and Jabalpur complement these large units by supporting ancillary production, manufacturing, and service industries. Fig. 5 and 6 highlights the number of large as well as medium scale industries in different districts.

These industries are strategically distributed to leverage local resources, labour availability, and connectivity, contributing significantly to the basin's industrial and socio-economic development. While the complete list of industries is provided in **Annexure Table A3 and A4**.



Graph 107 Distribution of Large-Scale Industries Across Districts

*(Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17)



Graph 108 Distribution of Medium-Scale Industries Across Districts

*(Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17)

In lower basin Large and medium industries are located mainly in Bharuch district. In Vagra and Dahej, Reliance Industries, OPaL and other big petrochemical companies operate. L&T Chemicals and Deepak Nitrite are also present in Vagra. Panoli has companies like Clariant, Atul Ltd. and GACL. Nabipur has Gujarat Fluorochemicals and Bharuch Enviro Infrastructure. Vilayat has Birla Cellulose and Sud Chemie. Palej has Aarti Industries. Ankleshwar has Atul Ltd., Sun Pharma and Intas Pharma. Jhagadia has PI Industries and Excel Industries. These large industries create a big share of wastewater and pollution load in the basin.

3.2.2 Status of Industrial Sewage Generation

Table 13 District Wise Status of Industrial Sewerage Waste Generation, Narmada Basin

District	Industrial Sewage (MLD)
Anuppur	3.90
Balaghat	16.00
Betul	51.99
Chhindwara	46.33
Damoh	0.00
Dindori	0.00
Hoshangabad	25.00
Jabalpur	8.00
Katni	0.14
Mandla	0.90
Narshimapur	1006.00
Raisen	13.80
Sagar	15.30
Seoni	9.50
Alirajpur	0.00
Barwani	1.00
Burhanpur	1.07
Dewas	4.50
Dhar	18.60

District	Industrial Sewage (MLD)
Harda	0.11
Jhabua	0.72
Khandwa	39.96
Khargone	22.00
Sehore	22.00
Bharuch	~500.00
Ankleshwar (Bharuch)	~450.00
Narmada (District)	~15.00

**Source: District Environment Plan, MPPCB (from Narmada River Basin Infrastructure and Planning Report)*

In specific context to Narmada River basin, amount of Industrial Sewage generation varies from district to district. **Table 13** provides district wise industrial waste generation in Narmada River basin. However, the statistics shows in the table includes value of the whole district even if a particular district's small portion falls under the basin.

The table on industrial sewage generation in the Narmada River Basin reveals a highly uneven pattern of wastewater discharge, directly linked to the degree of industrialization in different districts. At the top of the scale, Narshinghpur stands out with an exceptionally high industrial sewage load of around 1006 MLD, making it by far the largest single contributor within the basin. However, this district doesn't have presence of larger industrial areas in comparison to districts like Dhar which produce only 18 MLD industrial wastewater. Similarly, Bharuch district in Gujarat, well known as one of India's leading industrial hubs, generates enormous industrial wastewater, estimated between ~450 and ~500 MLD. Together, Narshinghpur and Bharuch account for the bulk of the basin's industrial sewage, clearly identifying them as critical hotspots of industrial wastewater pollution.

Other districts contribute moderately high loads, reflecting pockets of industrial development. Betul (51.99 MLD) and Chhindwara (46.33 MLD) show significant levels of wastewater generation, largely due to their industrial estates and medium-scale industries. Khandwa (39.96

MLD) also emerges as a major contributor, while Hoshangabad (25 MLD) and Khargone and Sehore (22 MLD each) reflect steady industrial growth alongside their agricultural bases. Districts like Dhar (18.6 MLD), Balaghat (16 MLD), and Sagar (15.3 MLD) add to the medium-level range of wastewater discharge, showing a mix of localized industries such as textiles, food processing, and small manufacturing.

In contrast, a group of districts generates only limited amounts of industrial sewage, indicating better water treatment facilities or low level of industrialisation. These include Jabalpur (8 MLD), Seoni (9.5 MLD), Raisen (13.8 MLD), and Mandla (0.9 MLD). The lowest values are reported from districts such as Barwani (1 MLD), Burhanpur (1.07 MLD), Jhabua (0.72 MLD), and Harda (0.11 MLD), while Damoh, Dindori, and Alirajpur show negligible or zero industrial sewage generation, reflecting their predominantly agrarian or tribal character with minimal industrial activity. A special case is Narmada district in Gujarat, where industrial sewage is estimated at ~15 MLD, reflecting growing industrialization along the river corridor.

Thus, the data highlights the concentrated nature of industrial pollution in the basin, where a few districts such as Narshinghpur and Bharuch dominate the wastewater load, while the majority contribute relatively little. This concentration creates uneven pressure on the river system highly industrialized zones face acute challenges of wastewater treatment, effluent management, and river pollution, whereas less industrialized districts are still at a stage where preventive measures can be taken before large-scale problems emerge. The situation underlines the need for district-specific strategies: advanced treatment plants, strict monitoring, and enforcement of effluent standards in high-load districts; and proactive planning with sustainable industrial practices in emerging zones.

In lower basin Bharuch–Ankleshwar belt is the main source of industrial wastewater in the lower basin. Together, these two areas generate about 231.201 MLD of wastewater. Out of this, Ankleshwar produces about 104.414 MLD and Bharuch about 126.787 MLD. Only about 41.962 MLD of this is treated and discharged into drains or rivers, while the rest is handled through reuse or common treatment plants. In individual estates, Ankleshwar produces about 62 MLD, Panoli about 14.5 MLD, Vilayat about 2 MLD, Dahej about 50 MLD, and Jhagadia about 100 MLD. Rajpipla-II produces only 0.50 MLD. Smaller districts generate very little wastewater. Chhota Udepur generates 0.108 MLD, and all of it is treated inside the industries. Rajpipla produces 2.546 MLD, which is also managed without discharge into rivers. There are five CETPs (Common Effluent Treatment Plant) in the Bharuch-Ankleshwar area, along with

many in-house treatment units. Compliance with standards is generally high, and the Gujarat Pollution Control Board carries out inspections and monitoring.

3.3 Agricultural Runoff and Water Quality

3.3.1 Fertilizer and Pesticide Use Patterns

Agriculture in the Narmada River Basin is highly intensive, with farmers cultivating crops such as wheat, rice, soybean, cotton, pulses, sugarcane, and horticultural produce. To sustain high yields, the use of chemical fertilizers (mainly urea, DAP, potash, and complex fertilizers) and pesticides has become widespread. Fertilizer consumption is particularly high in irrigated stretches of Madhya Pradesh and Gujarat, where canal irrigation and groundwater extraction support multiple cropping cycles. Along with fertilizers, pesticides, including insecticides, herbicides, and fungicides are extensively applied, especially in cash crop belts like cotton in the lower basin (Bharuch, Narmada, and adjoining regions) and soybean-growing areas in the upper and middle basin.

Nitrogen consumption (tons) across districts of the Narmada basin highlights sharp regional variations over time (**Fig. 6**). The highest levels are recorded in Hoshangabad (Narmadapuram), where consumption rose dramatically from 557.23 tons in 1970 to 87,117.01 tons in 2017, followed closely by Khargone/West Nimar with an increase from 2,125.17 tons to 74,408.48 tons. Other major consumption centers include Dhar and Vadodara (Baroda), which also show steep upward trends, crossing 40,000 and 39,000 tons respectively by 2017. In the mid-range, districts such as Khandwa (East Nimar), Sehore, Raisen, Narsinghpur, and Bharuch report values between 20,000 and 32,000 tons, indicating significant but comparatively moderate usage. At the lowest end, districts such as Shahdol and Bilaspur remain below 500 tons and 2 tons respectively, while others like Damoh and Sagar stay under 1,000 tons even in 2017. This contrast reflects a pattern where the central and agriculturally intensive districts of the basin have experienced sharp increases in fertilizer use, while peripheral and tribal-dominated regions continue to record minimal consumption (Since the figure shows historical fertilisers consumption, as a result, small portions of some then undivided districts—such as Durg, Bilaspur, Jalgaon, Damoh, and Panchmahal—are included in the analysis).

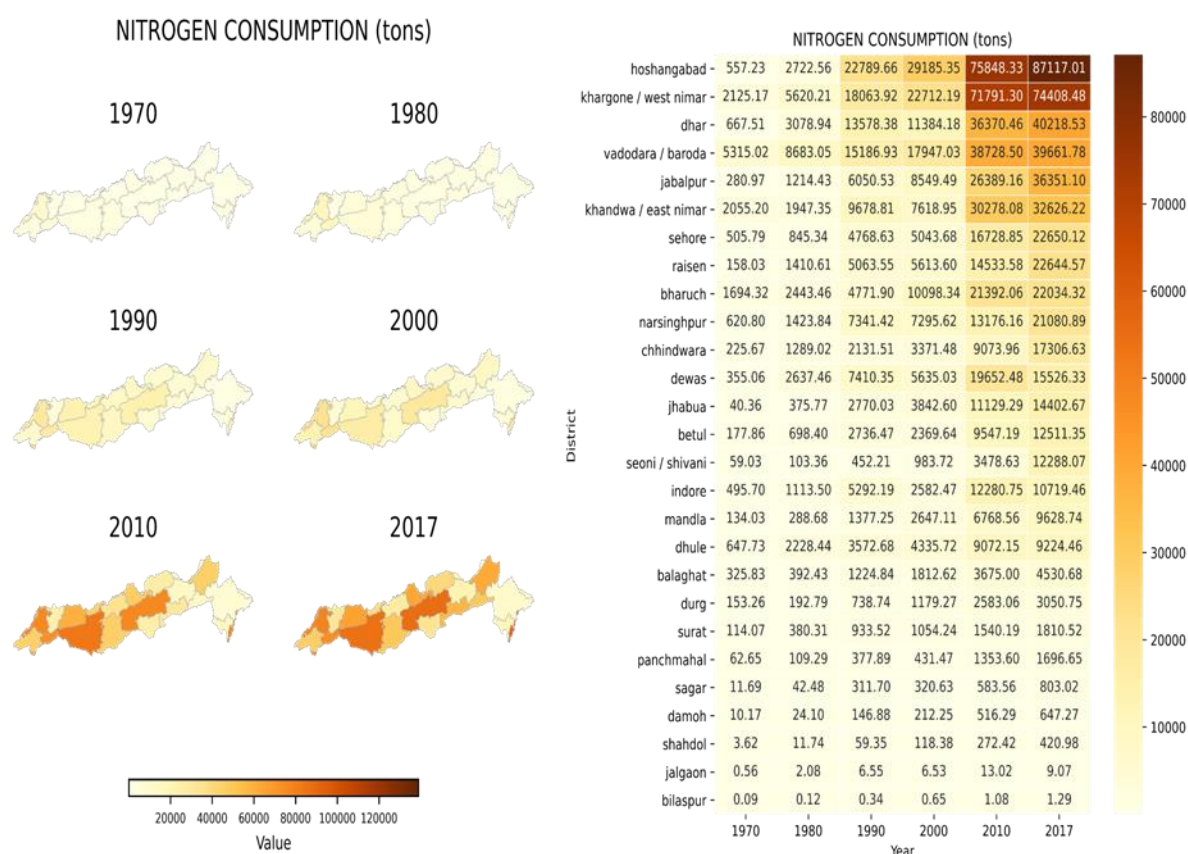


Figure 6 Use of Nitrogen in Narmada River Basin

In terms of phosphate-consumption Fig. 7 shows a pronounced increase in fertilizer use concentrated in the central Narmada belt. At the top end, Hoshangabad (Narmadapuram) rises from 209.58 tons (1970) to 50,706.60 tons (2017), and Khargone grows from 841.61 tons to 35,788.61 tons; Dhar also shows a large rise from 328.58 to 19,725.31 tons. A mid-range group includes Khandwa / East Nimar (105.91 → 13,243.30), Sehore (156.66 → 12,116.88), Raisen (249.49 → 10,163.77) and Bharuch (238.39 → 8,766.10), each reflecting substantial but lower consumption than the hotspots. At the low end are largely non-intensive or tribal/peripheral districts such as Bilaspur (0.03 → 0.52), Jalgaon (0.25 → 4.47) and Shahdol (6.26 → 234.10), which remain minimal consumers by 2017. Overall, the pattern highlights very large growth in phosphorus use in a few cores, irrigated districts (Hoshangabad, Khargone, Dhar), a broad middle band of significant users, and many fringe districts where inputs remain low.

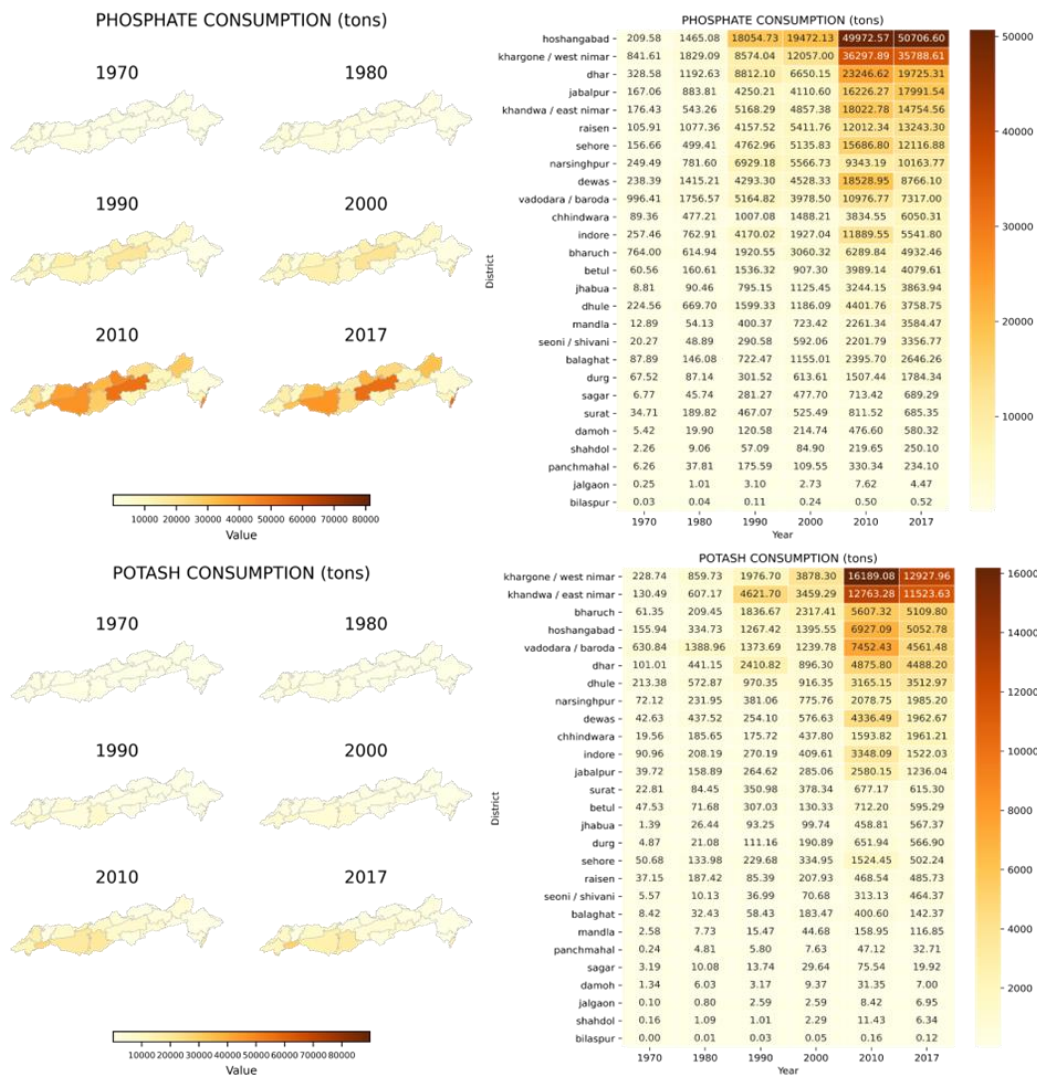


Figure 7 Potash and Phosphate Consumption, Narmada River Basin

Fig. 7 also shows the consumption of potash in the Narmada River Basin from 1970 to 2017. The largest users by 2017 are Hoshangabad, which increased from 228.74 tons in 1970 to 12,927.96 tons in 2017, and Khargone, which rose from 130.49 to 11,523.63 tons over the same period. A broad mid-range group includes Sehore (50.68 → 502 tons), Raizen (72.12 → 1,985.20 tons), and Bharuch (61.35 → 5,109 tons). Smaller but notable increases are seen in Mandla (2.28 → 117.85 tons) and Balaghat (8.42 → 142.37 tons).

In the Gujarat stretch of the Narmada basin, cotton, paddy, groundnut, and pulses are the dominant crops. These require substantial inputs of nitrogen (N), phosphorus (P), and potassium (K) fertilizers, which increases the risk of nutrient runoff into rivers during rainfall and irrigation return flows. Excess soluble nitrogen and phosphorus are readily mobilized during monsoon-induced surface runoff and subsurface flow, contributing to nutrient enrichment in the Narmada River and raising the potential for eutrophication.

Cotton cultivation in Gujarat also involves intensive insecticide and pesticide use. When applied shortly before or during heavy rains, these chemicals can be washed into the river through field runoff and drainage channels, creating pulses of agrochemical contamination.

Monitoring and estimation of pollution loads in Gujarat should consider:

- District-level crop areas and fertilizer/pesticide application rates.
- Local runoff coefficients and monsoon rainfall patterns to estimate pollutant transport.
- Timing of fertilizer and pesticide application relative to rainfall events.
- Validation using GPCB water quality data at key Narmada stations such as Bharuch, Garudeshwar, Chandod, and Zapor/NTPC.

Mitigation strategies include split-dose fertilizer application, vegetative buffer strips along drains, integrated pest management, and event-based monitoring at tributary confluences.

3.3.2 Seasonal Runoff Contribution

The Narmada River in Gujarat exhibits strong seasonal variability in flow and water quality, closely linked to monsoon rainfall and agricultural cycles.

1. Monsoon / Kharif Season (June–September)

Nearly 80–85% of the annual river flow occurs during the southwest monsoon, coinciding with the kharif cropping season. High-intensity rainfall generates significant surface runoff, resulting in elevated river discharge. This period is characterized by:

- **Peak pollutant transport** from agricultural fields (fertilizers, pesticides), untreated sewage, and eroded sediments.
- **Sharp increases** in turbidity, total dissolved solids (TDS), nutrients (nitrate, phosphate), and biochemical oxygen demand (BOD).
- **Episodic pollutant pulses** immediately after rainfall events, making **event-based sampling** (per storm or per day) more representative than monthly averages for capturing pollutant peaks.

2. Post-Monsoon / Rabi Season (October–March)

With the cessation of monsoon rainfall, river discharge gradually declines and becomes dominated by regulated reservoir releases and baseflow. Water quality during this season shows:

- **Stabilized turbidity and TDS**, with pollutant concentrations largely governed by point-source discharges (municipal wastewater, industrial effluents).
- Moderate nutrient levels from residual agricultural return flow after rabi sowing and irrigation cycles.
- Generally improved dissolved oxygen (DO) levels due to better dilution and lower organic load compared to monsoon peaks.

3. Summer / Pre-Monsoon Season (April–May)

This is the low-flow period of the river, marked by minimal rainfall and high evaporation losses. Key characteristics include:

- **Concentration of pollutants** (TDS, chloride, fluoride) due to reduced dilution capacity.
- **Elevated temperatures** can lead to localized drops in DO and higher biochemical activity.
- Greater vulnerability of aquatic ecosystems to even small pollutant loads because of limited flow.

Seasonal monitoring at key stations such as Garudeshwar has consistently demonstrated these patterns, underscoring the need for season-specific management interventions and **event-based monitoring during monsoon** to capture true pollutant load dynamics.

3.3.3 Impact on Water Quality and Environmental Risk

Water quality assessments in the Narmada Basin reveal exceedances of several critical parameters beyond WHO/IS standards. As highlighted in **Table 14**, Nitrate levels ranging from 45–90 mg/L, particularly in Omkareshwar and Bharuch, pose risks of drinking water contamination. Phosphate concentrations between 1.5–3.0 mg/L in Hoshangabad and Barwani indicate risks of eutrophication and aquatic toxicity. Pesticide residues, with 2-4 compounds detected in agricultural belts, highlight concerns over chronic human exposure and ecological stress. In addition, heavy metals such as zinc, nickel, and chromium have been found above permissible limits in river sediments across Madhya Pradesh and Gujarat, threatening aquatic ecosystems. These findings underscore the need for basin-wide pollution control and stricter water quality monitoring.

Table 14 Impact of Agrochemical Use on Water Quality

Parameter	Observed Level	Permissible Limit (WHO/IS)	Risk Zone (Basin)	Key Impacts
Nitrate (NO ₃ ⁻)	45–90 mg/L	<45 mg/L	Omkareshwar, Bharuch	Drinking water contamination
Phosphate (PO ₄ ³⁻)	1.5–3.0 mg/L	<1 mg/L	Hoshangabad, Barwani	Eutrophication, aquatic toxicity
Pesticide Residues	Detected compounds	(2–4NA)	Agricultural belts	Chronic exposure, health risks
Heavy Metals	Zn, Ni, Cr above limits	Varied	River sediments in MP/Guj	Aquatic ecosystem impact

**Source: CPCB Water Quality Report (2022), Singh et al. (2021)*

Water quality monitoring in Gujarat's Narmada basin (stations: Garudeshwar, Chandod, Zadeshwar/Bharuch, Zapor/NTPC) over the last decade shows:

BOD (Biochemical Oxygen Demand):

- Peaks during and after the monsoon, reflecting organic matter from agriculture and untreated or partially treated sewage.
- Ten-year data shows persistent high spikes with no significant long-term improvement.

COD (Chemical Oxygen Demand) & DO (Dissolved Oxygen):

- COD peaks often exceed BOD due to recalcitrant organics.
- Elevated COD corresponds with drops in DO, creating low-oxygen zones harmful to aquatic life.

Nutrients (Nitrogen & Phosphorus):

- Nitrate is transported in dissolved form; phosphorus is largely attached to sediment.
- Post-monsoon surges increase the risk of algal blooms.

TDS / Conductivity:

- Diluted during monsoon; increases in dry months, particularly downstream of irrigated areas or saline groundwater inflows.

Fecal Coliforms:

- Monsoon runoff carries animal and human waste, elevating coliform counts and posing public health risks.

Metals & Pesticide Residues:

- Heavy rainfall mobilizes metals and pesticides, especially near industrial zones or hazardous waste sites.
- Sediment monitoring is crucial to assess accumulation.

10-Year Summary – Gujarat Reach:

1. Monsoon-driven pollution dominates the annual cycle.
2. BOD, COD, nutrients, and coliforms consistently surge during rains.
3. Persistent high peaks indicate limited long-term improvement, pointing to the need for better fertilizer handling, erosion control, and sewage treatment.

3.4 Sewage and Other Pollution Sources in the Narmada River Basin, Gujarat

3.4.1 Untreated Sewage Discharges

A significant number of towns and drains along Narmada are discharging untreated sewage directly into the river. At Amarkantak, the origin of the Narmada, drains such as Gayatri, Savitri and Batrani Nallahs carry untreated sewage, since the proposed STP of 1.2 MLD remains under construction. Dindori faces a similar situation, with Sanbara, Samsan Ghat and city domestic drains releasing untreated waste as its 3.8 MLD plant is still incomplete. In Mandla, the situation is critical at Navghat (Tanki Ghat), where a 0.5 MLD STP is not functioning and raw sewage flows into the river; at Mandla Khai Nallah, a 7.75 MLD STP is under construction, but untreated water continues to discharge. Narmadapuram, with multiple drains like Kisanpura, Bhilpura, Korighat and Rajghat, has no functional STP, and untreated sewage flows directly into the river. Likewise, Kannod and Khategaon in Dewas district, as well as Nemawar, discharge untreated sewage in absence of treatment plants. Further downstream, Barwaha's

Nurani Masjid Nallah, Sanawad's Morghadi and Atmortakka Nallahs, Mandleshwar's Sonia Nagar and Gangajha drains, Maheshwar's Kamanighat, Bus Stand and Ward No.4 drains, and Badwani's Anjad and Tirchipulliya Nallahs are all releasing untreated wastewater into the Narmada, since their STPs are either under construction or only proposed. These locations represent the most serious pollution threats, with raw sewage flowing unchecked into the lifeline river.

Table 14 also shows that a few towns have made progress with sewage treatment but continue to discharge partially treated or untreated effluents. In Mandla's Ranrejghat, a 2 MLD plant is under construction, but sewage continues to enter the river untreated. Jabalpur, one of the largest contributors, has a mixed scenario: at Khandari Nallah in Lalpur and Tewar, STPs of 34 and 29 MLD are under construction, and about 60 MLD of sewage is only being managed through temporary bioremediation. At Shah Nallah near Gwarighat road, the small 0.55 MLD plant functions but still discharges partially into the river. Omkareshwar has four small STPs with a total capacity of 0.65 MLD that are operational, yet additional plants are under construction, and untreated sewage continues to reach Narmada. In Dhar, at Dharmapuri town, a 2 MLD treatment plant works and supplies treated water for gardening and agriculture, but the surplus treated water is ultimately discharged into the river. These examples show that partial interventions exist but are insufficient to bridge the large treatment gaps.

Apart from the wastewater discharge a few locations are treating wastewater completely and discharging in rivers or utilising for different purposes. In Jabalpur, at Moti Nallah near Gohalpur police station, a 32 MLD plant is operational, with treated water discharged into the Omti Nallah and then into the Pariyat and Hiran Rivers before meeting the Narmada. Similarly, the large Omti Nallah STP in Jabalpur treats 50 MLD, discharging treated effluents into connected rivers that ultimately join the Narmada. In Budhni, three small plants (2.2, 1.08 and 0.15 MLD) are functioning effectively, and their treated water is reused for gardening and construction, avoiding direct untreated discharge. These locations show the effectiveness of operational treatment facilities and provide a contrast to the widespread untreated discharges elsewhere

Urban centers along the Narmada River, particularly in the Bharuch-Ankleshwar region, often release untreated or partially treated sewage into the river. This contributes to elevated levels of Biochemical Oxygen Demand (BOD), nutrients like ammonia and nitrates, and coliform bacteria, adversely affecting water quality. The GPCB's Monthly Progress Report indicates a

total sewage generation of 4,003 Million Liters per Day (MLD), with an additional treatment capacity requirement of 337.56 MLD to meet the demand. Comparative analysis with nearby monitoring stations shows spikes in BOD levels, particularly during periods of heavy rainfall, indicating potential overflows or untreated discharges.

3.4.2 Non-Point & Industrial Pollution Sources

Industrial Stormwater (Ankleshwar-Jhagadia): During unseasonal rains, several factories in the Ankleshwar-Jhagadia industrial belt have been reported to discharge chemical-laced effluents directly into the Narmada River. This practice has led to the death of thousands of fish, highlighting the severe impact of industrial stormwater runoff on aquatic life.

Hazardous Waste Site (Jhagadia TSDF): The Jhagadia Hazardous Waste Treatment, Storage, and Disposal Facility (TSDF) has a capacity of 1 million Metric Tons (MT). While designed to contain leachate, extreme rainfall or leaks from the facility may release pollutants such as metals or organics into nearby drains that lead to the river.

Soil and Sediment Runoff: Agricultural runoff, particularly during heavy rains, washes soil and associated nutrients like phosphorus and pesticides into the river. This non-point source pollution contributes to nutrient loading and potential eutrophication of the river system.

4. Solid Waste Management

4.1 Quantity of Solid Waste Generation (Urban and Rural)

Solid waste management (SWM) across the Narmada River Basin districts shows wide variation in both current and projected waste generation, reflecting differences in urbanization, population density, and industrial activities. Based on secondary data analysis, the estimated municipal solid waste (MSW) generation in 2025 ranges from as low as 3.78 TPD in Seoni to as high as 313.42 TPD in Khargone, indicating stark contrasts in waste generation potential across the basin. By 2030, districts such as Dewas (222.20 TPD), Barwani (156.34 TPD), and Sehore (157.77 TPD) are projected to show significant growth in waste generation, attributed primarily to rapid urban expansion, increasing commercial activities, and demographic growth.

Major urban centers such as Indore (7.62 TPD as per current estimates), Jabalpur (112.50 TPD in 2030), and Hoshangabad (117.64 TPD in 2030) continue to emerge as critical nodes of solid waste generation. Indore, though reporting lower values in this dataset due to partial coverage of municipal boundaries, is well recognized as a benchmark city in India for successful waste segregation and processing. Other districts like Damoh are also projected to nearly double their waste generation from 57.47 TPD in 2025 to 70.86 TPD by 2030, highlighting the urgent need for strengthening collection and treatment systems as depicted in **Table 15**.

Table 15 Solid Waste Generation in the Narmada River Basin

Districts	Estimated Solid Waste Generation in 2025 (TPD)	Estimated Solid Waste Generation in 2030 (TPD)
Alirajpur	39.21	42.21
Anuppur*	25.98	27.34
Balaghat	5.55	5.87
Barwani	142.2	156.34
Betul	72.83	76.75
Burhanpur	10.6	11.4
Chindwara	37.65	40.24
Damoh	57.47	70.86
Dewas	191.48	222.2
Dhar	120.88	132.26
Dindori	6.63	-

Harda	20	-
Hosangabad	110.48	117.64
Indore	7.616	-
Jabalpur	106.1	112.5
Jhabua	18.95	21.28
Katni	65.49	70.9
Khandwa	108.32	-
Khargone	313.42	-
Mandla	13.5	-
Narshingpur	72.95	78.06
Raisen	124	96.53
Sagar	34	36
Sehore	122.67	157.77
Seoni	3.78	-
Umaria	12.51	14

*All the data filled according to the basin Portion.

Source: District Environmental Plan for different districts by MPPCB

It is also noteworthy that rural and tribal-dominated districts such as Alirajpur (39.21 TPD in 2025, 42.21 TPD in 2030), Anuppur (25.98 to 27.34 TPD), and Jhabua (18.95 to 21.28 TPD) have relatively low waste generation but face challenges in collection efficiency and scientific disposal. On the other hand, emerging urban industrial districts such as Dewas, Khargone, Raisen, and Sehore show rapid growth trends, which may intensify stress on existing landfill and dumping sites.

Overall, the projected increase in solid waste generation across the basin calls for proactive planning in waste collection, transportation, and scientific disposal facilities. Legacy waste management and compliance with Solid Waste Management Rules (2016) will be essential to mitigate risks of leachate contamination and pollution load into the Narmada and its tributaries. For granular tehsil-level estimates, readers are referred to **Annexure Table A5**, which provides detailed tabulations to support district-level planning and policy development.

The Lower Narmada Basin within Gujarat comprises a mix of large urban centers, medium-sized industrial towns, and small urban settlements distributed across the districts of **Surat, Bharuch, Dahod, Narmada, Panchmahal, Vadodara, and Chhota Udepur**(Table 16).

Together, these regions represent the terminal segment of the Narmada River basin, discharging into the Gulf of Khambhat. The population density and degree of urbanization vary widely, resulting in diverse waste generation characteristics.

According to the **Gujarat Pollution Control Board (GPCB)** and **Central Pollution Control Board (CPCB)**, the state as a whole generates approximately **9,800–10,800 tonnes of municipal solid waste per day (TPD)** (GPCB, Monthly Progress Reports 2023–24). Within this total, the lower basin districts contribute a significant fraction due to the presence of major urban centers such as Surat and Vadodara, as well as growing towns like Bharuch, Dahod, and Rajpipla in the Narmada district.

Surat District

Surat is Gujarat's largest generator of municipal solid waste and one of India's most advanced cities in terms of waste management systems. As per data published by the **Surat Municipal Corporation (SMC)**, the city generates approximately **2,000–2,200 TPD** of solid waste. The SMC manages door-to-door collection across all wards, using a fully mechanized fleet of compactors and transfer stations.

Over the last two years (Nov 2022–Jan 2024), the generation rate has shown a moderate increase of around **5–8%**, corresponding to urban population growth, higher per-capita waste generation, and improved waste collection efficiency. Surat operates large-scale processing facilities for segregation, composting, RDF preparation, and waste-to-energy projects under its Smart City initiatives.

Key features:

- Generation: ~2,000–2,200 TPD (SMC data, GPCB consolidated reports)
- Collection efficiency: ~100% (door-to-door coverage)
- Processing: 100% of collected waste through RDF, composting, and WtE routes
- Source: SMC Solid Waste Management Department, GPCB MPR 2023

Vadodara District

The **Vadodara Municipal Corporation (VMC)** is another major generator of waste in the lower Narmada region. As per VMC's solid waste management department reports, the city produces about **700–750 TPD** of municipal solid waste. Approximately 98% of waste is collected daily, supported by mechanized transfer and compactor vehicles.

Vadodara's waste consists largely of biodegradable materials (about 55–60%) and recyclables (25–30%), with the remainder being inert materials. Decentralized composting and biomethanation plants handle a portion of the biodegradable fraction.

Key features:

- Generation: ~700–750 TPD (VMC data)
- Collection efficiency: ~98%
- Processing: Composting and small biomethanation units; dry waste handled through MRFs
- Source: VMC Solid Waste Management Approach Report, 2024; GPCB PRS, 2023

Bharuch District

Bharuch, located on the Narmada River's lower reach, serves as an industrial and commercial hub with substantial municipal and industrial waste generation. The city's municipal solid waste generation is estimated at **80–120 TPD**, based on population size, industrial activity, and CPCB per-capita benchmarks. The **Bharuch Nagarpalika** has initiated door-to-door collection in all wards, though segregation at source remains partial.

Legacy waste at the old dumpsite is being addressed through biomining under the guidance of the GPCB. Plans are in progress to establish a **sanitary landfill and composting facility** under a public-private partnership (PPP) model.

Key features:

- Generation: ~80–120 TPD (estimated from CPCB benchmark and GPCB ULB category)
- Collection: >90% door-to-door coverage

- Processing: Ongoing composting pilot and planned RDF/WtE project under PPP
- Source: GPCB MPR 2023; CPCB State Inventory; Bharuch Nagarpalika DPR extracts

Dahod District

Dahod district is characterized by smaller urban bodies and semi-urban settlements. According to GPCB's monitoring records and district updates, Dahod's urban local bodies together generate approximately **50–70 TPD** of solid waste. Waste collection systems have been strengthened under the Swachh Bharat Mission, but complete segregation and treatment facilities are still under development.

The district has faced issues of **open dumping and legacy waste fires** at older sites (e.g., Devgadhi Baria), indicating the need for proper remediation and engineered disposal infrastructure.

Key features:

- Generation: ~50–70 TPD (estimated from CPCB benchmark and ULB records)
- Collection efficiency: ~85–90%
- Processing: Limited composting; no large-scale WtE facilities
- Source: GPCB Monthly Progress Reports, 2023; CPCB Urban Solid Waste Inventory

Narmada District (Rajpipla and Surrounding Towns)

The **Narmada district**, with **Rajpipla** as its headquarters, contributes a smaller yet important portion of the basin's total waste load. The municipal area and surrounding villages together generate roughly **20–30 TPD** of solid waste. Waste management operations are managed by the Rajpipla Nagar Palika, with support from district and state programs.

The majority of collected waste is disposed of at designated dump sites, with recent efforts underway to introduce small-scale composting and segregation at source. Legacy waste identification and biomining DPRs have been prepared under the supervision of GPCB.

Key features:

- Generation: ~20–30 TPD (Rajpipla and nearby GPs)
- Collection: Partial D2D coverage; manual collection in smaller wards
- Processing: Small composting pits; planned MRF development
- Source: GPCB River Basin Report, 2023; CPCB data portal

Panchmahal District

The **Panchmahal district**, including major urban centers such as Godhra and Halol, contributes an estimated **90–120 TPD** of municipal solid waste. Rapid urbanization and industrialization in Halol and nearby areas have increased per-capita waste generation. However, smaller municipalities and nagar panchayats within the district still rely on traditional collection systems and open dumping.

The district has been part of GPCB’s pilot initiatives for improved segregation and dry waste collection. Composting units have been set up in select wards through municipal and community partnerships.

Key features:

- Generation: ~90–120 TPD (estimated; CPCB benchmarks)
- Collection efficiency: 85–95% in major towns
- Processing: Limited composting; dry waste handled through manual sorting units
- Source: GPCB Annual Status Reports, 2023; CPCB National Inventory

Chhota Udepur District

Chhota Udepur represents the smallest waste-generating district within the Lower Narmada Basin. The district headquarters town produces approximately **15–25 TPD** of solid waste, primarily biodegradable in nature. Most waste is collected manually by the Nagar Palika and deposited at open dumps on the outskirts of the town.

The district's current focus is on setting up small decentralized composting units and improving segregation at source. GPCB has advised the district to identify land for a sanitary landfill and develop a micro-level solid waste management plan under the Swachh Bharat Mission (Urban).

Key features:

- Generation: ~15–25 TPD (estimated from CPCB per-capita norms)
- Collection: Partial coverage (70–80%)
- Processing: Small community compost pits; minimal dry waste recovery
- Source: GPCB Monthly Compliance Report, 2023; CPCB Solid Waste Inventory

Table 16 Summary of Waste Generation in Lower Narmada Basin (Gujarat)

District Major ULB	Estimated Waste Generation (TPD)	Key Features
Surat	2,000–2,200	Fully mechanized, 100% collection, WtE & RDF operational
Vadodara	700–750	Mechanized collection, composting and biomethanation in use
Bharuch	80–120	Industrial town, composting & PPP landfill proposed
Dahod	50–70	Smaller towns, partial segregation, legacy dump remediation
Narmada	20–30	Basic collection, composting planned
Panchmahal	90–120	Semi-urban, expanding collection & composting
Chhota Udepur	15–25	Small town, partial coverage, decentralized composting

*All the data filled according to the basin Portion. Source: District Environmental Plan for different districts by GPCB

Total Estimated Waste Generation in Lower Narmada Basin (Gujarat): ~3,000–3,300 TPD

- The **urban centers of Surat and Vadodara** dominate total waste generation, together accounting for nearly 85% of the basin’s waste load.
- Smaller districts such as **Narmada and Chhota Udepur** generate modest amounts but lack adequate processing infrastructure, requiring focused attention on decentralized solutions.
- **Bharuch and Panchmahal** represent growing industrial-urban regions where waste management systems are still evolving.
- **Dahod** requires urgent remediation of legacy sites and establishment of sanitary disposal facilities to prevent contamination of nearby water bodies.
- Continuous updating of ULB-level data through **GPCB’s Monthly Progress Reports** and **CPCB’s Swachh Bharat Mission dashboard** is essential to track trends between 2022 and 2024 and beyond.

4.1.1 Solid Waste Generation and Processing Status in Narmada River Basin Districts (Nov 2022 vs. Jan 2024)

Table 17 presents a comparative account of municipal solid waste generation, processing, and the corresponding processing gaps for districts falling under the Narmada River Basin. Data has been compiled for two reference periods – November 2022 and January 2024 – based on the Swachh Bharat Mission MIS (2022–23).

Key highlights:

- **Overall Improvement:** Most districts have shown significant progress in reducing the processing gap between 2022 and 2024.
- **High-Generation Districts:** Bhopal, Indore, and Jabalpur continue to be the largest waste generators, each exceeding 500 TPD, while smaller districts such as Dindori and Harda generate below 30 TPD.
- **Gap Reduction:** Districts like Sagar, Anuppur, Raisen, and Khandwa demonstrated

noticeable reduction in processing gaps, reflecting improved waste management efficiency.

- **Zero Gap Achievements:** By January 2024, most districts (e.g., Indore, Bhopal, Jabalpur, Katni, Dhar, Burhanpur, Mandla, Seoni, etc.) achieved near complete or complete processing, ensuring no backlog of unprocessed waste.
- **Sustained Performance:** Districts such as Alirajpur, Dewas, Khargone, Narsimhapur, and Chhindwara already had minimal or no processing gaps in 2022 and sustained this efficiency through 2024.

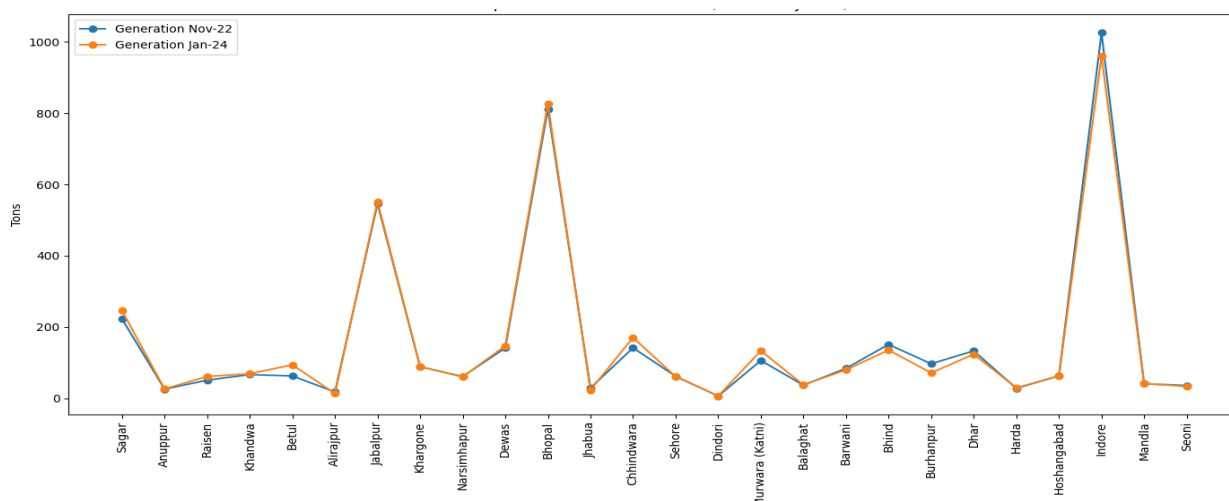
Table 17 Status of solid waste management district-wise

Sr. N.	District Name	Nov-22			District	Jan-24		
		Generation (Tons)	Processing (Tons)	Processing Gap		Generation (Tons)	Processing (Tons)	Processing Gap
1	Sagar	223.02	206.73	16.29	Sagar	245.97	233.49	12.48
2	Anuppur	25.5	14.99	10.51	Anuppur	26.21	18.39	7.83
3	Raisen	50.81	45.83	4.98	Raisen	61.07	56.53	4.54
4	Khandwa	66.4	62.1	4.3	Khandwa	68.77	64.85	3.92
5	Betul	62.61	61.13	1.48	Betul	93.92	91.66	2.27
6	Alirajpur	17.67	17.67	0	Alirajpur	13.32	13.16	0.15
7	Jabalpur	545.88	545.88	0	Jabalpur	551.85	551.75	0.1
8	Khargone	88.48	88.48	0	Khargone	88.59	88.52	0.07
9	Narsimhapur	60.9	60.74	0.15	Narsimhapur	60.02	59.96	0.06
10	Dewas	141.06	140.87	0.19	Dewas	146.91	146.86	0.05
11	Bhopal	811.72	811.72	0	Bhopal	826.65	826.61	0.04
12	Jhabua	28.7	28.7	0	Jhabua	23.17	23.14	0.03
13	Chhindwara	141.52	141.52	0	Chhindwara	170.32	170.31	0.01
14	Sehore	61.65	60.81	0.84	Sehore	60.87	60.87	0
15	Dindori	6.06	6.06	0	Dindori	6.44	6.44	0
16	Murwara (Katni)	105.88	105.88	0	Murwara (Katni)	133.23	133.23	0

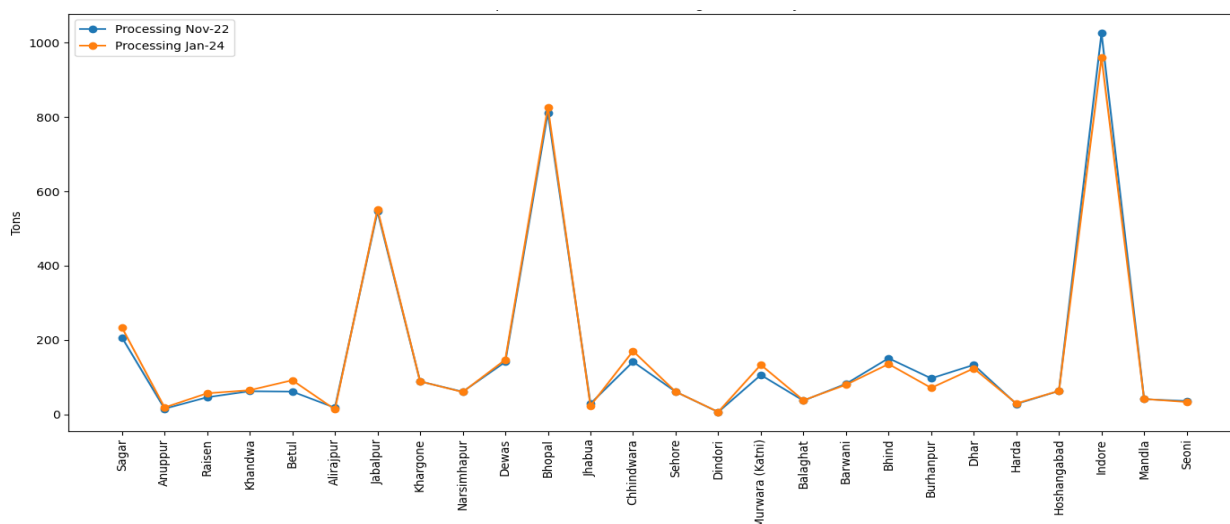
17	Balaghat	36.72	36.72	0	Balaghat	37.68	37.68	0
18	Barwani	83.86	82	1.86	Barwani	79.51	79.51	0
19	Bhind	150.9	150.9	0	Bhind	135.32	135.32	0
20	Burhanp	96.74	96.74	0	Burhanp	71.08	71.08	0
	ur				ur			
21	Dhar	132.96	132.96	0	Dhar	123.49	123.49	0
22	Harda	27.83	27.83	0	Harda	29.24	29.24	0
23	Hoshang	62.79	62.79	0	Hoshang	62.96	62.96	0
	abad				abad			
24	Indore	1025.89	1025.89	0	Indore	960.58	960.58	0
25	Mandla	40.71	40.71	0	Mandla	41.28	41.28	0
26	Seoni	35.78	35.78	0	Seoni	33.11	33.11	0

**Source: Presentation cum Compliance by Govt. of Madhya Pradesh in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016), National Green Tribunal, September 2025*

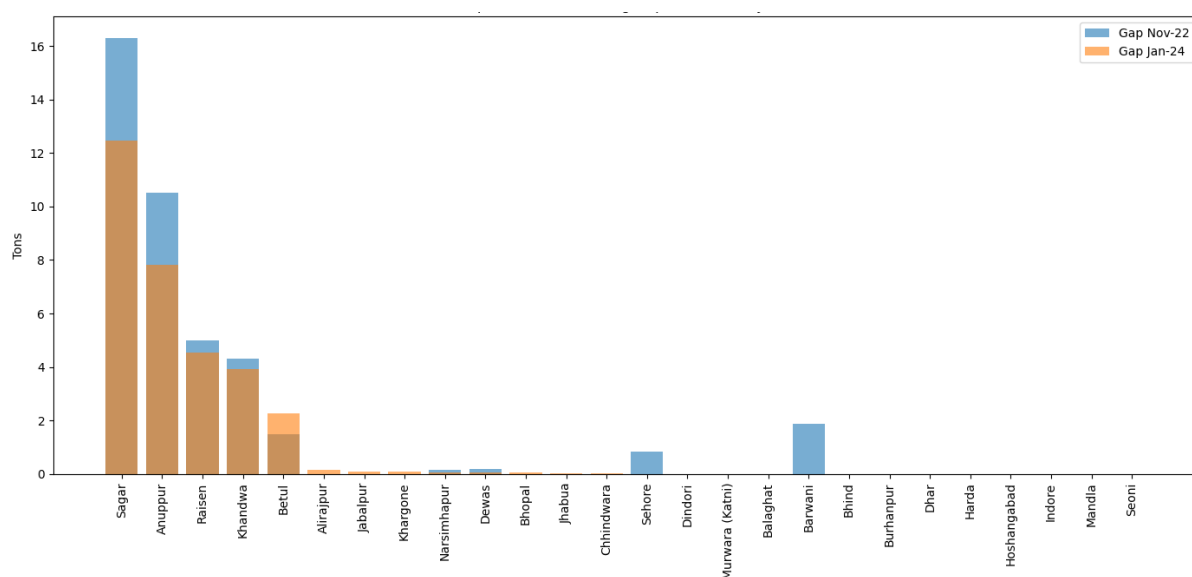
This progress highlights the effectiveness of localized solid waste management systems across the Narmada Basin and underscores the growing emphasis on scientific processing, segregation, and treatment of waste. **Garph. 109, 110 and 111** shows the comparative analysis of waste generation. The near elimination of processing gaps indicates substantial movement towards compliance with the Solid Waste Management Rules, 2016, and supports improved environmental health of the Narmada River Basin.



Graph 109 Comparison of Waste Generation (Nov 22 vs Jan 24)



Graph 110 Comparison of Waste Processing (Nov 22 vs Jan 24)



Graph 111 Comparison of Processing Gap (Nov 22 vs Jan 24)

To assess the recent trend of municipal solid-waste generation in the **Lower Narmada Basin (Gujarat)**, comparative estimates have been compiled for **November 2022** and **January 2024** using data from the **Gujarat Pollution Control Board (GPCB)**, **Central Pollution Control Board (CPCB)**, and available municipal-corporation sources (Surat Municipal Corporation and Vadodara Municipal Corporation). Where district-wise quantities were not published on GPCB/CPCB public dashboards, **CPCB per-capita generation benchmarks** (0.45–0.5 kg/person/day for municipal areas) and ULB classification have been used to derive conservative estimates.

Between November 2022 and January 2024, waste generation in most Lower-Basin districts has shown a **modest but consistent increase (4–8%)**, reflecting urban growth, improved collection coverage, and more accurate reporting by ULBs. Surat and Vadodara remain the two dominant contributors—together accounting for nearly **85 %** of the Basin’s total municipal solid waste given in **table 18**.

Table 18 Solid Waste Generation Comparison (Nov 2022 vs Jan 2024) for Lower Narmada Basin Districts, Gujarat.

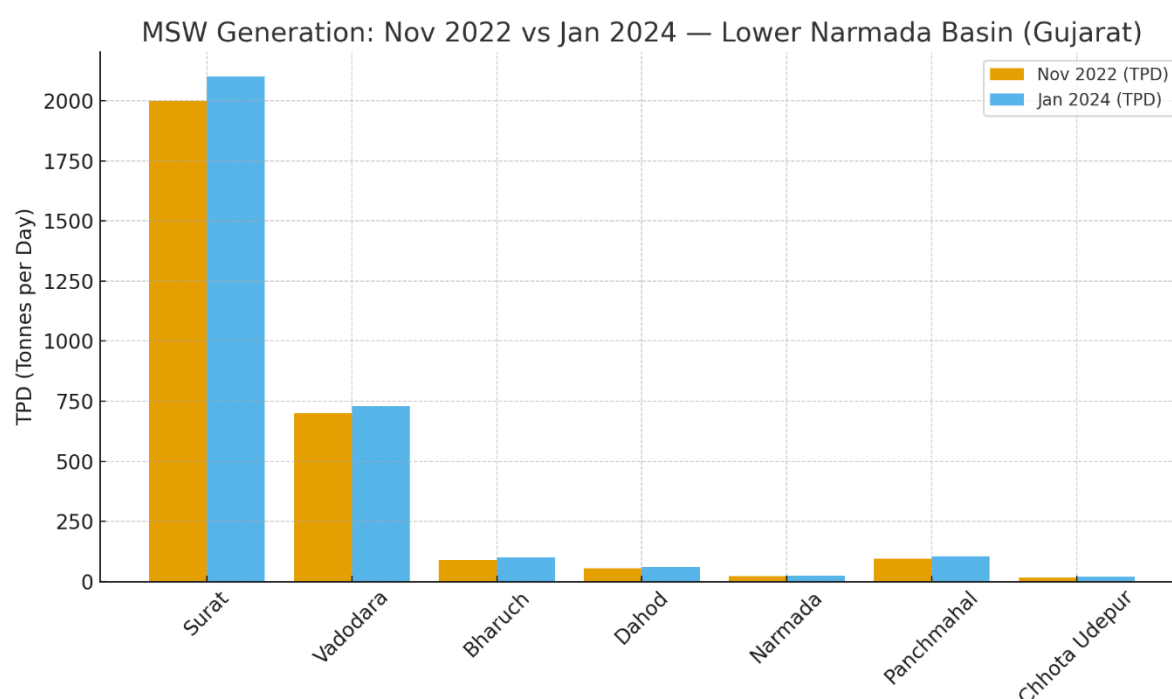
District / ULB	Nov 2022 (TPD)	Jan 2024 (TPD)	Change (TPD)	% Change	Remarks / Source
Surat	2,000	2,100	+100	+5.0 %	Reported by SMC & GPCB MPR (2023-24)
Vadodara	700	730	+30	+4.3 %	VMC Dept. Reports & GPCB State Data
Bharuch	90	100	+10	+11.1 %	Estimated (CPCB benchmark + ULB inputs)
Dahod	55	60	+5	+9.1 %	Estimated (CPCB benchmark; GPCB district notes)
Narmada	22	25	+3	+13.6 %	Estimated (ULB data / GPCB River Basin Report)
Panchmahal	95	105	+10	+10.5 %	Estimated (CPCB benchmark; ULB aggregates)
Chhota Udepur	17	20	+3	+17.6 %	Estimated (ULB inputs / GPCB 2023 MPR)

**(Sources: GPCB Monthly Progress Reports 2023-24; CPCB SWM Annual Report 2023; SMC & VMC ULB data; CPCB per-capita estimates for other districts.)*

District-wise Interpretation

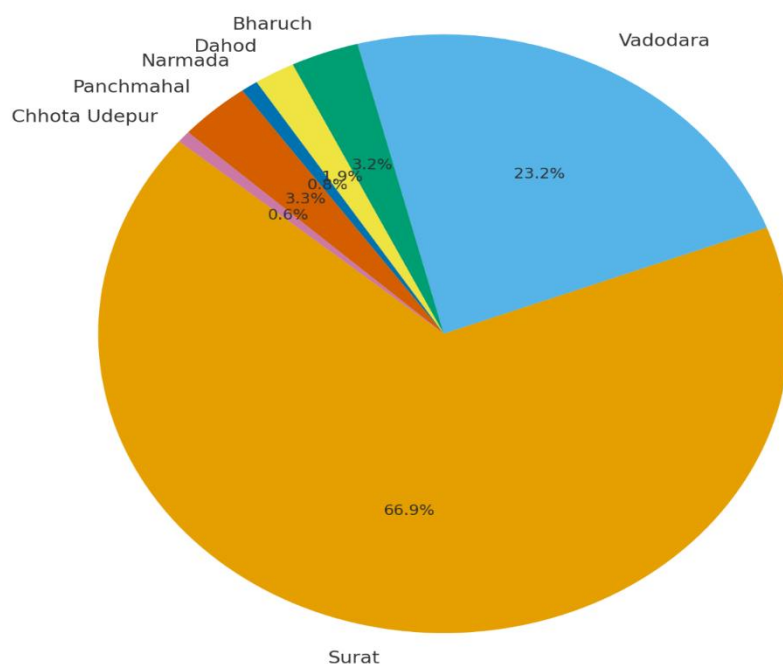
- **Surat:** Generation increased from ~2,000 TPD to ~2,100 TPD due to expansion of collection coverage and population growth. SMC’s mechanized system processes nearly 100 % of waste through composting, RDF and WtE facilities.

- **Vadodara:** Waste increased slightly (700 → 730 TPD), indicating stabilized generation with efficient collection. Composting and bimethanation plants are operational.
- **Bharuch:** A moderate rise (90 → 100 TPD) aligns with industrial growth and higher service coverage; a PPP sanitary landfill is under planning.
- **Dahod:** Incremental increase (55 → 60 TPD) attributed to improved reporting under Swachh Bharat Mission; processing facilities remain minimal.
- **Narmada (Rajpipla):** Small rise from 22 to 25 TPD; community-level composting and biomining proposals are under implementation.
- **Panchmahal:** Generation rose from 95 to 105 TPD following urban expansion in Godhra and Halol. Local composting and segregation programmes are being introduced.
- **Chhota Udepur:** Lowest absolute increase (17 → 20 TPD), yet highest percentage growth (≈18 %) due to improved door-to-door coverage and new municipal wards.



Graph 112 MSW Generation Comparison (Nov 2022 vs Jan 2024) — Lower Narmada Basin Districts
 *(Source: GPCB MPR 2023-24; CPCB benchmarks; ULB data)

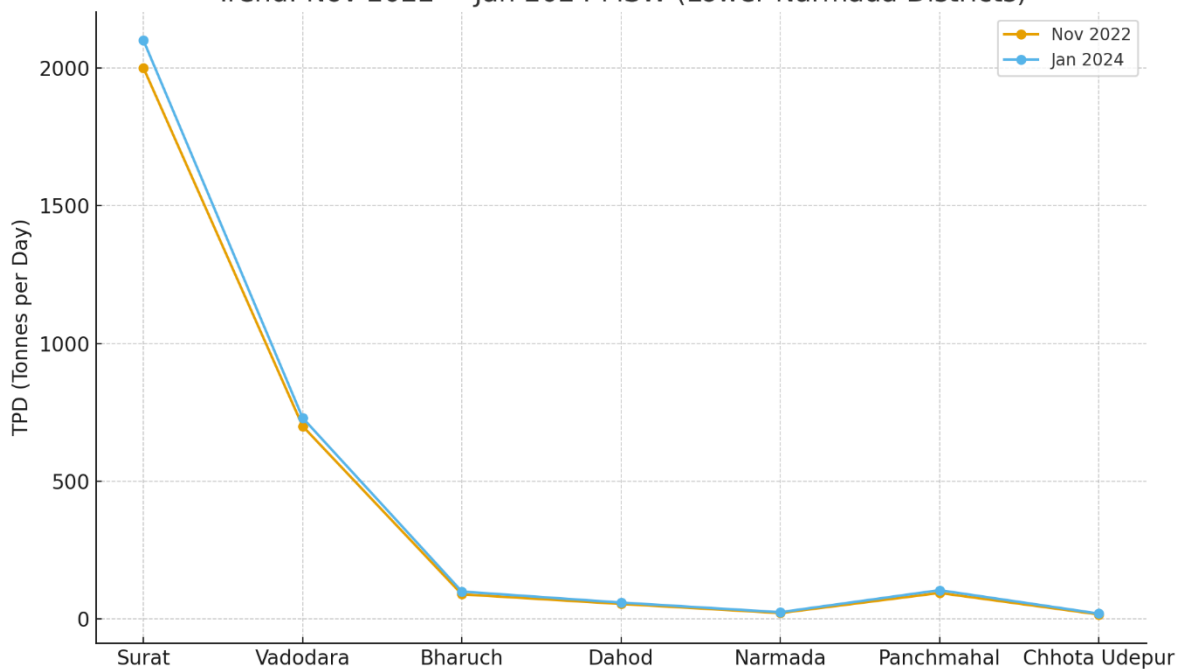
Share of MSW Generation by District — Jan 2024 (TPD)



Graph 113 Share of MSW Generation by District — Jan 2024

*Source: Compiled from GPCB MPR 2023-24 and ULB records

Trend: Nov 2022 → Jan 2024 MSW (Lower Narmada Districts)



Graph 114 Trend of MSW Generation (Nov 2022 → Jan 2024) — Lower Narmada Basin

*Source: GPCB MPR 2023-24; CPCB SWM Report 2023

- **Overall Increase:** Total MSW generation in the Lower Narmada Basin increased from approximately **2,980 TPD in Nov 2022 to 3,140 TPD in Jan 2024** ($\approx 5.4\%$ growth).
- **Dominant Cities:** Surat and Vadodara jointly contribute over **83 %** of the total basin waste, requiring sustained capacity for RDF and WtE operations.
- **Emerging Clusters:** Bharuch and Panchmahal are expanding industrial-urban centres with rising waste generation; they need cluster-level integrated facilities.
- **Smaller ULBs:** Narmada and Chhota Udepur have low absolute quantities but higher relative growth; decentralized composting and segregation should be prioritized.
- **Legacy Sites:** Dahod and Bharuch still rely partly on open dumping; remediation and sanitary-landfill development remain critical.

Between 2022 and 2024, Lower Narmada Basin districts recorded steady increases in municipal solid-waste generation. The trend underscores the urgency of scaling up processing infrastructure—particularly composting, MRFs, and RDF/WtE facilities—to maintain compliance with the **Solid Waste Management Rules, 2016**. Continuous monitoring through **GPCB Monthly Progress Reports** and **CPCB state inventories** will be vital for accurate trend assessment and basin-level planning.

4.2 Composition of Solid Waste

The composition of municipal solid waste (MSW) in the Narmada River Basin varies widely across urban local bodies (ULBs), reflecting differences in urbanization, industrial activity, consumption patterns, and the age of dumpsites. Due to limited availability of systematic datasets, composition analysis could be carried out for only four ULB sites Waraseoni (Balaghat), Pithampur (Dhar), Mandideep (Raisen), and Thandla (Jhabua). These datasets, though not fully uniform, provide a representative snapshot of the waste profile in different contexts: legacy sites dominated by inert and decomposed fractions, active dumpsites with a balanced mix of biodegradable and recyclable materials, and small-town sites with extremely high plastic and RDF fractions. Together, these results highlight the heterogeneity of solid waste streams across the basin and emphasize the need for site-specific waste management strategies including bio-mining, segregation at source, material recovery facilities (MRFs), and plastic waste channelization.

At the legacy waste site in Waraseoni (Balaghat district), the composition analysis reveals a predominance of sand and silt (56%) and decomposed/inert fractions (30.86%), which together account for nearly 87% of the total waste. This highlights the aged nature of the dumpsite, where active biodegradable fractions have already decomposed, leaving behind soil-like material and inert mass. Among the minor fractions, stone (8.14%) is notable, while plastic (2%), cardboard (1.2%), glass (0.8%), metals (0.6%), and cloth and rags (0.4%) constitute only a small share.

Overall, the waste profile at Waraseoni indicates a legacy-dominated dump with minimal recyclable or high-value fractions, underscoring the importance of bio-mining and safe disposal of the inert load to reduce long-term environmental impact.

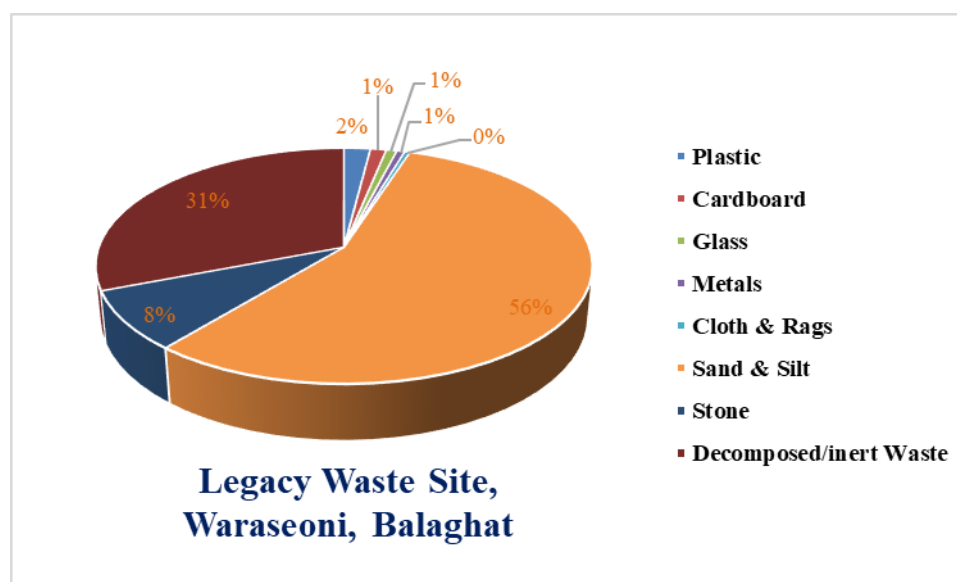


Figure 8 Legacy waste site in Waraseoni solid waste composition

**Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal*

The Pithampur dumpsite in Dhar district presents a mixed waste profile with natural soil (61.3%) and inert waste (14.9%) together accounting for more than three-fourths of the total composition. This indicates a significant proportion of non-recoverable fractions typical of open dumpsites. Among the recoverable materials, wood and miscellaneous waste (4.3%), cloths and rags (3.8%), and plastic waste (3.9%) form the largest components, suggesting moderate recycling potential. Smaller but notable fractions include paper (1.5%), glass and

metals (1.6%), synthesis waste (1.7%), food waste (2.9%), vegetable waste (1.7%), horticulture waste (1.3%), and rubber waste (1.1%).

Overall, the Pithampur composition reflects an active waste stream with a balance of inert and recyclable fractions, highlighting the need for segregation at source, material recovery facilities (MRFs), and controlled processing to reduce the burden on dumping grounds.

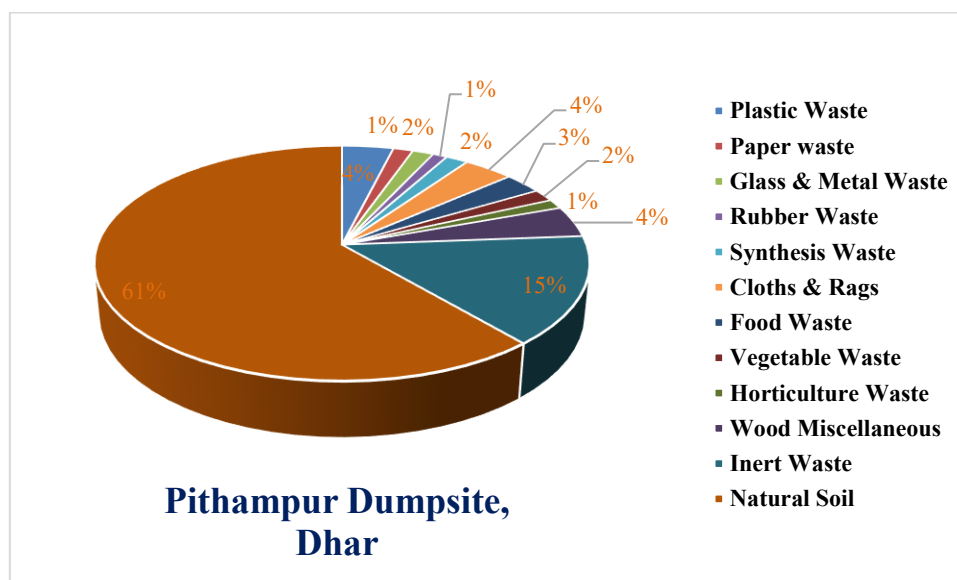


Figure 9 Pithampur dumpsite Dhar Solid waste composition

**Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal*

At the Mandideep dumpsite located behind HEG Industries in Raisen district, the composition analysis shows a relatively balanced distribution between biodegradable, recyclable, and inert fractions. Heavy fractions (24.50%), fine fractions (20.88%), and coarser fractions (15.54%) together dominate the profile, accounting for over 60% of the waste mass. These components represent the bulkier and degraded material typically found in industrial-adjacent dumpsites.

Biodegradable content is still significant, with biodegraded fractions (18.02%) highlighting ongoing organic degradation processes. Among recyclables, paper and cardboard (4.96%), glass (4.96%), and plastics/polythene/tetra packs (2.76%) are the main contributors. Smaller shares include horticulture and wood waste (1.24%), metals (0.67%), textiles (1.43%), and leather/rubber waste (5.05%), which collectively reflect the heterogeneous nature of municipal and industrial solid waste.

The results indicate that Mandideep’s dumpsite contains both recoverable fractions and a high load of inert and heavy materials, making it suitable for interventions such as bio-mining coupled with resource recovery to reduce landfill pressure.

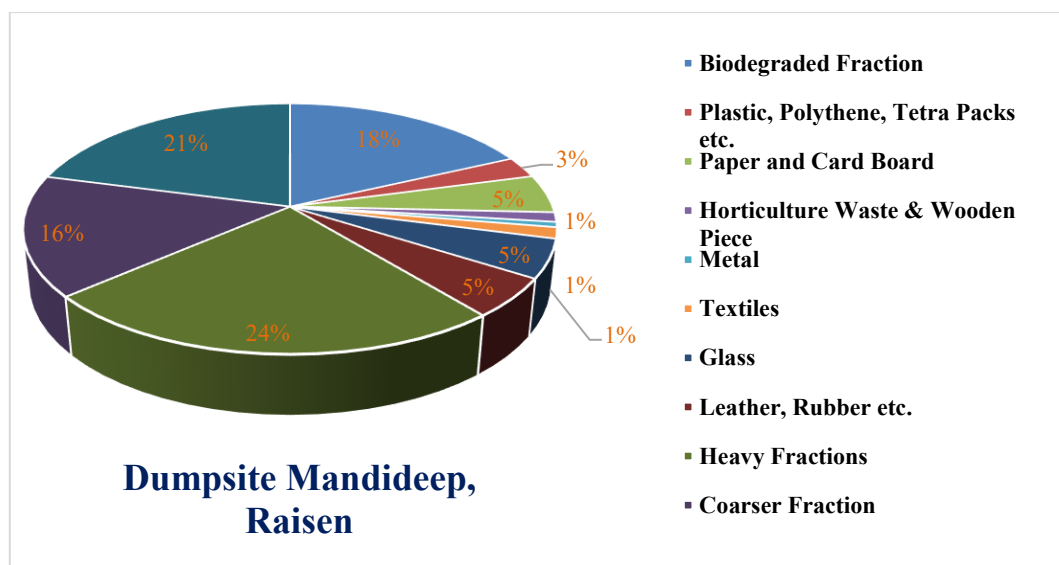


Figure 10 Mandideep dumpsite Raisen solid waste composition

**Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.*

The Thandla municipal council dumpsite in Jhabua district is overwhelmingly dominated by plastic, polythene, tetra packs, and textiles (92%), which together form the bulk of the waste stream. This extremely high proportion of refuse-derived fuel (RDF) material highlights growing dependence on plastic and synthetic packaging, even in smaller towns. Minor fractions include organic waste (3.5%), horticulture and wooden waste (1.5%), and inert material such as silt and debris (2%), while paper/cardboard (0.5%) and recyclable metals, glass, leather, and rubber (0.6%) are almost negligible.

The composition at Thandla reflects a plastic-heavy waste stream with very limited biodegradable or recyclable fractions, posing challenges for sustainable management. This underscores the urgent need for plastic waste segregation, RDF channelization to cement kilns or WtE plants, and awareness programs to reduce uncontrolled disposal.

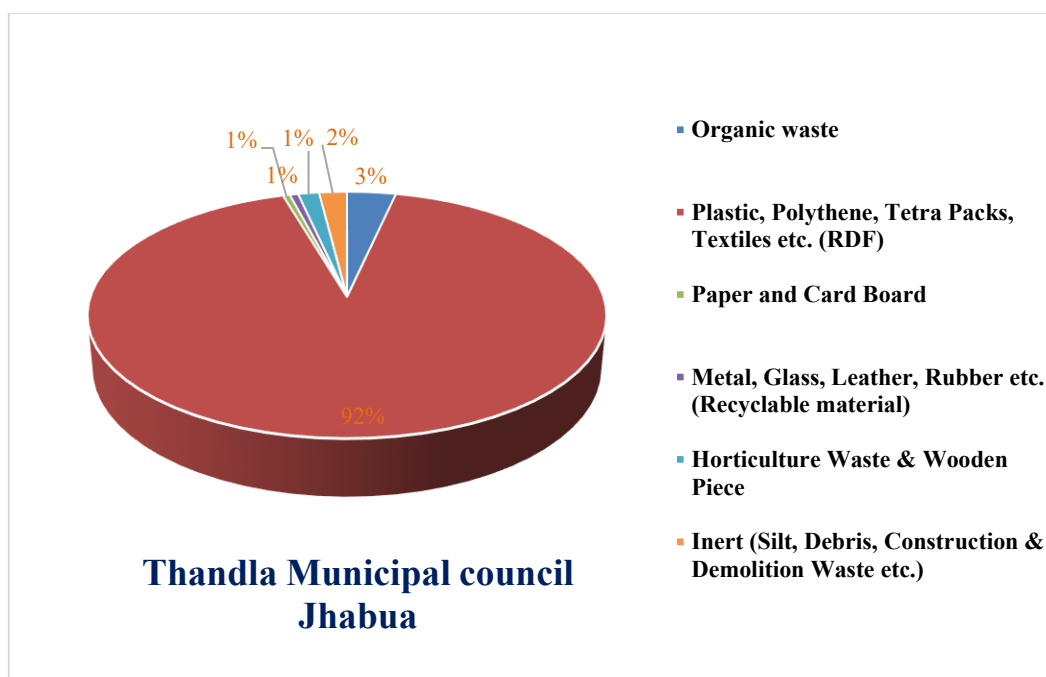


Figure 11 Thandla municipal council dumpsite Jhabua, solid waste composition.

Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.

The composition of municipal solid waste (MSW) in the **Lower Narmada Basin districts**—namely **Surat, Vadodara, Bharuch, Dahod, Narmada, Panchmahal, and Chhota Udepur**—reflects a mixture of residential, commercial, and institutional activities, varying with levels of urbanization and industrialization. Data and estimates have been compiled from **GPCB Monthly Progress Reports (2023–24)**, **CPCB Solid Waste Management Report (2023)**, and individual ULB reports (Surat and Vadodara).

Across Gujarat's ULBs, the waste stream typically consists of:

Table 19 Indicative composition of municipal solid waste in Lower Narmada Basin districts, Gujarat.

Component	Typical Share (% by weight)	Source / Remarks
Biodegradable (Wet waste)	45 – 60 %	Kitchen, vegetable, fruit and garden waste. Dominant in Surat & Vadodara due to dense residential zones.

<i>Component</i>	<i>Typical Share (% by weight)</i>	<i>Source / Remarks</i>
Dry Recyclables	25 – 35 %	<i>Paper, plastics, metals, and glass; proportion higher in industrial / commercial towns like Bharuch and Panchmahal.</i>
Inert & Construction Waste	10 – 20 %	<i>Soil, dust, debris; relatively high near road projects and urban expansion areas.</i>
Domestic Hazardous Waste	< 3 %	<i>Batteries, sanitary waste, small e-waste fractions; handled separately by ULBs and informal collectors.</i>

**(Sources: GPCB MPR 2023–24; CPCB SWM Annual Report 2023; Surat Municipal Corporation & Vadodara Municipal Corporation SWM Departments.)*

Surat:

Being a major urban and industrial hub, Surat’s waste stream contains roughly **50 % biodegradable**, **30 % dry recyclables**, and **20 % inert materials**. Markets and food establishments contribute large volumes of wet waste suitable for composting / biomethanation. SMC operates source segregation and RDF-feedstock preparation units.

Vadodara:

VMC reports a biodegradable share of **55–60 %**, supported by green-waste collection from gardens and households. Dry waste averages **30 %**, mainly paper, plastics, and packaging from residential colonies. Composting and small biomethanation plants handle segregated organic waste.

Bharuch:

As an industrial and trade-center district, Bharuch’s municipal waste shows higher **dry-fraction content (~35 %)** due to packaging and industrial canteen waste. Wet waste still accounts for nearly **50 %**, while **inerts (15 %)** arise from ongoing infrastructure works.

Dahod:

Predominantly residential with semi-urban characteristics; waste composition estimated at **55 % wet**, **30 % dry**, and **15 % inert**. Limited segregation leads to mixed streams; most recyclable recovery is by informal collectors.

Narmada (Rajpipla):

Smaller town areas show a biodegradable fraction exceeding **60 %**, mostly kitchen waste, while **dry recyclables (25 %)** come from local markets. Pilot composting sites are functional at ward level.

Panchmahal:

Urban areas such as Godhra and Halol generate mixed municipal and commercial waste—roughly **45 % wet**, **35 % dry**, and **20 % inert**. Plastic packaging and disposable materials have increased with industrial growth.

Chhota Udepur:

The smallest generator in the region; waste is largely biodegradable ($\approx$ **65 %**) with small portions of plastics and inerts. Composting and community pits are the preferred processing methods.

Seasonal and Event-Based Variation

- **Festivals (Navratri, Diwali, Ganesh Chaturthi):** Temporary surges of **15–25 %** in total waste, mainly from decorations, food, and plastics.
- **Monsoon season:** Higher moisture content and heavier biodegradable loads.
- **Industrial and market cycles:** Surat and Bharuch see fluctuating packaging waste from textile / chemical sectors.

Implications for Processing

1. High biodegradable content across all districts supports **composting and biomethanation** as primary wet-waste management routes.
2. Substantial dry fractions justify **Material Recovery Facilities (MRFs)** and **Refuse-Derived Fuel (RDF)** production, especially in Surat, Vadodara, and Bharuch.
3. Inert fractions require **engineered landfills or sanitary cells** to prevent leachate and riverine contamination.
4. Districts with smaller quantities (Narmada, Chhota Udepur) should promote **decentralized composting units** at ward / village cluster

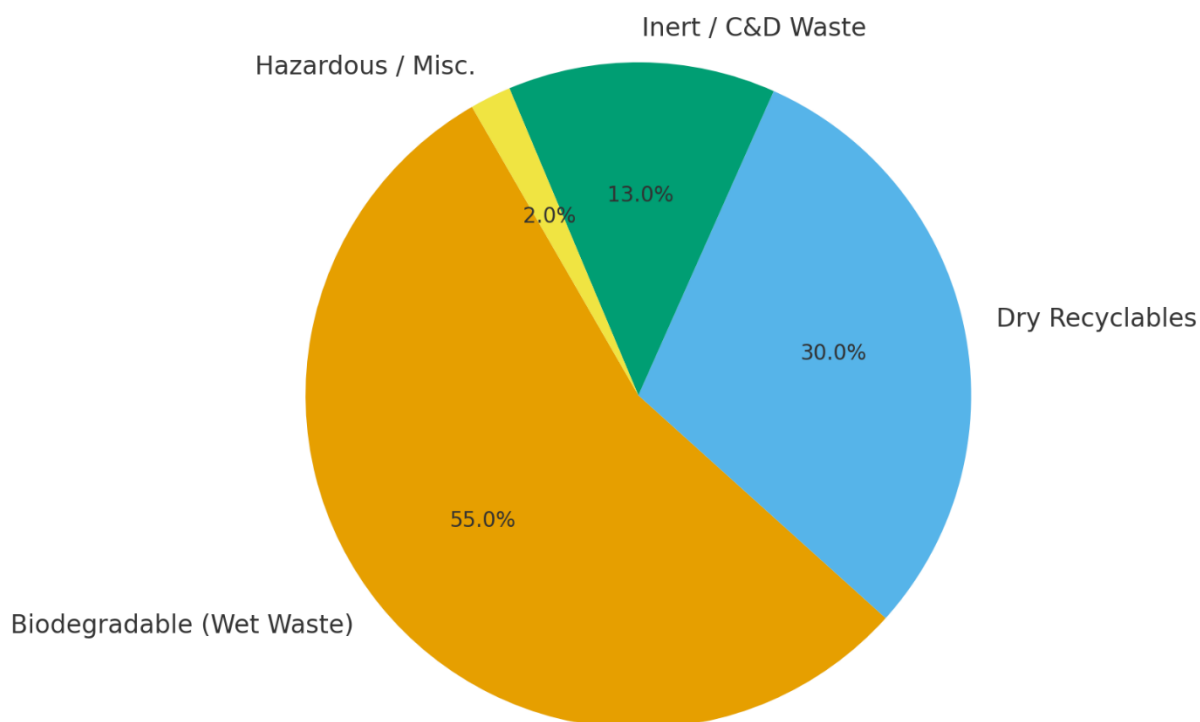


Figure 12 Typical Composition of Municipal Solid Waste in Lower Narmada Basin (Gujarat).

***Source:** Gujarat Pollution Control Board (GPCB) MPR 2023–24; Central Pollution Control Board (CPCB) SWM Annual Report 2023; Surat & Vadodara Municipal Corporation data.

- Waste composition remains broadly consistent with Gujarat’s state average, confirming the dominance of organic waste in total generation.
- Rising dry fractions in Bharuch and Panchmahal signal the need for formal recycling linkages and EPR-based plastic recovery.
- Segregation at source is improving in larger cities but remains limited in smaller ULBs.
- Seasonal peaks highlight the importance of flexible collection scheduling and temporary storage capacity.

4.3 Collection, Transportation, and Disposal Facilities

Solid waste management (SWM) in the districts of the Narmada Basin, Madhya Pradesh, has been documented through their respective District Environmental Plans (DEPs). The data tabulated in **Annexure Table A6 and A7** provides a comprehensive overview of the existing infrastructure, processes, and gaps across the waste management chain.

At the source level, segregation of dry and wet waste has been promoted, though the effectiveness varies across districts. Primary collection is generally carried out through waste

collection trolleys with separate compartments, enabling household-level segregation. In addition, mini collection trucks with separate compartments are deployed for secondary collection and transportation to processing facilities.

For wet waste management, two main pathways are observed: composting units and bio-methanation plants. Composting is widely adopted at both decentralized and centralized levels, while bio-methanation initiatives are present in selected districts but face challenges in scaling up due to technological and maintenance issues.

Dry waste management is primarily handled through Material Recovery Facilities (MRFs), where recyclable materials are sorted and sent for recycling. However, the coverage of MRFs remains limited in many districts, leading to inefficient recovery and leakage of recyclables into the landfill stream.

For non-recyclable waste, sanitary landfills are designated as the final disposal option. While some districts have operational landfills, others continue to face issues of inadequate infrastructure, scientific landfill development, and leachate management.

Across all stages like collection, transportation, and disposal, gaps and challenges remain significant. These include inadequate source segregation, limited capacity of collection fleets, insufficient infrastructure for composting and bio-methanation, underutilization of MRFs, and the need for scientifically designed and managed landfills. Strengthening institutional mechanisms, capacity building, and community participation are essential to improving SWM across the Narmada Basin.

The collection, transportation, and disposal systems for municipal solid waste (MSW) in the **Lower Narmada Basin**—covering the districts of **Surat, Vadodara, Bharuch, Dahod, Narmada, Panchmahal, and Chhota Udepur**—have undergone steady improvements in recent years under the **Solid Waste Management Rules, 2016**, and the **Swachh Bharat Mission (Urban)**. The **Gujarat Pollution Control Board (GPCB)** monitors performance indicators such as door-to-door collection coverage, segregation levels, processing capacity, and legacy waste remediation through monthly progress reports.

Collection Systems and Door-to-Door Coverage

All major urban centers in the lower basin have implemented **door-to-door (D2D) collection** of waste from households, commercial establishments, and institutions. The extent of coverage varies by ULB size and resources, as summarized below:

Table 20 Door-to-door collection and segregation status in Lower Narmada Basin districts.

District/ Major ULB	Door-to-Door Collection Coverage (%)	Segregation at Source (%)	Remarks / Status
Surat	100	90–95	Fully mechanized system with GPS-tracked vehicles; 100% ward coverage.
Vadodara	98	85–90	Well-established D2D collection with zonal transfer stations; segregation improving.
Bharuch	90	65–70	Partial segregation; coverage achieved in all wards. ULB planning cluster-level composting.
Dahod	85	50–60	Collection in all urban wards; segregation limited to select colonies.
Narmada (Rajpipla)	80	45–55	Manual collection in most wards; small pilot segregation programs active.

District/ Major ULB	Door-to-Door Collection Coverage (%)	Segregation at Source (%)	Remarks / Status
Panchmahal (Godhra, Halol)	88	60–65	Mechanized collection in major towns; smaller nagarpalikas rely on manual trolleys.
Chhota Udepur	70	40–50	Coverage in core areas; semi-urban and rural periphery still dependent on open points.

**(Sources: GPCB Monthly Progress Reports, 2023–24; CPCB SWM Annual Report, 2023; Surat and Vadodara Municipal Data.)*

Transportation and Transfer Infrastructure

Efficient transportation systems are critical to maintain uninterrupted waste movement from collection points to processing and disposal sites. Most large ULBs in the basin operate **mechanized vehicles** such as compactors, tippers, and closed-body trucks. Smaller towns continue to use tractors, tricycles, and open trolleys.

- **Surat:** Operates a fleet of over **650 GPS-enabled vehicles**, including compactors, refuse collectors, and road sweepers. Multiple **transfer stations** and a **centralized material recovery facility (MRF)** ensure uninterrupted waste flow to RDF and WtE plants.
- **Vadodara:** Maintains a fleet of **~350 vehicles** (compactors, mini-tippers, tricycles). **Six zonal transfer stations** and a **mechanical workshop** handle daily operations.
- **Bharuch and Ankleshwar:** Operate smaller vehicle fleets with mixed mechanized and manual systems. Waste from residential areas is transported to the designated landfill site at Bharuch.
- **Dahod, Panchmahal, and Narmada:** Use **tractor–trolley systems** with manual loading. Few compactors are available in district headquarters. Intermediate waste

collection centers (IWCCs) are planned to improve logistics.

- **Chhota Udepur:** Relies primarily on small trolleys and handcarts. Pilot mechanical collection vehicles have been introduced under SBM(U) funding.

Challenges:

- Lack of **covered vehicles** in smaller towns leads to spillage and littering along transport routes.
- **Limited transfer infrastructure** in remote areas increases operational costs and reduces collection efficiency.
- **Inconsistent segregation** at source results in mixed waste entering transport chains, limiting processing efficiency downstream.

Waste Processing and Disposal

Processing and disposal capacities vary significantly across the districts of the lower basin. Larger cities (Surat and Vadodara) have achieved full processing capacity, while smaller ULBs continue to depend on controlled or semi-controlled dump sites.

Table 21 Status of waste processing and disposal facilities in Lower Narmada Basin districts.

District / ULB	Primary Processing Method	Landfill / Dumpsite Location	Remarks
Surat	Composting, RDF, WtE	Khajod-Integrated Facility	Scientific landfill & large-scale RDF/WtE operations (Abellon, SMC).

District / ULB	Primary Processing Method	Landfill / Dumpsite Location	Remarks
Vadodara	Composting, Biomethanation	Moti Baug / Kapurai Landfill	Engineered landfill; biomethanation plant operational.
Bharuch	Composting (proposed), RDF (planned)	Near-Ankleshwar–Bharuch cluster	New landfill cell under PPP planning.
Dahod	Open dumping (partially remediated)	Devgadh, Baria dumpsite	Legacy waste remediation under progress.
Narmada (Rajpipla)	Manual composting	Near Rajpipla town	Small landfill; biomining DPR under preparation.
Panchmahal	Composting, manual segregation	Godhra outskirts	Semi-controlled landfill; leachate control required.
Chhota Udepur	Open dumping	Municipal boundary periphery	Site identified for future scientific landfill.

*(Sources: GPCB MPR 2023–24; CPCB SWM Report 2023; Surat & Vadodara Municipal Reports; Local ULB DPRs.)

Vehicle and Equipment Utilization

- **Surat:** Fully mechanized with real-time GPS tracking, route optimization, and automatic attendance monitoring for sanitation staff.

- **Vadodara:** Mechanized collection and transfer stations with periodic maintenance.
- **Medium ULBs (Bharuch, Panchmahal):** Semi-mechanized with scheduled collection timings.
- **Small ULBs (Narmada, Dahod, Chhota Udepur):** Manual systems; pressing need for compactors and covered vehicles.

GPCB has advised all ULBs to adopt “**Zero Open Transportation**” by 2030, with covered and leak-proof vehicles to prevent secondary pollution and littering.

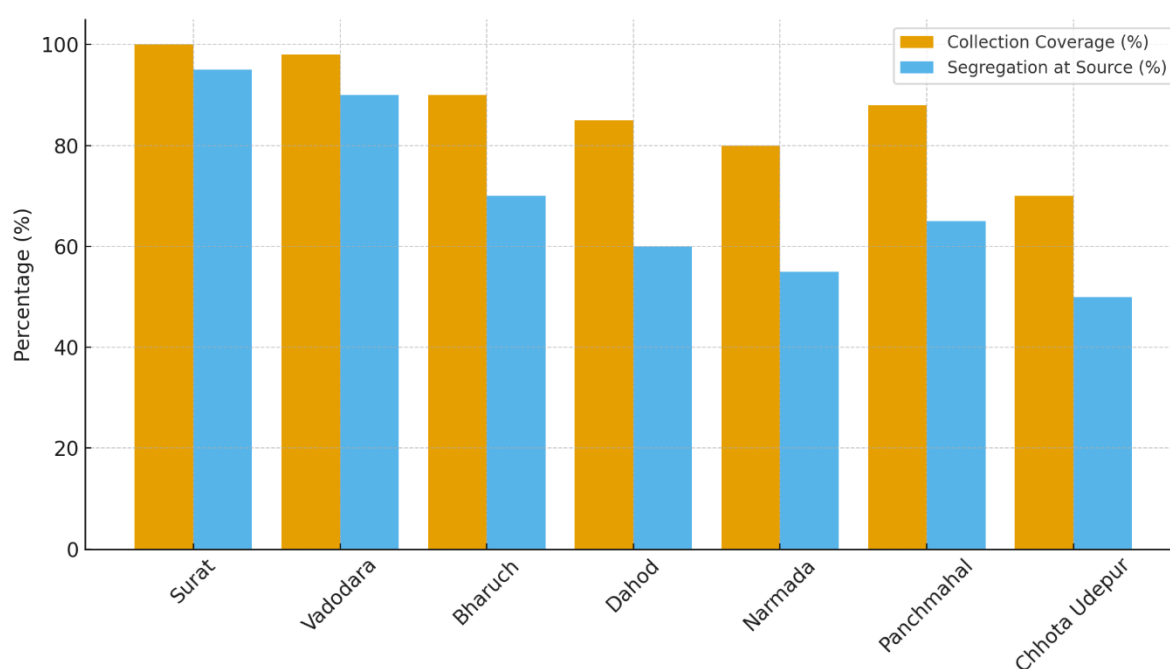


Figure 13 Waste Collection and Segregation Efficiency (Lower Narmada Basin, Gujarat).

**Source:* Gujarat Pollution Control Board (GPCB) MPR 2023-24; CPCB SWM Annual Report 2023; Surat and Vadodara Municipal Corporation data.

4.4 Status of Dumping Sites and Legacy Waste

4.4.1 Status Legacy Waste in Narmada Basin ULBs

Legacy waste management in the Narmada Basin reflects the intersection of historical urbanization trends, evolving regulatory frameworks, and current policy efforts aimed at environmental restoration. Each district and its Urban Local Bodies (ULBs) are grappling with accumulated waste generated over many years, necessitating a comprehensive strategy incorporating scientific remediation, monitoring, and stakeholder engagement. Under the Solid Waste Management Rules, 2016, and guidelines of the Central Pollution Control Board

(CPCB), ULBs are mandated to clear legacy waste through biomining, bioremediation, or scientific disposal. This report presents the status of dumping sites and legacy waste in the basin districts.

Table 22 for legacy waste status illustrates proactive steps taken by Madhya Pradesh authorities in a range of ULBs and detailed overview is tabulated in Annexure **Table A8**:

- ULBs such as Alirajpur, Bhavra, Jobat, Badwani, Rajpur, Multai, Betul, Dewas, Khategaon, Dhar (city), Kukshi, Manawar, Harda, Seoni Malwa, Itarsi, Narmadapuram, Mandideep, Bareli, Nasrullaganj, Jatara, Hatta, Balaghat, Mandla, Nainpur, and Pandhurna all reported as Work Started, indicating active legacy waste remediation primarily via biomining, bioremediation, or scientific disposal. This marks substantial government engagement aligning with the SWM Rules, 2016.
- ULBs such as Nepanagar, Sohagpur, Damoh, Ranapur, Pithampur, Mhowgaon, Petlawad, Chhanera, Bhikangaon, Betma, Kasrawad, and Junnardeo are flagged as Financial Evaluation Pending or Received at State SLTC (State Level Technical Cell) Pending. These delays signify ongoing processes of tendering, funding approval, and technical review. Such systemic bottlenecks highlight common governance challenges, calling for streamlined PPP (public-private partnership) models and state-level intervention.
- In some locations, Technical Pending status, such as Seoni with 1,57,664 MT, signals that remediation projects await necessary scientific and engineering clearance, reflecting the complexity of constructing effective legacy waste recovery infrastructure.

In the Narmada Basin, years of unregulated dumping created sprawling waste landscapes that threaten groundwater, soil morphology, and public health. The state's approach to management is rooted in

Table 22 Legacy waste in Narmada basin ULBs

District	Total Legacy Waste (MT)
Alirajpur	54,253
Barwani	64,617

District	Total Legacy Waste (MT)
Betul	1,50,363
Burhanpur	44,249
Dewas	1,15,072
Dhar	2,93,903
Harda	19,370
Hoshangabad (Narmadapuram)	2,60,629
Indore	2,99,206
Jabalpur	2,90,150
Raisen	1,00,053
Sehore	12,636
Sagar	26,446

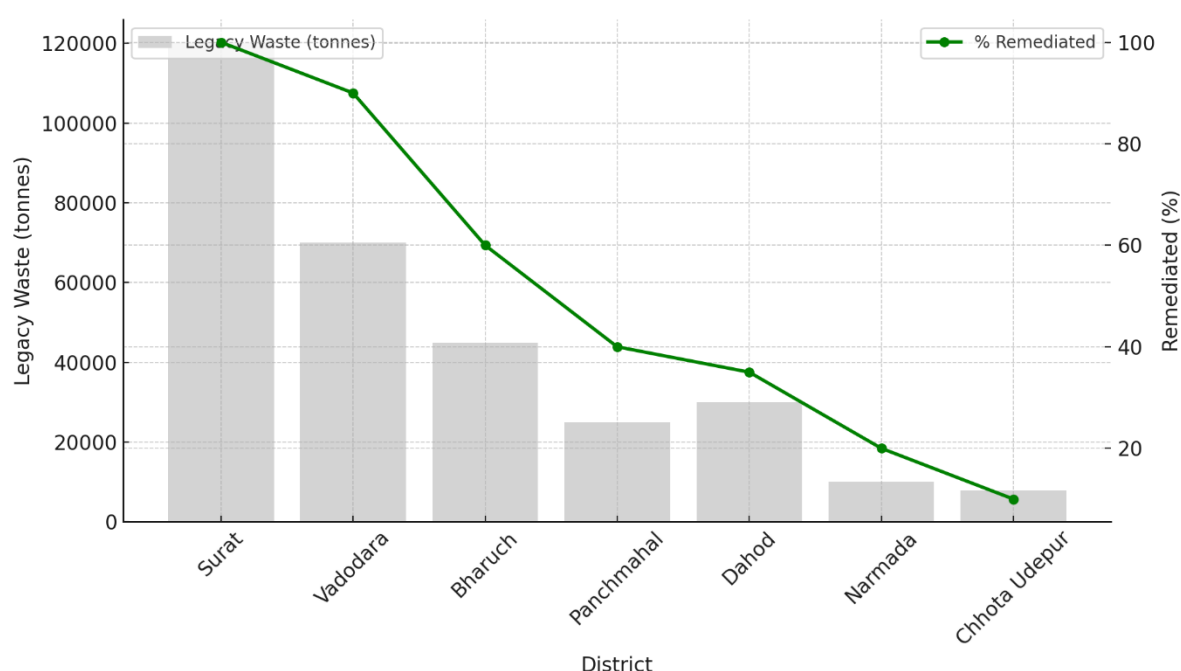
**Sources: Six Monthly Progress Report by Govt. of Madhya Pradesh in OA No. 606 of 2018 for State of Madhya Pradesh (Compliance of MSW Management Rules, 2016 and Other Environmental Issues), National Green Tribunal, September 2025*

- **Biomining and Resource Recovery:** Turning legacy waste into a resource stream, with staged excavation, mechanical separation, and recycling. Bioremediation techniques transform organic fractions into compost, while inert materials are disposed of scientifically.
- **Decentralized Processing:** Adoption of localized waste processing, with facilities constructed in moderate-sized towns, aligning with global circular economy models and the principles of Integrated Sustainable Waste Management (ISWM).
- **Stakeholder Inclusion:** Increasing involvement of local communities, NGOs, and authorized waste pickers, improving transparency and labor rights

Legacy waste refers to old, unsegregated municipal solid waste that has accumulated over the years in open dumpsites. The remediation of such sites is a major objective under the Solid Waste Management Rules, 2016, and is monitored by both GPCB and CPCB through monthly progress and tribunal compliance reports.

In the Lower Narmada Basin of Gujarat, nearly all ULBs have identified legacy dumpsites. The total quantity of legacy waste across the seven basin districts is estimated at $\approx 3.8 - 4.2$ lakh tonnes, of which a large share lies within the urban limits of Surat, Vadodara, and Bharuch.

Remediation is being carried out through biomining, resource recovery, and scientific capping under supervision of the GPCB, with financial support from the Swachh Bharat Mission (Urban) and state programs.



Graph 115 Legacy Waste Remediation Progress (Lower Narmada Basin, Jan 2024).

***Source:** Gujarat Pollution Control Board (GPCB) MPR 2023–24; CPCB SWM Annual Report 2023; ULB legacy waste remediation reports (Surat, Vadodara, Bharuch, Dahod, Panchmahal, Narmada, Chhota Udepur).

District-wise Status of Legacy Waste (2024)

Table 23 District-wise status of legacy waste remediation in Lower Narmada Basin ULBs (as of Jan 2024).

District / Major ULB	Estimated Legacy Waste (tonnes)	Remediated (%)	Method Used / Status	Remarks / Source
Surat	≈ 1,20,000 t	100 %	Legacy waste at Khajod site fully biomined and reclaimed. Recovered land used for WT E facilities.	SMC Report 2024; GPCB MPR 2023-24
Vadodara	≈ 70,000 t	90 %	Biomining at Kapurai VMC landfill under progress; residual inerts being capped scientifically.	SWM Dept 2024; GPCB Report 2023
Bharuch	≈ 45,000 t	60 %	Legacy site identified near Ankleshwar; biomining tender awarded; cluster-landfill project proposed.	Bharuch ULB DPR; GPCB MPR 2024
Dahod	≈ 30,000 t	35 %	Bioremediation initiated at Devgad Baria dumpsite after repeated fire incidents.	GPCB District Inspection Report 2023
Narmada (Rajpipla)	≈ 10,000 t	20 %	Site mapped; ULB preparing DPR for biomining and new sanitary landfill.	Rajpipla Nagar Palika Report; GPCB MPR 2023

Panchmahal (Godhra	& ≈ 25,000 t	40 %	Biomining ongoing at Godhra site; recovered inerts GPCB Regional being shifted to engineered Office 2024 cell.
Halol)			
Chhota Udepur	≈ 8,000 t	10 %	Initial survey completed; remediation proposal submitted to GPCB for approval. GPCB MPR 2024; ULB submission

**(Sources: GPCB Monthly Progress Reports 2023-24; CPCB SWM Annual Report 2023; ULB status updates.)*

4.4.2 Status of Dumping Sites in Narmada Basin ULBs

The status of municipal solid waste (MSW) management highlighted in **Table 18** in selected Urban Local Bodies (ULBs) that fall under the Narmada River Basin in Madhya Pradesh. Since these cities directly or indirectly influence the environmental health of the river system, the data serves as an important indicator of basin-level waste management progress.

Key Observations in the Narmada Basin

- **Zero Processing Gap Across ULBs**

All listed basin ULBs, including Bhopal, Indore, Jabalpur, and Sagar, demonstrate complete processing of generated waste in both November 2022 and the current reporting period. This ensures that untreated solid waste does not enter the natural environment and reduces the risk of dumping into Narmada's catchment areas.

- **Rising Waste Generation with Urbanization**

Growth in waste generation is visible in several ULBs:

- Indore increased from 984.6 TPD to 1192 TPD.
- Bhopal rose from 808.6 TPD to 823.3 TPD.

- Sagar and Chhindwara also show significant jumps.

While these increases reflect urban and economic expansion, they also emphasize the need for continued strengthening of waste management systems to safeguard riverine ecosystems.

Progress in Dumpsite Remediation

- Proper remediation of old dumpsites is particularly relevant in the Narmada Basin, as leachate from legacy waste poses a major contamination threat to groundwater and river flows.
- Complete remediation has been achieved in Bhopal, Indore, Jabalpur, Burhanpur, Chhindwara, and Sagar.
- Dewas and Khandwa, which were earlier at the tendering/DPR preparation stage, are now undergoing active remediation. This indicates progress toward basin-wide reduction of river pollution sources.

Indore and Bhopal as Flagship Models

Indore, with the largest daily waste generation (1192 TPD), and Bhopal, as the state capital, have achieved full processing with remediated dumpsites, positioning themselves as benchmarks for sustainable solid waste management within the Narmada Basin.

Basin-Level Significance

These advances in ULB waste management are highly significant for the Narmada River Basin's ecological integrity. By achieving 100% processing and accelerating dumpsite remediation, these cities are directly contributing to:

- Reduction of leachate contamination to river tributaries and groundwater.
- Prevention of open dumping, which often affects floodplains and riparian ecosystems along the basin.
- Establishment of replicable models for other basin towns and smaller ULBs yet to achieve full remediation.

Dhar Road Dumpsite Profile

The Dhar Road Dumpsite, located in Indore, Madhya Pradesh as depicted in **Table 18** (22.705°N, 75.88°E), spans approximately 15 acres and receives around 300 tons of mixed solid waste (MSW) and construction & demolition (C&D) debris daily. Managed by the Indore Municipal Corporation, this government-owned site remains active but lacks an immediate remediation plan. The dumpsite poses several environmental risks, including leachate generation and recurring fire incidents, primarily due to methane emissions from decomposing waste. Its proximity to the Gambhir River (about 3 km away) and a nearby industrial area further elevates the potential for groundwater and air pollution. Despite these challenges, no concrete steps have yet been initiated to mitigate the hazards associated with the site.

Table 24 Status of Dhar Road side dumpsite

Parameter	Details
Name	Dhar Road Dumpsite
Location	Dhar Road, Indore, Madhya Pradesh
Coordinates	22.705°N, 75.88°E
Area	15 acres
Waste Quantity (TPD)	300 tons/day
Waste Type	Municipal Solid Waste (MSW) and Construction & Demolition (C&D)
Ownership/Authority	Government – Indore Municipal Corporation
Status	Active
Key Issues	Leachate generation, Fire incidents
Nearby Receptors	Gambhir River (3 km), Industrial area nearby
Risk Highlight	Fire risk due to methane emissions

*Source: MPCB District environment plan

Table 25 Status of dumping sites in Narmada basin ULBs

S. N.	ULB Name	Generation (Tons, Nov-22)	Processing (Tons, Nov-22)	Processing Gap (Nov-22)	Dumpsite Status (Nov-22)	Generation (Current)	Processing (Current)	Processing Gap (Current)	Dumpsite Status (Current)
-------	----------	---------------------------	---------------------------	-------------------------	--------------------------	----------------------	----------------------	--------------------------	---------------------------

1	Bhopal	808.6	808.6	0	Remediated	823.3	823.3	0	Remediated
2	Burhanpur	86.97	86.97	0	Remediated	86.97	86.97	0	Remediated
3	Chhindwara	70.15	70.15	0	Remediated	96.90	96.90	0	Remediated
4	Dewas	103.67	103.67	0	Under Tendering	108.23	108.23	0	Remediation work under process
6	Indore	984.6	984.6	0	Remediated	1192.0	1192.0	0	Remediated
7	Jabalpur	506.19	506.19	0	Remediated	507.93	507.93	0	Remediated
8	Khandwa	42.67	42.67	0	DPR Prepared	46.30	46.30	0	Remediation work under process
13	Sagar	87.96	87.96	0	Remediated	112.29	112.29	0	Remediated

**Source: Presentation cum Compliance by Government of Madhya Pradesh in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016), submitted to the National Green Tribunal, September 2025)*

Uncontrolled dumping of solid waste has historically been one of the key environmental challenges in the **Lower Narmada Basin**, contributing to soil and groundwater contamination, odour nuisance, and visual degradation. Following the enforcement of the **Solid Waste Management Rules, 2016**, and directions from the **National Green Tribunal (O.A. No. 606/2018)**, all Urban Local Bodies (ULBs) in Gujarat have been directed by the **Gujarat Pollution Control Board (GPCB)** to identify, quantify, and remediate their existing dumpsites.

By **January 2024**, most ULBs in the lower basin have either (i) remediated old dumpsites through biomining, or (ii) initiated the development of **scientific landfills** for future disposal. The overall approach now focuses on transitioning from open dumping to **engineered, environmentally compliant landfill systems**.

District-Wise Status of Dumping Sites (as of Jan 2024)

Table 26 Status of existing dumping sites in Lower Narmada Basin districts (Jan 2024)

District/ Major ULB	Existing Dumpsite Location	Area (Ha)	Status (Jan 2024)	Remarks / Source
Surat	Khajod	80	Closed Remediated	Former dumpsite fully & reclaimed; site now hosts integrated SWM & WtE facility. (SMC Report, 2024)
Vadodara	Kapurai	75	Partially Under Remediation	Active; Biomining ongoing; residual inerts to be capped under GPCB supervision. (VMC SWM Dept, 2024)
Bharuch	Near Ankleshwar Industrial Cluster	25	Active (Controlled Dumping)	New sanitary landfill under design; partial waste containment practiced. (GPCB RO Bharuch, 2023)
Dahod	Devgadh Baria	18	Partially Biomining Initiated	Active; Dumpsite fenced; fire control and leachate management in progress. (GPCB MPR, 2023– 24)
Narmada (Rajpipla)	Near Rajpipla Town	10	Active (Uncontrolled)	Old dumpsite without liner; proposal submitted for new engineered landfill. (GPCB District Office, 2024)

Panchmahal (Godhra)	Godhra Outskirts	20	Under Remediation	Biomining in progress; inert capping planned post-screening. <i>(GPCB Regional Office, 2024)</i>
Chhota Udepur	Near Boundary	ULB 6	Active (Uncontrolled)	Dumpsite identified; DPR under preparation for cluster-level landfill. <i>(ULB Data, 2024)</i>

**(Sources: GPCB MPR 2023–24; CPCB SWM Annual Report 2023; ULB DPRs and local inspection reports.)*

Site Conditions and Environmental Risks

- **Leachate generation:** Sites at Vadodara, Bharuch, and Dahod show moderate leachate discharge due to mixed-waste accumulation and rainfall infiltration.
- **Groundwater contamination risk:** GPCB monitoring indicates increased conductivity and BOD levels near legacy sites, particularly in **Bharuch** and **Godhra**.
- **Fire hazard:** Seasonal fires during dry months are recurrent at Dahod and Panchmahal sites, attributed to methane generation.
- **Odour and visual impact:** Active dumping near Narmada and Chhota Udepur town limits causes odour nuisance and aesthetic degradation.

Upgradation and Remediation Initiatives

1. Scientific Landfill Development:

- Surat and Vadodara have already established or redeveloped engineered landfill facilities.
- Bharuch and Panchmahal are developing cluster-level sanitary landfills with GPCB approval.
- Narmada and Chhota Udepur are planning smaller capacity, lined landfill cells.

2. Biomining and Inert Capping:

- Biomining operations are underway at Vadodara, Panchmahal, and Dahod dumpsites.
- Recovered land is being converted into green zones or reused for SWM infrastructure.

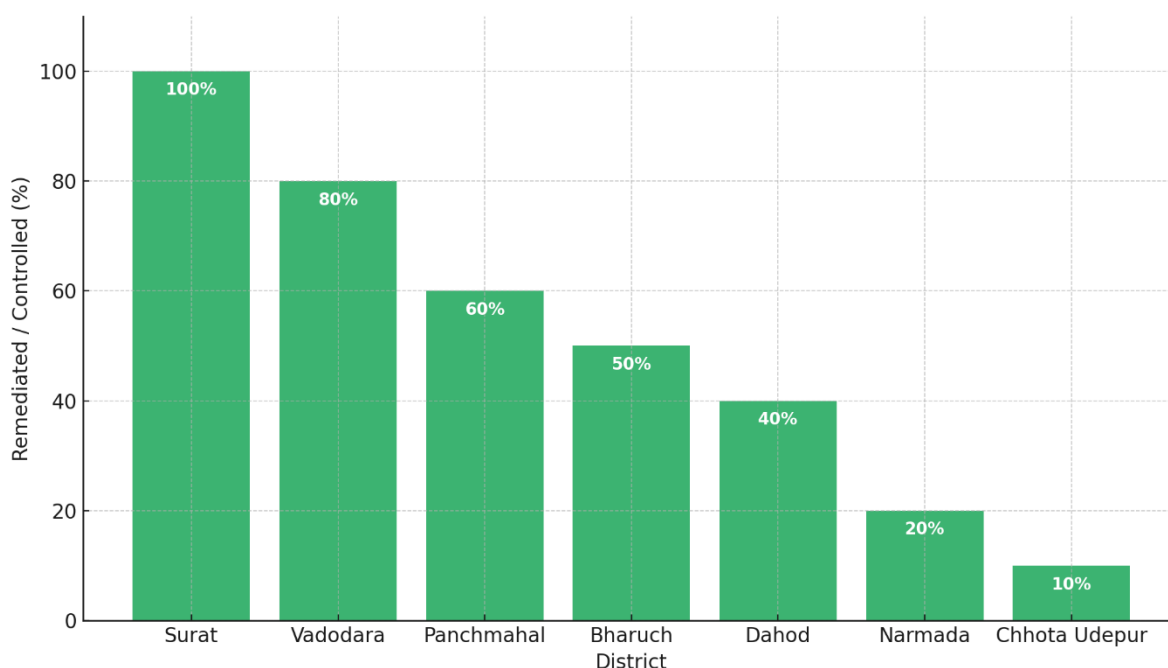
3. Environmental Safeguards:

- Fencing, leachate collection drains, and stormwater diversion structures installed at active sites.

- Periodic water quality monitoring conducted by GPCB to track improvement post-remediation.

4. Cluster Approach:

- GPCB has encouraged small ULBs (Narmada, Chhota Udepur, parts of Panchmahal) to adopt a **cluster-level approach** for shared processing and landfill facilities.



Graph 116 Status of Dumpsites in Lower Narmada Basin (Gujarat, 2024)

*(Source: GPCB MPR 2023–24; CPCB SWM Annual Report 2023; ULB Data)

4.5 Waste to Energy

4.5.1 Wet Waste Processing Facilities in Urban Local Bodies (ULBs) under Narmada River Basin

The Urban Local Bodies (ULBs) within the Narmada River Basin have established a diverse range of wet waste processing facilities aimed at ensuring scientific treatment of biodegradable waste. These facilities primarily include aerobic windrow composting, pit-based/vermi composting, anaerobic systems, and composting with/without Refuse-Derived Fuel (RDF) integration. The details are provided in **Annexure Table A9**.

Key Observations:

1. Technology Mix:

- A large number of ULBs rely on pit-based/vermi composting, which is suitable for small- to medium-scale towns such as Shahpura (Dindori), Bagli (Dewas), Kukshi (Dhar), and multiple towns in Raisen and Sehore districts.
- Major cities like Indore, Bhopal, Jabalpur, Sagar, Katni, Mandla, and Khandwa have adopted aerobic windrow composting systems, some integrated with RDF linkages, which ensure large-scale treatment of municipal wet waste.
- Select ULBs such as Bhopal (M Corp.), Hoshangabad (Seoni-Malwa), and Sagar (M Corp.) have operational anaerobic composting facilities, reflecting a move towards biogas/energy recovery-based solutions.

2. District-Level Distribution:

- Indore, Jabalpur, and Bhopal have the highest concentration of composting facilities, indicating their leadership in adopting decentralized waste management solutions.
- Sehore, Raisen, and Narsimhapur districts also demonstrate strong coverage, with multiple small-scale composting units spread across different ULBs, ensuring localized treatment.
- Smaller districts like Alirajpur, Anuppur, Dindori, Mandla, and Harda have fewer but strategically placed composting units tailored to their population and waste generation.

3. Integration with RDF:

- Cities such as Anuppur (Bijuri & Pasan), Barwani (Anjad, Niwali Burjurg, Thikari), Dhar (Kukshi), Mandla (Nainpur), Katni (Murwara), and Sagar (M Corp.) have facilities that integrate aerobic composting with RDF, thereby contributing to resource recovery and reduced landfill dependency.

4. Environmental Significance:

- The spread of composting facilities across the basin helps minimize the

quantum of biodegradable waste reaching dumpsites, thereby reducing leachate formation and methane emissions, which are critical for protecting the Narmada River and its tributaries from pollution.

- Decentralized composting also promotes circular economy practices by converting wet waste into compost, which can be utilized in agriculture and horticulture within the basin.

The Urban Local Bodies (ULBs) located within the **Lower Narmada River Basin of Gujarat** have developed a variety of **wet-waste processing systems** to ensure scientific treatment of biodegradable waste in compliance with the **Solid Waste Management Rules, 2016**. These facilities include **aerobic windrow composting, pit-based/vermi composting, biomethanation (anaerobic systems), and co-composting units with Refuse-Derived Fuel (RDF) integration** for recovery of resources and minimization of landfill loads.

Key Observations

1. Technology Mix

- **Small and Medium Towns (Bharuch, Dahod, Narmada, Panchmahal, Chhota Udepur):**

A majority of these ULBs rely on **pit-based or vermi composting** systems suited for low-volume waste (below 50 TPD). Such systems are operated at the ward or community level and maintained by local municipal staff. Example installations include **aerobic compost pits at Rajpipla (Narmada), community composting in Dahod, and market-waste pits in Godhra (Panchmahal)**.

- **Major Cities (Surat and Vadodara):**

These ULBs have adopted **large-scale aerobic windrow composting** integrated with **biomethanation and RDF linkages**, enabling full treatment of the city's organic waste.

- **Surat Municipal Corporation (SMC)** operates one of the largest composting and biomethanation facilities in western India, processing about 1,150 TPD of wet waste.
- **Vadodara Municipal Corporation (VMC)** manages composting and biogas

plants at **Kapurai and Moti Baug**, producing organic manure and electricity from biogas.

- **Emerging Technologies:**
Co-composting of sewage sludge and wet municipal waste has been initiated by **Panchmahal (Godhra)** and **Bharuch**, improving nutrient quality of compost and reducing sludge-disposal burdens.

2. District-Level Distribution

- **High Capacity Districts – Surat and Vadodara:**
These two districts account for nearly **85 % of total wet-waste processing capacity** in the Lower Basin. Surat alone treats 100 % of its wet waste, while Vadodara achieves ~87 %.
- **Moderate Coverage – Bharuch and Panchmahal:**
Both have functional composting units (30–55 TPD capacity) and are expanding decentralized systems under SBM(U).
- **Limited Coverage – Dahod, Narmada, and Chhota Udepur:**
These ULBs operate small community composters (≤ 10 TPD) with partial segregation; DPRs for cluster-level plants are under preparation with GPCB support.

3. Integration with RDF and Energy Recovery

- **Surat:** Integrated **composting + RDF + biomethanation** model at Khajod converts segregated organic waste to compost and biogas, while the residual fraction feeds the WtE plant.
- **Vadodara:** Biomethanation units produce biogas used for street-lighting and power generation; RDF from residual dry waste supports cement-industry co-processing.
- **Bharuch and Panchmahal:** Exploring RDF-linkages with nearby cement kilns to utilize non-recyclable combustible waste from composting facilities.

4. Environmental Significance

- **Reduction in Leachate and Methane Emissions:**

Enhanced composting coverage prevents deposition of organic waste in dumpsites, reducing leachate generation and greenhouse-gas emissions near the Narmada River floodplain.

- **Protection of Water Quality:**
Minimizing organic dumping decreases the biochemical oxygen demand (BOD) load in surface runoff, indirectly improving river-water quality.
- **Circular Economy Promotion:**
Decentralized composting enables recovery of organic manure for **urban landscaping, agriculture, and horticulture**, creating a sustainable waste-to-resource loop.
- **Community Participation:**
Surat, Vadodara, and Panchmahal ULBs have established citizen-involvement programs where residents deposit segregated wet waste at fixed times, ensuring cleaner collection streams.

4.5.2 Dry Waste Processing Facilities in ULBs under the Narmada River Basin

The Urban Local Bodies (ULBs) located within the Narmada River Basin have established Material Recovery Facilities (MRFs) for the management of dry waste. These facilities play a crucial role in segregation, recovery, and recycling of non-biodegradable fractions, thereby reducing the burden on dumpsites and minimizing environmental risks to the river system.

The dataset compiled from the Swachh Bharat Mission MIS (2022–23) highlights district-wise coverage of MRFs across municipalities, municipal corporations, cantonment boards, and nagar parishads. The facilities vary in terms of mechanization levels ranging from manual, semi-automated, to fully automated systems depending on waste generation volumes and operational capacities of respective ULBs. The details are provided in **Annexure Table A10**.

Key features of these facilities include:

- **Decentralized dry waste management** is aimed at source-level recovery.
- **Semi-automated and fully automated plants** in larger cities (e.g., Indore, Bhopal, Jabalpur, Khargone, Khandwa, Dhar, Jhabua), which improve efficiency and material

recovery rates.

- **Manual MRFs** in smaller towns and nagar parishads, ensuring local waste sorting and recycling linkages.
- Coverage across multiple districts such as Alirajpur, Anuppur, Balaghat, Barwani, Betul, Bhopal, Burhanpur, Chhindwara, Damoh, Dewas, Dhar, Dindori, Harda, Hoshangabad (Narmadapuram), Indore, Jabalpur, Jhabua, Khandwa, Khargone, Mandla, Katni, Narsinghpur, and Raisen all of which fall in the Narmada Basin.

This mapping provides an integrated picture of dry waste processing infrastructure, enabling better planning, monitoring, and policy alignment for river basin-based solid waste management. Strengthening these facilities, especially upgrading manual units to semi-automated ones will be critical to minimizing leakages of plastic and other dry waste into the Narmada River and its tributaries.



Figure 14 Material Recovery Facilities (MRF) & Composting Facilities in ULBs

The Urban Local Bodies (ULBs) across the **Lower Narmada River Basin of Gujarat** have developed a range of **dry waste processing systems** to manage recyclable and non-recyclable components of municipal solid waste in accordance with the **Solid Waste Management Rules, 2016**.

Dry waste in the region primarily comprises **plastics, paper, metals, glass, multilayered packaging (MLP), textiles, and inert fractions**.

Processing facilities include **Material Recovery Facilities (MRFs)**, **plastic baling and shredding units**, and **Refuse-Derived Fuel (RDF)** linkages with cement plants for energy recovery.

The specific details of dry-waste processing infrastructure are presented in *Annexure Table A10*.

Key Observations

1. Technology Mix

- **Material Recovery Facilities (MRFs):**

Almost all ULBs in the basin have operational MRFs or dry-waste sorting sheds.

- **Surat and Vadodara** operate **fully mechanized MRFs** with conveyor belts, segregation platforms, baling machines, and digital weighing systems.
- **Medium towns (Bharuch, Panchmahal, Dahod)** use **semi-mechanized MRFs** equipped with sorting tables and manual balers for plastics and paper.
- **Smaller ULBs (Narmada, Chhota Udepur)** depend on **manual sorting sheds** and private aggregators for recyclables.

- **Plastic Waste Management (PWM) Units:**

- **Surat** has a dedicated **plastic baling and recycling unit** integrated with its RDF plant at Khajod.
- **Vadodara** operates **plastic shredding and co-processing facilities**, sending shredded plastic to nearby cement industries under Extended Producer Responsibility (EPR) mechanisms.

- **RDF & Co-Processing Integration:**

Surat, Vadodara, and Bharuch ULBs forward non-recyclable combustible fractions (plastics, textiles, rubber) to **cement kilns** for RDF co-processing under GPCB authorization.

2. District-Level Distribution

- **Surat and Vadodara:**

Represent the most advanced ULBs in Gujarat, with **integrated MRF + RDF systems** operating at full efficiency.

Surat's RDF preparation and plastic recovery facility at **Khajod** is among the largest in the state, processing ~950–1,000 TPD of dry waste.

Vadodara's **Kapurai MRF** handles over 250 TPD of segregated dry waste.

- **Bharuch and Panchmahal:**

Moderate capacity ULBs with semi-mechanized segregation sheds and baling presses. Bharuch channels its non-recyclable waste to **Ambuja Cement (Kodinar)** under a co-processing agreement facilitated by GPCB.

- **Dahod, Narmada, and Chhota Udepur:**

Smaller ULBs with limited infrastructure—operating manual sorting platforms and basic plastic baling setups. GPCB has encouraged cluster-based MRF planning to optimize cost and improve recovery rates.

3. Integration with RDF and Resource Recovery

- **RDF Linkages:**

Surat's RDF production (~400 TPD) supplies nearby cement industries for thermal energy substitution. Vadodara also sends non-recyclable waste fractions to co-processing kilns.

- **Recycling Partnerships:**

ULBs collaborate with authorized **plastic recyclers and kabadiwala networks** to recover materials. Surat and Vadodara have digital tracking systems for recyclables.

- **EPR Implementation:**

GPCB facilitates partnerships between ULBs and **Producer Responsibility Organizations (PROs)** to channel multilayered plastic waste under the PWM Rules, 2022.

4. Environmental Significance

- **Reduction in Landfill Burden:**

Dry-waste segregation and RDF production significantly reduce the volume of residual waste reaching landfills. Surat and Vadodara collectively divert over **90 % of dry waste** from disposal.

- **Promotion of Circular Economy:**

Recyclable material recovery (plastics, paper, metals) feeds into local recycling industries, generating employment and conserving raw materials.

- **Energy Recovery:**

RDF utilization in cement kilns replaces coal and lowers CO₂ emissions, aligning with Gujarat's decarbonization goals.

- **Pollution Control:**

Proper dry-waste management mitigates open burning of plastics, a key factor in

urban air-quality improvement.

4.5.3 Waste-to-Energy Facilities in ULBs of the Narmada River Basin

Urban Local Bodies (ULBs) within the Narmada River Basin are increasingly adopting Waste-to-Energy (WtE) and Bio-CNG technologies to convert municipal solid waste into usable energy. These initiatives not only address the challenge of reducing landfill dependency but also contribute to renewable energy generation, climate mitigation, and circular economy goals.

The compiled data shown in **Table 27** from the Swachh Bharat Mission MIS (2022–23) and other government sources highlights major projects in Jabalpur, Indore, Rewa, and Bhopal, all of which fall within the Narmada Basin.

Key highlights include:

- **Jabalpur Waste-to-Energy Plant:** Located at Village Kathoda, this 11.5 MW incineration-based facility is one of the first large-scale operational WtE projects in Madhya Pradesh. It is operated by Jabalpur MSW Pvt. Ltd. and converts mixed waste into power, supplying electricity to the grid.
- **Indore Proposed WtE Plant:** Indore, a leader in waste management, has proposed a 6 MW incineration-based WtE plant, also under Jabalpur MSW Pvt. Ltd. This will further strengthen Indore's waste valorization infrastructure.
- **GOBARDhan Bio-CNG Plant, Indore:** Located at Kabitkhedi, these 20 MTPD Bio-CNG facilities, developed by Indore Municipal Corporation (IMC) in partnership with Mahindra, is Asia's largest Bio-CNG plant. It converts organic waste into compressed biogas, serving as a model for decentralized energy recovery.
- **Bhopal Biomethanation Plant:** Situated at Bittan Market, this 4–5 TPD plant is a smaller-scale biomethanation unit operated by Bhopal Municipal Corporation. It converts organic wet waste into biogas, demonstrating a localized, sustainable waste-to-energy solution.

Together, these initiatives illustrate how districts within the Narmada Basin are leveraging WtE and biomethanation technologies to reduce legacy waste burdens, cut greenhouse gas emissions, and create clean energy. Strengthening such infrastructure is crucial to ensure sustainable river basin-based waste management and to prevent untreated waste from reaching the Narmada River and its tributaries.

Table 27 List of waste-to-energy facilities under the Narmada River basin ULBs

Plant Name/Operator	Location	Capacity	Type/Details
Jabalpur Waste to Energy Plant	Village Kathoda, Jabalpur	11.5 MW	Incineration, operated by Jabalpur MSW Pvt. Ltd.
Indore Proposed WtE Plant	Indore	6 MW (planned)	Incineration, operated by Jabalpur MSW Pvt. Ltd.
GOBARDhan Bio-CNG Plant	Indore	20 MTPD (Bio-CNG)	Asia's largest, at Kabitkhedi, by IMC/Mahindra
Bhopal Biomethanation Plant	Bittan Market, Bhopal	4-5 TPD	Biogas from organic waste, BMC-ru

*Source: MPCB District environment plan

The **Lower Narmada Basin districts of Gujarat** have progressively adopted a mix of **composting, biomethanation, and waste-to-energy (WtE)** technologies to manage municipal solid waste scientifically and minimize landfill dependency. Under the **Solid Waste Management Rules, 2016**, and the **Swachh Bharat Mission (Urban)**, the **Gujarat Pollution Control Board (GPCB)** has facilitated establishment of both centralized and decentralized processing facilities, focusing on energy recovery from organic and high-calorific waste fractions.

The region's two largest ULBs — **Surat** and **Vadodara** — have already operationalized integrated waste-to-energy and composting systems, while mid-sized ULBs such as **Bharuch** and **Panchmahal** are in advanced planning or implementation phases.

Key Observations

1. Technology Mix and Facility Types

- **Composting Plants:**

Aerobic windrow and vermi-composting are the dominant technologies across smaller ULBs. Surat and Vadodara use large-scale windrow composting integrated with biomethanation.

- **Biomethanation Facilities:**

- **Surat:** Operates biomethanation units at Khajod and other zones, generating **biogas for electricity and transport fuel**.
- **Vadodara:** Runs biomethanation plants at Kapurai and Moti Baug, producing methane for energy and digestate for composting.

- **Waste-to-Energy (WtE) Facilities:**

- **Surat:** Hosts one of Gujarat's largest **WtE plants** (operated by Abellon Clean Energy), converting RDF and organic fractions into electricity (~1,000 TPD capacity).
- **Vadodara:** Has a smaller WtE pilot linked to RDF from its MRF and composting facility.
- **Bharuch:** DPR prepared for RDF-based WtE cluster project; co-processing linkages in progress with nearby cement kilns.

- **Decentralized Units:**

Dahod, Panchmahal, Narmada, and Chhota Udepur ULBs rely primarily on small-scale composting and limited biogas units, suitable for community and institutional waste.

2. Integration and Energy Recovery

- **Surat Model:**

The integrated waste-processing complex at **Khajod** processes both wet and dry waste streams, generating RDF, compost, and renewable energy. Electricity produced from the WtE plant is supplied to the local grid under a **Power Purchase Agreement (PPA)** with the state discom.

- **Vadodara Model:**

Combines composting and biomethanation with RDF generation; excess biogas used for lighting and heating applications.

- **Cluster Planning:**
Bharuch and Panchmahal districts are finalizing DPRs for shared RDF and WtE facilities serving nearby small ULBs (e.g., Ankleshwar, Godhra, Halol).
- **Byproduct Utilization:**
Compost and digestate are distributed to local farmers and used in urban landscaping; RDF fractions are co-processed in cement plants.

3. Environmental and Economic Benefits

- **Landfill Diversion:**
Over 80 % of the total waste generated in Surat and Vadodara is processed through composting, biomethanation, or WtE systems, drastically reducing landfill demand.
- **Energy Generation:**
WtE and biomethanation plants contribute to renewable energy generation, supporting Gujarat's clean energy targets.
- **Emission Reduction:**
Controlled anaerobic digestion and energy recovery significantly reduce methane and carbon dioxide emissions compared to open dumping.
- **Circular Economy:**
The recovered compost and energy strengthen circular economy linkages, promoting resource efficiency and sustainable development.

4.5.4 Status of Waste-to-Energy and Composting Facilities in Narmada River Basin Districts

The Narmada River Basin districts are actively progressing toward integrated solid waste management with a mix of composting units and waste-to-energy (WtE) facilities. The compiled data shown in **Table 21** reflects the status of projects across multiple clusters, highlighting their processing capacity, project developers, and timelines.

Salient Findings:

1. Operational Energy Plant

Jabalpur Cluster: With an estimated waste generation of 450 TPD, this cluster has already commissioned a large-scale Energy-based Waste-to-Energy plant. The project, developed by

M/s Essel InfraProjects Ltd, demonstrates the viability of energy recovery from mixed waste in the basin.

2. Projects under Environmental Clearance and Construction

Sagar Cluster: Covering 11 ULBs with 185 TPD waste generation, a composting project is under review for Environmental Clearance, with M/s Ramky Enviro Engineers Ltd as the developer. Katni Cluster: With 92 TPD waste generation, construction has already started on a composting facility by M/s Ramky Enviro Engineers Ltd, and the plant was targeted to be operational by June 2017.

3. Projects under Bidding and Concession Agreement Stages

Bhopal Cluster: Encompassing 8 ULBs and generating 1065 TPD waste, an energy-based facility is proposed. A concession agreement has been signed with M/s Essel Infra Projects Ltd, and the DPR preparation is underway. Indore Cluster: With 1004 TPD waste from 8 ULBs, the energy project is currently under bidding, with the concession agreement targeted for mid-March 2017. Hoshangabad, Burhanpur, and Dewas Clusters: Together generating nearly 491 TPD of waste (164 TPD, 176 TPD, and 151 TPD, respectively), these composting projects are under bidding, with concession agreements expected by April 2017.

4. Clusters in FSR and Bid Preparation Stage

Chhindwara (173 TPD), Betul (89 TPD), Balaghat (97 TPD), Narsinghpur (556 TPD), and Pithampur (241 TPD) have projects in the Feasibility Study Report (FSR) stage, with consultants such as M/s Royal HaskoningDHV and M/s Ecoprop-Meghraj engaged. Bid publication was targeted for March–April 2017.

Table 28 Status of waste-to-energy facilities under the Narmada River basin ULBs

S. No	Name of Cluster	No. of ULBs	Est. Waste Generation (TPD)	Type of Processing	Status	Remarks	Timeline
1	Sagar	11	185	Compost	Under EC	M/s Ramky Enviro	Environmental Clearance of project

						Engineers Ltd	
2	Katni	5	92	Compost	Constructio n Started	M/s Ramky Enviro Engineers Ltd	Plant operational by June 2017
3	Jabalpur	1	450	Energy	Commissio ned	M/s Essel InfraProje cts Ltd	-
4	Bhopal	8	1065	Energy	Concession agreement Signed	M/s Essel InfraProje cts Ltd	DPR Preparation by Concessionai re is underway
6	Indore	8	1004	Energy	Under Bidding	Concessio naire agreemen t signing is targeted	Concession agreement to be signed by Mid March 2017
8	Hoshang abad	14	164	Compost	Under Bidding	Concessio naire agreemen t signing is targeted	Concession agreement to be signed by 30th April 2017
9	Burhanp ur	10	176	Compost	Under Bidding	Concessio naire agreemen t signing is targeted	Concession agreement to be signed by 30th April 2017
10	Dewas	24	151	Compost	Under Bidding	Concessio naire	Concession agreement to

						agreement be signed by the signing 30th April is targeted 2017
16	Chhindwara	20	173	Compost	FSR Prepared. Bid documents to be prepared	FSR Consultant - M/s Royal Haskonin gDHV Bids expected to be published by 31st March 2017
17	Betul	8	89	Compost	FSR Prepared. Bid documents to be prepared	FSR Consultant - M/s Royal Haskonin gDHV Bids expected to be published by 31st March 2017
22	Balaghat	13	97	Compost	FSR Prepared. Bid documents to be prepared	FSR Consultant - M/s Ecoprop-Meghraj Bids expected to be published by 30th April 2017
23	Narsinghpur	15	556	Compost	FSR Prepared. Bid documents to be prepared	FSR Consultant - M/s Ecoprop-Meghraj Bids expected to be published by 30th April 2017
24	Pithampur	24	241	Compost	FSR Prepared. Bid documents to be prepared	FSR Consultant - M/s Ecoprop-Meghraj Bids expected to be published by 30th April 2017

District-Level Facility Status for Lower Narmada Basin (as of Jan 2024)

Table 29 Status of composting, biomethanation, and waste-to-energy facilities in Lower Narmada Basin ULBs (as of Jan 2024).

District	Facility Type	Capacity (TPD)	Operational Status (Jan 2024)	Remarks / Source
Surat	Composting, Biomethanation, RDF, WtE	1,150 (Wet), 1,000 (WtE)	Fully Operational	Integrated facility at Khajod; 100% waste processed scientifically. (SMC, 2024)
Vadodara	Composting, Biomethanation, RDF (Pilot)	400 (Wet), 50 (RDF)	Partially Operational	Composting + biomethanation at Moti Baug and Kapurai; RDF pilot under trial. (VMC, 2024)
Bharuch	Composting, (Planned), processing	RDF 55 (Wet), Co-35 (Dry)	DPR Prepared	Cluster RDF/WtE project in design; small composting functional. (GPCB, 2023–24)
Panchmahal (Godhra, Halol)	Composting, composting	Co-58 (Wet)	Operational	Market-waste composting in Godhra; RDF linkages under study. (GPCB RO Vadodara, 2024)
Dahod	Community Composting, Biomethanation (Pilot)	33 (Wet)	Limited Operations	Pilot biogas plants; decentralized composters. (GPCB, 2023–24)
Narmada (Rajpipla)	Pit/Vermi Composting	15 (Wet)	Functional (Small Scale)	Manual composting; DPR for biomethanation

District	Facility Type	Capacity (TPD)	Operational Status (Jan 2024)	Remarks / Source
				pending. (<i>Rajpipla ULB, 2024</i>)
Chhota Udepur	Pit Composting	12 (Wet)	Functional (Small Scale)	Decentralized composting in core wards; limited infrastructure. (<i>ULB Data, 2024</i>)

*(Sources: GPCB Monthly Progress Reports 2023–24; CPCB SWM Annual Report 2023; Surat & Vadodara Municipal Reports 2024; ULB DPRs.)

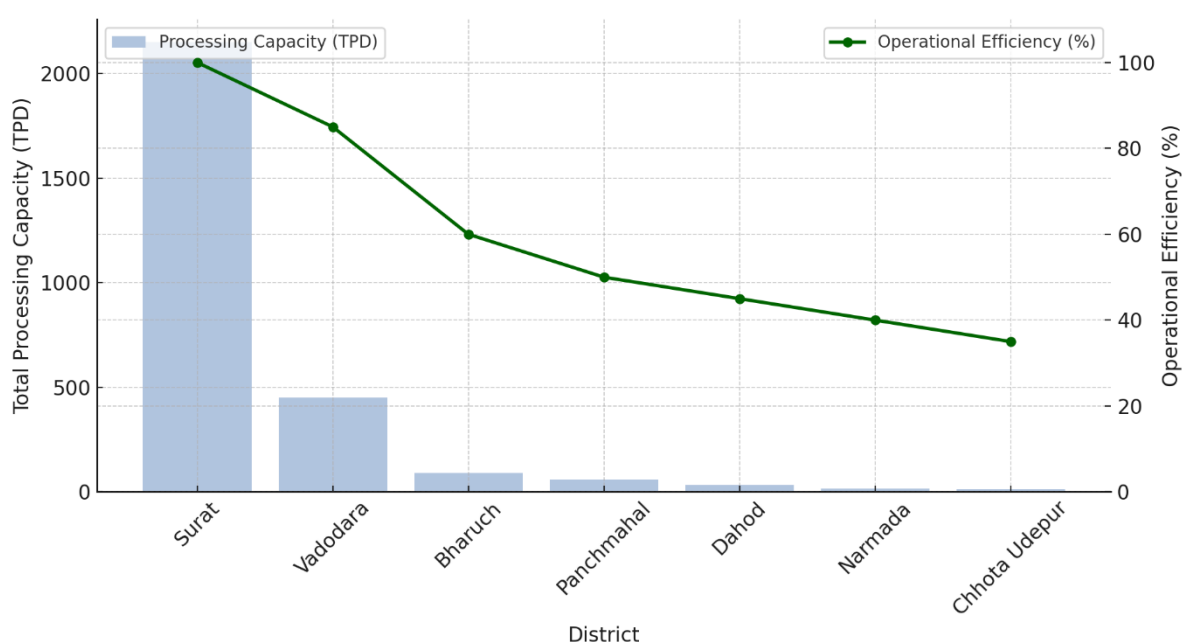


Figure 15 Status of Waste-to-Energy and Composting Facilities in Lower Narmada Basin (Jan 2024)

*(Source: GPCB MPR 2023–24; CPCB SWM Annual Report 2023; Surat & Vadodara Municipal Data)

This figure can compare total processing capacity (TPD) of “Composting + Biomethanation + WtE” across districts — showing Surat and Vadodara as fully operational leaders, followed by Bharuch and Panchmahal (developing), and smaller towns (Dahod, Narmada, Chhota Udepur) at early stages.

4.5.6 Successful Models of Solid Waste Management (Case Studies)

4.5.6.1 Solid Waste Management Initiatives in Narmada Basin Cities

Cities located within the Narmada River Basin have emerged as frontrunners in sustainable solid waste management (SWM) through innovative practices, decentralized models, and technology-driven systems as depicted in **Table 30**. These initiatives not only address urban waste challenges but also contribute to safeguarding the ecological health of the Narmada Basin.

Indore has established itself as the national benchmark in SWM by achieving 100% door-to-door collection, six-way segregation at source, and scientific processing through bio-CNG and composting plants. The city has also completed dumpsite reclamation and integrated GPS-based logistics for efficiency. Indore's recognition as India's Cleanest City for seven consecutive years (2017–2023) reflects its replicable model, which is now being adopted nationally and globally. Jabalpur has pioneered waste-to-energy generation (11.5 MW), alongside integrated landfill management, composting, and Material Recovery Facilities (MRFs). Its success in energy recovery has earned it a place among India's best SWM practices, providing a scalable solution for medium-sized cities.

Bhopal, the state capital within the basin, focuses on compost pelletization and decentralized organic waste recycling, reducing dependence on centralized dumping. The city has achieved a five-star Garbage-Free rating and reclaimed major dumpsites, reinforcing its sustainable urban waste approach. Sagar has implemented a cluster-based SWM model with decentralized composting and landfill improvement. Its approach has been highlighted at the CII Waste Exchange as a replicable cluster success story, demonstrating effective resource utilization across multiple towns. Dindori and Mandsaur have adopted compost pelletization and localized organic waste recycling systems, fostering the growth of a regional compost market. These practices not only reduce landfill pressure but also promote circular economic linkages in agriculture using organic soil conditioners.

The **Lower Narmada Basin** in Gujarat demonstrates a range of **successful and replicable models of solid waste management (SWM)**, combining innovation, public participation, and robust infrastructure.

While **Surat and Vadodara** stand out as model cities with integrated systems covering collection, segregation, processing, and energy recovery, smaller towns such as **Bharuch and**

Panchmahal (Godhra) are making notable progress through cluster-based and decentralized initiatives.

These examples highlight scalable approaches that align with the **Solid Waste Management Rules, 2016**, and contribute to improving the environmental health of the Narmada River Basin.

Table 30 Solid waste management initiatives in Narmada basin cities

City/Location	Model Key Features	References
Indore	100% door-to-door waste collection, 6-way segregation at source, bio-CNG & composting plants, complete dumpsite reclamation, GPS-tracked logistics	<u>India's Cleanest City (2017-2023), global replication, multiple national awards</u> <u>simcindore.mp+5</u>
Jabalpur	Integrated landfill, waste-to-energy facility, composting units, MRFs	<u>Waste-to-energy: 11.5 MW, national SWM best practice</u> <u>abp.championsofchange+2</u>
Bhopal	Compost pelletization, decentralized organic waste recycling	<u>Five-star Garbage-Free city rating, major dumpsite reclaimed</u> <u>gov.eletsonline+1</u>
Sagar	Decentralized composting, landfill improvement, cluster-based SWM	<u>Cluster success recognized by CII Waste Exchange</u> <u>ciiwasteexchange</u>
Dindori, Mandasaur	Compost pelletization, decentralized organic waste recycling	<u>Regional compost market, decreased landfill burden</u> <u>pelletmill+1</u>

*Source: MPCB District environment plan

4.5.6.2 Indore - A Model for Municipal Solid Waste Management

Indore has been recognized as the Cleanest City of India for seven consecutive years, setting a benchmark for sustainable urban solid waste management practices. The city currently generates approximately 1,192 metric tonnes (MT) of municipal solid waste (MSW) per day, which is managed through a combination of centralized and decentralized processing systems as depicted in **Table 31**.

Table 31 Waste generation and processing mechanisms in Indore

S.N.	Type of Waste	Quantity Generated (TPD)	Processing Mechanism	Functional Processing Capacity (TPD)
1	Wet Waste	692	Centralized	550
			Decentralized	205
2	Dry Waste	485	Centralized	400 + 200
			Decentralized	44
3	Sanitary Waste	11.5	Centralized	20
4	Domestic Hazardous & E-Waste	3.4	Centralized	55
5	Construction & Demolition (C&D)	64.6	Centralized	100
6	Landfill Residuals	36	Centralized	600,000
Total	—	1,192	—	1,474

*Source: MPCB District environment plan

Indore's waste processing capacity (1,474 TPD) exceeds its current waste generation (1,192 TPD), ensuring efficient management with surplus capacity for future growth. The city follows six-way waste segregation at source, ensuring high efficiency in recycling and recovery. Centralized and decentralized models operate in synergy, with bio-CNG, composting, MRFs, and sanitary waste processing units. Scientific landfill management and C&D waste recycling facilities significantly reduce environmental impacts.

Waste Generation in Indore and segregation – 6 Bin Concept

Indore has adopted an innovative 6-bin concept as shown in Fig. 16, for municipal waste segregation, enabling efficient management of the city's total daily waste generation of 1192 tons per day (TPD). This approach supports comprehensive resource recovery and safe disposal, setting a benchmark for urban waste handling in India.

- **Wet Waste (692 TPD):** Organic and biodegradable waste, such as kitchen scraps and leftovers, forms the largest fraction and is directed towards composting and biogas

production.

- **Non-Plastic Waste (304 TPD):** Dry recyclable materials like paper, cardboard, metals, and glass are collected separately, enhancing recycling outcomes.
- **Plastic Waste (179 TPD):** Segregated plastics are sent to recycling units or co-processing facilities, reducing landfill pressure and plastic pollution.
- **E-Waste (2.1 TPD):** Discarded electronics and gadgets, although a small stream, require special treatment to manage hazardous substances and recover valuable materials.
- **Sanitary Waste (11.5 TPD):** Items like diapers and sanitary products are isolated to prevent contamination of other streams and enable appropriate disposal.
- **Domestic Hazardous Waste (3.4 TPD):** Household hazardous items like paints, chemicals, batteries are separated to avoid environmental harm and support safe treatment

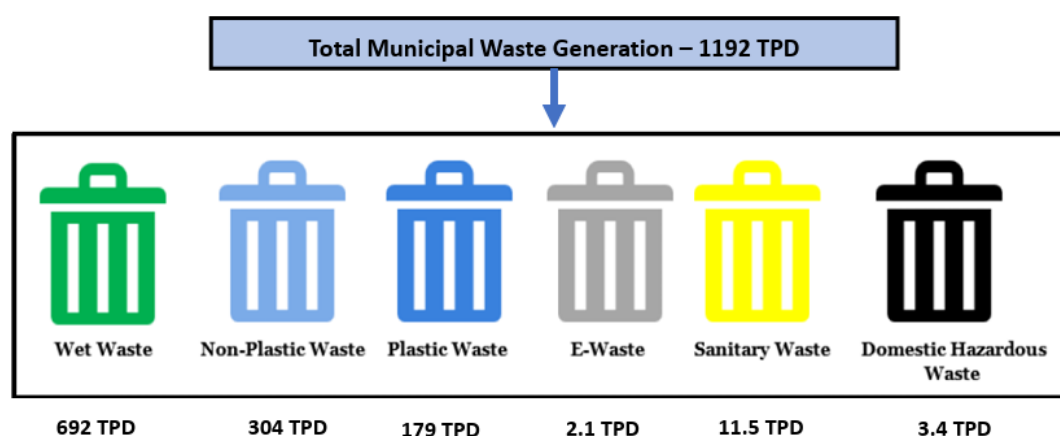


Figure 16 Six bin concepts for waste generation and segregation in Indore

Indore's 6-bin system improves recycling rates, reduces landfill burden, and minimizes environmental and health risks by ensuring that each waste category is processed via its most sustainable pathway. This practice enhances urban sustainability and can serve as a model for other Indian cities, particularly in ecologically sensitive areas such as the Narmada River Basin.

D2D Collection Fleet Expansion and Technological Advancement in Indore

Indore has achieved a transformative upgrade in municipal solid waste management through D2D (Door-to-Door) Collection Fleet Expansion and Technological Advancement, ensuring 100% service coverage in all city wards with a robust, technologically integrated system as shown in **Fig. 17, 18 and 19**.

Fleet Expansion and 100% Coverage

- The city deploys a comprehensive fleet that includes:
 - 575 D2D vehicles dedicated to daily household waste collection.
 - 74 Bulk Waste vehicles for institutional and commercial establishments.
 - 8 Meat Collection vehicles for specialized organic waste streams.
 - 15 C&D (Construction & Demolition) vehicles to manage construction debris.
 - 85 Garden Waste vehicles for horticultural waste.
 - 148 Litter Bin/Road Sweeping waste vehicles to maintain street cleanliness.

This diverse and specialized vehicle fleet ensures that every type of municipal waste is collected efficiently, covering all sources and all wards without lapse.

Technological Integration and Automation

The system leverages an end-to-end automated service delivery mechanism, utilizing GIS mapping and digital command centers to route vehicles, monitor collection progress, and track operational performance in real time. Centralized control rooms oversee fleet operations, guaranteeing prompt response and continuous optimization of collection schedules and resources.

Impact of Technological Advancement

Integrated technologies enable precision routing, high accountability, and adaptive resource allocation, reducing missed pickups and maximizing efficiency. Real-time dashboards foster transparency, data-driven decision-making, and rapid resolution of service bottlenecks.

Innovations Enhancing Segregation

Vehicles are customized for multi-bin waste segregation (wet, dry, plastic, hazardous, sanitary, and e-waste), supporting Indore's pioneering 6-bin strategy and facilitating downstream recycling and treatment. Seamless linkage between collection and processing ensures minimal waste leakage and maximizes resource recovery across the urban landscape.

Indore's integrated fleet and smart management platform set a national benchmark, driving efficiency, accountability, and sustainability in urban waste management while protecting river basin health



Figure 17 D2D collection fleet expansion and technological advancement

Indore has undergone a remarkable transformation, converting its legacy dumpsite into a vibrant city forest through phased bioremediation and ecological restoration initiatives

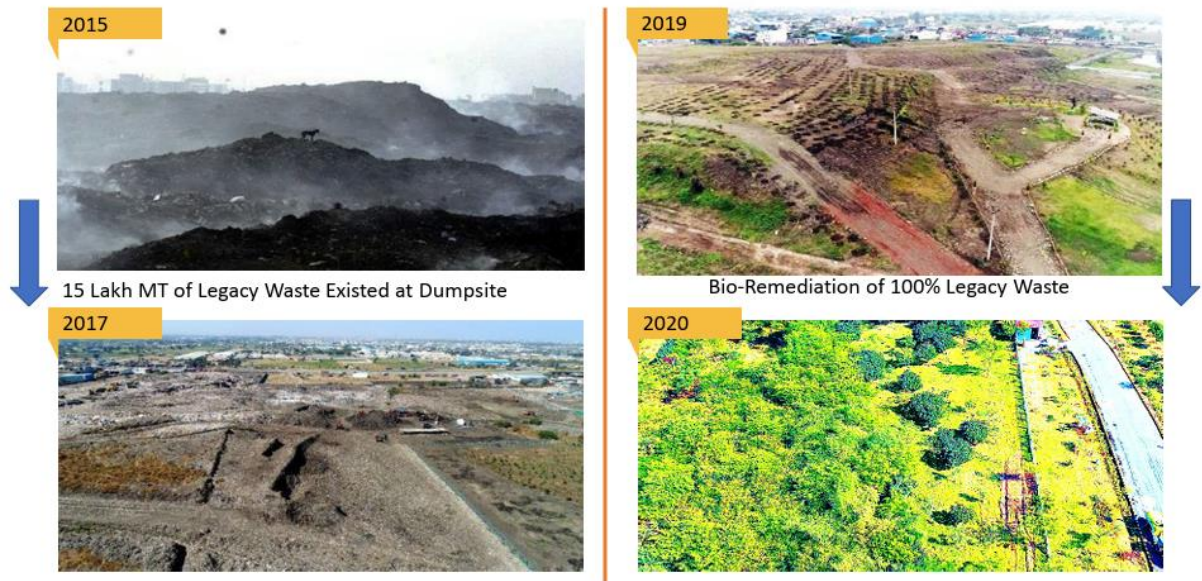


Figure 18 Indore bioremediation transformation into city forest (2015 to 2020)

- Reclamation of over 100 acres land worth Rs 300 Cr.
- Transformation into City Forest – 1.50 lac trees planted



Figure 19 Indore bioremediation transformation into city forest (2022 to present)

4.5.6.3 Dumpsite Remediation of Legacy Waste – Bhanpur Khanti (Bhopal)

Bhanpur Khanti in Bhopal exemplifies large-scale dumpsite remediation and land reclamation, showcasing effective management of legacy municipal waste and urban restoration as shown in **Fig. 20**.

Before Remediation

The site contained around 7.23 lakh tons of legacy waste, spread across a vast 37-acre area. This unremediated dumpsite posed significant environmental and health risks due to uncontrolled dumping, leachate contamination, and air pollution.

After Remediation

21 acres of land, valued at Rs. 85 crores, have been successfully reclaimed for productive urban use, unlocking economic and social benefits for Bhopal. 16 acres of land have been scientifically capped, preventing further soil, air, and groundwater pollution through engineered containment and restoration measures.



Figure 20 Transformation of Bhanpur Khanti Dumpsite

4.5.6.4 Best Practice Integrated Solid Waste Management Project at Jabalpur

Jabalpur's integrated solid waste management system is showcased by its Waste to Energy Plant at Katonda, a benchmark initiative developed under the Public-Private Partnership (PPP) model as shown in **Fig. 21**.

Katonda Waste to Energy Plant – Key Features

The plant boasts an installed capacity of 11.5 MW, converting municipal solid waste into electricity and significantly reducing landfill dependency. Every day, the facility processes 550 tons of municipal waste collected from 16 Urban Local Bodies (ULBs) across the region, making it a central hub for inter-city waste management. Participation in the PPP model has driven technological efficiency and financial sustainability, fostering a collaborative approach between Jabalpur Municipal Corporation and private sector expertise.

Environmental and Social Impact

The systematic treatment and energy recovery from waste have led to enhanced urban cleanliness, helping to mitigate environmental hazards caused by unprocessed garbage and open dumping. Reduced waste volumes, improved hygiene, and organized disposal practices have collectively transformed local environmental conditions in Jabalpur and its satellite ULBs. Jabalpur's Katonda Waste to Energy project stands out as a best practice example, demonstrating the effectiveness of integrated solid waste systems and PPP models in scaling up urban sustainability and resource recovery efforts.

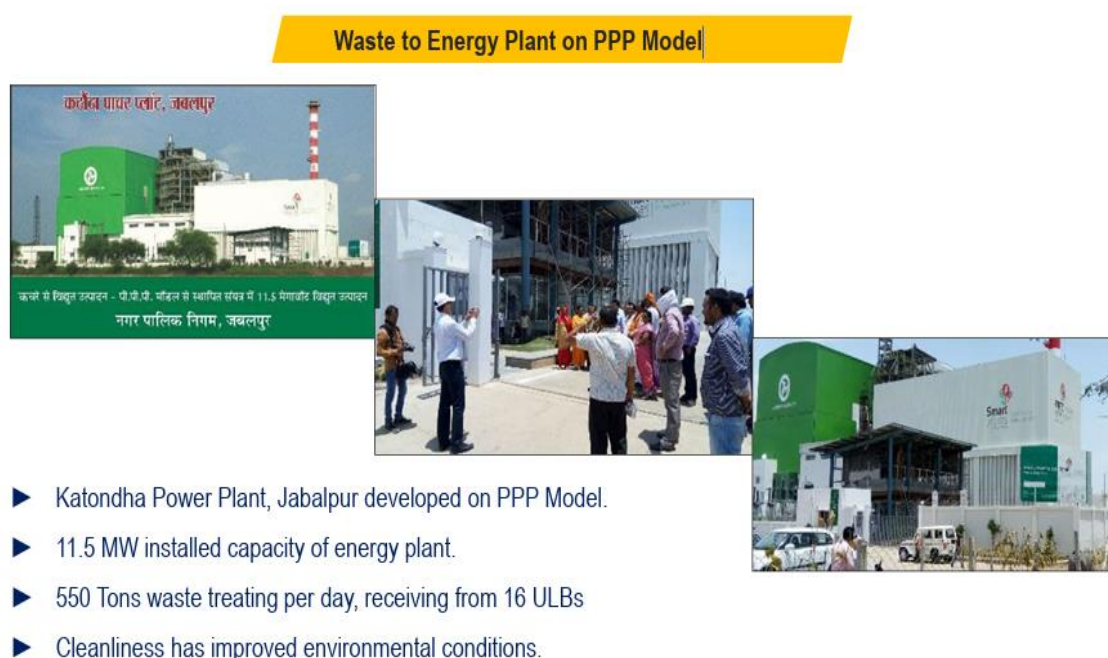


Figure 21 Integrated Solid Waste Management Project at Jabalpur

4.5.6.5 Best Practice Integrated Solid Waste Management Project, Kanti Cluster

The Kanti Cluster Integrated Solid Waste Management Project represents a model of comprehensive waste management infrastructure, combining advanced processing, composting, and landfill operations to sustainably handle municipal waste across regional urban centers as shown in **Fig. 22**.

Waste Processing Infrastructure

- **Katni MSW Plant:** Central to the cluster, this facility is landscaped and engineered for high-volume municipal solid waste (MSW) intake, ensuring clean and organized site operations.
- **Trommel:** Automated trommel screens are used for mechanical segregation, efficiently separating recyclables, organics, and inert fractions for downstream processing.
- **Aeration Pit:** Managed pits facilitate controlled aeration of organic waste, accelerating decomposition and enhancing compost quality through aerobic microbial action.



Figure 22 Waste Processing Plant & Scientific Landfill of Integrated Solid Waste Management, Kanti cluster

Specialized Treatment Facilities

- Composting: Organic fractions are converted into nutrient-rich compost, supporting soil health and local agriculture while diverting biodegradable waste from landfills.
- Animal Carcass Incinerator: Dedicated incinerators safely process animal remains, minimizing biohazard and odour risks for the surrounding community.

Scientific Landfill Operations

- Scientific Landfill: Residual inert and non-recyclable wastes are securely contained in engineered landfills, with layers and leachate management systems that shield soil and groundwater from contamination, ensuring compliance with environmental norms

4.5.6.6 Best Practice - Integrated Solid Waste Management Project – Sagar Cluster

The Sagar Cluster Integrated Solid Waste Management Project showcases an exemplary model of regional waste processing as shown in **Fig. 23**, combining mechanized segregation and scientific landfill solutions to address municipal solid waste challenges comprehensively.

Waste Processing Plant Components



Figure 23 Waste Processing Plant & Scientific Landfill photos of Integrated Solid Waste Management, Sagar cluster

Sagar MSW Plant Overview & Entry:

The project features a well-structured campus with modern access and secure perimeter, facilitating organized intake and processing of municipal waste from surrounding areas. The main plant building and expansive shed support scalable operations, sheltering advanced processing equipment and ensuring all-weather handling of waste materials.

Mechanized Segregation

Trommel: Automated trommel machinery is deployed for bulk mechanical separation, sorting recyclables and compostable fractions from incoming waste with high efficiency.

Waste Pit: Carefully engineered waste pits ensure safe temporary storage and pre-processing before onward movement to treatment or disposal units.

4.5.6.7 Surat – A Model for Integrated Waste Management

Background:

Surat Municipal Corporation (SMC) has emerged as one of India's leading ULBs in solid waste management, maintaining 100% door-to-door collection, segregation, processing, and scientific disposal. The city's integrated system supports daily handling of **over 2,100 TPD** of municipal solid waste.

Key Features:

- **100% Door-to-Door Collection:** GPS-tracked, covered vehicles operate across all 100 wards.
- **Segregation at Source:** Maintained at **>95 %**, facilitated through community awareness and enforcement drives.
- **Wet Waste Processing:**
 - **Composting & Biomethanation Units (1,150 TPD)** producing compost and biogas.
 - **Refuse-Derived Fuel (RDF) and WtE Facility (1,000 TPD)** operational at Khajod.
- **Scientific Landfill:** Legacy waste fully remediated; reclaimed land redeveloped into the **Integrated Solid Waste Management Complex**.

- **Public–Private Partnership (PPP):** The WtE plant, operated by Abellon Clean Energy, generates electricity supplied to the local grid.
- **Digital Monitoring:** Integrated MIS dashboard monitors vehicle routes, waste tonnage, and plant performance in real time.

Impact:

- Zero open dumping or burning within city limits.
- Over **90 % diversion from landfill**, positioning Surat as a **Zero Waste City**.
- Model replication initiated by other ULBs through the Gujarat Urban Development Department.

Source: Surat Municipal Corporation (SMC), GPCB MPR 2023–24, CPCB SWM Annual Report 2023.

4.5.6.8 Vadodara – Integration of Composting, Biomethanation, and RDF

Background:

The **Bharuch-Ankleshwar** urban cluster, part of the Lower Narmada Basin, generates approximately **90–100 TPD** of waste. Given the proximity of industrial areas and cement plants, Bharuch has adopted a **cluster-based approach** to waste processing and disposal.

Key Features:

- **Composting and RDF Planning:**
 - DPR prepared for **cluster-level RDF-based Waste-to-Energy facility** serving Bharuch, Ankleshwar, and nearby towns.
- **Segregation Improvements:**
 - ULB-level initiatives increased segregation to **~70 %** through awareness campaigns.
- **Co-Processing:**
 - Non-recyclable waste supplied to **Ambuja Cement (Kodinar)** for RDF utilization.
- **Legacy Waste Remediation:**
 - **60 %** of old dumpsite remediated by early 2024 under GPCB monitoring.

Impact:

- Reduction in open dumping and improved waste utilization.
- Demonstration of effective industrial–municipal cooperation for RDF usage.
- Serves as a **replicable model for small industrial towns** in the basin.

Source: GPCB RO Bharuch Reports 2023–24; CPCB SWM Annual Report 2023.

4.5.8.9 Panchmahal (Godhra) – Decentralized and Market Waste Composting

Background:

The Panchmahal district (notably **Godhra and Halol**) generates about **105 TPD** of waste, with organic waste comprising over 50 %. The district’s ULBs have implemented decentralized composting and market-waste management programs.

Key Features:

- **Decentralized Composting:** Ward-level composting units and market compost pits operational.
- **Segregation Drives:** Conducted in schools and residential areas with community engagement.
- **Co-Composting:** Integration of organic waste and sewage sludge under pilot initiatives.
- **ULB Partnerships:** Support from GPCB and SBM(U) for technical guidance and equipment supply.

Impact:

- Reduced waste transportation costs by treating waste at source.
- Improved compost utilization in local agriculture and municipal landscaping.
- Enhanced citizen participation in waste segregation and composting.

Source: GPCB RO Vadodara, Panchmahal ULB DPRs, 2023–24.

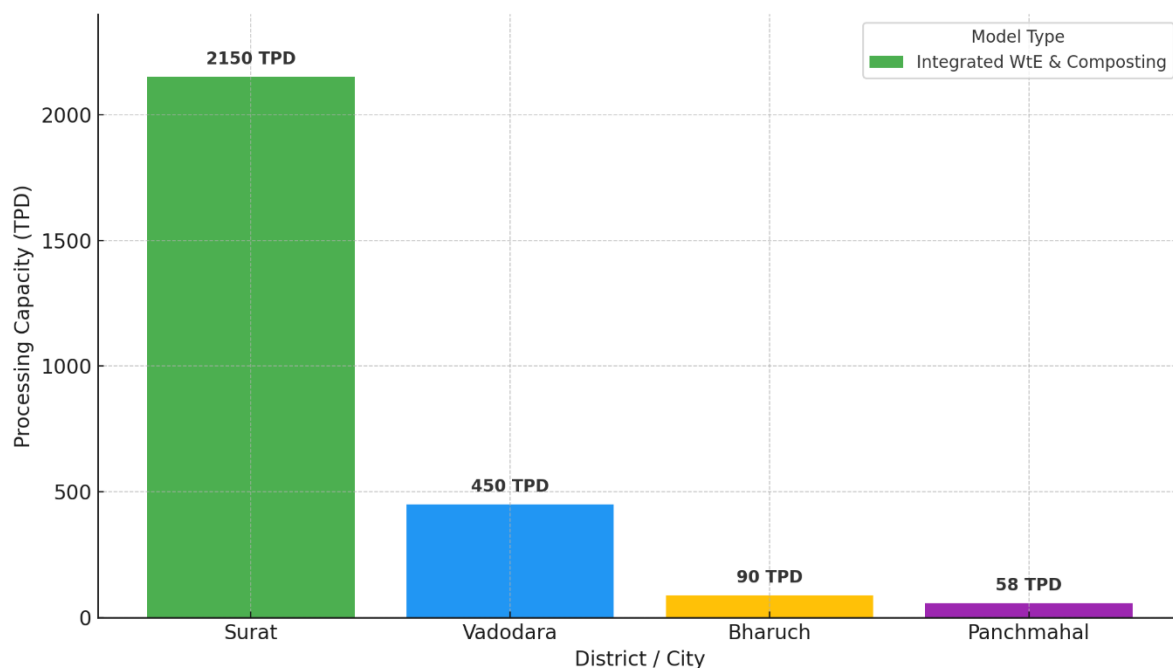


Figure 24 Successful Solid Waste Management Models in Lower Narmada Basin (Gujarat).

***Source:** Gujarat Pollution Control Board (GPCB) MPR 2023–24; CPCB SWM Annual Report 2023; Surat, Vadodara, Bharuch, and Panchmahal ULB Reports 2024

This figure can summarize key model types (Integrated, Cluster-Based, and Decentralized) with their respective districts — Surat (Integrated WtE), Vadodara (Biomethanation + RDF), Bharuch (Cluster RDF), Panchmahal (Decentralized Composting).

5. Wastewater Treatment Facilities

5.1 Sewage Treatment Plants (STPs)

5.1.1 Operational Sewage Treatment Plants

The Narmada River Basin districts collectively operate a limited number of sewage treatment plants (STPs), with wide variations in installed capacity which is depicted in **Table 32**. The highest coverage is observed in Jabalpur, which has 11 operational STPs with a cumulative capacity of 154.15 MLD, reflecting its large urban population and central role as a basin city. Other districts with moderate treatment capacity include Dewas (38 MLD across 3 plants), Burhanpur (24 MLD across 2 plants), and Dhar (12 MLD with 1 plant). Smaller districts such as Sehore (15.28 MLD across 3 STPs), Khandwa (17.95 MLD across 2 STPs), and Khargone (17.6 MLD with 1 STP) provide localized coverage but at much lower scales. Meanwhile, rural and semi-urban districts like Anuppur (1.2 MLD), Dindori (3.85 MLD), Mandla (10 MLD), and Narsinghpur (9 MLD) operate only single or dual units, with very limited treatment capacity compared to wastewater generation.

Overall, the existing treatment infrastructure is concentrated in a few urban centers, leaving large portions of the basin underserved. This uneven distribution underscores the urgent need for capacity augmentation, operational efficiency improvements, and expansion of decentralized wastewater treatment systems. For detailed information on STP locations, average utilization, and wastewater reuse practices, refer to **Annexure Table A16**.

Table 32 Operational Sewage Treatment Plants in the Narmada River Basin

Districts	Number of Operational STPs	Cumulative Capacity (MLD)
Anuppur	1	1.2
Burhanpur	2	24
Dewas	3	38
Dhar	1	12
Dindori	1	3.85

Jabalpur	11	154.15
Khandwa	2	17.95
Khargone	1	17.6
Mandla	1	10
Narsinghpur	2	9
Sehore	3	15.28
Bharuch	1	29.3
Narmada	1	5.5
Chhota udepur	1	5

**Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.*

The Lower Narmada basin contains a mix of municipal and institutional STPs serving urban centres, industrial townships and tourist facilities. Rajpipla is served by a 5.5 MLD SBR-based STP located near Chitrawadi (commissioned 2024 under GWSSB) and treats sewage from the town's partially developed network with some reuse for non-potable purposes. Ankleshwar GIDC has a dedicated 22.5 MLD STP serving the industrial estate. Chhotaudepur operates a 5.0 MLD STP (conveyance constrained by rising-main problems) adequate for the town's ~4.0 MLD sewage generation. Bharuch city has an integrated sewerage system divided into five zones with an ultimate STP capacity of 29.3 MLD, fed by ~118 km of pipelines, 5,910 manholes and 166 vertical drops. Smaller institutional STPs include the 500 KLD STP at Ekta Nagar (Kevadia township) and a ~400 KLD STP at the Sardar Patel Zoological Park (Jungle Safari). Rural areas in the lower basin lack centralized STP coverage and rely predominantly on on-site systems (septic tanks and leaching pits).

5.1.2 Treatment Technologies Adopted in STPs & Treated Effluent Quality Parameters

STPs in the Narmada River Basin use varied technologies, with SBR (15 plants) being the most common due to its high efficiency and compact footprint, though it requires skilled operation.

Phytorid systems (7 plants) provide a low-energy, eco-friendly option suitable for decentralized setups, while oxidation ponds (2 plants) are simple and low-cost but land-intensive. A single unit applies Devorts electrochemical technology, mainly for industrial wastewater. For 5 STPs, technology details are unavailable, reflecting reporting gaps. Overall, SBR dominates in urban centers, while Phytorid and oxidation ponds serve smaller towns as depicted in **Table 33**.

Table 33 Wastewater treatment technologies adopted in STPs

Technology	No. of STPs based in this technology	Procedure	Features
SBR (Sequential Batch Reactor)	15	Operates in cycles: Fill, react (aeration), Settle, Decant, and Idle. Treats wastewater in batches with aeration and sedimentation.	Batch process; high efficiency; small footprint; requires skilled operation.
Oxidation Pond	2	Large shallow basins where sunlight, algae, and microorganisms work together to degrade organic matter.	Simple, low-cost, effective in sunny climates; requires a large land area.
Phytorid	7	Constructed wetland system using specific aquatic plants and soil media to naturally treat wastewater.	Eco-friendly; low energy; enhances biodiversity; suitable for decentralized systems.
Devorts	1	Uses electrolysis or electrochemical methods for pollutant removal, often in industrial wastewater.	Chemical-free; high removal efficiency for specific pollutants; cost-effective for industrial use.
Not Available	5	Not applicable	Not applicable

** Source: Hon'ble National Green Tribunal Principal Bench. (2024, March 14). Order no. 606/2018*

Measured effluent quality where data is available shows effective treatment performance for several plants. In **Rajpipla**, the influent sewage reaching the STP(5.5 MLD SBR) shows variable and at times high pollutant loads, with BOD ranging from 24 to 138 mg/L, COD up to 435 mg/L, TSS(Total Suspended Solids) often exceeding 370 mg/L, and Total Nitrogen between 10 and 16 mg/L, with pH slightly acidic at 5.8–6.1. The treated effluent consistently complies with standards, with BOD reduced to less than 6.5 mg/L, COD to under 20 mg/L, and TSS below 10 mg/L. Nutrient removal is effective, with Total Nitrogen around 6 mg/L, NH₄-N less than 3.6 mg/L, and phosphorus at 0.8 mg/L. Pathogen removal is also significant, with fecal coliforms reduced to 70 MPN/100 ml, pH stabilized around 7.1–7.5, and residual chlorine maintained between 0.3 and 0.9 mg/L.

Bharuch's STP system with Sequential batch Reactor (SBR) uses automated dosing and ORP/DO-based disinfection and monitoring. The outlet water sample from the Bharuch city STP test result shows that the pH of the water is 8.26, which is within the normal range. The BOD is 17 mg/L and COD is 51 mg/L, which means some organic matter is still present. Fecal coliform count is 920 MPN/100ml, showing that bacteria are present in the treated water. Suspended solids are low at 8 mg/L. The nutrients measured were 1.89 mg/L of total nitrogen and 0.138 mg/L of total phosphate. Overall, the STP is reducing pollution load, but the presence of coliform and BOD indicates the water still needs care before reuse or safe discharge.

Sludge generated by municipal STPs in the lower basin is handled using conventional thickening and dewatering processes. Rajpipla, Chhota udepur and Bharuch plants employ sludge thickening and dewatering (Bharuch also uses centrifugation) with subsequent safe disposal.

5.1.3 Under-Construction STPs

Several sewage treatment plants (STPs) are under construction across the Narmada River Basin, indicating ongoing efforts to bridge the gap between sewage generation and treatment. The largest upcoming capacity is in Hoshangabad (21 MLD), followed by Barwani (20.3 MLD across 4 units) and Khargone (15.99 MLD across 4 units), which together represent major capacity additions in the basin. Districts such as Mandla (9.5 MLD), Jabalpur (8.8 MLD), and Jhabua (8 MLD) also show significant expansions. Smaller-scale plants are being developed in Alirajpur (3.5 MLD), Dhar (11.18 MLD), and Sehore (5.2 MLD), while rural or semi-urban districts like Betul, Burhanpur, Seoni, and Narsinghpur are constructing plants of less than 2 MLD each (**Table 34**).

Overall, under-construction STPs add a cumulative capacity of over 100 MLD, which will substantially improve wastewater treatment once commissioned. However, the progress of construction, scheme-wise funding details, and expected completion timelines vary widely as shown in **Table 26**. For location-specific data, physical progress, reuse provisions, and scheme details, refer to **Annexure Table A17**.

Table 34 Under construction Sewage Treatment Plants in the Narmada River Basin

District	Number of Under Construction STPs	Cummulative Capacity (MLD)
Alirajpur	2	3.5
Balaghat	1	1.5
Barwani	4	20.3
Betul	2	1.6
Burhanpur	2	1.7
Chhindwara	1	3
Dhar	2	11.18
Hoshangabad	1	21
Jabalpur	2	8.8
Jhabua	2	8
Khargone	4	15.99
Mandla	2	9.5
Narsinghpur	1	1
Raisen	1	2.7
Seoni	1	1

Sehore	2	5.2
Bharuch	1	14

**Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.*

Ankleshwar has a municipal 14 MLD STP that receives sewage from an extensive underground drainage network of 146 km length. 90% work has been completed and it will be ready by 31 December 2025.

5.2 Common Effluent Treatment Plants (CETPs) and Final Effluent Treatment Plant (FETPs)

The Narmada River Basin is supported by a limited number of Common Effluent Treatment Plants (CETPs) given in **table 35**. They are designed to handle industrial wastewater collectively from small and medium enterprises. At present, there are three operational CETPs with a cumulative capacity of 9.1 MLD, along with two additional CETPs under construction/proposed with a combined capacity of 0.6 MLD.

Overall, while the running CETPs are functional and compliant, their scale is relatively modest compared to the growing industrial effluent load in the basin. With increasing industrialization in clusters such as Pithampur, Mandideep, and Sanwer Road, there is a pressing need to expand CETP capacity, ensure regular monitoring, and promote Zero Liquid Discharge (ZLD) practices to minimize untreated effluent discharge into the Narmada and its tributaries.

Table 35 Status of CETPs in the Narmada River Basin

Town / Location	No. of Industries	Industrial Discharge (MLD)	Status of ETPs	Status of CETPs	Capacity (MLD)
Govindpura					
Industrial Area, Bhopal	26	~0.2	Running	Existing	0.2

Sanwer					
Road,	168	~4.0	Running	Existing	4
Indore					
Burhanpur	51	~4.5	Running	Existing	4.5
Meghnagar,	-	-	-	Existing	0.4
Jhabua					
Rangwasa,	-	-	-	Under	0.2
Indore				Construction	

Source: Government of Madhya Pradesh. (2023, October). National Mission for Clean Ganga: Monthly Progress Report in NGT Case No. 673 of 2018 (in compliance to NGT order dated 24.09.2020).

The lower Narmada basin's industrial clusters are served by several CETPs/FETPs of varying capacities and by a regional deep-sea disposal network given in **table 36**. Ankleshwar GIDC operates a CETP (Enviro Technology Ltd.) whose capacity has been increased from 1.8 MLD to 2.2 MLD and discharges treated effluent to the sea via underground pipeline. Panoli GIDC hosts a CETP (Panoli Enviro Technology Ltd.) with planned upgradation from 2.04 MLD to 2.664 MLD, with deep-sea disposal. Dahej GIDC has a 40 MLD CETP proposed and is linked to a regional submarine disposal arrangement; the GIDC has developed a 90 MLD wastewater collection and disposal pipeline that connects Vilayat pumping station to Dahej for deep-sea discharge. The Vilayat area is explicitly connected to the 90 MLD collection/disposal scheme serving Dahej/PCPIR. Jhagadia GIDC is served by a FETP (Narmada Clean Tech) with an effluent handling cap not to exceed 35 MLD, discharging via underground pipeline to marine disposal. Bharuch GIDC (Saykha) is estimated to host a 40 MLD CETP linked into the same 90 MLD pipeline network. GIDC's 90 MLD disposal pipeline has an overall length of about 40 km from Vilayat pumping station to Dahej, and includes onshore and offshore sections from the Dahej final pumping station in deep sea. For parts of Narmada district such as Rajpipla and Tilakwada GIDCs, no common CETP capacity data is available in the supplied dataset and industries in these areas are understood to operate individual ETPs (Effluent Treatment Plant).

Table 36 Industrial clusters and CETP in lower Narmada basin gujarat

Industrial Cluster / Area	CETP / FETP Operator	Capacity (MLD)	Disposal Method / Notes
Ankleshwar GIDC	Enviro Technology Ltd.	2.2	Discharges via underground pipeline to sea; capacity increased from 1.8 MLD
Panoli GIDC	Panoli Enviro Technology Ltd.	2.04	Deep-sea disposal
Dahej GIDC	GIDC	40 (proposed)	Linked to 90 MLD wastewater collection and deep-sea disposal pipeline
Vilayat Area	GIDC	Linked to 90 MLD	Part of 90 MLD collection/disposal scheme serving Dahej/PCPIR
Jhagadia GIDC	Narmada Clean Tech (FETP)	35	Effluent via underground pipeline to marine disposal
Bharuch GIDC (Saykha)	GIDC	40 (estimated)	Linked to same 90 MLD deep-sea disposal pipeline

Source: EC compliance for the period of Apr22 to Sep22 ETL and dahej effluent infrastructure and pre-feasibility report for proposed expansion of existing common effluent treatment plant (cetp) operated by m/s. panoli enviro technology ltd. (petl) (common effluent treatment plant)

Sludge and solid residuals from CETPs/FETPs in the lower basin are dewatered and managed through authorized hazardous-waste facilities. **M/s. Bharuch Enviro Infrastructure Ltd. (BEIL)** with facilities in both **GIDC Ankleshwar** and **GIDC Dahej**, and **M/s. Shesh Enviro Infra Pvt. Ltd.** at **Saykha** (Dahej–Vagra belt). CETP/FETP sludge is typically dewatered on-site and transported to these TSDF facilities for final disposal. In addition, common biomedical waste handling in the region is supported by facilities such as **Globe Biocare (Plot No.144/B, GIDC Ankleshwar)**. For industries in locations without common CETPs (e.g., Rajpipla, Tilakwada), individual ETP sludges are handled locally and dispatched to authorized disposal sites as per the operating company arrangements.

6. Solid and Hazardous Waste Disposal

6.1 Hazardous waste

6.1.1 Hazardous Waste Generation

Hazardous waste generation in the Narmada River Basin is concentrated in a few industrially active districts. Dhar (29,210.15 MT/annum), Sehore (9,335.93 MT/annum), Sagar (6,322.21 MT/annum), Dewas (6,597.18 MT/annum), and Indore (5,724.57 MT/annum) are the major contributors, reflecting their dense industrial presence. Significant recyclable or utilizable fractions are reported from Dhar (21,144.57 MT), Dewas (4,931.36 MT), and Sagar (over 5,800 MT), highlighting strong recovery potential.

Smaller volumes are observed in districts like Betul (32.33 MT), Harda (136.40 MT), and Umari (32.48 MT), while some tribal-dominated areas, such as Dindori, report negligible or no hazardous waste. A few districts, including Burhanpur (4,503.9 MT) and Khargone (2,701.4 MT), show large utilizable fractions, indicating scope for resource recovery through co-processing or industrial reuse.

Overall, hazardous waste generation as shown in **Table 37** is unevenly distributed, with industrial hubs accounting for most of the load. Strengthening recycling and utilization pathways, along with strict compliance with TSDF and co-processing norms, will be essential to minimize environmental risks.

Table 37 Hazardous Waste Generation in the Narmada River Basin

Districts	Quantity of HW in the district (MT/Annum)	Quantity of Recyclable/utilizable HW (MT/Annum)
Alirajpur	0.05	Recyclable – 0.05
Anuppur*	604.27	Recyclable – 35.89, Utilizable - 565.07
Balaghat	37.47	Recyclable – 37.47
Barwani	314.26	Recyclable- 35.92, Utilizable- 152.86

Betul	32.33	Recyclable-21.01, Utilizable-1.54
Burhanpur	4503.9	Recyclable- 0.11, Utilizable- 4478.00
Chindwara	1186	Recyclable – 154
Damoh	42.64	Recyclable – 42.64
Dewas	6597.18	Recyclable – 4931.36
Dhar	29,210.15	Recyclable – 21,144.57
Dindori	0	0
Harda	136.404	Recyclable-0.9421, Utilizable0.135
Hosangabad	66.89	Recyclable – 43.30, Utilizable – 19.98
Indore	5724.57	-
Jabalpur	2566.19	Recyclable –2466.38
Jhabua	1184.53	Recyclable- 960.55, Utilizable- 100.91
Katni	101.8	Recyclable- 101.51, Utilizable- 0
Khandwa	62.93	Recyclable- 30.67, Utilizable- 0.00
Khargone	2701.4	Recyclable- 1059.20, Utilizable- 1155.90
Mandla	10.967	Recyclable – 0.86
Narshingpur	262.62	Recyclable – 10.18
Raisen	5711.8	Recyclable – 119.05

Sagar	6322.21	Recyclable – 304.08, Utilizable – 5805.10
Sehore	9335.93	Recyclable-350.03, Utilizable HW- 5.14
Seoni	0	Recyclable – 8.25
Umaria	32.48	Recyclable -17.22

**All the data filled according to the basin Portion.*

Source: Madhya Pradesh Pollution Control Board. (2025). District Environment Plans

Bharuch district is one of the biggest hazardous waste-producing areas in Gujarat. About 2,380 industries in the district generate nearly 12,53,485 metric tons of hazardous waste every year. Out of this, around 1,04,525 MT/year is incinerable, 5,92,019 MT/year is landfillable, and 5,56,941 MT/year is recyclable or usable again. Ankleshwar, being a large industrial area, produces a major part of this waste. In Narmada district (Rajpipla), 18 industries generate about 312 MT/year of hazardous waste, of which 269.70 MT/year is incinerable and 42.303 MT/year is landfillable, with no recyclable waste. In Chhota Udepur district, 19 industries produce around 11.095 MT/year of hazardous waste, including 0.11 MT/year incinerable, 14.51 MT/year landfillable, and 953.352 MT/year recyclable or reusable waste. As below table shows the hazardous waste in lower basin.

Table 38 Hazardous waste generated in lower basin

District	No. of Industries	Total Waste (MT/year)	Incinerable (MT/year)	Landfillable (MT/year)	Recyclable/Utilizable (MT/year)
Bharuch	2,380	12,53,485	1,04,525	5,92,019	5,56,941
Narmada	18	312	269.70	42.303	0
Chhota Udepur	19	11.095	0.11	14.51	953.352

**Source: GPCB District environment plan*

6.1.2. Hazardous Waste Management

Hazardous waste management in the Narmada Basin is primarily facilitated through the **Treatment, Storage and Disposal Facility (TSDF) located in Dhar district**, which caters to

most of the basin districts. The district-level review identified from District Environmental Plans, presented in **Annexure Table A11**, highlights both progress and existing gaps.

A number of districts such as **Betul, Dindori, Chhindwara, Dewas, Dhar, Jabalpur, Mandla, Narsinghpur, Raisen, Sehore, and Seoni** have established collection centers and are effectively linked to the Dhar TSDF, with minimal or no gaps reported. In these districts, hazardous waste management is operational, and institutional mechanisms are relatively strong.

On the other hand, several districts including **Anuppur, Barwani, Burhanpur, Damoh, Hoshangabad (Narmadapuram), Jhabua, Katni, Umaria, and West Nimar** lack dedicated collection centres. The District Environmental Plans (DEPs) for these regions identify the urgent need to establish proper collection facilities and strengthen linkages with TSDF Dhar. In some districts such as **Damoh and Hoshangabad**, it is specifically recommended that Urban Local Bodies (ULBs) be integrated into the TSDF network to ensure compliance with hazardous waste rules.

In a few cases, hazardous waste generation has been assessed, such as in **Balaghat and Sagar**, reflecting either low industrial activity or minimal hazardous waste generation in those districts. However, continuous monitoring and preparedness are emphasized to manage future risks.

Common challenges noted across the basin include the **need for training and capacity-building of personnel**, provision of adequate protective equipment, and improved awareness of safe handling practices. While the Dhar TSDF provides a central facility for disposal, the absence of decentralized collection and pre-treatment infrastructure in many districts poses a risk of improper handling at local levels.

Thus, while progress has been made in select districts, hazardous waste management in the Narmada Basin remains uneven. Strengthening collection networks, capacity building, and ensuring effective TSDF linkages across all districts are critical steps for a more robust and compliant hazardous waste management framework.

Bharuch district has strong systems for handling hazardous waste. The district has 3 Treatment, Storage, and Disposal Facilities (TSDFs) and 6 Secured Landfill Sites (SLFs). No contaminated sites have been found so far. In Ankleshwar, industries have their own storage areas and are connected with TSDFs through the Gujarat Pollution Control Board's online

XGN system. Workers receive regular training, and compliance is strictly checked to ensure proper waste disposal.

In Narmada district, there are no TSDFs, so hazardous waste from 18 industries is sent to other districts for safe disposal. A new Material Recovery Facility (MRF) is planned for domestic hazardous waste management by December 2022, and worker training is also planned by that time. In Chhota Udepur district, industries are linked to state-level TSDFs through the XGN system. Although the district has no local disposal site, all waste is properly handled and sent to authorized facilities. No contaminated sites have been reported in the area.

6.2 Biomedical waste

6.2.1. Biomedical Waste Generation

Biomedical waste generation in the Narmada River Basin as shown in **Table 39** is highest in Jabalpur (1,899 kg/day), followed by Khargone (368.97 kg/day), Khandwa (297.47 kg/day), and Dewas (273.33 kg/day), where treatment facilities manage the full load. Most districts show near-complete treatment, reflecting adequate capacity. However, Burhanpur is a major concern, generating 260.62 kg/day but treating only 3.62 kg/day, indicating critical infrastructure and compliance gaps. Smaller districts like Dindori (52 kg/day) and Harda (85 kg/day) generate less waste, all fully treated. Overall, while most areas show satisfactory performance, isolated deficits highlight the need for stricter monitoring and expanded treatment facilities.

Overall, while most districts demonstrate adequate biomedical waste treatment capacity, isolated cases of non-compliance highlight the urgent need for capacity enhancement, strict monitoring, and improved coordination between healthcare institutions and authorized Common Biomedical Waste Treatment Facilities (CBWTFs). Addressing such gaps will be crucial to minimize the risk of infectious waste contaminating soil, water, and the broader environment in the basin.

Table 39 Biomedical Waste Generation in the Narmada River Basin

Districts	Quantity of Biomedical Waste generated (Kg/day)	Quantity of biomedical waste treated per day (Kg/day)
Alirajpur	190	190

Anuppur*	87.49	87.49
Balaghat	172	172
Barwani	184.59	184.59
Betul	243.55	243.55
Burhanpur	260.62	3.62
Chindwara	268.68	239.17
Damoh	111.84	111.84
Dewas	273.33	273.33
Dhar	201.1	201.1
Dindori	52	52
Harda	85	85
Hosangabad	310.6	310.6
Indore	29	-
Jabalpur	1899	1703
Jhabua	207.3	207.3
Katni	219.05	219.05
Khandwa	297.47	297.47
Khargone	368.97	368.97
Mandla	115	115
Narshingpur	120	120
Raisen	95.35	95.35

Sagar	271	296.35
Sehore	190.17	190.17
Seoni	130	128
Umaria	96.65	96.65

*All the data filled according to the basin Portion.

Source: Madhya Pradesh Pollution Control Board. (2025). District Environment Plans

In Bharuch district, about 27,200 kilograms of biomedical waste is generated every day from 827 authorized healthcare units, which include 292 bedded and 688 non-bedded facilities. In Ankleshwar, the same waste is collected from hospitals and clinics across the district. In Narmada district, Rajpipla generates around 180–200 kilograms of biomedical waste per day. In Chhota Udepur district, about 250 kilograms of biomedical waste is generated daily from 98 bedded and 142 non-bedded facilities(**Table 40**). Overall, all districts produce biomedical waste regularly from hospitals, clinics, and other medical centers.

Table 40 Biomedical Waste generated in lower basin

District	No. of Facilities	Waste Generated (kg/day)	Facility Type
Bharuch	827 (292 bedded + 688 non-bedded)	27,200	Hospitals & clinics
Narmada (Rajpipla)	-	180–200	Hospitals & health centers
Chhota Udepur	240 (98 bedded + 142 non-bedded)	250	Healthcare units

*Source: GPCB District environment plan

6.2.2. Biomedical Waste Management

Biomedical waste management (BMWM) across the Narmada Basin districts of Madhya Pradesh shows comparatively better institutional coverage than other waste streams such as C&D or plastic waste, primarily due to the presence of Common Biomedical Waste Treatment Facilities (CBWTFs) across the state. Almost all districts are covered by regional CBWTFs, ensuring treatment and disposal through incineration, autoclaving, or deep burial in accordance

with the Biomedical Waste Management Rules, 2016. The district level review is presented in **Annexure Table A12**.

District-level monitoring committees have been formed in all districts, which is a positive step towards compliance and oversight. Districts such as Betul, Burhanpur, Chhindwara, Damoh, Dewas, Dhar, Dindori, Harda, Hoshangabad (Narmadapuram), Indore, Jabalpur, Jhabua, Katni, Mandla, Narsinghpur, Raisen, Sagar, Sehore, Umaria, and West Nimar report no major gaps in biomedical waste management. Treatment and disposal are being carried out through functional CBWTFs located either within the district or in nearby districts, and committees are active in monitoring compliance.

However, specific gaps have been noted in certain districts:

- Alirajpur and Barwani – Require periodic awareness programs and inventory updates to ensure accurate reporting of biomedical waste generation and treatment.
- Anuppur and Balaghat – Need full implementation of barcoding for biomedical waste tracking from generation to disposal.
- Seoni – Requires strengthening of the barcoding system for biomedical waste to achieve full efficiency.

These gaps primarily relate to operational aspects (such as inventory management, barcoding, and awareness) rather than infrastructure deficiencies. The presence of CBWTFs has ensured that biomedical waste is not left untreated in any of the districts.

Indore, Jabalpur, and Bhopal divisions host major CBWTFs that also cater to adjoining districts, ensuring regional coverage. For instance, Indore's Hoswin Incinerator Pvt. Ltd. manages biomedical waste from multiple western districts, while Medisure Incinerators in Khandwa cater to southern districts such as Burhanpur and West Nimar. Similarly, Seoni and Umaria CBWTFs serve adjoining tribal and forest-dominated districts.

In summary, biomedical waste management in the Narmada Basin is relatively robust, with full coverage by CBWTFs and functioning monitoring committees. The critical areas requiring attention include strict enforcement of barcoding, timely inventory updates, and regular awareness programs for healthcare facilities and local bodies. Compared to other waste streams, biomedical waste management demonstrates a higher level of compliance and

institutionalization, though operational efficiency needs continuous improvement. Bharuch district has an organized biomedical waste management system. A Common Biomedical Waste Treatment and Disposal Facility (CBWTF) named *Globe Bio Care* at Ankleshwar manages all biomedical waste from the district. The facility has an incinerator of 100 kg/hr, an autoclave of 50 kg/hr, and a shredder of 50 kg/hr. All waste generated is treated properly. However, barcoding for waste tracking has not started yet, though GPS has been installed in collection vehicles.

In Ankleshwar, the *Globe Bio Care* facility plays a key role in treating biomedical waste for the entire district. Barcoding is yet to be used and is planned by June 2022. Training and awareness for healthcare staff are being conducted. Wastewater from hospitals goes into municipal drains, and a new Sewage Treatment Plant (STP) is planned by December 2022. In Rajpipla, biomedical waste is sent to *Globe Bio Care* in Bharuch since there is no local facility. About 180–200 kg/day of waste is treated daily. Only 10% of healthcare units use barcoding, but full implementation is planned by December 2023. Regular awareness and training programs are held, and wastewater is managed through septic tanks until improvements are completed. In Chhota Udepur district, biomedical waste is sent to the *Samvedana BMW Incinerator* at Halol, where 100% of the waste is treated safely. Barcoding is already in use, but continuous staff training and strict monitoring are needed to ensure full compliance.

6.3 E-Waste

6.3.1. E-Waste Generation

E-waste generation in the Narmada River Basin depicted in **Table 41** reflects the rapid growth of electronics consumption and urbanization. The largest share is concentrated in Indore (51,115.53 MT/annum), which alone accounts for the majority of the basin's total, owing to its status as a metropolitan hub and industrial center. Other notable contributors include Sagar (133.50 MT/annum), Jabalpur (115.07 MT/annum), Dewas (89.92 MT/annum), and Katni (75.60 MT/annum), all of which have expanding urban populations and active commercial sectors.

Medium-scale quantities are reported from districts such as Damoh (65.53 MT), Jhabua (66.62 MT), Raisen (59.71 MT), and Narsinghpur (58.90 MT), while smaller volumes are observed in Barwani (77.20 MT), Sehore (77.35 MT), and Khargone (21.60 MT). Tribal and rural-

dominated districts such as Dindori (6.61 MT), Balaghat (10.25 MT), and Anuppur (38 MT) generate relatively low e-waste due to lower levels of electronic penetration.

The data indicate a highly uneven distribution, with Indore emerging as the critical hotspot requiring advanced collection, segregation, and recycling facilities. For smaller districts, decentralized collection and integration into state-level recycling frameworks will be essential to prevent unscientific disposal and associated risks of soil and groundwater contamination.

Table 41 Electronic Waste Generation in the Narmada River Basin

Districts	Present Status (MT/Annum)
Alirajpur	41.7
Anuppur*	38
Balaghat	10.25
Barwani	77.2
Betul	11.9
Burhanpur	43.4
Chindwara	11.58
Damoh	65.53
Dewas	89.92
Dhar	28.56
Dindori	6.61
Harda	-
Hosangabad	11.73
Indore	51115.53
Jabalpur	115.07

Jhabua	66.62
Katni	75.6
Khandwa	18.73
Khargone	21.6
Mandla	-
Narshingpur	58.9
Raisen	59.71
Sagar	133.5
Sehore	77.35
Seoni	18.25
Umaria	39.59

**All the data filled according to the basin Portion.*

Source: Madhya Pradesh Pollution Control Board. (2025). District Environment Plans

In Bharuch district, about 4,480 metric tons of e-waste is generated every year. The amount is quite large because of the growing use of electrical and electronic items. In Ankleshwar, e-waste generation is also high due to its industrial activities. Rajpipla and Chhota Udepur have smaller towns, but e-waste is still produced from households, offices, and shops. However, the exact quantity in these two districts has not been recorded yet. Overall, all four districts generate e-waste, but proper systems to handle it are not yet in place.

6.3.2. E-Waste Management

E-waste management across the Narmada Basin is still at a nascent stage compared to biomedical and hazardous waste management. While a few districts such as Chhindwara, Raisen, Sehore, and Indore have established e-waste collection centres and partial linkages with authorized recyclers/dismantlers, the majority of districts continue to face gaps in infrastructure and stakeholder integration. As identified in the District Environmental Plans of different districts, these challenges remain a critical concern requiring systematic intervention

In most districts, including Alirajpur, Anuppur, Damoh, Dindori, Harda, Hosangabad, Jhabua, Katni, Mandla, Narshingpur, Sagar, Seoni, Umaria, and West Nimar, no collection centres are available, and urban local bodies (ULBs) are not linked with authorized recyclers. The absence of formal collection mechanisms often leads to informal or illegal dismantling and recycling practices, posing serious environmental and health risks.

Some districts such as Balaghat, Jabalpur, and Mandla report informal linkages with recyclers at Indore district, but these are limited and not backed by structured agreements. Districts like Barwani, Betul, Burhanpur, Dewas, Dhar, Khandwa, and West Nimar have established collection points but lack proper channelization of e-waste to authorized recyclers/dismantlers, reflecting a partial compliance gap.

Chhindwara stands out as a relatively better-performing district, with two collection points and functional linkages, reporting no major gaps. Similarly, Raisen and Sehore have established collection centres and demonstrated effective linkage with recyclers, showing that with institutional support, e-waste management systems can be developed successfully.

The key challenges identified include:

- Lack of collection centres in most districts.
- Absence of formal agreements between ULBs and authorized recyclers/dismantlers.
- Need for awareness campaigns to regulate illegal dismantling and recycling.
- Inadequate infrastructure investment for collection, storage, and channelization.

Overall, e-waste management in the Narmada Basin requires urgent strengthening through the establishment of dedicated collection centres in every district, formal linkages with authorized recyclers, and enhanced public and institutional awareness. In contrast to biomedical waste, which is largely covered through Common Biomedical Waste Treatment Facilities (CBWTFs), and hazardous waste, which is managed with the support of the Dhar TSDF, e-waste continues to remain the weakest component of the basin's waste management framework. The district-level situation, highlighting existing facilities and identified gaps, is presented in **Annexure Table A13**.

At present, there are no e-waste collection centers or authorized recyclers in Bharuch district. The district plans to create an inventory and set up collection points by December 2022, along

with public awareness programs. In Ankleshwar, no collection centers exist yet, but work is planned to identify bulk e-waste generators and prepare an inventory by December 2022. Collection points and awareness drives will follow, and the informal sector will be included in the system.

In Rajpipla, there are also no recyclers or collection centers. The Nagarpalika plans to open collection points by August 2022 and connect with authorized recyclers. Awareness programs will be done with help from local offices and communities. In Chhota Udepur, no inventory or facility exists yet. The district plans to prepare an e-waste inventory, set up collection centers at municipal and block levels, and link them with recyclers in nearby districts. Awareness activities will also be started to stop unsafe handling of e-waste by informal workers.

6.4 Plastic waste

6.4.1. Plastic Waste Generation

Plastic waste generation in the Narmada River Basin shows in **Table 42** distinct variations across districts, primarily influenced by levels of urbanization, industrial activity, and consumption patterns. Districts such as Khargone (11.30 TPD in 2025), Mandla (21.23 TPD in 2025), and Dewas (7.12 to 8.06 TPD by 2030) emerge as major contributors, reflecting both urban expansion and increased packaging-related waste streams. Similarly, Dhar (6.10 to 6.70 TPD) and Hoshangabad (3.74 to 3.98 TPD) indicate steady growth trends. In contrast, smaller and rural districts such as Seoni (0.01 TPD), Balaghat (0.30 TPD), and Anuppur (0.52 to 0.55 TPD) generate relatively negligible amounts but face persistent challenges of improper disposal and collection inefficiencies.

The uneven growth of plastic waste across the basin underscores the need for differentiated management strategies: high-generation urban centers require advanced material recovery facilities (MRFs) and recycling infrastructure, while low-generation rural areas demand improved door-to-door collection and awareness programs to curb open dumping and burning. For granular, tehsil-level estimates and projections of plastic waste generation, refer to **Annexure Table A5**.

Table 42 Plastic Waste Generation in the Narmada River Basin

Districts	Estimated Generation in 2025 (TPD)	Estimated Generation in 2030 (TPD)
Alirajpur	0.78	0.85
Anuppur*	0.52	0.55
Balaghat	0.3	0.3
Barwani	2.83	3.1
Betul	1.96	2
Burhanpur	0.5	0.6
Chindwara	0.98	1.04
Damoh	2.79	2.9
Dewas	7.12	8.06
Dhar	6.1	6.7
Dindori	0.3	-
Harda	1.26	-
Hosangabad	3.74	3.98
Indore	0.56	-
Jabalpur	2.13	2.25
Jhabua	1.01	1.11
Katni	1.31	1.42
Khandwa	3.94	-
Khargone	11.3	-

Mandla	21.23	-
Narshingpur	2.06	2.13
Raisen	2.62	2.81
Sagar	0.67	0.71
Sehore	2.7	3.44
Seoni	0.01	-
Umaria	0.6	0.7

**All the data filled according to the basin Portion.*

Source: Madhya Pradesh Pollution Control Board. (2025). District Environment Plans

In Bharuch district, a large amount of plastic waste is generated. Bharuch Nagarpalika produces about 7 tons per day (TPD), and Ankleshwar Nagarpalika generates around 3 TPD. At the panchayat level, about 480 kilograms per day of plastic waste is produced, and gram panchayats together generate around 11 kilograms per day. In Narmada district, Rajpipla Nagarpalika produces about 2 TPD of plastic waste. At the block and taluka level, generation is around 150 kilograms per day, and another 150 kilograms per day is produced by 222 gram panchayats across 5 tehsils. In Chhota Udepur district, the quantity is much smaller, around 0.040 TPD, mostly from Chhota Udepur town. Overall, plastic waste generation in all four districts varies, but proper systems for collection and recycling are still developing.

Table 43 Plastic Waste in each Nagarpalika in lower basin

Nagarpalika	Plastic Waste (TPD)
Bharuch	7
Ankleshwar	3
Narmada (Rajpipla)	2
Chhota Udepur	0.040

**Source: GPCB District environment plan*

6.4.2. Plastic Waste Management

The analysis of plastic waste management across the districts of the Narmada Basin in Madhya Pradesh highlights uneven progress in door-to-door collection, establishment of plastic waste collection centers, and availability of disposal or recycling facilities.

Door-to-door collection of plastic waste has been initiated in most districts, with full coverage reported in Barwani, Betul, Chhindwara, Dhar, Harda, Hoshangabad (Narmadapuram), Indore, Jabalpur, Jhabua, Katni, Narsinghpur, Raisen, Sagar, and parts of West Nimar. However, in several districts like Anuppur, Damoh, Dewas, Sehore, and Seoni, door-to-door collection remains partial or inadequate, indicating a need for strengthening local level implementation and improving coverage efficiency.

Plastic waste collection centers exist only in select districts, often concentrated at district headquarters (e.g., Betul, Chhindwara, Katni, Harda, and Hoshangabad). In rural and smaller ULBs, the absence of collection points continues to hinder effective channelization of waste for recycling or co-processing. Similarly, disposal and recycling facilities are not uniformly distributed. While Indore, Jabalpur, Dhar, Jhabua, and Hoshangabad have integrated solutions through cement co-processing, waste-to-energy plants, or recycling facilities, many districts such as Alirajpur, Mandla, Narsinghpur, and Seoni continue to rely on basic collection without proper disposal mechanisms.

Districts like Harda, Hoshangabad, Indore, Sagar, Jhabua, and Umaria report minimal or no gaps, reflecting relatively advanced systems of collection and recycling. In contrast, districts including Anuppur, Balaghat, Mandla, Sehore, and West Nimar require urgent establishment of plastic waste collection centers and processing facilities. In many cases, informal recycling practices are prevalent, but integration of the informal sector with formal waste management structures remains limited.

The major challenges identified across districts include partial or absent door-to-door collection, lack of decentralized collection centers, inadequate plastic waste processing or recycling units, and insufficient community awareness. Additionally, the enforcement of Extended Producer Responsibility (EPR) and integration of NGOs or private operators into the plastic waste value chain are limited in scope.

In summary, while certain urban centers in the Narmada Basin have made commendable progress by linking plastic waste management with recycling, co-processing, and energy recovery, large parts of the basin still face systemic gaps. Strengthening collection networks, establishing district-level processing units, and promoting sustainable recycling practices are critical for effective plastic waste management across the basin. The district-level situation, highlighting existing facilities and identified gaps, is presented in **Annexure Table A14**. In Bharuch district, door-to-door collection of plastic waste is about 74% in panchayats and

almost complete in nagarpalikas, but segregation and transfer systems still need improvement. Bharuch Nagarpalika has started setting up material recovery facilities, while Ankleshwar, Jambusar, and Amod have not yet begun. In Ankleshwar, the collection of plastic waste has started but is not fully covered. The waste is taken directly to the dumping site because a plastic waste collection center is not yet built. A new center is planned once land is allotted, with completion targeted by December 2022. Awareness programs are being planned by the Nagarpalika and district offices to promote recycling and proper disposal.

In Narmada district, plastic waste management in Rajpipla is still developing. Door-to-door collection of dry waste, including plastic, is 0% at present, but the Nagarpalika has achieved full coverage within the town. In villages, only about 50% of waste is collected. Plans are made to build compost pits, segregation sheds, and collection centers at the block level with the help of the Statue of Unity authority by March 2024. In Chhota Udepur district, door-to-door collection is not fully working, and organized systems like material recovery facilities are missing. Plastic waste centers are few, and public awareness about recycling is very low. The district's action plan includes 100% door-to-door collection, new MRFs, and links with recyclers and cement plants for safe plastic co-processing.

6.5 C&D WASTE

6.5.1. C&D Waste Generation

Construction and Demolition (C&D) waste generation across the Narmada River Basin shown in **Table 32** districts is projected to rise moderately between 2025 and 2030, reflecting steady urban growth and infrastructure expansion. Districts such as Dewas (49.42 TPD in 2025, 54.99 TPD in 2030), Barwani (19.45 to 21.39 TPD), and Hoshangabad (17.16 to 18.20 TPD) stand out as major contributors due to rapid real estate and industrial activity. Comparatively, smaller and rural-dominated districts such as Seoni (0.50 TPD), Balaghat (0.54-0.57 TPD), and Dindori (0.80 TPD) show very low C&D waste loads. This variation highlights the need for targeted strategies urban clusters will require dedicated C&D recycling facilities, while rural areas may benefit from decentralized reuse and backfilling practices. For detailed tehsil-level estimates and projected trends, refer to **Annexure Table A5**.

Table 44 C&D Waste Generation in the Upper and middle Narmada River Basin

Districts	Estimated C& D Waste Generation (TPD) in 2025	Estimated C&D Waste Generation (TPD) in 2030
Alirajpur	5.37	5.78
Anuppur*	3.56	3.75
Balaghat	0.54	0.57
Barwani	19.45	21.39
Betul	10.96	11.57
Burhanpur	2.07	2.22
Chindwara	5.66	5.97
Damoh	7.88	9.41
Dewas	49.42	54.99
Dhar	16.56	18.12
Dindori	0.802	-
Harda	2.09	-
Hosangabad	17.16	18.2
Indore	0.5	-
Jabalpur	14.5	15.5
Jhabua	4.8	5.32
Katni	9	9.7
Khandwa	6.4	-
Khargone	2.03	-

Mandla	2.95	-
Narsinghpur	10.85	11.35
Raisen	13.78	14.78
Sagar	4.6	4.9
Sehore	16.47	21.25
Seoni	0.5	
Umaria	2.45	2.68

**All the data filled according to the basin Portion.*

Source: Madhya Pradesh Pollution Control Board. (2025). District Environment Plans

In Bharuch district, a large amount of Construction and Demolition (C&D) waste is generated. Bharuch Nagarpalika produces about 5 tons per day (TPD), while Ankleshwar Nagarpalika generates around 0.5 TPD. In Narmada district, Rajpipla Nagarpalika also produces about 0.5 TPD of C&D waste. In Chhota Udepur district, around 0.5 metric tons per day of C&D waste is generated. C&D Waste Generated by each Nagarpalika shown in **table 45**. The quantity of waste varies, but most of these areas do not have proper recycling systems in place yet.

Table 45 C&D Waste generated in each Nagarpalika in lower basin

Nagarpalika	Waste Generated (TPD)
Bharuch	5
Ankleshwar	0.5
Narmada (Rajpipla)	0.5
Chhota Udepur	0.5

**Source: GPCB District environment plan*

6.5.2. C&D Waste Management

The review of C&D waste management in the Narmada Basin districts highlights an overall lack of comprehensive systems for collection, recycling, and reuse of construction and demolition waste, though a few districts have made noteworthy progress.

In districts such as Harda, Jabalpur, Raisen, Sagar, Mandla, Narsinghpur, and West Nimar, partial or complete systems for collection and reuse are operational, and in some cases recycling facilities exist. For instance, Jabalpur and Harda demonstrate good practices with available recycling units and use of recycled material in non-structural concrete, road filling, and low-lying area restoration, leaving little to no gap in implementation. Similarly, Sagar has both collection and recycling systems, indicating compliance with C&D waste management rules.

Conversely, many districts such as Alirajpur, Anuppur, Barwani, Burhanpur, Dindori, Jhabua, Katni, Dhar, and Dewas show significant gaps, with either no facilities for collection and recycling or reliance on unscientific practices like dumping into low-lying areas. In several ULBs, separate collection exists but without supporting recycling plants, leading to improper disposal. Alirajpur and Burhanpur represent typical cases where waste is dumped without treatment, indicating urgent need for processing facilities.

Chhindwara stands out as one of the few districts with a private third-party operator (M/s Dadaji Traders) engaged in recycling activities, though such facilities are restricted to urban centers and not yet extended to rural areas. Hoshangabad (Narmadapuram) shows partial progress, with separate collection in multiple ULBs, but lacks adequate recycling facilities, limiting the scientific use of collected material.

Districts such as Balaghat, Mandla, Seoni, and Narsinghpur report very low generation of C&D waste, which has led to a perception that dedicated recycling plants are not required. In these cases, waste is reused locally for filling or minor construction activities. However, as urbanization accelerates, even these districts may require systematic solutions in the future.

The most common practice across the basin remains dumping of C&D waste into low-lying areas or reuse in landfilling and road works, often without scientific monitoring. Dedicated recycling plants are absent in the majority of districts, and even where collection is available, segregation and utilization of recycled products are limited.

Key challenges include the absence of dedicated recycling units, lack of decentralized collection points, limited awareness among local bodies, and insufficient integration of C&D waste into circular economy approaches such as paver block production and road construction.

In summary, only a handful of districts like Harda, Jabalpur, Raisen, and Sagar show relatively strong systems with minimal gaps, while the majority of the Narmada Basin districts continue to face critical deficiencies in C&D waste management. Establishing decentralized collection centers, setting up recycling facilities, and ensuring the use of recycled products in local construction are essential steps to bridge these gaps and promote sustainable management of construction waste. The district-level situation, highlighting existing facilities and identified gaps, is presented in **Annexure Table A15**.

At present, Bharuch and Ankleshwar Nagarpalikas do not have any recycling facilities. Bharuch Nagarpalika has prepared a Detailed Project Report for a C&D waste treatment and recycling facility at Saykha GIDC, which was planned for completion by June 2022. Bharuch has also started using C&D waste for road pavements and aims to fully implement this by June 2024. In Ankleshwar, C&D waste is collected by vans, but there is no fixed site for deposition or recycling. Due to the small amount of waste, no recycling plant is planned yet, but the waste is used for landfilling and road filling.

In Narmada district, Rajpipla Nagarpalika does not have any recycling facility or deposition point. It plans to identify deposition sites and start collecting user fees for C&D waste by December 2022. Reuse of C&D waste in construction is not happening yet, but may start in the future. In Chhota Udepur district, recycling facilities are available, but waste is not properly separated at the source, and deposition points are missing. The district plans to set up common deposition sites, use recycled waste for paving and rural roads, and increase public awareness through regular campaigns.

7. Pollution Load Assessment

7.1 Domestic Pollution Load Status

Domestic sewage is one of the major sources of pollution in the Narmada Basin, with most districts reporting either the absence of sewage treatment plants (STPs) or insufficient treatment capacity. The compiled data highlights that while the generation of sewage is significant, the treatment infrastructure remains grossly inadequate, leading to large volumes of untreated wastewater entering rivers, lakes, and streams.

Districts such as Alirajpur, Anuppur, Betul, Harda, Narmadapuram, Seoni, and Umaria report 100% untreated sewage, indicating no treatment facilities despite moderate to high generation levels. Similarly, Jhabua, Dhar, Khandwa, and Barwani discharge almost their entire sewage load untreated, contributing to heavy riverine pollution. In Balaghat, Mandla, and Narsinghpur, negligible treatment exists, often through isolated units with very low capacity.

Urban centers like Indore, Jabalpur, Dewas, Sagar, Sehore, and Raisen have partial sewage treatment infrastructure, but treatment capacity lags behind sewage generation. For example, Indore generates 475 MLD of sewage, but only a part is treated despite the presence of multiple STPs. Jabalpur generates 230 MLD, but with 60% untreated sewage, major gaps persist. Similarly, Dewas and Sagar have STPs, but treatment percentages remain low.

The requirement of new STPs is universally high, with most districts needing between 5 to 15 additional plants to handle existing and future loads. A few districts, such as Burhanpur and Sehore, show partial progress with one STP operational, but untreated sewage still exceeds 50%.

Key challenges observed across the basin include:

- High percentage of untreated sewage (70–100% in most districts).
- Insufficient STP capacity, often limited to large cities only.
- Direct discharge of sewage into rivers, drains, and lakes, aggravating water pollution.
- Inadequate sewerage networks, with only a few ULBs having partial underground systems.

Domestic sewage is one of the main sources of pollution in the Lower Narmada Basin. The amount of wastewater depends on how much water is supplied, how many houses are connected to the drainage system, and whether sewage is treated before being released. The basin includes big towns like Bharuch, Ankleshwar, Rajpipla, and Chhotaudepur, along with special areas such as Ekta Nagar and many rural villages(**Table 46**).

In Bharuch, the sewage generation is the highest in the basin. The city produces about 28 MLD of sewage every day. The sewage is collected through a long underground network divided into five zones. The sewage is treated at the STP, which has a design capacity of 29.3 MLD. But due to lack of proper drainage network and not covering whole city some waste remains untreated. But major part of the sewage generated in Bharuch can be treated before discharge.

In Ankleshwar, the sewage generation is about 7.5 MLD. The city has developed a strong sewerage network of around 146 km, which covers most of the residential and commercial areas. A sewage treatment plant with a capacity of 14 MLD is under construction. This is almost double the present sewage generation, which means the plant has enough capacity to handle future growth.

In Rajpipla, the sewage generation is about 4.5 MLD. Earlier, a large part of this sewage was discharged directly into the Karjan River, which polluted the water. Now, this practice has stopped. A 5.5 MLD SBR-based STP was commissioned in 2024 near Chitrawadi by GWSSB. At present, treated water from the plant is reused for gardening, road cleaning, and other non-drinking purposes.

In Chhotaudepur, the total sewage generation is about 4.0 MLD. The town has an STP of 5.0 MLD capacity, which is enough to treat all the sewage. The system faces some problems in the rising main pipeline, but overall, no untreated sewage is being discharged into natural lakes or rivers.

In Ekta Nagar, which includes Kevadia township and the Jungle Safari area, there are two small STPs. The Kevadia township has a 500 KLD STP, while the Jungle Safari has a 400 KLD STP. Together, these treat about 0.9 MLD of sewage. During peak tourist seasons, however, some wastewater may bypass the plants and enter small drains.

In rural areas, most households use septic tanks, leaching pits, or open/covered drains. Piped sewer systems are very limited, and most villages do not have treatment plants. Some larger

villages and small towns may have partial networks or small FSTPs, but data on exact sewage generation is not available.

When we combine the sewage generated by the main towns and special facilities, the Lower Narmada Basin produces around 44.9 MLD of domestic sewage every day. The total installed treatment capacity in these towns is about 54.7 MLD, which is enough to cover the reported load. However, untreated sewage may still enter rivers and drains from rural areas and during peak tourist periods.

Table 46 Domestic Sewage Generation and Treatment in Lower Narmada Basin

Town/Area	Sewage Generated	STP Capacity	Status / Notes
Bharuch	28 MLD	29.3 MLD	5 sewer zones, network of 118 km, effluent meets standards
Ankleshwar	7.5 MLD	14.0 MLD	146 km network, 90% complete, plant commissioned in 2021
Rajpipla	4.5 MLD	5.5 MLD	New SBR plant (2024), reuse for gardening and cleaning
Chhotaudepur	4.0 MLD	5.0 MLD	Adequate capacity, no untreated discharge
Ekta Nagar + Safari	0.9 MLD	0.9 MLD	Small STPs, bypass possible during tourist peak
Rural Basin	Not available	Not available	Mostly septic tanks and drains, few small FSTPs

*Source: GPCB District environment plan

In conclusion, domestic sewage management in the Narmada Basin is highly inadequate, with a critical gap between generation and treatment. Unless significant investments are made in

STPs, sewerage networks, and decentralized treatment facilities, untreated sewage will remain a major environmental and public health concern in the basin. The district-level situation, highlighting existing facilities and identified gaps, is presented in **Annexure Table A18**.

7.2 Industrial Pollution Load Status

Industrial wastewater generation in the Narmada Basin varies significantly across districts, reflecting the uneven distribution of industries. The data reveals that while some districts have substantial industrial discharge, others report negligible or no industrial wastewater.

Districts such as Anuppur (3.9 MLD, 346 industries), Balaghat (16 MLD, 30 industries), Burhanpur (1.07 MLD, 75 industries), Indore (7.3 MLD, 385 industries), Jabalpur (8 MLD, 30 industries), Sehore (19.2 MLD, 24 industries), and West Nimar (22.28 MLD, 17 industries) show a clear presence of industrial activities. In these districts, most industries are reported to be meeting environmental standards, and a few, like Burhanpur and Indore, have operational Common Effluent Treatment Plants (CETPs).

On the other hand, Narsinghpur (1006 MLD, 15 industries) and Sagar (15,360 KLD, 7 industries) stand out due to very high volumes of wastewater generation relative to the number of discharging units, indicating large-scale industries with significant effluent discharge. Similarly, Khandwa (39.6 MLD, 6 industries) and Chhindwara (46.43 MLD) also highlight concentrated industrial discharge.

Several districts such as Alirajpur, Damoh, Dindori, Mandla, and Harda report zero or negligible wastewater discharge, reflecting either a lack of major industrial units or effective zero liquid discharge practices. However, in some cases (e.g., Harda – 110 KLD, Jhabua – 720 KLD, Umaria – 489 KLD, Katni – 140 KLD), discharge values are recorded in kiloliters per day, which, though smaller than MLD figures, still indicate localized industrial activity.

A critical observation is the lack of CETPs across most districts. Except for Indore, Burhanpur, and Jhabua, where one CETP exists, most districts have no common treatment facilities, compelling industries to rely on individual Effluent Treatment Plants (ETPs). While compliance with standards is generally reported as satisfactory in many districts, the absence of CETPs raises concerns about long-term monitoring, enforcement, and sustainability of wastewater treatment. The district-level situation, highlighting existing facilities and identified gaps, is presented in **Annexure Table A19**.

The Lower Narmada Basin has some of the most industrialized areas of Gujarat. Bharuch and Ankleshwar together form one of the largest industrial hubs in India. These areas have many chemical, pharmaceutical, textile, pulp & paper, and petrochemical industries. Because of this, they generate a very large amount of industrial wastewater every day.

The total industrial wastewater generation in the basin is about 233.9 MLD which is shown in **table 47**. Most of this comes from Bharuch and Ankleshwar, which together produce around 231.2 MLD. Out of this, about 41.96 MLD is treated within industries and discharged into rivers or drains, while the rest is either reused by industries or treated in Common Effluent Treatment Plants (CETPs).

Other districts like Narmada (Rajpipla) and Chhota Udepur have very few industries and their wastewater load is very small. Rajpipla generates about 2.55 MLD, and Chhota Udepur only 0.11 MLD. In both these places, wastewater is treated inside the industries and not released into rivers.

Most industries in the basin follow the discharge rules given by the Gujarat Pollution Control Board (GPCB). Online systems are also available for complaints and regular inspections are carried out.

The largest contribution to industrial wastewater load comes from the chemical and pharmaceutical sector, followed by textile, pulp & paper, and petrochemical industries. Smaller contributions come from agro-based and engineering industries.

Table 47 Industrial Wastewater Load in Lower Narmada Basin

District / Area	Wastewater Generated	Discharge Status	Notes
Bharuch Region	126.787 MLD	Part discharged (treated), rest managed in CETPs	Major hub – Dahej, Vilayat
Ankleshwar Region	104.414 MLD	Part discharged (treated), rest managed in CETPs	Major hub – Ankleshwar, Panoli
Rajpipla (Narmada)	2.546 MLD	No untreated discharge	43 industries discharge wastewater

Chhota Udepur	0.108 MLD	No untreated discharge	Only 5 industries discharge wastewater
		About 41.96 MLD	
Total Basin	233.855 MLD	treated and discharged, reused or sent to CETPs	Majority load from Bharuch– Ankleshwar

*Source: GPCB District environment plan

In conclusion, industrial wastewater in the Narmada Basin shows a mixed picture: while some districts like Indore, Burhanpur, and Sehore demonstrate structured wastewater management, others, especially with high discharge volumes like Narsinghpur and Sagar, highlight the urgent need for strengthening treatment infrastructure, setting up CETPs, and ensuring strict compliance monitoring to prevent industrial pollution of the Narmada River system. In lower basin most of industrial waste generated in Bharuch district which is treated in CETP and discharged deep sea.

8.Summary And Conclusion

Summary

The Narmada River Basin covers 38 districts in Madhya Pradesh, Gujarat, Maharashtra, and Chhattisgarh. It gives water to more than 20 million people. The river is very important for farming, industries, and nature.

This study looks at how polluted the river is. It studies waste from homes, industries, and farms. The data is taken from CPCB, MPPCB, GPCB, NGT, and district reports.

Most of the pollution comes from domestic sewage. Many towns do not have enough sewage treatment plants. In towns like Amarkantak, Mandla, Narmadapuram, Barwani, and Jhabua, most sewage goes into the river without cleaning. Even in big cities like Indore and Jabalpur, treatment plants do not work at full capacity. In cities like Bharuch, Rajpipla and Chhota udepur treatment plant is available with more than the sewage generated by city but drainage network does not cover whole areas of city. More small and local treatment systems are needed.

In the lower basin, water is clean at Panetha and Zonor, okay at Garudeshwar, but polluted at Zadeshwar. The main reason for this pollution is waste from towns and industries. Cities like Bharuch and Ankleshwar have sewage treatment plants, but some waste water still goes into the river without treatment. Small towns like Rajpipla and Chhota Udepur have small systems and cause less pollution.

Industrial waste also adds to pollution. Some districts like Indore, Burhanpur, and Sehore have common treatment plants (CETPs) for factory waste. But others like Narsinghpur and Sagar have only individual treatment plants (ETPs), which are not well checked. In the lower basin, industries in Bharuch, Ankleshwar, Dahej, Panoli, and Jhagadia make large amounts of wastewater. Most of this comes from chemical and medicine factories. It is treated in common plants and sent through a deep-sea disposal through pipeline.

Agricultural runoff contributes significantly to nutrient and pesticide loading. Elevated nitrate (45–90 mg/L) and phosphate levels (1.5–3.0 mg/L) in districts like Omkareshwar, Hoshangabad, and Barwani indicate risks of eutrophication and water contamination. Heavy metals such as zinc, nickel, and chromium in sediments further highlight ecological stress.

Solid waste management shows progress in larger cities like Indore, Bhopal, and Jabalpur, where biomining, composting, and waste-to-energy facilities are operational. However, smaller

and tribal districts face challenges in collection, segregation, and scientific disposal. Biomedical and hazardous waste management has improved with centralized facilities, but e-waste management remains underdeveloped, with few collection centers and weak stakeholder linkages.

Solid waste is another problem, especially in big towns of lower basin. Bharuch and Ankleshwar make the most garbage. Waste separation and composting are happening in some areas but not everywhere. Hospital and hazardous waste are managed better, but plastic and e-waste still need more care.

Overall, the Narmada River faces pollution from many sources domestic sewage, industries, farms waste. Better treatment, strict rules, and public awareness are needed to keep the river clean.

Conclusion

The Narmada River Basin faces serious pollution problems. The main reasons are sewage from towns, waste from industries, farm chemicals, and poor waste management in small towns. Some big programs like **AMRUT** and **Namami Gange** are helping to build new treatment plants. But there are still many problems. Many towns do not have full sewer lines. Some treatment plants do not work properly. Rules are not always followed, and people's participation is still low.

To clean the river, a **combined effort** is needed. The government, industries, and people must all work together. More money should be spent on new and better treatment plants. Small towns should use simple local systems for wastewater treatment. Industries must treat all their waste before releasing it. All waste should be checked and managed properly.

The Narmada River is becoming more polluted each year. Water is still clean in some areas, but near big towns and industries, it is polluted. People should separate waste at home. Treated water should be reused for gardens or industries.

Saving the Narmada River is very important for people, animals, and nature. The government should check the water quality often. With strong action and teamwork, the Narmada can become clean again and continue to support life and development in the region.

REFERENCES

1. Directorate of Industries, Govt. of India (2017). *Industrial profile of Madhya Pradesh 2016–17*. Development Commissioner (MSME), Ministry of MSME. Retrieved from https://dcmsme.gov.in/dips/state_wise_profile_16-17/Madhya%20Pradesh%20profile%2016-17.pdf
2. Government of Madhya Pradesh (2025). *Monthly Progress Report (MPR) – Madhya Pradesh, June 2025*. Atal Mission for Rejuvenation and Urban Transformation (AMRUT). National Mission for Clean Ganga (NMCG). Retrieved from https://nmcg.nic.in/writereaddata/fileupload/ngtmpr/30_Madhya%20Pradesh%20MPR%20June%202025.pdf
3. Central Pollution Control Board (CPCB). (n.d.). *List of NWMP monitoring network*. Retrieved 2025, from https://cpcb.nic.in/wqm/WQMN_list.pdf
4. Madhya Pradesh Pollution Control Board (MPPCB). (2023). *District Environmental Plans (DEPs): Alirajpur, Anuppur, Balaghat, Barwani, Betul, Burhanpur, Chhindwara, Damoh, Dewas, Dhar, Dindori, Harda, Hoshangabad (Narmadapuram), Indore, Jabalpur, Jhabua, Katni, Mandla, Narsinghpur, Raisen, Sagar, Sehore, Seoni, Umaria, and West Nimar*. Retrieved from <https://www.mppcb.mp.gov.in>
5. Government of Madhya Pradesh. (2023, October). *National Mission for Clean Ganga: Monthly Progress Report in NGT Case No. 673 of 2018 (in compliance to NGT order dated 24.09.2020)*. https://nmcg.nic.in/writereaddata/ngtmpr/20_Madhya%20Pradesh%20-%20MPR%20Oct%202023.pdf
6. Government of Madhya Pradesh. (2024). *Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues)*. NGT, https://www.greentribunal.gov.in/sites/default/files/news_updates/ACTION%20TAKEN%20REPORT%20BY%20GOVT.%20OF%20MADHYA%20PRADESH%20IN%20OA%20NO.%20606%20of%202018%20%28COMPLIANCE%20OF%20MSW%20MGT%20RULES%2C%202016%20AND%20OTHER%20ENVIRONMENTAL%20ISSUES%29.pdf
7. Government of Madhya Pradesh. (n.d.). *Presentation cum compliance in OA No. 606*

- (*Compliance of MSW Management Rules, 2016*). National Green Tribunal.
[https://www.greentribunal.gov.in/sites/default/files/news_updates/PRESENTATION%20CUM%20COMPLIANCE%20BY%20GOVT.%20OF%20MADHYA%20PRADESH%20IN%20OA%20NO.%20606%20of%202018%20\(COMPLIANCE%20OF%20MSW%20RULES%2C%202016\).pdf](https://www.greentribunal.gov.in/sites/default/files/news_updates/PRESENTATION%20CUM%20COMPLIANCE%20BY%20GOVT.%20OF%20MADHYA%20PRADESH%20IN%20OA%20NO.%20606%20of%202018%20(COMPLIANCE%20OF%20MSW%20RULES%2C%202016).pdf)
8. Government of Madhya Pradesh. (n.d.). *Six monthly progress report in OA No. 606 for Madhya Pradesh (Compliance of MSW Management Rules, 2016)*. National Green Tribunal.
[https://www.greentribunal.gov.in/sites/default/files/news_updates/SIX%20MONTHLY%20PROGRESS%20REPORT%20BY%20GOVT.%20OF%20MADHYA%20PRADESH%20IN%20OA%20NO.%20606%20of%202018%20\(COMPLIANCE%20OF%20MSW%20RULES%2C%202016\).pdf](https://www.greentribunal.gov.in/sites/default/files/news_updates/SIX%20MONTHLY%20PROGRESS%20REPORT%20BY%20GOVT.%20OF%20MADHYA%20PRADESH%20IN%20OA%20NO.%20606%20of%202018%20(COMPLIANCE%20OF%20MSW%20RULES%2C%202016).pdf)
9. Madhya Pradesh Pollution Control Board. (2023). *Solid waste management annual report 2022-23*.
https://www.mppcb.mp.gov.in/PdfView.aspx?h=SOLID%20WASTE%20MANAGEMENT%20Annual%20Report%2022-23&pdf=Letter_No._2195_Dt._29.8.2023.pdf
10. Central Pollution Control Board. (n.d.). *Report by CPCB in OA No. 606 (Compliance of Municipal Solid Waste Management Rules, 2016)*. National Green Tribunal.
[https://www.greentribunal.gov.in/sites/default/files/news_updates/REPORT%20BY%20CPCB%20IN%20OA%20NO.%20606%20of%202018%20\(COMPLIANCE%20OF%20MUNICIPAL%20SOLID%20WASTE%20RULES%2C%202016\).pdf](https://www.greentribunal.gov.in/sites/default/files/news_updates/REPORT%20BY%20CPCB%20IN%20OA%20NO.%20606%20of%202018%20(COMPLIANCE%20OF%20MUNICIPAL%20SOLID%20WASTE%20RULES%2C%202016).pdf)
11. Government of Madhya Pradesh. (n.d.). *Presentation by Chief Secretary, Govt. of Madhya Pradesh in OA No. 606 (Compliance of Municipal Solid Waste Management Rules, 2016) NGT*.
[https://www.greentribunal.gov.in/sites/default/files/news_updates/PRESENTATION%20BY%20CS%2C%20GOVT.%20OF%20MP%20IN%20OA%20NO.%20606%20\(COMPLIANCE%20OF%20MSW%20RULES%2C%202016\).pdf](https://www.greentribunal.gov.in/sites/default/files/news_updates/PRESENTATION%20BY%20CS%2C%20GOVT.%20OF%20MP%20IN%20OA%20NO.%20606%20(COMPLIANCE%20OF%20MSW%20RULES%2C%202016).pdf)
12. Madhya Pradesh Pollution Control Board. (2023). *Solid waste management annual report 2022-23*.
https://www.mppcb.mp.gov.in/PdfView.aspx?h=SOLID%20WASTE%20MANAGEMENT%20Annual%20Report%2022-23&pdf=Letter_No._2195_Dt._29.8.2023.pdf

ANNEXURE

Table A - 1 List of National Water Quality Monitoring Stations in Narmada Basin, Madhya Pradesh

S.No.	Water Quality Station Code	Station Name	State	Water Body Type	Frequency	Latitude	Longitude
1	44	NARMADA AT SETHANIGHAT	Madhya Pradesh	River	Monthly	22.762685	77.71649
2	45	NARMADA AT MANDLESHWAR	Madhya Pradesh	River	Monthly	22.177393	75.659194
3	1234	NARMADA AT HOSHANGABAD U/S	Madhya Pradesh	River	Monthly	22.756614	77.690036
4	1235	NARMADA AT HOSHANGABAD D/S	Madhya Pradesh	River	Monthly	22.756584	77.687182
5	1236	CHOTA TAWA BEFORE CONFL. WITH RIVER NARMADA	Madhya Pradesh	River	Monthly	22.562555	77.978245
6	1239	NARMADA AT BADWANI	Madhya Pradesh	River	Monthly	22.037618	74.901793
7	1240	NARMADA AT NARSINGHPUR	Madhya Pradesh	River	Monthly	22.948622	79.190171
8	1241	NARMADA AT MANDLA NEAR ROAD BRIDGE	Madhya Pradesh	River	Monthly	22.598638	80.371449
9	1242	NARMADA NEAR SOURCE AT AMARKANTAK	Madhya Pradesh	River	Monthly	22.67244	81.759518

		CHAMBAL AT					
10	1365	NAGDA U/S (WATER INTAKE POINT)	Madhya Pradesh	River	Monthly	22.428806	75.508689
11	1366	CHAMBAL AT NAGDA D/S	Madhya Pradesh	River	Monthly	22.428791	75.508684
12	1430	NARMADA AT D/S OF OMKARESHWAR	Madhya Pradesh	River	Monthly	22.243694	76.150756
13	1431	NARMADA AT MAHESHWAR	Madhya Pradesh	River	Monthly	22.169213	75.583684
14	1613	KOLAR DAM WATER SUPPLY INTAKE WELL, DISTT. SEHORE	Madhya Pradesh	River	Monthly	22.960571	77.346276
15	2099	NARMADA LALPUR, JABALPUR	Madhya Pradesh	River	Monthly	23.124913	79.774671
16	2100	GOUR RIVER BHOGA DOOR, JABALPUR	Madhya Pradesh	River	Monthly	23.117239	79.99781
17	2106	NARMADA AT NEMAWAR	Madhya Pradesh	River	Monthly	22.496589	76.980555
18	2112	NARMADA NEAR MORTAKKA BRIDGE, BADWAH	Madhya Pradesh	River	Monthly	22.226413	76.045534
19	2113	NARMADA NEAR PUNASA DAM, PUNASA	Madhya Pradesh	River	Monthly	22.283823	76.471463
20	2114	KUNDA AT KHARGONE	Madhya Pradesh	River	Monthly	22.168429	75.723814

21	2123	NARMADA AT KORIGHAT, HOSHANGABAD	Madhya Pradesh	River	Monthly	22.761672	77.7185
22	2127	DENWA NEAR SARNI, ROAD BRIDGE	Madhya Pradesh	River	Monthly	22.124707	78.03034
23	2128	KHANDARI RESERVOIR WATER OFF TAKE POINT	Madhya Pradesh	Lake	Monthly	23.160981	80.017522
24	2129	PERIAT TANK NEAR WSS INTAKE POINT, JABALPUR	Madhya Pradesh	Tank	Monthly	23.235908	80.126153
25	2134	NAGCHUN TALAB AT KHANDWA	Madhya Pradesh	Lake	Monthly	21.86345	76.322945
26	2143	WELL AT KATHODA, JABALPUR	Madhya Pradesh	Ground Water	Half-Yearly	23.227505	79.915977
27	2144	WELL AT MADAI GRAM, JABALPUR	Madhya Pradesh	Ground Water	Half-Yearly	23.220203	79.990963
28	3296	Choral Dam Near Indore	Madhya Pradesh	Lake	Monthly	22.415255	75.756472
29	3299	Hanumantal, Jabalpur	Madhya Pradesh	Pond	Monthly	23.181769	79.937793
30	3300	Ranital Talab, Jabalpur	Madhya Pradesh	Pond	Monthly	23.173288	79.918843
31	3305	Banjar at Malanjkhanda Near W/S Intake Point, Jabalpur	Madhya Pradesh	River	Monthly	22.049425	80.716076

32	3308	Chambal at Jawnpawa (Origin Point), Indore	Madhya Pradesh	River	Monthly	22.429328	75.508403
33	3321	Narmada at Amarkantak (Origin Point), Rewa / Narmada at Dindori, U/S Intake Point	Madhya Pradesh	River	Monthly	22.672558	81.766411
34	3322	Narmada at Mandla, Near Shamshanghat, Jabalpur	Madhya Pradesh	River	Monthly	22.598282	80.370419
35	3323	Narmada at Lalpur, Near W/S Intake Point, Jabalpur	Madhya Pradesh	River	Monthly	23.124518	79.775616
36	3324	Narmada at Panchwatighat, Before Mixing with Bawanganga River, Jabalpur	Madhya Pradesh	River	Monthly	23.133109	79.800745
37	3325	Narmada at Sarswatighat, After Mixing with Bawanganga River, Jabalpur	Madhya Pradesh	River	Monthly	23.133838	79.800558
38	3334	Tube Well Near Bus Stand, Betul	Madhya Pradesh	Ground Water	Half-Yearly	21.900458	77.89643
39	3335	Tube Well / Hand Pump at Industrial Area Rechhai, Jabalpur	Madhya Pradesh	Ground Water	Half-Yearly	23.220624	79.976447
40	3822	Ground water near Bus Stand, Dist Narsingpur	Madhya Pradesh	Ground Water	Half-Yearly	22.94411	79.208599

41	3827	Ground water, Vill Imalia, Dist Jabalpur	Madhya Pradesh	Ground Water	Half-Yearly	23.111594	79.917595
42	3828	Hiran, NH-7 Road Bridge, Sihora, Dist Jabalpur	Madhya Pradesh	River	Monthly	23.277289	79.649662
43	3829	Hiran, NH-12 Road Bridge, Vill Vikrampur, Dist Narsingpur	Madhya Pradesh	River	Monthly	23.277283	79.648869
44	3830	Gokulpur Talab, Dist Jabalpur	Madhya Pradesh	Pond	Monthly	23.197346	79.986861
45	3847	Narmada River at Dharampuri	Madhya Pradesh	River	Monthly	22.146769	75.346636
46	3849	Sitapat Talab, Dhar	Madhya Pradesh	Pond	Monthly	22.549469	75.309064
47	4577	Narmada at Chandanghat near Road Bridge	Madhya Pradesh	River	Monthly	22.88105	81.303172
48	4578	Narmada at Puskar Dam, Amarkantak	Madhya Pradesh	River	Monthly	22.67693	81.755469
49	4579	Narmada at Dindori (D/S after mixing to city domestic, near Jogotikaria Ghat)	Madhya Pradesh	River	Monthly	22.941913	81.077261
50	4580	Narmada near Bhairav Temple at Shhastradhar, Mandla	Madhya Pradesh	River	Monthly	22.631124	80.3371
51	4581	Narmada near Road Bridge (D/S Bargi Dam), Jabalpur	Madhya Pradesh	River	Monthly	22.940994	79.923146

52	4582	Narmada at Jamtara, near Railway Bridge	Madhya Pradesh	River	Monthly	23.100994	79.966769
53	4583	Narmada near Road Bridge, Jhansinghat, Jabalpur	Madhya Pradesh	River	Monthly	23.109371	79.586374
54	4584	Narmada NH-44 near Road Bridge at Barman, Jabalpur	Madhya Pradesh	River	Monthly	23.030243	79.019407
55	4585	Barmanghat, 100 m DS of Mainghat	Madhya Pradesh	River	Monthly	23.029888	79.019363
56	4586	Narmada NH-44 near Road Bridge, Jhikoli, Narsingpur	Madhya Pradesh	River	Monthly	23.003329	79.192231
57	4587	Narmada near Road Bridge, Sandia	Madhya Pradesh	River	Monthly	22.910862	78.352357
58	4588	Narmada at U/S of Village Jait	Madhya Pradesh	River	Monthly	22.862205	77.916949
59	4589	Narmada at D/S of Village Jait	Madhya Pradesh	River	Monthly	22.862872	77.914305
60	4590	Narmada U/S before confluence of River Tawa, near Bandrabhan	Madhya Pradesh	River	Monthly	22.803009	77.777088
61	4591	Narmada D/S after confluence of River Tawa, Ramnagar	Madhya Pradesh	River	Monthly	22.795902	77.761918
62	4592	Narmada at Budhnighat, Budhni	Madhya Pradesh	River	Monthly	22.766248	77.687799
63	4593	Narmada at D/S of Textile Unit,	Madhya Pradesh	River	Monthly	22.751431	77.609669

		Village Holipura, Budni					
64	4594	Narmada at before confluence of River Jamner, Vill. Nemawar	Madhya Pradesh	River	Monthly	22.496059	76.979299
65	4595	Narmada 500 m D/S near Jain Mandir, Vill. Nemawar	Madhya Pradesh	River	Monthly	22.497439	76.97156
66	4596	Narmada at Hanuwantia, Dist. Khandwa	Madhya Pradesh	River	Monthly	22.145048	76.587208
67	4597	Narmada at U/S Omkareshwar Dam, Khandwa	Madhya Pradesh	River	Monthly	22.245668	76.150971
68	4598	Narmada River at Dhareshwar	Madhya Pradesh	River	Monthly	22.196054	75.915996
69	4599	Narmada near W/S intake Jallod	Madhya Pradesh	River	Monthly	22.158717	75.690453
70	4600	Narmada River at Shahastradhara (Jalkoti)	Madhya Pradesh	River	Monthly	22.162334	75.545073
71	4601	Narmada at Khalghat	Madhya Pradesh	River	Monthly	22.157587	75.446411
72	4602	Narmada River at Semalda, U/S of Barwani	Madhya Pradesh	River	Monthly	22.094917	75.045778
73	4603	Narmada at Koteswar	Madhya Pradesh	River	Monthly	22.081378	74.789532
74	4605	River Narmada at Kakrana, interstate boundary, Alirajpur	Madhya Pradesh	River	Monthly	22.052205	74.46991

75	4613	Hiran River after mixing Pariyat River, near Road Bridge, Vill. Ganiyari, Jabalpur	Madhya Pradesh	River	Monthly	23.102058	79.980953
76	4614	Suptal, Jabalpur	Madhya Pradesh	Lake	Monthly	23.156502	79.900679
77	4615	Ganga Sagar, Jabalpur	Madhya Pradesh	Lake	Monthly	23.156789	79.910135
78	4616	Banjar River near Road Bridge, Vill. Kharradhar, Malanjkhanda	Madhya Pradesh	River	Monthly	22.021068	80.712809
79	4617	Wainganga River near Vill. Budunda, Malanjkhanda, Dist. Balaghat	Madhya Pradesh	River	Monthly	22.049905	80.716074
80	4618	Ground Water at Industrial Area, Maneri, Dist. Mandla	Madhya Pradesh	Ground Water	Half-Yearly	23.108477	80.252455
81	4619	Dejla Dewda, Khargone	Madhya Pradesh	River	Monthly	21.593227	75.62365
82	4621	Nanpur Fata Dam, Alirajpur	Madhya Pradesh	River	Monthly	22.291974	74.580008
83	4627	Hand Pump, District Hospital, Hoshangabad	Madhya Pradesh	Ground Water	Half-Yearly	22.757891	77.718766
84	4630	Hand Pump, District Hospital, Betul	Madhya Pradesh	Ground Water	Half-Yearly	21.901733	77.895572

*Source: CPCB, List of NWMP Monitoring Network, https://cpcb.nic.in/wqm/WQMN_list.pdf

Table A - 2 MPPCB Water Quality Monitoring Stations in Narmada River Basin, Madhya Pradesh (2019–2025)

S. No	District	City/Town	Name of Monitoring Station	Water Quality Category					
				2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
1	Anuppur	Amarkantak	Narmada River at Amarkantak origin point, Dist. Anuppur	A	A	A	A	A	A
2	Anuppur	Amarkantak	Narmada River at Puskar Dam, Amarkantak	A	A	A	A	A	B
3	Anuppur	Amarkantak	Narmada River at Kapildhara	A	A	A	A	A	A
4	Dindori	Dindori	Narmada River at Chandanghat near Road Bridge, Dist. Dindori	A	A	A	A	A	A
5	Dindori	Dindori	Narmada River at Dindori Up Stream	A	A	A	A	A	A
6	Dindori	Dindori	Narmada River at Dindori Down Stream	A	A	A	A	A	A
7	Mandla	Mandla	Narmada River near Shamshanghat, Mandla	A	A	A	A	A	A
8	Mandla	Mandla	Narmada River near Road Bridge, Mandla	A	A	A	A	A	A
9	Mandla	Mandla	Narmada River at Bhairav Temple,	A	A	A	A	A	A

			Shhastradhar, Mandla						
			Narmada River near Road						
10	Jabalpur	Jabalpur	Bridge (Down Stream of Bargi Dam), Jabalpur	A	A	A	A	A	A
			Narmada River at Jamtara, near Railway Bridge						
11	Jabalpur	Jabalpur		A	A	A	A	A	A
			Narmada River near Water Supply Intake Point, Lalpur						
12	Jabalpur	Lalpur		A	A	A	A	A	A
			Narmada River at Tilwaraghat						
13	Jabalpur	Jabalpur		A	A	A	A	A	A
			Narmada River at Panchwatighat, Bheraghat						
14	Jabalpur	Jabalpur		A	A	A	A	A	A
			Narmada River at Sarswastighat after mixing Bawanganga River						
15	Jabalpur	Jabalpur		A	A	A	A	A	A
			Narmada near Road Bridge at Jhansinghat, Jabalpur						
16	Jabalpur	Jabalpur		A	A	A	A	A	A
			Narmada N.H.44 near Road						
17	Narsinghpur	Narsinghpur	Bridge at Barman, Jabalpur	A	A	A	A	A	A
			Narmada Barmanghat, 100						
18	Narsinghpur	Narsinghpur		A	A	A	A	A	A

			mts D/s						
			Mainghat						
19	Narsinghpur	Narsinghpur	Narmada S.H.44 near Road Bridge, Jhilkoli, Narsinghpur	A	A	A	A	A	A
20	Narmadapuram	Sandia	Narmada River near Road Bridge, Sandia, Narmadapuram	B	B	A	A	A	A
21	Sehore	Jait	Narmada River at Up Stream of Village Jait, Sehore	B	B	A	A	A	A
22	Sehore	Jait	Narmada River at Down Stream of Village Jait, Sehore	B	B	A	A	A	A
23	Sehore	Sehore	Narmada River at Shahaganj Guest House, Sehore	B	B	B	A	B	A
24	Sehore	Sehore	Narmada Up Stream before Confluence of River Tawa near Bandrabhan, Sehore	B	B	B	A	B	A
25	Sehore	Ramnagar	Narmada Down Stream after Confluence of River Tawa, Ramnagar	B	B	B	A	B	A
26	Sehore	Budhni	Narmada Budhnighat, Budhni	B	B	B	A	B	A

27	Narmadapuram	Narmadapuram	Narmada Korighat, Hoshangabad	B	B	B	A	B	A
28	Narmadapuram	Narmadapuram	Narmada Sethanighat, Hoshangabad	B	B	B	A	B	A
29	Narmadapuram	Narmadapuram	Narmada River 100 m Down Stream after SPM Nalla	B	B	B	A	B	A
30	Sehore	Budhni	Narmada at Down Stream of Textile Unit, Village Holipura, Budhni	B	B	B	A	B	A
31	Dewas	Nemawar	Narmada at before Confluence of River Jamner, Village Nemawar	A	A	A	A	A	A
32	Dewas	Nemawar	Narmada at Water Supply Intake Point, Nemawar	A	A	A	A	A	A
33	Dewas	Nemawar	Narmada 500 m Down Stream near Jain Mandir, Village Nemawar	A	A	A	A	A	A
34	Khandwa	Hanuwantia	Narmada at Hanuwantia, Dist. Khandwa	A	A	A	A	A	A
35	Khandwa	Punasa	Narmada River at Punasa Dam	A	A	A	A	A	A

			Narmada at Up						
36	Khandwa	Omkareshwar	Stream, Omkareshwar Dam	A	A	A	A	A	A
37	Khandwa	Omkareshwar	Narmada at Omkareshwar Down Stream	A	A	A	A	A	A
38	Khargone	Barwaha	Narmada, Barwaha near Mortakka Bridge	A	A	A	A	A	A
39	Khargone	Dhadeshwar	Narmada River at Dhadeshwar	A	A	A	A	A	A
40	Khargone	Jallod	Narmada near Water Supply Intake, Jallod	A	A	A	A	A	A
41	Khargone	Mandleshwar	Narmada River at Mandleshwar Down Stream	A	A	A	A	A	A
42	Khargone	Maheshwar	Narmada River at Maheshwar Down Stream	A	A	A	A	A	A
43	Khargone	Maheshwar	Narmada River at Shahastradhara (Jalkoti)	A	A	A	A	A	A
44	Dhar	Dhar	Narmada at Khalghat	A	A	A	A	A	A
45	Dhar	Dharamपुरी	Narmada at Dharamपुरी	A	A	A	A	A	A
46	Dhar	Dharamपुरी	Narmada River at Dharamपुरी Down Stream	A	A	A	A	A	A
47	Barwani	Barwani	Narmada River at Semalda Up	A	A	A	A	A	A

			Stream of Barwani						
			Narmada River						
48	Barwani	Barwani	at Rajghat, Barwani	A	A	A	A	A	A
			Narmada River						
49	Dhar	Nisarpur	at Koteswar, Nisarpur, Dhar	A	A	A	A	A	A
			Narmada River						
50	Alirajpur	Kakrana	at Kakrana	A	A	A	A	A	A

Source: NGT Report OA No. 71/2023

(https://www.greentribunal.gov.in/sites/default/files/news_updates/report%20in%2071-2023_30_10_23.pdf#page=847.13);

MPPCB, Narmada Water Quality Report (Yearly) (<https://www.mppcb.mp.gov.in/narmada-report16-17-17-18eng.aspx>)

Table A - 3 List of Large-Scale Industries in Narmada Basin, Madhya Pradesh

S.No	Industry / Company	District
1	M/s H.J.I.Division of Orient Paper Mill, Amlai	Anuppur
2	M/s Ramnik Power and Allies Pvt. Ltd.	Balaghat
3	Sarandi Baraseoni	Balaghat
4	M/s Hindustan Copper Limited, Malajkhand	Balaghat
5	M/s Manganese Ore India Ltd, Barveli	Balaghat
6	M/s Hindustan Liver Limited	Chhindwara
7	M/s Raymond Limited, Growth Center, Borgaon	Chhindwara
8	ACC, Cement Ltd., Kymore	Katni
9	Svil Mines Ltd., Village Gudri Kuteshwer, Berhi	Katni
10	ACE Refractory Ltd., Gudri	Katni
11	Averest Asbestos Industries Ltd., Kymore	Katni
12	Ordnance Factory, Katni	Katni
13	M/s Simplex Tubes Pvt. Ltd., Growth Centre, Maneri	Mandla

14	M/s P.S.L. Engineering Works Pvt. Ltd., Growth Centre, Maneri	Mandla
15	M/s M.P. Glychem Industries Ltd., Vill. Dobhi T. Niwas	Mandla
16	M/s Hindustan Petroleum Corp. Ltd., Growth Centre, Maneri	Mandla
17	M/s Primo Pick-N-Pack Pvt Unit No. 01, Growth Centre, Maneri	Mandla
18	M/s Primo Pick-N-Pack Pvt Unit No. 02, Growth Centre, Maneri	Mandla
19	M/s Agrawal Indotax Ltd., Vill. Block – Chhaigaon Makhan	Khandwa
20	M/s Navalsingh Sahkari Shakkari Karkhana Mariyadit	Burhanpur
21	Diamond Cement Factory Pvt. Ltd., Nursingharh	Damoh
22	Diamond Cement Factory Pvt. Ltd., Imlai	Damoh
23	Gas Filling Plant, Bhitoni, Teh. Shahpura	Jabalpur
24	Heavy Vehicle Factory	Jabalpur
25	Ordnance Factory, Khamaria	Jabalpur
26	Telecom Factory	Jabalpur
27	Grey Iron Foundry	Jabalpur
28	Gun Carriage Factory	Jabalpur
29	M/s Udaipur Beverage Ltd., Richhai	Jabalpur
30	M/s Vardhman Fabric	Sehore
31	M/s Trident Ltd, Hoshangabad Rd., Budni	Hoshangabad
32	M/s Trident Ltd, Unit 2 (Yarn Divn), Vill Barkhedhi	Hoshangabad
33	M/s S.E.L Manufacturing Co. Ltd., Vill Mehtwada, Teh. Jawer	Sehore
34	M/s Eknath Solvent Extraction, Gram Pidgoan	Harda
35	M/s SKG Solvent (Closed)	Harda
36	M/s Nikil Sugar Mill, Gram Baranga	Harda
37	M/s Dhanlaxmi Solvent Extraction, Gram Palasner	Harda
38	M/s Mahakaushal Sugar & Power Industries Ltd., Bachyi Tehsil	Narsinghpur
39	M/s Kareli Sugar Mill Pvt Ltd., Gadawara Road	Narsinghpur
40	M/s Ruchi Soya Industries Ltd., Pipariya Road, Gadawara	Narsinghpur
41	M/s Ruchi Soya Industries Ltd., Pipariya Road, Gadawara, Unit No 2	Narsinghpur

42	M/s Shakti Sugar Mill Pvt Ltd., Gram Kaundya, Teh. Gadarwara	Narsinghpur
43	M/s BLA Power Industries Ltd., Gangai, Teh Gadarwara	Narsinghpur
44	M/s National Steel & Agro Ind., Lebad	Dhar
45	M/s R.S.A.L. Steel & Agro Ind., Lebad	Dhar
46	M/s Bajrang Agro (Food & Oil Div.) Ind. Pvt. Ltd., Lebad	Dhar
47	M/s Great Geliyan Ltd., Lebad	Dhar
48	M/s PEB Steel Lloyd (I) Pvt. Ltd., Plot No. 3-B, I/A, Kheda	Dhar
49	M/s Allcargo Global Logistics Ltd., Plot No-13-B I/A, Kheda	Dhar
50	M/s Dhar Cement Ltd., Ghandwani	Dhar

*Source: *INDUSTRIAL PROFILE OF MADHYA PRADESH 2016 – 17*,
https://dcmsme.gov.in/dips/state_wise_profile_16-17/Madhya%20Pradesh%20profile%2016-17.pdf

Table A - 4 List of Medium Scale Industries in Narmada Basin, Madhya Pradesh

S.No	Industry Name	Location	District
1	Bhansali Engineering Polymers Limited	Satnur, Sounsar	Chhindwara
2	P.B.M Polytex Limited	Growth center, Borgaon, Sounsar	Chhindwara
3	Hanuman Komokroystic Limited	Growth center, Borgaon, Teh.- Sounsar	Chhindwara
4	Suryawanshi Sparring Mills	Gram Rajna, Teh. Pandurna	Chhindwara
5	Niket Industries	A.K.V.N., Borgaon, Teh.- Saunsar	Chhindwara
6	Vindhyachal Synthetic Pvt. Limited	A.K.V.N., Borgaon, Teh-Saunsar	Chhindwara
7	Ranbo Webers & Processor Pvt. Limited	A.K.V.N., Borgaon	Chhindwara
8	J.C.O. Gas Pipe Limited	A.K.V.N., Borgaon, Teh- Saunsar	Chhindwara
9	Ojaswi Marble & Granite Pvt Ltd	Hardua	Katni
10	Alied Minerals & Pvt. Ltd.	Katni	Katni
11	Dabur India Ltd Company	Katni	Katni
12	Sheela Agro Pvt Ltd	Katni	Katni
13	Fair Food Overseas Pvt Ltd	Katni	Katni
14	Bhandari Foils Pvt. Ltd. (Unit No.2, Tube Div.)	Plot No. D1-D4, No.1	Dewas

15	Prestige Feed Mills Pvt. Ltd.	49-C, Industrial Area No. 2	Dewas
16	Bhandari Foils and Tubes Ltd. (Unit No.1)	Plot No. D1-D4, Industrial Area No.1	Dewas
17	Shiva Automotive Pvt. Ltd.	37c-38 D, Industrial Area No. I	Dewas
18	Sanghavi Foods Pvt. Ltd.	Vill.- Amona, Teh.- Dewas	Dewas
19	Bhaskar Ex-Oils Ltd.	Vill.-Chotamalsapura, P.O.-Siya, Teh.- Dewas	Dewas
20	Prestige Foods Ltd.	6-B, Industrial Area No.3	Dewas
21	Gabriel India Ltd.	Industrial Area No.3	Dewas
22	B E Commercial Unit of Eicher Motor Ltd.	Dewas	Dewas
23	Prestige Soya Industries (Prop. Prestige Food Ltd.)	48-B, Industrial Area No. 3	Dewas
24	Fluidomat Pvt. Ltd.	7c 8j Industrial Area No-1	Dewas
25	Gajra Differential Gears Pvt. Ltd. (Formerly Gajra Transmission & Differential Gears Ltd.)	Gram - Lohar Pipliya, Teh.- Dewas	Dewas
26	M.P. Wind Farms	Jamgod Rani Hills	Dewas
27	Premier Proteins Ltd.	47-B/48-B, Industrial Area No.1	Dewas
28	Hind Filter Ltd.	1/A & 8/A, Industrial Area No. 1	Dewas
29	S & H Gears Pvt. Ltd.	Station Road	Dewas
30	G.G Automotive Gears Ltd.	2-A, Industrial Area No. 1	Dewas
31	Premier Industries (India) Ltd (Formerly Premier Extraction Pvt. Ltd.)	45/47-A, Industrial Area No.1	Dewas
32	Kuber Lighting Pvt. Ltd. (Formerly Kalpana Industries (Indore) Pvt. Ltd.)	Industrial Area No.3	Dewas
33	Premier Nutrition Ltd. (Prop Premier Industries (India) Pvt. Ltd.)	55/56-A, Industrial Area No.1	Dewas
34	S. Kumar Ltd. (Formerly Skm Fabrics)	Industrial Area No.2	Dewas
35	Shubham Powgen (Prop Gwalior Chemical Industries Ltd.)	Jamgod Rani Hills	Dewas
36	Caparo Tubes Steel	Dewas	Dewas
37	Petogan Labs Ltd	Dewas	Dewas

38	Krishana Foods Products	Siya	Dewas
39	National News Print & Paper Mills Ltd.	Nepa Nagar	Burhanpur
40	Narendra Cott Fibers Industries	Khasra No. 218/1, 2, 3, Alabagh	Burhanpur
41	Burhanpur Tapti Mills Ltd.	Lalbag	Burhanpur
42	M/s KEC International	Gram Devri, Panagar	Jabalpur
43	M/s Narmada Gelatines Ltd	Vill Meerganj, Bheraghat Rd.	Jabalpur
44	M/s Oxilari Production Centre	Indl Area, Ricchai	Jabalpur
45	M/s Commercial Engg & Body building	Indl Area, Ricchai	Jabalpur
46	M/s Om Agro Product	Gram Bheeta, Teh. Shahpura	Jabalpur
47	M/s Jabalpur Polytex Pvt Ltd	Indl Area, Ricchai	Jabalpur
48	M/s Carborandum Universal Ltd	Indl Estate Adhartal	Jabalpur
49	M/s Pragya Energy Pvt Ltd	Indl Area, Ricchai	Jabalpur
50	M/s Nai Dunia	Indl Area, Ricchai	Jabalpur
51	M/s Narsing Extruction allied product Pvt Ltd	Khamtra, Teh. Narsinghpur	Narsinghpur
52	M/s Narmada Sugar Pvt Ltd	Paudar, Post Sali Chauka, Teh. Gadarwara	Narsinghpur
53	M/s Dhanlaxmi Solvex Pvt Ltd	Baghwar Road, Teh. Kareli	Narsinghpur
54	M/s D & H India Pvt. Ltd.	Sejwaya	Dhar
55	M/s Tirupati Starch & Chemicals	Lebad, Sejwaya	Dhar
56	M/s Manoj Surgical Industries Ltd.	Plot no. 106, Sector-III Pithampur	Dhar
57	M/s Pithampur Poly Products Ltd.	Plot no. 115, Sector-III Pithampur	Dhar
58	M/s Labh Ganga Marketing Pvt. Ltd.	Sector-III Pithampur	Dhar
59	M/s Badve Engg. Ltd.	Plot no. 161, Sector-III Pithampur	Dhar
60	M/s Maya Spinners Pvt. Ltd.	Plot no. 21, Sector-III Pithampur	Dhar
61	M/s Signet Industries Ltd.	Sector-III Pithampur	Dhar
62	M/s Healthico Pharma Pvt. Ltd.	Sector-III Pithampur	Dhar
63	M/s Alpa Laboratories Ltd.	Sector-III Pithampur	Dhar
64	M/s P.J. Beverajes Ltd.	Plot no. 76, Sector-III Pithampur	Dhar
65	M/s Vertax Techno Soft Ltd.	Plot no. 809A to 809E, Sector-III Pithampur	Dhar
66	M/s Jaideep International	Plot No. 1, I/A, Kheda	Dhar

67	M/s Saral Logistics Systems Pvt. Ltd.	Plot No-1-C, Industrial Area, Kheda	Dhar
68	M/s Manish Agrotech Pvt. Ltd.	Plot No-90-91, Industrial Area, Kheda	Dhar
69	M/s Pratibha Sentex Ltd	Plot No-05, Industrial Area, Kheda	Dhar
70	M/s Trans World Textiles P.Ltd	Plot No-13-A, Industrial Area, Kheda	Dhar
71	M/s Mahle Migma P.Ltd.	Plot No. 9, 10 & 11, I/A, Kheda	Dhar
72	M/s Dabur India Ltd.	Plot No. 86-A, I/A, Kheda	Dhar

Source: INDUSTRIAL PROFILE OF MADHY PRADESH 2016 – 17,

https://dcmsme.gov.in/dips/state_wise_profile_16-17/Madhya%20Pradesh%20profile%2016-17.pdf

Table A - 5 Solid, Plastic and C&D waste generation in Narmada river Basin, Madhya Pradesh

Districts	Block(ULB)	Solid waste		Plastic waste		C&D Waste	
		Estimated Solid Waste Generation in 2025 (TPD)*	Estimated Solid Waste Generation in 2030 (TPD)*	Estimated Generation in 2025 (TPD)	Estimated Generation in 2030 (TPD)	Estimated C& D Waste Generation (TPD) in 2025	Estimated C&D Waste Generation (TPD) in 2030
Alirajpur	Alirajpur	17.02	18.32	0.34	0.37	2.33	2.51
	Bhavara	10.2	10.98	0.2	0.22	1.4	1.5
	Jobat	11.99	12.91	0.24	0.26	1.64	1.77
		39.21	42.21	0.78	0.85	5.37	5.78
Anuppur*	Pushprajgarh	25.98	27.34	0.52	0.55	3.56	3.75
Balaghat	Baihar	5.55	5.87	0.3	0.3	0.54	0.57
Barwani	Barwani	21.55	23.7	0.43	0.47	2.95	3.24

	Niwali	15.61	17.16	0.31	0.34	2.13	2.35
	Pansemal	18.02	19.81	0.36	0.39	2.46	2.71
	Rajpur	25.24	27.75	0.5	0.55	3.45	3.8
	Sendhwa	42.07	46.25	0.84	0.92	5.76	6.33
	Thikiri	19.71	21.67	0.39	0.43	2.7	2.96
		142.2	156.34	2.83	3.1	19.45	21.39
Betul*	Amla	10	10.5	0.5	0.5	2.36	2.49
	Betul	19.63	20.7	0.39	0.41	2.69	2.84
	Bhaisdehi	14.93	15.74	0.3	0.31	2.04	2.16
	Multai	14.89	15.7	0.5	0.5	2.04	2.15
	Shahpur	13.38	14.11	0.27	0.28	1.83	1.93
		72.83	76.75	1.96	2	10.96	11.57
Bhopal	Huzur						

Burhanpur	Nepanagar	10.6	11.4	0.5	0.6	2.07	2.22
Chindwara*	Amarwara	17.14	18.09	0.34	0.36	2.35	2.48
Whole district	Junnardev	7	7.9	0.37	0.39	1.46	1.54
	Tamiya	13.51	14.25	0.27	0.29	1.85	1.95
		37.65	40.24	0.98	1.04	5.66	5.97
Damoh	Jabera	20.19	24.9	0.98	0.4	2.77	3.41
	Tendukheda	17.09	21.06	0.83	0.34	2.34	2.89
		37.28	45.96	1.81	0.74	5.11	6.3
Dewas	Bagli	33.36	42.61	0.67	0.85	3.77	4.81
	Dewas	110.62	119.1	5.5	6	40.41	43.51
	Kannod	26.37	33.57	0.53	0.67	2.9	3.69
	Khategaon	21.14	26.92	0.42	0.54	2.34	2.98
		191.48	222.2	7.12	8.06	49.42	54.99

Dhar	Dhar	14.31	15.65	0.7	0.8	1.96	2.14
	Dharampuri	18.37	20.1	0.9	1	2.52	2.75
	Gandhwani	21.2	23.19	1.1	1.2	2.9	3.18
	Kukshi	11.2	12.26	0.6	0.6	1.53	1.68
	Manawar	19.32	21.14	1	1.1	2.65	2.9
	Sardarpur	36.48	39.92	1.8	2	5	5.47
		120.88	132.26	6.1	6.7	16.56	18.12
Dindori	Dindori	4.19		0.19		0.8	
	Shahpura	2.44		0.11		0.002	
		6.63		0.3		0.802	
Khandwa	Harsud	16.52		0.5		3.5	
	Khandwa	45.97		2.07		0.9	
	Pandhana	45.83		1.37		2	
		108.32		3.94		6.4	

Harda	Harda	18		0.5		0.08	
	Khirkiya	0.5		0.5		2	
	Timurni	1.5		0.26		0.01	
		20		1.26		2.09	
Hosangabad	Babai						
	Bankheri	15.12	16.03	0.3	0.32	2.07	2.2
	Hoshangabad	12.73	13.49	0.25	0.27	1.74	1.85
	Itarsi	33	35.5	1.67	1.77	6.55	6.94
	Piparia	14.48	15.35	0.82	0.87	1.98	2.1
	Seonimalwa	20.22	21.44	0.4	0.43	2.77	2.94
	Sohagpur	14.93	15.83	0.3	0.32	2.05	2.17
		110.48	117.64	3.74	3.98	17.16	18.2
Indore	Mhow	7.616		0.56		0.5	

Jabalpur	Jabalpur	19.7	20.9	0.39	0.42	2.7	2.9
	Kundam	14.3	15.1	0.29	0.3	2	2.1
	Majholi	18.8	19.9	0.38	0.4	2.6	2.7
	Patan	15.6	16.6	0.31	0.33	2.1	2.3
	Shahpura	21.4	22.7	0.43	0.45	2.9	3.1
	Shihora	16.3	17.3	0.33	0.35	2.2	2.4
		106.1	112.5	2.13	2.25	14.5	15.5
Jhabua	Jhabua	14	15.8	0.72	0.79	2.8	3.1
	Ranapur	4.95	5.48	0.29	0.32	2	2.22
		18.95	21.28	1.01	1.11	4.8	5.32
Katni	Bahuriband	25.85	27.98	0.52	0.56	3.5	3.8
	Dhimarkhera	23.11	25.02	0.46	0.5	3.2	3.4
	Rathi	16.53	17.9	0.33	0.36	2.3	2.5
		65.49	70.9	1.31	1.42	9	9.7

Mandla	Bichhiya	2.35		1.83		0.54	
	Mandla			7.7		2.3	
	Nainpur	9.65		11.2		0.05	
	Niwas	1.5		0.5		0.06	
		13.5		21.23		2.95	
Narshingpur	Gadarwara	16	17	0.8	0.8	3.12	3.3
	Gotegaon	20.1	21.28	0.4	0.43	2.8	2.9
	Kareli	14.6	15.46	0.29	0.31	2	2.1
	Narshingpur	18.25	19.32	0.37	0.39	2.5	2.6
	Tendukheda	4	5	0.2	0.2	0.43	0.45
		72.95	78.06	2.06	2.13	10.85	11.35
Raisen	Bareli	12	13	0.6	0.7	1.19	1.28
	Begamganj	14.13	15.16	0.28	0.3	2.99	3.21
	Gairatganj	13.42	14.4	0.26	0.28	1.94	2.08

Goharganj							
	Raisen	16	17	0.8	0.8	3.04	3.26
	Silvani	17.11	18.36	0.34	0.36	2.78	2.98
	Udaipura	17.34	18.61	0.34	0.37	1.84	1.97
		124	96.53	2.62	2.81	13.78	14.78
Sagar	Deori	19	20	0.38	0.4	2.6	2.8
	Kesli	15	16	0.29	0.31	2	2.1
		34	36	0.67	0.71	4.6	4.9
Sehore	Ashta	37.75	49.13	0.75	0.98	5.17	6.73
	Budhni	15.68	20.41	0.31	0.41	2.15	2.8
	Ichawar	19.14	24.92	0.38	0.5	2.62	3.41
	Nasrullaganj	8.67	9.39	0.43	0.47	0.85	0.92
	Sehore	41.43	53.92	0.83	1.08	5.68	7.39
		122.67	157.77	2.7	3.44	16.47	21.25

Seoni	Ghansaur						
	Lakhnadon	3.78		0.01		0.5	
Umaria	Bandhogarh*						
	Whole						
	District	12.51	14	0.6	0.7	2.45	2.68
West Nimar (Khargone)	Barwah	58.4		1.75		0.02	
	Bhagwanpura	38.6		1.16			
	Bhikangaon	35.11		1.05		0.46	
	Jhirniya	40.35		1.21			
	Kasrawad	43.99		1.32		0.5	
	Khargone	41.03		3.13		0.53	
	Maheshwar	39.25		1.18		0.52	

Seagon	16.69	0.5	
	313.42	11.3	2.03

*Source: MPCB District environment plan

Table A - 6 District-wise Status of Solid Waste Collection and Segregation Facilities in Narmada Basin, Madhya Pradesh

Collection of solid waste							
S. No.	Districts	Block (ULB)	Segregation (Dry and wet) At Source	Waste Collection trolleys with separate Compartments	Mini Collection Trucks with separate compartments	Gaps and Challenges	References
		Alirajpur	89%	3	6	required 3 more mini collection trucks.	
1	Alirajpur	Bhavara	100%	1	1	2 More mini collection trucks are required.	https://www.mppcb.mp.gov.in/pr oc/Tech/DEP%20Alirajpur.pdf
		Jobat	70%	1	1	2 More mini collection trucks are required.	

2	Anuppur (Whole District)	District	NA	NA	NA	More Collection trolleys are required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	Baihar	100%	22	sufficient	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
		Barwani	90%	Sufficient	9	6 more no mini collection trucks are required	
		Niwali	50%	2	3	4 more collection trolleys and 3 mini collection trucks are required.	
		Pansemal	50%	4	3	4 more collection trolleys and 3 mini collection trucks are required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Badwani_2023.pdf
4	Barwani	Rajpur	50%	2	3	4 more collection trolleys and 2 mini collection trucks are required.	
		Sendhwa	90%	Sufficient	9	6 more no mini collection trucks are required	
		Thikiri	50%	2	3	4 more collection trolleys and 3 mini collection trucks are required.	

5	Betul (whole District)	Amla	Partial	No Information	No Information	Smaller vehicles available with the ULBs' are required to make several trips for transportation of waste and At least 02 Waste transfer points to be developed in Betul Municipal Corporation	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
		Betul					
		Bhaisdehi					
		Multai					
6	Burhanpur	Shahpur	50%	8	1	8 Collection trolleys and 1 mini truck is required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
		Nepanagar					
7	Chindwara (whole District)	Amarwara	Partial	Sufficient in ULBs	Sufficient in ULBs	Mini Collection Trucks with separate compartments to be procured with 60:40 ratio along with 03 additional compartments for bio-medical, sanitary waste and domestic hazardous wastes.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
		Junnardev					
		Tamiya					
8	Damoh	Jabera	NA	NA	NA	NA	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
		Tendukheda	100%	1	2	10 Collection trolleys and 1 mini truck is required.	
9	Dewas	Bagli	50%	Sufficient	2	No Gap Identified	

		Dewas	100%	180	56	15 Collection trolleys and 10 mini truck are required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
		Kannod	50%	Sufficient	3	No Gap Identified	
		Khategaon	50%	Sufficient	5	No Gap Identified	
		Dhar					
		Dharampuri					
10	Dhar (whole District)	Gandhwani	Partially	Yes available	Yes available	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf
		Kukshi					
		Manawar					
		Sardarpur					
11	Dindori (whole District)	Dindori	Partial	Required	Required	More collection trolleys with compartments and mini collection trucks are required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DINDORI_2023.pdf
		Shahpura					
		Harsud	50%	2	3	8 Collection trolleys and 7 mini truck are required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
12	Khandwa	Khandwa	Partial	50	1	8 collection trolleys with compartments and 6 mini collection trucks are required.	

		Pandhana	50%	2	3	8 collection trolleys with compartments and 3 mini collection trucks are required.	
		Harda	100%	34 DTDC vehicles	5	No Gap Identified	
13	Harda	Khirkhya	100%	2 DTDC vehicles	5	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf
		Timurni	100%	2 DTDC vehicles	5	No Gap Identified	
		Babai	80%	2	3	1 collection trolleys with compartments and 2 mini collection trucks are required.	
14	Hosangabad	Bankheri	80%	1	5	3 collection trolleys with compartments and 5 mini collection trucks are required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Hoshangabad&pdf=/proc/Tech/Narmadapuram_2023.pdf
		Hoshangabad	88%	40	7	No Gap Identified	
		Itarsi	85%	5	1	4 collection trolleys with compartments and 1 mini collection trucks are required.	

		Piparia	85%	3	5	5 more no of mini collection trucks and 3 more waste collection trolleys are required	
		Seonimalwa	85%	3	5	3 collection trolleys with compartments and 5 mini collection trucks are required.	
		Sohagpur	70%	3	15	No Gap Identified	
15	Indore	Mhow	75%	Sufficient	Sufficient	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Indore&pdf=/proc/Tech/INDORE_2023.pdf
		Jabalpur	100%	Sufficient	Sufficient	No Gap Identified	
		Kundam	100%			No Gap Identified	
16	Jabalpur	Majholi	100%	Sufficient	Sufficient	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur&pdf=/proc/Tech/Jabalpur_2023.pdf
		Patan	100%	Sufficient	Sufficient	No Gap Identified	
		Shahpura	100%	Sufficient	Sufficient	No Gap Identified	
		Shihora	100%	Sufficient	Sufficient	No Gap Identified	
17	Jhabua (whole District)	Jhabua				No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
		Ranapur	Partial	Sufficient	Sufficient	No Gap Identified	
18		Bahuriband	Partially		50	No Gap Identified	

19	Katni (whole District)	Dhimarkhera		Sufficient in all ULBs		No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
		Rathi				No Gap Identified	
	Mandla	Bichhiya	100%	Sufficient	Sufficient	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf
		Mandla	100%	Sufficient	Sufficient	No Gap Identified	
		Nainpur	100%	Sufficient	Sufficient	No Gap Identified	
		Niwas	100%	Sufficient	Sufficient	No Gap Identified	
		Gadarwara	100%	8	Sufficient	No Gap Identified	
		Gotegaon	100%	7 WC trolleys and 21 Tricycle	Sufficient	No Gap Identified	
20	Narshingpur	Kareli	100%	Sufficient	Sufficient	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGHPUR_2023.pdf
		Narshingpur	100%	11 WC trolleys and 50no Tricycle	Sufficient	No Gap Identified	
		Tendukheda	100%	3 WC trolleys and 2 no Tricycle	Sufficient	02 no. of Waste Collection Trolleys and 05 Tricycle/Handcart are required.	
21	Raisen	Bareli	100%	3	6	No Gap Identified	

		Begamganj	100%	0	9	2 collection trolleys with compartments are required.	
		Gairatganj	100%	6	2	No Gap Identified	
		Goharganj	NA	NA	NA	NA	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf
		Raisen	90%	3	13	2 collection trolleys with compartments and 6 mini collection trucks are required.	
		Silvani	100%	4	3	No Gap Identified	
		Udaipura	100%	5	3	No Gap Identified	
22	Sagar	Deori	100%	Sufficient	Sufficient	No Gap Identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sagar&pdf=/proc/Tech/Sagar_2023.pdf
		Kesli	NA	NA	NA	NA	
		Ashta	80%	Sufficient	Sufficient	No Gap Identified	
		Budhni	100%	Sufficient	Sufficient	No Gap Identified	
23	Sehore	Ichawar	YES	Yes	YES	NA	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHORE_2023.pdf
		Nasrullaganj	80%	1	5	3 collection trolleys with compartments and 5 mini collection trucks are required.	

		Sehore	70%	5	1	6 collection trolleys with compartments and 4 mini collection trucks are required.	
24	Seoni	Ghansaur	45%	20%	10%	Enough vehicles are required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
		Lakhnadon	50%	Sufficient	Sufficient	No Gap Identified	
25	Umaria	Bandhogarh	Partial	Insufficient	Insufficient	More collection trolleys with separate compartments are needed at all 4 ULBs in district	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Umaria&pdf=/proc/Tech/UMARIYA_2023.pdf
		Barwah	50%	2	1	2 collection trolleys and 2 mini collection trucks are required	
		Bhagwanpura	NA	NA	NA	NA	
		Bhikangaon	50%	1	0	2 collection trolleys and 1 mini collection trucks are required	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Damoh.pdf
26	West Nimar	Jhirniya	NA	NA	NA	NA	
		Kasrawas	50%	2	Not required	No Gap Identified	
		Khargone	100%	Sufficient	4	No Gap Identified	
		Maheshwar	50%	2	Not required	No Gap Identified	
		Seagon	NA	NA	NA	NA	

*Source: MPCB District environment plan

Table A - 7 District-wise Status of Solid Waste Treatment and Disposal Facilities in Narmada Basin, Madhya Pradesh

Waste Treatment and Disposal								
S.No.	Districts	Block (ULB)	Wet waste: Composting	Bio methanation	Dry waste: Material recovery Facility (MRF)	Non-Recyclable Waste: Sanitary Landfill	Gaps and Challenges	References
1	Alirajpur	Alirajpur	Composting units with 5.19 tonnes capacity	No	1 MRF partially Functioning	Not available	100% wet waste disposal by composting and Need of Sanitary Landfill	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Alirajpur.pdf
		Bhavara	Not available	No	Not available	Not available	Treatment and disposal facilities are required.	
		Jobat	Not available	No	Not available	Not available	Construction of proper composting facility, MRF and	

							sanitary landfill required	
2	Anuppur (Whole District)	District	Composting wet organic waste mixed with cow dung to create manure.	No	MRF facility exists	Available	required training to site workers to utilize these facilities in full efficiency	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	Baihar	Not available	Not available	Not available	Not available	Construction of proper composting facility, MRF and sanitary landfill required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
4	Barwani	Barwani	110composting pits are available	Not available	MRF facility available	Not available	Required sanitary landfill for non-recyclable waste	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Badwani_2023.pdf
		Niwali	Not available	Not available	Not available	Not available	Construction of proper composting facility, MRF and sanitary landfill required	

5	Betul (Whole District)	Pansemal	Not available	Not available	Not available	Not available	Construction of proper composting facility, MRF and sanitary landfill required	
		Rajpur	Not available	Not available	Not available	Not available	Construction of proper composting facility, MRF and sanitary landfill required	
		Sendhwa	110composting pits are available	Not available	MRF facility available	Not available	Required sanitary landfill for non-recyclable waste	
		Thikiri	Not available	Not available	Not available	Not available	Construction of proper composting facility, MRF and sanitary landfill required	
		Amla						
		Betul	Not available	Not available	Only Betul and Multai have the MRF	Not available	Composting and MRF facility is required and 02 SLF to be	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
		Bhaisdehi						

		Multai			facility available		developed to cater the command areas.	
		Shahpur						
6	Burhanpur	Nepanagar	Not Available	Not available	Not available	Not available	Construction of proper composting facility, MRF and sanitary landfill required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
		Amarwara					Proposal to be set up at the site identified and allocated for MSW disposal. In villages NADEP compost, community soak pits etc. has been constructed for waste management	
7	Chindwara (Whole District)	Junnardev	Available at all ULB levels	Not available	Available at all ULBs	Not available at any of the ULBs		https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
		Tamiya						
		Jabera	Not available	Not available	Not Available	Not available		
8	Damoh	Tendukheda	Not Available	Not Available	Not Available	Not Available	Construction of proper composting facility, MRF and	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf

						sanitary landfill required	
		Bagli	Not Available	Not Available	Not Available	Not Available	required proper composting facility, MRF and sanitary landfill
9	Dewas	Dewas	Not Available	Not Available	2 MRFs available	Not Available	Construction of proper composting facility and sanitary landfill required https://www.mppcb.mp.gov.in/ PdfView.aspx?h=Dewas&pdf= /proc/Tech/DEWAS_2023.pdf
		Kannod	Not Available	Not Available	Not Available	Not Available	required proper composting facility, MRF and sanitary landfill
		Khategaon	Not Available	Not Available	Not Available	Not Available	required proper composting facility, MRF and sanitary landfill
10	Dhar (Whole District)	Dhar Dharampuri	Not Available	Not Available	MRF facility	Not Available all ULBs has dump sites	Construction of proper composting https://www.mppcb.mp.gov.in/ PdfView.aspx?h=Dhar&pdf=/p roc/Tech/DHAR_2023.pdf

		Gandhwani			available in		facility and sanitary	
		Kukshi			all ULBs		landfill required	
		Manawar						
		Sardarpur						
11	Dindori (Whole District)	Dindori	Available	Not available	MRF facility available in all ULBs	Available at all ULBs	Functional Upgradation in Technologies are required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DINDORI_2023.pdf
		Shahpura						
		Harsud	Not available	Not available	Not available	Not available	required proper composting facility, MRF and sanitary landfill	
12	Khandwa	Khandwa	8 Nos of composting pits are available	Not available	MRF facility is Available	Not Available	Sanitary landfill facility is required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
		Pandhana	Not available	Not available	Not available	Not available	required proper composting facility, MRF and sanitary landfill	

		Harda	Yes available	Not available	Yes, temporary facility is available	Not available	composting facility Upgradation, MRF and sanitary landfill facility required	
13	Harda	Khirkiya	Yes available	Not available	Yes available	Not available	MRF and sanitary landfill facility required	https://www.mppcb.mp.gov.in/ PdfView.aspx?h=Harda&pdf= proc/Tech/harda_2023.pdf
		Timurni	Not available	Not available	Yes, temporary facility is available	Not available	required proper composting facility, MRF and sanitary landfill	
14	Hosangabad	Babai	Available	Not available	MRF available but Refused derive Facility (RDF) not available	Not available	Composting technology Upgradation and sanitary landfill required	https://www.mppcb.mp.gov.in/ PdfView.aspx?h=Hoshangabad &pdf=/proc/Tech/Narmadapur am_2023.pdf
		Bankheri	Available	Not available	MRF available but Refused derive	Proposed but not available	Sanitary landfill required	

					Facility (RDF) not available		
						Composting technology	
		Hoshangabad	Yes Available	Not available	MRF available	Not available	Upgradation and sanitary landfill required
		Itarsi	Yes available	Not available	Available	Not available	Need of sanitary Landfill
		Piparia	Available	Not available	Available	Not available	Need of sanitary Landfill
		Seonimalwa	Available	Not available	Yes Available	Not available	sanitary Landfill required
		Sohagpur	Yes available	Not available	Available	Not available	Need of sanitary Landfill
15	Indore	Mhow	Adequate capacity composting pits are available	Not available	03 different grades MRF machines available	Not available	Construction of sanitary landfills is required.
							https://www.mppcb.mp.gov.in/ PdfView.aspx?h=Indore&pdf= /proc/Tech/INDORE_2023.pdf

16	Jabalpur	Jabalpur	Sent to waste to energy plant	Not available	available	available	No Gap
		Kundam	Not available	Not available	Not Available	Not available	
		Majholi	Not available	Not Available	not available	Not available	Wet, dry and non- recyclable waste treatment and disposal facility is required
		Patan	Composting pits are installed	Not Available	yes available	Not available	Sanitary landfill is required for inert and non-recyclable wastes
		Shahpura	Not available	not available	not available	Not available	Wet, dry and non- recyclable waste treatment and disposal facility is required
		Shihora	linked with MSW processing Plant	Not available	available	Sent to waste to energy plant	No Gap

[https://www.mppcb.mp.gov.in/
PdfView.aspx?h=Jabalpur&pd
f=/proc/Tech/Jabalpur_2023.p
df](https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur&pdf=/proc/Tech/Jabalpur_2023.pdf)

			located at Katni					
17	Jhabua (Whole District)	Jhabua	No bulk wet waste generators in the district	Not available	MRF facility is available in all ULBs	No sanitary landfill available in district	Composting Facility and sanitary landfill required in the district.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
		Ranapur						
18	Katni (Whole District)	Bahuriband				All ULBs have a single sanitary landfill site at Katni MSW plant	Composting facility is required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
		Dhimarkhera	Not available	Not available	MRF facility is available in all ULBs			
		Rathi						
19	Mandla	Bichhiya	Not available	Not available	Not available	Not available	Wet, dry and non-recyclable waste treatment and disposal facility is required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf
		Mandla	Not available	Not available	Available	Not available	Composting Facility and sanitary landfill required.	

		Nainpur	Not available	Not available	1 MRF is available	Not available	Composting Facility and sanitary landfill required.
		Niwas	Not available	Not available	1 MRF is available	Not available	Composting Facility and sanitary landfill required.
		Gadarwara	Yes available	Not available	Yes available	Not available	Sanitary Landfill is required
		Gotegaon	Not available	Not available	available	Not available	composting facility Upgradation, MRF and sanitary landfill facility required
20	Narshingpur	Kareli	Not available	Not available	Not available	Not available	composting facility Upgradation, MRF and sanitary landfill facility required
		Narshingpur	Yes available	Not available	available	Not available	Sanitary Landfill is required
		Tendukheda	Not available	Not available	Not available	Not available	composting facility Upgradation, MRF

https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGH PUR_2023.pdf

							and sanitary landfill facility required	
		Bareli	Yes available	Not available	Available	Not available	Sanitary landfill is required	
		Begamganj	Yes available	Not available	Not available	Not available	Sanitary landfill is required	
		Gairatganj	Yes available	Not available	Available	Not available	Sanitary landfill is required	
21	Raisen	Goharganj						https://www.mppcb.mp.gov.in/PdfView.aspx?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf
		Raisen	Yes available	Not available	Available	Not available	Sanitary landfill is required	
		Silvani	Yes available	Not available	Not available	Not available	MRF and sanitary landfill required.	
		Udaipura	Yes available	Not available	Yes available	Not available	Sanitary landfill is required	
		Deori	Yes available	Not available	MRF is available	Yes available	No Gap	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sagar&pdf=/proc/Tech/Sagar_2023.pdf
22	Sagar	Kesli	Not available	Not available	Not Available	Not available	Not available	

23	Sehore	Ashta	Not available	Not available	Not available	Not available	Composting Facility, MRF and sanitary landfill required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHORE_2023.pdf
		Budhni	Not available	Not available	Available	Not available	Composting Facility and sanitary landfill required.	
		Ichawar						
		Nasrullaganj	Not available	Not available	Available	Not available	sanitary landfill required.	
		Sehore	Not available	Not available	Available	Not available	Composting Facility and sanitary landfill required.	
24	Seoni	Ghansaur	Not available	Not available	Not available	Not available	Composting Facility MRF and sanitary landfill required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
		Lakhnadon	Not available	Not available	Not available	Not available	Composting Facility MRF and	

							sanitary landfill required.	
25	Umaria	Bandhogarh (Whole District)	Available in all ULBs	Not available	MRF facilities exist in all ULBs of district	Yes available	Facilities need upgradation	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Umaria&pdf=/proc/Tech/UMARIYA_2023.pdf
		Barwah	Not available	Not available	Not available	Not available	Composting Facility, MRF and sanitary landfill are required.	
		Bhagwanpura	Not available	Not available	Not Available	Not available	Not available	
26	West Nimar	Bhikangaon	Not available	Not available	Not Available	Not Required	Composting Facility and MRF facility required.	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Damoh.pdf
		Jhirmiya	Not available	Not available	Not Available	Not available	Not available	
		Kasrawas	Not available	Not available	Not available	Not required	Composting Facility, MRF and sanitary landfill are required.	

Khargone	Yes available	Not available	2 MRF Available	Not available	sanitary landfill required.
Maheshwar	Not available	Not available	Not Available	Not available	Composting Facility, MRF and sanitary landfill are required.
Seagon	Not available	Not available	Not Available	Not available	Not available

*Source: MPCB District environment plan

Table A - 8 Comprehensive overview of legacy waste dumpsites and projected timelines for complete processing in Narmada basin ULBs

District	Tehsil / ULB	Legacy Waste (MT)	Status	Date
Alirajpur	Alirajpur	24,858	Work Started	25-Jan
Alirajpur	Bhavra	11,091	Work Started	25-Mar
Alirajpur	Jobat	18,304	Work Started	25-Mar
Barwani	Badwani	37,141	Work Started	25-Mar
Barwani	Rajpur	27,476	Work Started	25-Mar
Betul	Multai	16,405	Work Started	24-Nov
Betul	Betul	1,33,958	Work Started	25-Mar
Burhanpur	Nepanagar	44,249	Financial Evaluation Pending	25-May
Dewas	Dewas	90,183	Work Started	25-Jan
Dewas	Khategaon	24,889	Work Started	25-Jan
Dhar	Dhar	1,55,065	Work Started	24-Nov
Dhar	Kukshi	43,830	Work Started	25-Mar
Dhar	Manawar	59,292	Work Started	25-Mar
Dhar	Pithampur	35,716	Received at State SLTC Pending	25-May
Harda	Harda	14,678	Work Started	24-Nov
Harda	Timarni	4,692	Received at State SLTC Pending	25-May
Hoshangabad (Narmadapuram)	Seoni Malwa	21,207	Work Started	25-Mar
Hoshangabad (Narmadapuram)	Itarsi	31,438	Work Started	25-Mar
Hoshangabad (Narmadapuram)	Betul (part)	1,33,958	Work Started	25-Mar
Hoshangabad (Narmadapuram)	Narmadapuram	62,296	Work Started	25-Mar
Hoshangabad (Narmadapuram)	Sohagpur	11,730	Financial Evaluation Pending	25-May

Indore	Dhar (repetition but basin linked)	1,55,065	Work Started	24-Nov
Indore	Barwaha	54,268	Work Started	25-Jan
Indore	Mandleshwar	13,630	Work Started	25-Jan
Indore	Maheshwar	42,680	Work Started	25-Jan
Indore	Alirajpur (repeated, basin linked)	24,858	Work Started	25-Jan
Indore	Jhabua	63,283	Work Started	25-Mar
Indore	Thandla	20,583	Work Started	25-Mar
Indore	Ranapur	36,988	Financial Evaluation Pending	25-May
Indore	Pithampur	35,716	Received at State SLTC Pending	25-May
Indore	Mhowgaon	20,383	Received at State SLTC Pending	25-May
Indore	Petlawad	31,489	Received at State SLTC Pending	25-May
Indore	Chhanera	24,172	Received at State SLTC Pending	25-May
Indore	Bhikangaon	35,332	Received at State SLTC Pending	25-May
Indore	Betma	28,913	Received at State SLTC Pending	25-May
Indore	Kasrawad	17,641	Received at State SLTC Pending	25-May
Jabalpur	Balaghat	65,761	Work Started	25-Mar
Jabalpur	Mandla	60,901	Work Started	25-Jan
Jabalpur	Nainpur	17,440	Work Started	25-Jan
Jabalpur	Pandhurna	45,739	Work Started	25-Mar
Jabalpur	Junnardeo	8,645	Received at State SLTC Pending	25-May

Jabalpur	Seoni	1,57,664	Technical Pending	25-Jul
Mandla	Mandla (district-level overlap with Jabalpur)	60,901	Work Started	25-Jan
Raisen	Mandideep	83,617	Work Started	24-Nov
Raisen	Bareli	16,436	Work Started	25-Jan
Sehore	Nasrullaganj	12,636	Work Started	25-Mar
Sagar	Jatara	10,358	Work Started	24-Nov
Sagar	Hatta	6,799	Work Started	25-Jan
Sagar	Damoh	9,289	Financial Evaluation Pending	25-May
Sagar	Badagaon	1,958	Received at State SLTC Pending	25-May

Sources: Six Monthly Progress Report by Govt. of Madhya Pradesh in OA No. 606 of 2018 for State of Madhya Pradesh (Compliance of MSW Management Rules, 2016 and Other Environmental Issues), National Green Tribunal, September 2025

Table A - 9 Wet waste processing facilities under the Narmada River basin ULBs

S.No	District Name	ULB Code	ULB Name	Plant Type	To	Sub Category
1	Alirajpur	802433	Alirajpur (M)	Waste Compost	To	Aerobic (Windrow Compost) Without RDF
2	Alirajpur	802432	Jobat (NP)	Waste Compost	To	Aerobic (Windrow Compost) Without RDF
3	Anuppur	802421	Amarkantak	Waste Compost	To	Pit-Based / Vermi
4	Anuppur	802419	Anuppur (M)	Waste Compost	To	Aerobic (Windrow Compost) Without RDF

5	Anuppur	900691	Bangawan (Rajnagar)	Waste To Compost	Pit-Based / Vermi
6	Anuppur	802416	Bijuri (M)	Waste To Compost	Aerobic (Windrow Compost) With RDF
7	Anuppur	900692	Dola	Waste To Compost	Pit-Based / Vermi
8	Anuppur	900693	Dumarkachar	Waste To Compost	Pit-Based / Vermi
9	Anuppur	802420	Jaithari	Waste To Compost	Aerobic (Windrow Compost) Without RDF
10	Anuppur	802417	Kotma	Waste To Compost	Aerobic (Windrow Compost) Without RDF
11	Anuppur	802418	Pasan (M)	Waste To Compost	Aerobic (Windrow Compost) With RDF
12		900156	Shadhora	Waste To Compost	Pit-Based / Vermi
13	Balaghat	802398	Baihar (NP)	Waste To Compost	Pit-Based / Vermi
14		802397	Balaghat (M)	Waste To Compost	Pit-Based / Vermi
15	Barwani	802286	Anjad (NP)	Waste To Compost	Aerobic (Windrow Compost) With RDF
16		802285	Badwani	Waste To Compost	Pit-Based / Vermi
17		900677	Niwali Burjurg	Waste To Compost	Aerobic (Windrow Compost) With RDF
18		802289	Pansemal (NP)	Waste To Compost	Aerobic (Windrow Compost) Without RDF
19		802287	Rajpur (M)	Waste To Compost	Pit-Based / Vermi

20		802291	Sendhwa (M)	Waste To Aerobic (Windrow Compost Compost) Without RDF
21		900678	Thikari	Waste To Aerobic (Windrow Compost Compost) With RDF
22	Betul	802340	Amla	Waste To Pit-Based / Vermi Compost
23		802334	Athner (NP)	Waste To Pit-Based / Vermi Compost
24		802335	Betul (M)	Waste To Aerobic (Windrow Compost Compost)
25	Betul	802336	Betul-Bazar (NP)	Waste To Pit-Based / Vermi Compost
26		802333	Bhainsdehi (NP)	Waste To Pit-Based / Vermi Compost
27		802339	Multai (M)	Waste To Pit-Based / Vermi Compost
28		900703	Shahpur	Waste To Pit-Based / Vermi Compost
29	Bhopal	802311	Berasia	Waste To Pit-Based / Vermi Compost
30		802312	Bhopal (M Corp.)	Waste To Aerobic (Windrow Compost Compost) Without RDF
31		802312	Bhopal (M Corp.)	Waste To Anaerobic Without Compost RDF
32	Burhanpur	802439	Burhanpur (M Corp.)	Waste To Aerobic (Windrow Compost Compost) Without RDF
33		802441	Nepanagar	Waste To Pit-Based / Vermi Compost
34	Chhindwara	802377	Amarwara (M)	Waste To Pit-Based / Vermi Compost

35		802380	Jamai (M)	Waste To Pit-Based / Vermi Compost
36	Damoh	802167	Damoh	Waste To Pit-Based / Vermi Compost
37		802168	Tendukheda (NP)	Waste To Pit-Based / Vermi Compost
38	Dewas	802254	Bagli (NP)	Waste To Pit-Based / Vermi Compost
39		802248	Dewas (M Corp.)	Waste To Aerobic ... (incomplete in your text)
40	Dewas	802248	Dewas (M Corp.)	Waste To Aerobic (Windrow Compost) Without RDF
41		802249	Kannod (NP)	Waste To Aerobic (Windrow Compost) With RDF
42		802251	Kantaphod (NP)	Waste To Pit-Based / Vermi Compost
43		802253	Karnawad	Waste To Pit-Based / Vermi Compost
44		802256	Khategaon (NP)	Waste To Pit-Based / Vermi Compost
45		802244	Tonk Khurd (NP)	Waste To Pit-Based / Vermi Compost
46	Dhar	802257	Badnawar (NP)	Waste To Pit-Based / Vermi Compost
47		802264	Dahi (NP)	Waste To Aerobic (Windrow Compost) Without RDF
48		802266	Dhamnod (NP)	Waste To Pit-Based / Vermi Compost
49		802260	Dhar (M)	Waste To Pit-Based / Vermi Compost

50		802267	Dharampuri (NP)	Waste To Pit-Based / Vermi Compost
51		802263	Kukshi (NP)	Waste To Aerobic (Windrow Compost) With RDF
52		802265	Manawar (M)	Waste To Pit-Based / Vermi Compost
53		802262	Mandav (NP)	Waste To Pit-Based / Vermi Compost
54		802261	Pithampur (M)	Waste To Aerobic (Windrow Compost) Without RDF
55		802258	Rajgarh (NP)	Waste To Pit-Based / Vermi Compost
56		802259	Sardarpur	Waste To Aerobic (Windrow Compost) Without RDF
57	Dindori	802371	Dindori (NP)	Waste To Pit-Based / Vermi Compost
58	-	802370	Shahpura (NP)	Waste To Pit-Based / Vermi Compost
59	Harda	802342	Harda (M)	Waste To Aerobic (Windrow Compost) Without RDF
60	Harda	802341	Khirkiya	Waste To - Compost
61	Harda	900704	Sirali	Waste To Pit-Based / Vermi Compost
62	Harda	802343	Timarni (NP)	Waste To Pit-Based / Vermi Compost
63	Hoshangabad	802347	Babai (NP)	Waste To Pit-Based / Vermi Compost
64	Hoshangabad	802346	Hoshangabad (M)	Waste To Pit-Based / Vermi Compost

65	Hoshangabad	802345	Itarsi (M)	Waste To Compost	Pit-Based / Vermi
66	Hoshangabad	802350	Pachmarhi Cantt	Waste To Compost	Aerobic (Windrow Compost) Without RDF
67	Hoshangabad	802349	Pipariya (M)	Waste To Compost	Pit-Based / Vermi
68	Hoshangabad	802344	Seoni-Malwa (M)	Waste To Compost	Anaerobic Without RDF
69	Hoshangabad	802348	Sohagpur (NP)	Waste To Compost	Pit-Based / Vermi
70	Hoshangabad	900171	Vankhedi	Waste To Compost	Aerobic (Windrow Compost) Without RDF
71	Indore	802270	Betma	Waste To Compost	Pit-Based / Vermi
72	Indore	802269	Depalpur (NP)	Waste To Compost	Pit-Based / Vermi
73	Indore	802271	Hatod (NP)	Waste To Compost	Pit-Based / Vermi
74	Indore	802273	Indore	Waste To Compost	Aerobic (Windrow Compost) Without RDF
75	Indore	802273	Indore	Waste To Compost	Aerobic (Windrow Compost) Without RDF
76	Indore	802273	Indore	Waste To Compost	Aerobic (Windrow Compost) Without RDF
77	Indore	802273	Indore	Waste To Compost	Aerobic (Windrow Compost) Without RDF

78	Indore	802273	Indore	Waste To Aerobic (Windrow Compost Compost) Without RDF
79	Indore	802273	Indore	Waste To Aerobic (Windrow Compost Compost) Without RDF
80	Indore	802273	Indore	Waste To Aerobic (Windrow Compost Compost) Without RDF
81	Indore	802273	Indore	Waste To Aerobic (Windrow Compost Compost) Without RDF
82	Indore	802273	Indore	Waste To Aerobic (Windrow Compost Compost) Without RDF
83	Indore	802273	Indore	Waste To Aerobic (Windrow Compost Compost) Without RDF
84	Indore	802273	Indore	Waste To Aerobic (Windrow Compost Compost) Without RDF
85	Indore	802275	Mhow Cantt	Waste To Aerobic (Windrow Compost Compost) Without RDF
86	Indore	802276	Mhowgaon (NP)	Waste To Pit-Based / Vermi Compost
87	Indore	802274	Rau (NP)	Waste To Pit-Based / Vermi Compost
88	Indore	802268	Runji-Gautampura (NP)	Waste To Pit-Based / Vermi Compost
89	Indore	802272	Sanwer	Waste To Pit-Based / Vermi Compost

90	Jabalpur	802363	Barela (NP)	Waste To Pit-Based / Vermi Compost
91	Jabalpur	802362	Bhedaghat	Waste To Aerobic (Windrow Compost) Without RDF
92	Jabalpur	802360	Jabalpur Cantt	Waste To Aerobic (Windrow Compost) Without RDF
93	Jabalpur	802357	Katangi (NP)	Waste To Pit-Based / Vermi Compost
94	Jabalpur	802356	Manjholi_J	Waste To Pit-Based / Vermi Compost
95	Jabalpur	802364	Panagar	Waste To Pit-Based / Vermi Compost
96	Jabalpur	802358	Patan_MP	Waste To Pit-Based / Vermi Compost
97	Jabalpur	802359	Shahpur Bhitoni	Waste To Pit-Based / Vermi Compost
98	Jhabua	802429	Jhabua (M)	Waste To Pit-Based / Vermi Compost
99	Jhabua	900157	Meghnagar	Waste To Pit-Based / Vermi Compost
100	Jhabua	802428	Petlawad (NP)	Waste To Pit-Based / Vermi Compost
101	Jhabua	802430	Ranapur	Waste To Pit-Based / Vermi Compost
102	Jhabua	802427	Thandla (NP)	Waste To Pit-Based / Vermi Compost
103	Khandwa	802434	Chhanera (NP)	Waste To Pit-Based / Vermi Compost
104	Khandwa	802435	Khandwa (M Corp.)	Waste To Aerobic (Windrow Compost) Without RDF

105	Khandwa	802437	Mundi (NP)	Waste To Pit-Based / Vermi Compost
106	Khandwa	802436	Omkareshwar (NP)	Waste To Pit-Based / Vermi Compost
107	Khandwa	802438	Pandhana (NP)	Waste To Pit-Based / Vermi Compost
108	Khargone	802278	Barwaha (M)	Waste To Pit-Based / Vermi Compost
109	Khargone	802283	Bhikangaon (NP)	Waste To Pit-Based / Vermi Compost
110	Khargone	900679	Bistaan	Waste To Pit-Based / Vermi Compost
111	Khargone	900158	Karahi Padliya Khurd	Waste To Pit-Based / Vermi Compost
112	Khargone	802282	Kasrawad	Waste To Pit-Based / Vermi Compost
113	Khargone	802284	Khargone (M)	Waste To Aerobic (Windrow Compost) Without RDF
114	Khargone	802280	Maheshwar (NP)	Waste To Pit-Based / Vermi Compost
115	Khargone	802281	Mandleshwar	Waste To Pit-Based / Vermi Compost
116	Khargone	802279	Sanawad (M)	Waste To Pit-Based / Vermi Compost
117	Mandla	802374	Bamhani (NP)	Waste To Pit-Based / Vermi Compost
118	Mandla	802375	Bhua Bhicchhia	Waste To Pit-Based / Vermi Compost
119	Mandla	802373	Mandla	Waste To Pit-Based / Vermi Compost
120	Mandla	802376	Nainpur	Waste To Aerobic (Windrow Compost) With RDF

121	Mandla	802372	Niwas	Waste To Pit-Based / Vermi Compost
122	Murwara (Katni)	802351	Katni	Waste To Aerobic (Windrow Compost) With RDF
123	Narsimhapur	900163	Chichli	Waste To Pit-Based / Vermi Compost
124	Narsimhapur	802366	Gadarwara (M)	Waste To Pit-Based / Vermi Compost
125	Narsimhapur	802365	Gotegaon (NP)	Waste To Aerobic (Windrow Compost) Without RDF
126	Narsimhapur	802368	Kareli (M)	Waste To Pit-Based / Vermi Compost
127	Narsimhapur	802367	Narsinghpur	Waste To Pit-Based / Vermi Compost
128	Narsimhapur	900159	Sainkheda	Waste To Pit-Based / Vermi Compost
129	Narsimhapur	900162	Salichouka (Bawaikalan)	Waste To Aerobic (Windrow Compost) Without RDF
130	Narsimhapur	802369	Tendukheda (NP)	Waste To Pit-Based / Vermi Compost
131	Raisen	802330	Baadi	Waste To Pit-Based / Vermi Compost
132	Raisen	802329	Bareli	Waste To Pit-Based / Vermi Compost
133	Raisen	802325	Begamganj (M)	Waste To Pit-Based / Vermi Compost
134	Raisen	802324	Gairatganj (NP)	Waste To Pit-Based / Vermi Compost
135	Raisen	802327	Mandideep	Waste To Pit-Based / Vermi Compost

136	Raisen	802328	Obedullaganj (NP)	Waste To Pit-Based / Vermi Compost
137	Raisen	802323	Raisen (M)	Waste To Pit-Based / Vermi Compost
138	Raisen	802322	Sanchi (NP)	Waste To Pit-Based / Vermi Compost
139	Raisen	802331	Silwani	Waste To Pit-Based / Vermi Compost
140	Raisen	802326	Sultanpur_R	Waste To Pit-Based / Vermi Compost
141	Raisen	802332	Udaipura	Waste To Pit-Based / Vermi Compost
142	Sagar	802159	Sagar (M Corp.)	Waste To Anaerobic With Compost RDF
143	Sagar	802160	Sagar Cantt (CB)	Waste To Pit-Based / Vermi Compost
144	Sehore	802315	Ashta (M)	Waste To Pit-Based / Vermi Compost
145	Sehore	802320	Budni (NP)	Waste To Pit-Based / Vermi Compost
146	Sehore	802318	ICHHAWAR	Waste To Pit-Based / Vermi Compost
147	Sehore	802317	Jawar (NP)	Waste To Pit-Based / Vermi Compost
148	Sehore	802316	Kothri (NP)	Waste To Pit-Based / Vermi Compost
149	Sehore	802319	Nasrullaganj (NP)	Waste To Pit-Based / Vermi Compost
150	Sehore	802321	Rehti (NP)	Waste To Pit-Based / Vermi Compost
151	Sehore	802314	Sehore (M)	Waste To Pit-Based / Vermi Compost

152	Sehore	900127	Shahganj Parishad	Nagar	Waste To Compost	Pit-Based / Vermi
153	Seoni	802394	BARGHAT		Waste To Compost	Aerobic (Windrow Compost) Without RDF
154	Seoni	900688	CHAPARA		Waste To Compost	— (Sub Category missing in data provided)
155	Seoni	900689	KEWLARI		Waste To Compost	Pit-Based / Vermi
156	Seoni	802392	LAKHNADOAN		Waste To Compost	Pit-Based / Vermi
157	Seoni	802393	Seoni (M)		Waste To Compost	Pit-Based / Vermi

Source: Solid Waste Management Annual Report 2022-23, Madhya Pradesh Pollution Control Board, Letter No. 2195 Dt. 29.8.2023

Table A - 10 List of dry waste processing facilities under the Narmada River basin ULBs

S.No	District Name	ULB Code	ULB Name	Plant Type	Sub Category
1	Alirajpur	802433	Alirajpur (M)	Material Recovery Facility (MRF)	Semi Automated
2		802432	Jobat (NP)	Material Recovery Facility (MRF)	Manual
3	Anuppur	802421	Amarkantak	Material Recovery Facility (MRF)	Manual

4	802419	Anuppur (M)	Material Recovery Facility (MRF)	Manual
5	900691	Bangawan (Rajnagar)	Material Recovery Facility (MRF)	Manual
6	802416	Bijuri (M)	Material Recovery Facility (MRF)	Manual
7	900692	Dola	Material Recovery Facility (MRF)	Manual
8	900693	Dumarkachar	Material Recovery Facility (MRF)	Manual
9	802420	Jaithari	Material Recovery Facility (MRF)	Manual
10	802417	Kotma (M)	Material Recovery Facility (MRF)	Manual
11	802418	Pasan (M)	Material Recovery Facility (MRF)	Manual

12	Balaghat	802398	Baihar (NP)	Material Recovery Facility (MRF)	Manual
13		802397	Balaghat (M)	Material Recovery Facility (MRF)	Semi Automated
14	Barwani	802286	Anjad (NP)	Material Recovery Facility (MRF)	Manual
15		802285	Barwani (M)	Material Recovery Facility (MRF)	Semi Automated
16		900677	Niwali Burjurg	Material Recovery Facility (MRF)	Manual
17		802288	Palsud (NP)	Material Recovery Facility (MRF)	Manual
18		802289	Pansemal (NP)	Material Recovery Facility (MRF)	Manual
19		802287	Rajpur (NP)	Material Recovery Facility (MRF)	Manual

20		802291	Sendhwa (M)	Material Recovery Facility (MRF)	Manual
21	Betul	802340	Amla	Material Recovery Facility (MRF)	Manual
22		802335	Betul (M)	Material Recovery Facility (MRF)	Manual
23		802336	Betul-Bazar (NP)	Material Recovery Facility (MRF)	Manual
24		802333	Bhainsdehi (NP)	Material Recovery Facility (MRF)	Manual
25		802339	Multai (M)	Material Recovery Facility (MRF)	Manual
26		900703	Shahpur	Material Recovery Facility (MRF)	Manual
27	Bhopal	802311	Berasia (M)	Material Recovery Facility (MRF)	Manual

28		802312	Bhopal (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
29		802312	Bhopal (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
30		802312	Bhopal (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
31		802312	Bhopal (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
32		802312	Bhopal (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
33		802312	Bhopal (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
34	Burhanpur	802439	Burhanpur (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
35		802441	Nepanagar (M)	Material Recovery Facility (MRF)	Manual

36		802440	Shahpur (NP)	Material Recovery Facility (MRF)	Manual
37	Chhindwara	802377	Amarwara (M)	Material Recovery Facility (MRF)	Manual
38		802379	Chaurai Khas (M)	Material Recovery Facility (MRF)	Manual
39		802386	Chhindwara	Material Recovery Facility (MRF)	Semi Automated
40		802380	Junnardeo	Material Recovery Facility (MRF)	Semi Automated
41	Damoh	802167	Damoh (M)	Material Recovery Facility (MRF)	Manual
42		802164	Hatta (M)	Material Recovery Facility (MRF)	Manual
43		802166	Hindoriya	Material Recovery Facility (MRF)	Manual

44		900165	Patera	Material Recovery Facility (MRF)	Manual
45		802165	Pathariya	Material Recovery Facility (MRF)	Manual
46		802168	Tendukheda (NP)	Material Recovery Facility (MRF)	Manual
47	Dewas	802254	Bagli (NP)	Material Recovery Facility (MRF)	Manual
48		802245	Bhaurasa (NP)	Material Recovery Facility (MRF)	Manual
49		802248	Dewas (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
50		802255	Hatpipliya	Material Recovery Facility (MRF)	Manual
51		802249	Kannod (NP)	Material Recovery Facility (MRF)	Manual

52	802251	Kantaphod (NP)	Material Recovery Facility (MRF)	Manual
53	802253	Karnawad	Material Recovery Facility (MRF)	Manual
54	802256	Khategaon	Material Recovery Facility (MRF)	Manual
55	802256	Khategaon (NP)	Material Recovery Facility (MRF)	Manual
56	802250	Loharda (NP)	Material Recovery Facility (MRF)	Manual
57	900166	Nemawar	Material Recovery Facility (MRF)	Manual
58	802247	Pipalrawan (NP)	Material Recovery Facility (MRF)	Manual
59	802252	Satwas (NP)	Material Recovery Facility (MRF)	Manual

60		802246	Sonkatch (NP)	Material Recovery Facility (MRF)	Manual
61		802244	Tonk Khurd (NP)	Material Recovery Facility (MRF)	Manual
62	Dhar	802257	Badnawar (NP)	Material Recovery Facility (MRF)	Semi Automated
63		802264	Dahi (NP)	Material Recovery Facility (MRF)	Manual
64		802266	Dhamnood (NP)	Material Recovery Facility (MRF)	Manual
65		802260	Dhar (M)	Material Recovery Facility (MRF)	Semi Automated
66	Dhar	802263	Kukshi (NP)	Material Recovery Facility (MRF)	Semi Automated
67	Dhar	802265	Manawar (M)	Material Recovery Facility (MRF)	Manual

68	Dhar	802262	Mandav (NP)	Material Recovery Facility (MRF)	Manual
69	Dhar	802261	Pithampur (M)	Material Recovery Facility (MRF)	Semi Automated
70	Dhar	802258	Rajgarh (NP)	Material Recovery Facility (MRF)	Manual
71	Dhar	802259	Sardarpur (NP)	Material Recovery Facility (MRF)	Fully Automated
72	Dindori	802371	Dindori (NP)	Material Recovery Facility (MRF)	Manual
73	Dindori	802370	SHAHPURA_D	Material Recovery Facility (MRF)	Manual
74	Harda	802341	KHIRKIYA	Material Recovery Facility (MRF)	Manual
75	Harda	900704	SIRALI	Material Recovery Facility (MRF)	Manual

76	Harda	802343	Timarni (NP)	Material Recovery Facility (MRF)	Manual
77	Hoshangabad	802347	Babai (NP)	Material Recovery Facility (MRF)	Manual
78	Hoshangabad	802346	Hoshangabad (M)	Material Recovery Facility (MRF)	Semi Automated
79	Hoshangabad	802345	Itarsi (M)	Material Recovery Facility (MRF)	Manual
80	Hoshangabad	802350	PACHMARHI CANTT	Material Recovery Facility (MRF)	Manual
81	Hoshangabad	802349	Pipariya (M)	Material Recovery Facility (MRF)	Manual
82	Hoshangabad	802344	SEONI MALWA	Material Recovery Facility (MRF)	Manual
83	Hoshangabad	802348	Sohagpur (NP)	Material Recovery Facility (MRF)	Manual

84	Hoshangabad	900171	Vankhedi	Material Recovery Facility (MRF)	Manual
85	Indore	802270	BETMA	Material Recovery Facility (MRF)	Manual
86	Indore	802269	Depalpur (NP)	Material Recovery Facility (MRF)	Manual
87	Indore	802271	Hatod (NP)	Material Recovery Facility (MRF)	Manual
88	Indore	802273	INDORE	Material Recovery Facility (MRF)	Manual
89	Indore	802273	INDORE	Material Recovery Facility (MRF)	Manual
90	Indore	802273	INDORE	Material Recovery Facility (MRF)	Manual
91	Indore	802273	INDORE	Material Recovery Facility (MRF)	Semi Automated

92	Indore	802273	INDORE	Material Recovery Facility (MRF)	Fully Automated
93	Indore	802273	INDORE	Material Recovery Facility (MRF)	Manual
94	Indore	802277	Manpur (NP)	Material Recovery Facility (MRF)	Manual
95	Indore	802275	Mhow Cantt (CB)	Material Recovery Facility (MRF)	Manual
96	Indore	802276	Mhowgaon (NP)	Material Recovery Facility (MRF)	Manual
97	Indore	802274	Rau (NP)	Material Recovery Facility (MRF)	Manual
98	Indore	802268	Runji-Gautampura (NP)	Material Recovery Facility (MRF)	Manual
99	Indore	802272	SANWER	Material Recovery Facility (MRF)	Manual

100	Shahdol	900694	Bakho	Material Recovery Facility (MRF)	Manual
101	Jabalpur	802363	Barela (NP)	Material Recovery Facility (MRF)	Manual
102	Jabalpur	802363	BERALA_M	Material Recovery Facility (MRF)	Manual
103	Jabalpur	802362	BHEDAGHAT	Material Recovery Facility (MRF)	Manual
104	Jabalpur	802360	JABALPUR CANTT	Material Recovery Facility (MRF)	Manual
105	Jabalpur	802357	Katangi (NP)	Material Recovery Facility (MRF)	Manual
106	Jabalpur	802356	MANJHOLI_J	Material Recovery Facility (MRF)	Manual
107	Jabalpur	802364	PANAGAR	Material Recovery Facility (MRF)	Manual

108	Jabalpur	802358	PATAN_MP	Material Recovery Facility (MRF)	Manual
109	Jabalpur	802359	SHAHPUR BHITONI	Material Recovery Facility (MRF)	Manual
110	Jhabua	802429	Jhabua (M)	Material Recovery Facility (MRF)	Manual
111	Jhabua	900157	MEGHNAGAR	Material Recovery Facility (MRF)	Fully Automated
112	Jhabua	802428	Petlawad (NP)	Material Recovery Facility (MRF)	Manual
113	Jhabua	802430	RANAPUR	Material Recovery Facility (MRF)	Manual
114	Jhabua	802427	Thandla (NP)	Material Recovery Facility (MRF)	Manual
115	Khandwa	802434	Chhanera (NP)	Material Recovery Facility (MRF)	Manual

116	Khandwa	802435	Khandwa (M Corp.)	Material Recovery Facility (MRF)	Semi Automated
117	Khandwa	802437	Mundi (NP)	Material Recovery Facility (MRF)	Fully Automated
118	Khandwa	802436	Omkareshwar (NP)	Material Recovery Facility (MRF)	Manual
119	Khandwa	802438	PANDHANA	Material Recovery Facility (MRF)	Manual
120	Khargaon	802278	Barwaha (M)	Material Recovery Facility (MRF)	Manual
121	Khargaon	802283	Bhikangaon (NP)	Material Recovery Facility (MRF)	Manual
122	Khargaon	900679	BISTAAN	Material Recovery Facility (MRF)	Semi Automated
123	Khargaon	900158	KARAH PADLIYA KHURD	Material Recovery Facility (MRF)	Manual

124	Khargaon	802282	KASRAWAD	Material Recovery Facility (MRF)	Manual
125	Khargaon	802284	KHARGONE	Material Recovery Facility (MRF)	Semi Automated
126	Khargaon	802280	Maheshwar (NP)	Material Recovery Facility (MRF)	Manual
127	Khargaon	802281	MANDLESHWAR	Material Recovery Facility (MRF)	Manual
128	Khargaon	802279	SANAWAD	Material Recovery Facility (MRF)	Semi Automated
129	Mandla	802374	Bamhani (NP)	Material Recovery Facility (MRF)	Manual
130	Mandla	802375	BHUA BHICHHIA	Material Recovery Facility (MRF)	Manual
131	Mandla	802373	Mandla (M)	Material Recovery Facility (MRF)	Manual

132	Mandla	802376	Nainpur (M)	Material Recovery Facility (MRF)	Manual
133	Mandla	802372	NIWAS	Material Recovery Facility (MRF)	Manual
134	Murwara (Katni)	802351	KATNI	Material Recovery Facility (MRF)	Semi Automated
135	Narsimhapur	900163	CHICHLI	Material Recovery Facility (MRF)	Manual
136	Narsimhapur	802366	Gadarwara (M)	Material Recovery Facility (MRF)	Semi Automated
137	Narsimhapur	802365	Gotegaon (NP)	Material Recovery Facility (MRF)	Manual
138	Narsimhapur	802368	Kareli (M)	Material Recovery Facility (MRF)	Manual
139	Narsimhapur	802367	NARSINGHPUR	Material Recovery Facility (MRF)	Semi Automated

140	Narsimhapur	900159	SAINKHEDA	Material Recovery Facility (MRF)	Manual
141	Narsimhapur	900162	Salichouka (Bawaikalan)	Material Recovery Facility (MRF)	Manual
142	Narsimhapur	802369	Tendukheda (NP)	Material Recovery Facility (MRF)	Manual
143	Raisen	802330	BAADI	Material Recovery Facility (MRF)	Manual
144	Raisen	802329	Baraily (NP)	Material Recovery Facility (MRF)	Manual
145	Raisen	802325	Begamganj (M)	Material Recovery Facility (MRF)	Manual
146	Raisen	802324	Gairatganj (NP)	Material Recovery Facility (MRF)	Manual
147	Raisen	802327	MANDIDEEP	Material Recovery Facility (MRF)	Manual

148	Raisen	802328	Obedullaganj (NP)	Material Recovery Facility (MRF)	Manual
149	Raisen	802323	Raisen (M)	Material Recovery Facility (MRF)	Manual
150	Raisen	802322	Sanchi (NP)	Material Recovery Facility (MRF)	Manual
151	Raisen	802331	SILWANI	Material Recovery Facility (MRF)	Manual
152	Raisen	802326	Sultanpur (NP)	Material Recovery Facility (MRF)	Manual
153	Raisen	802332	UDAIPURA	Material Recovery Facility (MRF)	Manual

Source: Solid Waste Management Annual Report 2022-23, Madhya Pradesh Pollution Control Board, Letter No. 2195 Dt. 29.8.2023

Table A - 11 District-wise Status of Hazardous Waste Management in Narmada Basin, Madhya Pradesh

Hazardous Waste Management						
S.No.	Districts	Block (ULB)	Collection center	Location of treatment and Disposal	Gaps and Challenges	References
1	Alirajpur	Alirajpur Bhavara Jobat	yes Available	TSDf facility Dhar district.	Required skills training for the personnel working in the sites	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Alirajpur.pdf
2	Anuppur (Whole District)	District	Not Available	TSDf facility Dhar district.	Required skills training for the personnel working in the sites	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	Baihar	Not required	TSDf facility Dhar district.	01 TSDf is proposed in the Mandla district at Industrial Area Maneri	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
4	Barwani	Barwani Niwali Pansemal	Not Available	TSDf facility Dhar district.	persons working in the site with appropriate training, equipment and the information necessary to ensure their safety	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Barwani_2023.pdf

		Rajpur				
		Sendhwa				
		Thikiri				
		Amla				
		Betul		TSDF facility Dhar		
5	Betul (Whole District)	Bhaisdehi	Available	district.	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
		Multai				
		Shahpur				
6	Burhanpur	Nepanagar	Not available	TSDF, Dhar district	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
		Amarwara				
7	Chindwara (Whole District)	Junnardev	1 Collection center in district	TSDF, Dhar district	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
		Tamiya				
8	Damoh	Tendukheda	Not available	Not linked with the facilities of TSDF	ULBs are required to be linked with TSDF and Waste be sent as per rule	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
		Bagli				
9	Dewas	Dewas	Yes Available	TSDF, Dhar district	No gap identified	

		Kannod				https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
		Khategaon				
		Dhar				
		Dharamपुरi				
10	Dhar (Whole District)	Gandhwani	Yes Available	TSDF, Dhar district	In all ULBs collection units must be established.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf
		Kukshi				
		Manawar				
		Sardarpur				
		Dindori				
11	Dindori (Whole District)	Shahpura	Yes Available	TSDF, Dhar district	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DIN DORI_2023.pdf
		Harsud				
12	Khandwa	Khandwa	Not Available	TSDF, Dhar District	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
		Pandhana				
		Harda				
13	Harda	Khirkiya	yes Available	TSDF, Dhar District	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf
		Timurni				

		Babai				
		Bankheri				
		Hoshangabad				
14	Hosangabad	Itarsi	No Collection Center available	TSDF, Dhar District	Collection centers are required to be established	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Hoshangabad&pdf=/proc/Tech/Narmadapuram_2023.pdf
		Piparia				
		Seonimalwa				
		Sohagpur				
15	Indore	Mhow	Not Available	TSDF, Dhar District	Not Available	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Indore&pdf=/proc/Tech/INDORE_2023.pdf
		Jabalpur				
		Kundam				
16	Jabalpur	Majholi	Yes available	TSDF, Dhar District	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur&pdf=/proc/Tech/Jabalpur_2023.pdf
		Patan				
		Shahpura				
		Shihora				
17		Jhabua	Not available	TSDF, Dhar District		

	Jhabua (Whole District)	Ranapur			Hazardous waste collections are required in ULBs	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
18	Katni (Whole District)	Bahuriband Dhimarkhera Rathi Bichhiya	Not available	TSDF, Dhar District	Collection centers are required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
19	Mandla	Mandla Nainpur Niwas Gadarwara Gotegaon	Yes available	TSDF, Dhar District	No gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf
20	Narshingpur	Kareli Narshingpur Tendukheda Bareli	Yes available	TSDF, Dhar District	No gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGHPUR_2023.pdf
21	Raisen	Begamganj Gairatganj Goharganj	Directly linked with TSDF	TSDF, Dhar District	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf

		Raisen				
		Silvani				
		Udaipura				
22	Sagar	Deori	Not required	TSDF Dhar	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sagar&pdf=/proc/Tech/Sagar_2023.pdf
		Kesli				
		Ashta				
		Budhni				
23	Sehore	Ichawar	Yes available	TSDF, Dhar District	No gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHORE_2023.pdf
		Nasrullaganj				
		Sehore				
		Ghansaur				
24	Seoni	Lakhnadon	Yes available	TSDF Dhar	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
		Bandhogarh				
25	Umaria	(Whole District)	Not available	TSDF Dhar	Collection centers need to be established	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Umaria&pdf=/proc/Tech/UMARIYA_2023.pdf
		Barwah				
26	West Nimar	Bhagwanpura	Not available	TSDF, Dhar	No Gap identified	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Khargone.pdf
		Bhikangaon				

Jhirniya
Kasrawas
Khargone
Maheshwar
Seagon

*Source: MPCB District environment plan

Table A - 12 District-wise Status of Biomedical Waste Management in Narmada Basin, Madhya Pradesh

S.No	Districts	Block (ULB)	Biomedical Waste Management			References
			Treatment and disposal at	District level monitoring Committee	Gaps and Challenges	
1	Alirajpur	Alirajpur Bhavara Jobat	CBWTF Indore	Yes formed	Inventory Updation required. Awareness programs should be runned time to time	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Alirajpur.pdf
2	Anuppur (Whole District)	District	CBWTF Umari	Yes Formed	Barcoding on biomedical waste generated is to be implemented	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf

3	Balaghat	Baihar	CBMWTF Seoni	Yes Formed	Barcoding on biomedical waste generated is to be implemented	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
4	Barwani	Barwani Niwali Pansemal Rajpur Sendhwa Thikiri Amla Betul	CBWTFs Facility established in Indore Dist. M/s. Hoswin Incinerator Pvt. Ltd	Yes Formed	Regularly awareness programs is to be runned by the authority.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Badwani_2023.pdf
5	(Whole District)	Bhaisdehi Multai Shahpur	CBWTF operating in Sehore	Yes formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
6	Burhanpur	Nepanagar	CBWTF M/s. Medisure Incinerators Dist-Khandwa	Yes Formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
7		Amarwara Junnardev	CBWTF Seoni District	Yes Formed	No Gaps identified	

	Chindwara (Whole District)	Tamiya				https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
8	Damoh	Tendukheda	CBWTF situated at Satna/Sagar District	Yes formed	No Gaps identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
9	Dewas	Bagli Dewas Kannod Khategaon Dhar Dharampuri	CBWTF Ujjain District. (M/s. Hostech Eco Management Pvt Ltd)	Yes Formed	No Gaps identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
10	Dhar (Whole District)	Gandhwani Kukshi Manawar Sardarpur	M/s Hoswin Incinerator Private Limited, Indore.	Yes Formed	No Gaps identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf
11	Dindori (Whole District)	Dindori Shahpura	CBWTF Facility at Dindori	Yes Formed	No Gaps identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DINDORI_2023.pdf
12	Khandwa	Harsud		Yes Formed	No Gap identified	

		Khandwa	M/s. Medisure Incinerators, Village Pandhana	Bilankheda, Dist. Khandwa		https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
13	Harda	Harda Khirkiya Timurni Babai Bankheri Hoshangabad	Not Found	Yes Formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf
14	Hosangabad	Itarsi Piparia Seonimalwa Sohagpur	Not Found	Yes	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Hoshangabad&pdf=/proc/Tech/Narmadapuram_2023.pdf
15	Indore	Mhow	M/s. Hoswin incinerator at Sanwer Road CBWTF Indore	Yes Formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Indore&pdf=/proc/Tech/INDORE_2023.pdf
16	Jabalpur	Jabalpur		Yes Formed	No Gap identified	

		Kundam				
		Majholi				
		Patan	CBWTF Jabalpur District			https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur&pdf=/proc/Tech/Jabalpur_2023.pdf
		Shahpura				
		Shihora				
17	Jhabua (Whole District)	Jhabua	M/s Hoswin Incinerator Private Limited, Indore.	Yes formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
		Ranapur				
18	Katni (Whole District)	Bahuriband				
		Dhimarkhera	CBWTF Jabalpur	Yes formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
		Rathi				
		Bichhiya				
19	Mandla	Mandla	CBWTF Seoni	Yes Available	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf
		Nainpur				
		Niwas				
		Gadarwara				
20	Narshingpur	Gotegaon	CBWTF Narshingpur	Yes Available	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGHPUR_2023.pdf
		Kareli				
		Narshingpur				

		Tendukheda				
		Bareli				
		Begamganj				
		Gairatganj				
21	Raisen	Goharganj	CBWTF, Raisen	Yes Formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf
		Raisen				
		Silvani				
		Udaipura				
22	Sagar	Deori	CBWTF Sagar	yes formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sagar&pdf=/proc/Tech/Sagar_2023.pdf
		Kesli				
		Ashta				
23	Sehore	Budhni	(M/s Environment Protection Corporation, Sehore	Yes Available	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHO RE_2023.pdf
		Ichawar				
		Nasrullaganj				
		Sehore				
24	Seoni	Ghansaur	CBWTF Seoni	yes formed	Bar code system is to be implemented with full efficiency	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
		Lakhnadon				

25	Umaria	Bandhogarh (Whole District) Barwah	CBWTF Umaria	yes formed	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Umaria&pdf=/proc/Tech/UMARIYA_2023.pdf
26	West Nimar	Bhagwanpura Bhikangaon Jhirniya Kasrawas Khargone Maheshwar Seagon	CBWTFs Facility established in Khandwa Dist. M/s. Medisure Incinerators, Village-Bilankheda,	Yes formed	No Gap identified	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Khargone.pdf

*Source: MPCB District environment plan

Table A - 13 District-wise Status of E-Waste Management in Narmada Basin, Madhya Pradesh

S.No.	Districts	Block (ULB)	E-Waste Management			References
			Collection Points	Linkage among Stakeholders to channelize	Gaps and challenges	

recycling and dismantling						
1	Alirajpur	Alirajpur Bhavara Jobat	Not available	Not linked	E Waste Collection Centres are required and linkage with recyclers and dismantlers are required.	https://www.mppcb.mp.gov.in/proc/Tech/h/DEP%20Alirajpur.pdf
2	Anuppur (Whole District)	District	Not available	Not linked	E Waste Collection Centres are required and linkage with recyclers and dismantlers are required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	Baihar	Not available	Linkage with recyclers at Indore District	Awareness regarding regulation of illegal recycling/dismantling is to be done	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
4	Barwani	Barwani Niwali Pansemal Rajpur Sendhwa Thikiri	Available	Not linked	ULBs are required to be linked with authorized recyclers/Dismantlers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Badwani_2023.pdf
5		Amla		Not linked		

		Betul	01 collection			
	Betul (Whole District)	Bhaisdehi Multai Shahpur	points at District head quarter Betul		ULBs are required to be linked with authorized recyclers/Dismantlers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
6	Burhanpur	Nepanagar	Available at district Headquarter	Not Linked	ULBs are required to be linked with authorized recyclers/Dismantlers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
7	Chindwara (Whole District)	Amarwara Junnardev Tamiya	02 collection points at District head quarter Chhindwara.	Linked	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
8	Damoh	Tendukheda	Not available	Not linked	ULBs are required to be linked with the authorized recyclers/ dismantlers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
9	Dewas	Bagli Dewas Kannod Khategaon	Yes available	20%v linked	ULBs are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
10		Dhar Dharampuri	3 collection centers are	Not Linked		

		Gandhwani	available in			
	Dhar (Whole District)	Kukshi Manawar Sardarpur	District		ULBs are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf
11	Dindori (Whole District)	Dindori Shahpura	Not available	Not linked	ULBs are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DIN_DORI_2023.pdf
12	Khandwa	Harsud Khandwa Pandhana	Yes Available	Not Linked	ULBs are required to be linked with authorized recyclers/Dismantlers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
13	Harda	Harda Khirkhya Timurni	Not available	Not linked	Collection Centers are required and "ULBs are required to be linked with authorized recyclers/Dismantlers."	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf
14	Hosangabad	Babai Bankheri Hoshangabad Itarsi Piparia	Not available	Not linked	E-Waste Collection Point/collection centers required to be Established and ULBs required to be linked with authorized Recyclers/ Dismantlers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Hoshangabad&pdf=/proc/Tech/Narmadapuram_2023.pdf

		Seonimalwa				
		Sohagpur				
15	Indore	Mhow	4 E waste collection points available	Only Indore Municipal Corporation has linkage with the Recyclers/ Dismantlers	ULBs are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Indore&pdf=/proc/Tech/INDORE_2023.pdf
		Jabalpur				
		Kundam				
16	Jabalpur	Majholi	Yes available	E-waste generation, already linkage with recyclers at indore district.	ULB are required to be linked with authorized recyclers/ Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur&pdf=/proc/Tech/Jabalpur_2023.pdf
		Patan				
		Shahpura				
		Shihora				
		Jhabua			E-Waste Collection Point/collection centers required to be Established and ULBs required to be linked with authorized Recyclers/ Dismantlers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
17	Jhabua (Whole District)	Ranapur	No e waste Collection center	Not linked		
		Bahuriband				
18		Dhimarkhera	Not Available	Not linked	ULB's is required to hand over the e-waste collected in their jurisdiction to the	

	Katni (Whole District)	Rathi			MPPCB authorized Ewaste recycler or dismantle by executing an MOU with the recycler/dismantler.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
19	Mandla	Bichhiya Mandla Nainpur Niwas Gadarwara Gotegaon	Not available	Linkage with authorized recyclers at Indore District	1 E waste collection point and recycling is required and ULBs are required to be linked with authorized dismantler and recyclers.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf
20	Narshingpur	Kareli Narshingpur Tendukheda Bareli Begamganj	Not available	Linkage with authorized recyclers at Indore District	ULB are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGHPUR_2023.pdf
21	Raisen	Gairatganj Goharganj Raisen Silvani	2 Collection points are available within the district	Linked with the authorized recyclers.	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf

		Udaipura				
22	Sagar	Deori	Not available	Not linked	ULB are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sagar&pdf=/proc/Tech/Sagar_2023.pdf
		Kesli				
		Ashta				
23	Sehore	Budhni	1 Collection center available	Linkage with authorized recycler/Dismantler	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHORE_2023.pdf
		Ichawar				
		Nasrullaganj				
		Sehore				
24	Seoni	Ghansaur	Not available	already linkage with recyclers at Indore district	Collection centers are required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
		Lakhnadon				
25	Umaria	Bandhogarh (Whole District)	Not available	Not linked	ULB are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Umaria&pdf=/proc/Tech/UMARIYA_2023.pdf
		Barwah				
26	West Nimar	Bhagwanpura	Yes available	No linked	ULBs are required to be linked with authorized recyclers/Dismantlers	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Khargone.pdf
		Bhikangaon				
		Jhirniya				
		Kasrawas				

Khargone

Maheshwar

Seagon

*Source: MPCB District environment plan

Table A - 14 District-wise Status of Plastic Waste Management in Narmada Basin, Madhya Pradesh

Plastic waste Management							
S.No.	Districts	Block (ULB)	Door to Door Collection	Plastic waste collection centers	Plastic waste Disposal Facility	Gaps and Challenges	References
1	Alirajpur	Alirajpur	Partial	1 plastic waste collection centre is established at the MRF.	Not available	100% Door to Door collection of PW, recycling and disposal facility is also required	https://www.mppcb.m.p.gov.in/proc/Tech/DEP%20Alirajpur.pdf
		Bhavara	Partial	Not Available	Not available	100% Door to Door collection of PW, recycling and disposal facility is also required	
		Jobat	Partial	Not Available	Not Available	100% Door to Door collection of PW, recycling	

						and disposal facility is also required	
2	Anuppur (Whole District)	District	Not Available	Not Available	Not Available	PW recycling facility is needed, educational awareness needed, and PW collection centers required.	https://www.mppcb.mpp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	Baihar	full	Not Available	Not Available	01 Plastic Waste Collection centers is required.	https://www.mppcb.mpp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT2023.pdf
		Barwani	100%	Collected door to door	Available	No gap identified	
		Niwali	100%	Not available	Not Available	Plastic waste collection and disposal centers are required.	
4	Barwani	Pansemal	100%	Not available	Available	PW collection needs to be opened	https://www.mppcb.mpp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Badwani_2023.pdf
		Rajpur	100%	Not available	Available	PW collection needs to be opened	
		Sendhwa	100%	Not available	Not Available	Plastic waste collection and disposal centers are required.	

		Thikiri	100%	Not available	Not Available	Plastic waste collection and disposal centers are required.	
		Amla					
5	Betul (Whole District)	Betul				Collection of plastic waste from Panchayat level to be established.	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
		Bhaisdehi	100%	Available only at district headquarter	Available		
		Multai					
		Shahpur					
6	Burhanpur	Nepanagar	100%	Not required	Not available	Access to plastic waste disposal facility shall be provided	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
		Amarwara				Transfer points and material recovery facility to be developed at identified locations. and Involvement of informal sector /NGO shall be taken up.	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
7	Chindwara (Whole District)	Junnardev	100%	Yes, only at district head quarter.	Yes available		
		Tamiya					

8	Damoh	Tendukheda	100%	Not Required	Not available	Plastic waste collection and disposal centers are required.	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
		Bagli	100%	Not Required	Not Available	PW collection and disposal centers are required.	
9	Dewas				Plastic Waste is being recycled to		
		Dewas	Not Available	Not Available	Use in Road Making and Co Processing in Cement Plants	Door-to-door collection and collection centers are required.	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
		Kannod	100%	Not Required	Not Available	PW collection and disposal centers are required.	
10	Dhar (Whole District)	Khategaon	100%	Not Required	Not Available	Plastic waste collection and disposal centers are required.	
		Dhar Dharampuri	100%	Yes available	Plastic waste sent to	ULB's and village panchayat must collect the	https://www.mppcb.m.p.gov.in/PdfView.aspx

		Gandhwani			cement plant for co processing	plastic waste generated by their jurisdiction	?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf
		Kukshi					
		Manawar					
		Sardarpur					
		Dindori					https://www.mppcb.mpp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DINDORI_2023.pdf
11	Dindori (Whole District)	Shahpura	Partial	Not available	District has access to PW recycling.	Door-to-door collection and collection centers are needed	
		Harsud	100%	Not required	Not Available	PW Collection center and disposal facility is required	https://www.mppcb.mpp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
12	Khandwa	Khandwa	100%	Available	Not Available	PW disposal facility is required	
		Pandhana	100%	Not required	Not Available	PW disposal facility is required	
		Harda	100%	Available	Yes available	No gap identified	https://www.mppcb.mpp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf
13	Harda	Khirkiya	100%	Available	Yes available	No gap identified	
		Timurni	100%	Available	Yes available	No gap identified	
14	Hosangabad	Babai	100%	Yes available	Recycling facility is available	No gap identified	https://www.mppcb.mpp.gov.in/PdfView.aspx?h=Hoshangabad&pdf

		Bankheri	100%	Available at MRF centers	Available in District Headquarters	No gap identified	=/proc/Tech/Narmadapuram_2023.pdf
		Hoshangabad	100%	Yes available	Recycling facility is available	No gap identified	
		Itarsi	90%	Collection at MRF centers	Available in District Headquarters	No gap identified	
		Piparia	100%	Yes available	District has adequate No. of Plastic Waste recycling /utilization or disposal units	No gap identified	
		Seonimalwa	100%	Collection at MRF centers	Available in District Headquarters	No gap identified	
		Sohagpur	100%	Yes available	Available	No gap identified	
15	Indore	Mhow	100%	Yes, Collected in MRFs	disposal done by the help of dust cleaner,	No gap identified	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Indore&pdf=/proc/

				shredding, bailing		Tech/INDORE_2023.pdf
16	Jabalpur	Jabalpur	100%	Available	Plastic waste is disposed to Waste to Energy plant	01 Plastic Waste Collection centers is required.
		Kundam	Not Available	Not Available	Not Available	Not Available
		Majholi	100%	Yes available	Plastic waste is disposed to Waste to Energy plant	01 Plastic Waste Collection centers is required.
		Patan	100%	Available	Plastic waste is disposed to Waste to Energy plant	01 Plastic Waste Collection centers is required.
		Shahpura	100%	Available	Plastic waste is disposed to Waste to Energy plant	01 Plastic Waste Collection centers is required.

[https://www.mppcb.m
p.gov.in/PdfView.aspx
?h=Jabalpur&pdf=/pro
c/Tech/Jabalpur_2023.
pdf](https://www.mppcb.m
p.gov.in/PdfView.aspx
?h=Jabalpur&pdf=/pro
c/Tech/Jabalpur_2023.
pdf)

		Shihora	100%	Available	Plastic waste is disposed to Waste to Energy plant	01 Plastic Waste Collection centers is required.	
17	Jhabua (Whole District)	Jhabua Ranapur	100%	Available	Yes, Plastic waste send to cement plant for coprocessing	No gap identified	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
18	Katni (Whole District)	Bahuriband Dhimarkhera Rathi	100%	Not available	Yes, District has access to PW recycling / utilization or disposal facilities	Collection centers are required	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
19	Mandla	Bichhiya	100%	Not available	Not Available	Transfer points and material recovery facility to be developed at identified locations. and Involvement of informal sector /NGO shall be taken up.	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf

		Mandla	100%	Not available	Not Available	01 Plastic Waste Collection centers is required.	
		Nainpur	100%	Not available	Not Available	PW processing facility is required	
		Niwas	100%	Not available	Not Available	01 Plastic Waste Collection centers is required.	
		Gadarwara	100%	Available	Not Available	PW processing unit is required	
		Gotegaon	100%	Not Available	Not Available	PW processing unit is required	
20	Narshingpur	Kareli	100%	Not Available	Not Available	PW processing unit is required	<a href="https://www.mppcb.m
p.gov.in/PdfView.aspx
?h=Narsinghpur&pdf=
/proc/Tech/NARSING
HPUR_2023.pdf">https://www.mppcb.m p.gov.in/PdfView.aspx ?h=Narsinghpur&pdf= /proc/Tech/NARSING HPUR_2023.pdf
		Narshingpur	100%	Not Available	Not available	PW processing unit is required	
		Tendukheda	100%	Not Available	Not Available	PW processing unit is required	
21	Raisen	Bareli	100%	Available	Not Available	No gap identified	<a href="https://www.mppcb.m
p.gov.in/PdfView.aspx">https://www.mppcb.m p.gov.in/PdfView.aspx
		Begamganj	100%	Available	Available	No gap identified	<a href="https://www.mppcb.m
p.gov.in/PdfView.aspx">https://www.mppcb.m p.gov.in/PdfView.aspx

					District has adequate	?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf
		Gairatganj	80%	Available	No. of Plastic Waste recycling / utilization or disposal units	Door to door collection efficiency require to be 100%
		Goharganj	Not Available	Not Available	Not Available	Not Available
		Raisen	90%	Available	District has adequate No. of Plastic Waste recycling / utilization or disposal units	Door to door collection efficiency require to be 100%
		Silvani	80%	Not available	Available	Plastic Waste Collection centers are required.
		Udaipura	100%	Not available	Available	Plastic Waste Collection centers is required.
22	Sagar	Deori	100%	Yes available	Yes available	No gap identified

		Kesli	Not Available	Not Available	Not Available	Not Available	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Sagar&pdf=/proc/Tech/Sagar_2023.pdf
		Ashta	95%	Not available	Not Available	Plastic Waste Collection centers and processing units is required.	
		Budhni	Not Available	Not Available	Not Available	Not Available	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHORE_2023.pdf
23	Sehore	Ichawar	Not Available	Not Available	Not Available	Not Available	
		Nasrullaganj	100%	Local body setup own MRF facility	Not Available	Plastic Waste processing units are required.	
		Sehore	60%	1 available	Plastic waste is segregated and sell to recyclers	Plastic Waste Collection centers and processing units is required.	
		Ghansaur	85%	Not available	Not Available	Plastic Waste door to door collection, Collection centers and processing units are required	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
24	Seoni	Lakhnadon	100%	Not available	Not Available	Plastic Waste Collection centers and	

						processing units are required	
25	Umaria	Bandhogarh (Whole District)	Partial	Available at local level	District has access to PW recycling through the Material recovery facility established at dump sites of every ULB in district	No gap identified	https://www.mppcb.m.p.gov.in/PdfView.aspx?h=Umaria&pdf=/proc/Tech/UMARIYA_2023.pdf
		Barwah	100%	Not required	Not Available	01 Plastic Waste Collection centers is required.	
26	West Nimar	Bhagwanpura	Not Available	Not Available	Not Available	Not Available	https://www.mppcb.m.p.gov.in/proc/Tech/DEP%20Khargone.pdf
		Bhikangaon	100%	Not required	Not Available	Plastic Waste processing units are required	
		Jhirniya	Not Available	Not Available	Not Available	Not Available	

Kasrawas	100%	Not required	Not Available	Plastic Waste processing units are required
Khargone	100%	Yes available	Not Available	Plastic Waste processing units are required
Maheshwar	100%	Not required	Not Available	Plastic Waste processing units are required
Seagon	Not Available	Not Available	Not Available	Not Available

*Source: MPCB District environment plan

Table A - 15 District-wise Status of C&D Waste Management in Narmada Basin, Madhya Pradesh

C&D Waste Management							
S.No.	Districts	Block (ULB)	Collection and deposition facility	Recycling Facility	Use of recycled waste	Gaps and Challenges	References
1	Alirajpur	Alirajpur	Available	Not available	C&D waste is being used to dump in the low-lying areas.	C&D waste processing unit required	
		Bhavara	C&D waste is collected separately and is used to dump in low lying areas,	Not available	C&D waste is being used to dump in the low-lying areas.	Not available	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Alirajpur.pdf
		Jobat	Separate collection is done deposition points are not required	Not Available	C&D waste is being used to dump in the low-lying areas.	C&D waste recycling facility is required.	

2	Anuppur (Whole District)	District	Not Available	Not Available	Currently, C&D waste is mostly utilized by the local bodies for facilitating different repair and construction works	C&D waste collection and recycling facilities are required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	Baihar	Not required	Not required	Not required	Quantity of C&D waste generator is very less	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
4	Barwani	Barwani	Not Available	Not Available	No Information	Separate collection of C&D waste, establishment of C&D waste collection & recycling facility is required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Barwani_2023.pdf
		Niwali	Not Available	Not Available	No Information	C&D waste recycling	

						facility to be established.	
		Pansemal	Not Available	Not Available	No Information	C&D waste process plant to be required.	
		Rajpur	Not Available	Not Available	No Information	C&D waste process plant to be required.	
		Sendhwa	Not Available	Not Available	No Information	C&D waste process plant to be required.	
		Thikiri	Not Available	Not Available	No Information	C&D waste process plant to be required.	
5	Betul (Whole District)	Amla Betul Bhaisdehi	Yes, available in all ULBs	Not Available	Not available	Local C&D waste management plans can be	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&p

		Multai				integrated to develop common collection and recycling facilities	df=/proc/Tech/Betu_1_23.pdf
		Shahpur					
6	Burhanpur	Nepanagar	Not available	Not available	C&D waste is being used to dump in the low-lying areas	C&D waste processing plant to be required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
7	Chindwara (Whole District)	Amarwara Junnardev Tamiya	Available at all ULBs	YES M/s dadaji traders sonpur teh. + dist. Chhindwara third party contract.	Not available	C&D waste processing plants are also to be established in village area	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
8	Damoh	Tendukheda	Available	Not available	Dumped to low lying areas	C&D waste processing plant required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
9	Dewas	Bagli	Not Available	Not required	Dumped to low lying areas	C&D waste processing	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/Dewas_2023.pdf

		Dewas	1 collection facility is available	Not available	Proposed to make paver blocks	plant required. required new 02collection point in city and C&D Waste recycling facility required01 No.	w.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
		Kannod	Not Available	Not required	Dumped to low lying areas	C&D waste processing plant required.	
		Khategaon	Not Available	Not required	Dumped to low lying areas	C&D waste processing plant required.	
10	Dhar (Whole District)	Dhar Dharamपुरी Gandhwani Kukshi	Available in each UIBs	Not Available	Filling Low lying areas	C&D waste processing plant required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf

		Manawar					
		Sardarpur					
		Dindori					https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DINDORI_2023.pdf
11	Dindori (Whole District)	Shahpura	Not available	Not Available	Utilized by local bodies for facilitating different repair and construction works.	C&D waste processing plant required.	
						C&D waste recycling facilities be established.	
		Harsud	Not required	Not required	Dumped in low lying areas	and C&D waste processing Plant required.	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
12	Khandwa					C&D waste recycling facilities be established.	
		Khandwa	Not Available	Not available	Dumped in low lying areas	and C&D waste processing	

						Plant required.
						C&D waste recycling facilities be established.
		Pandhana	Not Required	Not required	Dumped in low lying areas	and C&D waste processing Plant required.
		Harda	Yes Available	Yes available	Collected and stored C&D waste is reused for filling of low-lying areas and road filling.	No Gap identified
13	Harda	Khirkiya	Yes Available	Yes available	Collected and stored C&D waste is reused for filling of low-lying areas and road filling.	No Gap identified
		Timurni	Yes Available	Yes available	Collected and stored C&D waste is reused for filling of low-lying areas and road filling.	No Gap identified

https://www.mppcb.mp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf

14	Hosangabad	Babai	Provided at dump site	Not Available	Used for Construction activities	C&D waste processing facility is required
		Bankheri	Not available	Not Available	Used for construction activities in PM Awas Yojna	C&D waste recycling facility available within Municipal Council
		Hoshangabad	Separate collection point of C&D Waste Provided at dump site	Not Available	Collected and stored C&D waste is reused for filling of land restoration, road & PMAY construction activities.	C&D waste recycling facility available within Municipal Council
		Itarsi	Provided at dump site	Not Available	Used for Construction activities	C&D waste processing facility is required
		Piparia	Provided at dump site	Not Available	Used for Construction activities	C&D waste processing

https://www.mppcb.mp.gov.in/PdfView.aspx?h=Hoshangabad&pdf=/proc/Tech/Narmadapuram_2023.pdf

						facility is required	
		Seonimalwa	Separate collection point of C&D Waste Provided at dump site	Not Available	Used for Construction activities	C&D waste processing facility is required	
		Sohagpur	Not available	Not Available	Collected and stored C&D waste is reused for filling of land restoration, road & PMAY construction activities.	C&D waste collection and processing facility is required	
15	Indore	Mhow	Not required	Not required	Not required	C&D waste collection and processing facility is required	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Indore&pdf=/proc/Tech/INDORE_2023.pdf
16	Jabalpur	Jabalpur	Available	Yes available	Used in nonstructural concrete	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur
		Kundam	Not available	Not available	Not available	Not available	

		Majholi	Not required Collection is very less	Not required	Not required	No Gap identified	&pdf=/proc/Tech/J abalpur_2023.pdf
		Patan	Not required small town	Not required	Not required	No Gap identified	
		Shahpura	Not required	Not required	Not required	No Gap identified	
		Shihora	Not required	Not required	Not required	No Gap identified	
17	Jhabua (Whole District)	Jhabua Ranapur	Not available	Not available	Filling into low lying areas	C&D waste processing unit is required	https://www.mppcb .mp.gov.in/PdfVie w.aspx?h=Jhabua& pdf=/proc/Tech/JH ABUA_2023.pdf
18	Katni (Whole District)	Bahuriband Dhimarkhera Rathi	Not available	Not available	ULB's are currently collecting the C& D waste is Filling into Low lying areas in the district.	C&D waste processing unit is required	https://www.mppcb .mp.gov.in/PdfVie w.aspx?h=Katni&p df=/proc/Tech/KAT NI_2023.pdf
19	Mandla	Bichhiya Mandla	Not required	Not required	Not required	No Gap identified	https://www.mppcb .mp.gov.in/PdfVie w.aspx?h=Mandla

		Nainpur	Not required	Not required	Not required	No Gap identified	&pdf=/proc/Tech/MANDLA.pdf
		Niwas	Not required	Not required	Not required	No Gap identified	
		Gadarwara	Not required	Not required	Not required	No Gap identified	
		Gotegaon	Not required	Not required	Not required	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGHPU_R_2023.pdf
20	Narshingpur	Kareli	Not required	Not required	Not required	No Gap identified	
		Narshingpur	Not required	Not required	Not required	No Gap identified	
		Tendukheda	Not required	Not required	Not required	No Gap identified	
		Bareli	Yes available	Not required	Collected and stored C&D waste is reused for filling of land and PMAY construction activities.	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf
21	Raisen	Begamganj	Not available	Not required	Collected and stored C&D waste is reused for filling of	No Gap identified	

				land and PMAY construction activities.		
		Gairatganj	Yes available	Not required	Collected and stored C&D waste is reused for filling of land and PMAY construction activities.	No Gap identified
		Goharganj	Not available	Not available	Not available	Not available
		Raisen	Available	Not required	Collected and stored C&D waste is reused for filling of land and PMAY construction activities.	No Gap identified
		Silvani	Available	Not required	Collected and stored C&D waste is reused for filling of land and PMAY construction activities.	No Gap identified
		Udaipura	Provided at Dump sites	Not required	Collected and stored C&D waste is reused for filling of land and PMAY construction activities.	No Gap identified
22	Sagar	Deori	available	Yes available	available	No Gap identified
		Kesli	Not available	Not available	Not available	Not available

<https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sagar&p>

		Ashta	Available	Collected and disposed to low lying area	Not required	No Gap identified	
		Budhni	Not available	Not available	Not available	Not available	
		Ichawar	Not available	Not available	Not available	Not available	
23	Sehore	Nasrullaganj	Separate collection point for C&D Waste- Provided at dump site	Not required	Collected and stored C&D waste is reused for filling of land and PMAY construction activities.	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHORE_2023.pdf
		Sehore	Available	No facility exists	Yes	C&D recycling facility needed	
		Ghansaur	Not required	Not required	Not required	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
24	Seoni	Lakhnadon	Not required	Not required	Not required	No Gap identified	

25	Umaria	Bandhogarh (Whole District)	Not available	Not available	C&D waste is mostly utilized by the local bodies for facilitating different repair and construction works	No Gap identified	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Umaria&pdf=/proc/Tech/UMARIYA_2023.pdf
		Barwah	Not required	Not required	Not required	No Gap identified	
		Bhagwanpura	Not available	Not available	Not available	Not available	
		Bhikangaon	Not required	Not required	Not required	No Gap identified	
		Jhirniya	Not available	Not available	Not available	Not available	
26	West Nimar	Kasrawas	Not required	Not required	Not required	No Gap identified	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Khargone.pdf
		Khargone	Available	Yes available	C&D waste is mostly utilized by the local bodies for facilitating different repair and construction works	No Gap identified	
		Maheshwar	Not required	Not required	Not required	No Gap identified	
		Seagon	Not available	Not available	Not available	Not available	

*Source: MPCB District environment plan

Table A - 16 Operational STPs details in Narmada Basin, Madhya Pradesh

<u>S.No.</u>	District	ULBs	Existing STP Capacity (MLD)	Average Capacity Being Utilized (MLD)	Reuse (MLD)	Operational Status of STP
1	Anuppur	Amarkantak	1.2	0.365	0.08	World Bank, Operational July 2024
2	Burhanpur	Burhanpur (M Corp.)	3	2.3	0	Operational
3	Burhanpur	Burhanpur (M Corp.)-Boharda	21	14	5.94	AMRUT, Operational Mar 2021
4	Dewas	Dewas (M Corp.) - Balawli	12	2.32	0.7	Operational
5	Dewas	Dewas (M Corp.) - Nanda Nagar, Mendki	14	5.44	1.24	UDISSMT, Operational Dec 2020
6	Dewas	Dewas (Saroday Nagar)	12	12	0	UDISSMT, Operational Oct 2021

						World Bank, Operational
7	Dhar	Dharampuri	12	1.01	0.2	April 2022
8	Dindori	Dindori	3.85	1.155	0.35	Vishesh Nidhi
		Jabalpur (M Corp.) -				Unoperational, diverted to
9	Jabalpur	Kathonda	50	0	0	32 MLD STP Kathonda
		Jabalpur (M Corp.) -				
10	Jabalpur	Gwarighat	0.55	0.24	0.1	Operational
		Jabalpur (M Corp.) -				
11	Jabalpur	Gulatual	0.5	0.12	0.02	Operational
		Jabalpur (Zone				AMRUT, Operational Nov
12	Jabalpur	II Kathonda)	25.51	25.51	15	2022
		Jabalpur Zone				
		three and four at				AMRUT, Operational Sep
13	Jabalpur	Lalpur	48	34.22	27	2024

						AMRUT, Operational Mar
14	Jabalpur	Jabalpur Zone 5	29	19.88	20	2024
		Bhedaghat				World Bank, Operational
15	Jabalpur	(Panchwat)	0.43	0.38	0.37	Dec 2023
		Bhedaghat				World Bank, Operational
16	Jabalpur	(Lameta)	0.12	0.084	0.08	Dec 2023
		Bhedaghat				World Bank, Operational
17	Jabalpur	(Gopalpur)	0.22	0.147	0.08	Dec 2023
		Bhedaghat				World Bank, Operational
18	Jabalpur	(Bhopurda)	0.06	0.043	0.014	Dec 2023
		Bhedaghat				World Bank, Operational
19	Jabalpur	(Lametaghat)	0.06	0.043	0.014	Dec 2023
		Omkareshwar				
		(Near				Vishesh Nidhi,
20	Khandwa	Viradharsham)	11.5	0.385	0.1	Operational July 2024
		Omkareshwar				
		(Trenching				Vishesh Nidhi,
21	Khandwa	Ground, Village)	6.45	3.1	0.9	Operational Oct 2024

		Khargone (M)-				AMRUT, Operational
22	Khargone	Ward No.14	17.6	4	3.52	May 2021
23	Mandla	Mandla	10	3.41	0.12	Operational
		Narsingpur (Dedwara				
24	Narsinghpur	Region, Near WTP)	6.25	1.875	0.56	KFW
		Narsingpur				
25	Narsinghpur	(Naktua Region, Near Jail)	2.75	0.825	0.25	KFW
		Budhni (Near Dindayal				
26	Sehore	Convent School)	2.2	1.38	1	Vishesh Nidhi, Operational July 2022
		Budhni (Vill-				
27	Sehore	Mana)	1.08	0.69	0.45	Vishesh Nidhi, Operational July 2022
		Sehore (M) -				
28	Sehore	Junia Wada	12	0.67	0.17	AMRUT, Operational 2018

Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.

Table A - 17 Under Construction STPs details in Narmada Basin, Madhya Pradesh

S. No.	Name of Location	Capacity of Physical	Status	Completion	Scheme			
District	ULBs	Plant (MLD)	Progress (%)	Timeline				
1	Alirajpur	Jobat (NP)	Jobat (NP)	0.5	0	Work in progress	Mar-26	SBM 2.0
2	Alirajpur	Alirajpur (M)	Alirajpur (M)	3	0	Work in progress	Dec-25	SBM 2.0
3	Balaghat	Baihar (NP)	Baihar (NP)	1.5	0	Work in progress	Feb-26	SBM 2.0
4	Barwani	Badwani	Bheelkhera	9	92	Work in progress	Jun-25	
			Village					
5	Barwani	Sendhwa	Semalia	8.5	72	Work in progress	Sep-25	
6	barwani	Thikari	Thikari	2	0	Work in progress	Feb-26	
7	Barwani	Palsud (NP)	Palsud (NP)	0.8	0	Work in progress	Jan-26	SBM 2.0
8	Betul	Shahpur (NP)	Shahpur (NP)	1	0	Work in progress	Mar-26	SBM 2.0
9	Betul	Bhanishedi	Bhanishedi	0.6	0	Work in progress	Mar-26	SBM 2.0
10	Burhanpur	Nepanagar	Nepanagar	1	0	Work in progress	Feb-26	SBM 2.0
11	Burhanpur	Nepanagar	Nepanagar	0.7	0	Work in progress	Feb-26	SBM 2.0

12	Chhindwara	Jamai (NP)	Jamai (NP)	3	0	Work in progress	Feb-26	SBM 2.0
13	Dhar	Manavar	Manavar	4.5	0	Work in progress	Dec-25	SBM 2.0
14	Dhar	Dhamnod	Dhamnod	6.68	67	Work in progress	Sep-26	ADB
15	Hoshangabad	Hoshangabad	Kishanpur, Rashulia	21	33	Work in progress	Dec-25	
16	Jabalpur	Panagar (M)	Panagar (M)	3.8	0	Work in progress	Mar-26	SBM 2.0
17	Jabalpur	Jabalpur (Civil Lines)	Civil Line	5	2	Work in progress	Jun-25	
18	Jhabua	Jhabua	Jhabua	5	0	Work in progress	Dec-25	SBM 2.0
19	Jhabua	Jhabua	Jhabua	3	0	Work in progress	Aug-26	AMRUT 2.0
20	khargone	Maheshwar	Maheshwar	4.91	48	Contract Terminated on 06.09.2024; rebid for balance work	Dec-25	
21	Khargone	Mandleshwar	Mandleshwar	3	86	Work in progress	Jun-25	Vishesh Nidhi

22	Khargone	Sanavad	Sanavad	5.58	42	Work in progress	Jun-25	ADB
23	Khargone	Kasrawad (NP)	Kasrawad (NP)	2.5	0	Work in progress	Feb-26	SBM 2.0
24	Mandla	Mandla	Ward No. 23	7.75	80	Work in progress	May-25	
25	Mandla	Mandla	Village Gaunjhi	1.75	80	Work in progress	Jun-25	
26	Narsinghpur	Gotegaon (NP)	Gotegaon (NP)	1	0	Work in progress	Mar-26	SBM 2.0
27	Raisen	Obedullaganj (NP)	Obedullaganj (NP)	2.7	0	Work in progress	Feb-26	SBM 2.0
28	Seoni	Lakhandon (NP)	Lakhandon (NP)	1	0	Work in progress	Mar-26	SBM 2.0
29	Sehore	Nasrullaganj	Ward No.6	4.2	99	Contract Terminated on 06.09.2024; rebid for balance work	Jun-25	World Bank
30	Sehore	Ichhawar (NP)	Ichhawar (NP)	1	0	Work in progress	Mar-26	SBM 2.0

*Source: MPCB District environment plan

Table A - 18 District-wise Status of Domestic Sewage Status in Narmada Basin, Madhya Pradesh

Domestic Sewage Status										
S.No.	Districts	No of bore wells	Nalas/ Drain s/Cree ks meeti ng Rivers	Quantity of domestic sewage generate d (MLD)	Perce n tage of untreat ed sewage (%)	Quant ity of sewage e flowin g into lakes	No of STPs avail able	No of STPs requi red	No of ULBs have partial undergr ound sewerage network	References
1	Alirajpur	NA	225	8.4	100	NA	0	2	1	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Alirajpur.pdf
2	Anuppur (Whole District)	2096	3	20	100	NIL	0	10	1	https://www.mppcb.mp.gov.in/Pdf/View.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	2991	6	19	100	0.1 KLD	0	8	Nil	https://www.mppcb.mp.gov.in/Pdf/View.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
4	Barwani	16360	23	12.312	100	NA	0	2	1	https://www.mppcb.mp.gov.in/Pdf/View.aspx?h=Barwani&pdf=/proc/Tech/Badwani_2023.pdf

5	Betul (Whole District)	NA	10	80	100	NIL	0	7	2	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
6	Burhanpur	NA	55	27.76	74.78	0	1	None	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
7	Chindwara (Whole District)	NA	10	100	50%	0	0	6	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
8	Damoh	NA	17	136	-	Nil	0	6	6	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
9	Dewas	NA	56	29	83	Nil	3	1	13	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
10	Dhar	NA	407	47.4	98.86	NIL	1	10	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf
11	Dindori	NA	6	3.49	-	0	0	2	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DINDORI_2023.pdf
12	Khandwa	7678	NIL	23.64	97.25	NIL	NIL	1	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%

										20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
13	Harda	1056	Nil	11.8	100	NIL	0	4	0	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf
14	Narmadapuram	5800	4	35	100	NIL	Nil	8	8	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Hoshangabad&pdf=/proc/Tech/Narmadapuram_2023.pdf
15	Indore	NA	56	475	0	Nil	11	0	None	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Indore&pdf=/proc/Tech/INDORE_2023.pdf
16	Jabalpur	8832	9	230	60	NIL	4	18	2	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur&pdf=/proc/Tech/Jabalpur_2023.pdf
17	Jhabua	NA	275	1.8	100	NIL	0	2		https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
18	Katni	1235	3	20	NIL	Nil	NIL	4	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
19	Mandla	3180	16	10.25	95	NIL	1	6	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf

20	Narshingpur	4738	10	30	100	1 MLD	NIL	10	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGHPUR_2023.pdf
----	-------------	------	----	----	-----	-------	-----	----	-----	---

*Source: MPCB District environment plan

Table A - 19 District-wise Status of Industrial Wastewater Status in Narmada Basin, Madhya Pradesh

Industrial Wastewater Status							References
S.No.	Districts	Quantity of industrial wastewater (MLD)	No of Industries discharging wastewater	No of Industries meeting Standards	Common Effluent Treatment Facilities		
1	Alirajpur	0	0	All	NIL		https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Alirajpur.pdf
2	Anuppur (Whole District)	3.9	346	346	NIL		https://www.mppcb.mp.gov.in/PdfView.aspx?h=Anuppur&pdf=/proc/Tech/DEP%20Anuppur.pdf
3	Balaghat	16	30	30	NIL		https://www.mppcb.mp.gov.in/PdfView.aspx?h=Balaghat&pdf=/proc/Tech/BALAGHAT_2023.pdf
4	Barwani	1	3	3	NIL		https://www.mppcb.mp.gov.in/PdfView.aspx?h=Barwani&pdf=/proc/Tech/Badwani_2023.pdf

5	Betul (Whole District)	51.99	0	NA	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Betul&pdf=/proc/Tech/Betul_23.pdf
6	Burhanpur	1.07	75	75	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Burhanpur&pdf=/proc/Tech/BURHANPUR_2023.pdf
7	Chindwara (Whole District)	46.433	NIL	100	Nil	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Chhindwara&pdf=/proc/Tech/CHHINDWARA_2023.pdf
8	Damoh	0	0	11	None	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Damoh&pdf=/proc/Tech/Damoh_2023.pdf
9	Dewas	4.696	0	437	NA	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dewas&pdf=/proc/Tech/DEWAS_2023.pdf
10	Dhar	18.6	NIL	NIL	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dhar&pdf=/proc/Tech/DHAR_2023.pdf
11	Dindori	0	0	All	Nil	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Dindori&pdf=/proc/Tech/DINDORI_2023.pdf
12	Khandwa	39.601	6	6	NA	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Khandwa%20(East%20Nimar)&pdf=/proc/Tech/Khandwa_2023.pdf
13	Harda	110KLD	0	11	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Harda&pdf=/proc/Tech/harda_2023.pdf
14	Narmadapuram	25	Nil	NA	Nil	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Hoshangabad&pdf=/proc/Tech/Narmadapuram_2023.pdf
15	Indore	7.3	385	385	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Indore&pdf=/proc/Tech/INDORE_2023.pdf

16	Jabalpur	8	30	30	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jabalpur&pdf=/proc/Tech/Jabalpur_2023.pdf
17	Jhabua	720 KLD	0	0	1	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Jhabua&pdf=/proc/Tech/JHABUA_2023.pdf
18	Katni	140 KLD	NIL	40	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Katni&pdf=/proc/Tech/KATNI_2023.pdf
19	Mandla	0.9	NIL	10	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Mandla&pdf=/proc/Tech/MANDLA.pdf
20	Narshingpur	1006	15	15	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Narsinghpur&pdf=/proc/Tech/NARSINGHPUR_2023.pdf
21	Raisen	13.082	Nil	381	Nil	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Raisen&pdf=/proc/Tech/Raisen_2023.pdf
22	Sagar	15360 KLD	7	7	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sagar&pdf=/proc/Tech/Sagar_2023.pdf
23	Sehore	19.2	24	24	Nil	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Sehore&pdf=/proc/Tech/SEHORE_2023.pdf
24	Seoni	10.5	7	7	Nil	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Seoni&pdf=/proc/Tech/Seoni_2023.pdf
25	Umaria	489 KLD	0	220	NIL	https://www.mppcb.mp.gov.in/PdfView.aspx?h=Umari&pdf=/proc/Tech/UMARIYA_2023.pdf
26	West Nimar	22.28	17	17	Nil	https://www.mppcb.mp.gov.in/proc/Tech/DEP%20Khangone.pdf

*Source: MPCB District environment plan



© cNarmada, cGanga and NRCD, 2025

